Configuration Summary

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DHW ANALYSIS - CONFIGURATION SUMMARY
Project: oisst_regression
Generated: 2026-10-18 14:07:40
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-- Data Preparation --

  input_dir.............................. /g/data/ia39/aus-ref-clim-data-nci/oisst/data/yearly/
  file_pattern........................... oisst-avhrr-v02r01_{year}.nc
  year_range............................. (1981, 2024)
  variable............................... sst
  detrend_method......................... global_linear
  standardise_method..................... global
  fill_method............................ diffusion
  fill_diffusion_iterations.............. 500
  fill_fallback_value.................... 0
  downscale.............................. True
  target_lat_size........................ 128
  target_lon_size........................ 256
  temporal_average....................... none
  subset_region.......................... False
  lat_bounds............................. None
  lon_bounds............................. None
  diagnostic_year........................ 2020
  diagnostic_pdf......................... diagnostics/data_preparation_diagnostics_128x256.pdf
  output_dir............................. data/processed_128x256/
  save_intermediates..................... True
  downscaled_file........................ sst_downscaled.nc
  temporally_averaged_file............... sst_temporally_averaged.nc
  detrended_file......................... sst_detrended.nc
  standardised_file...................... sst_standardised.nc
  filled_file............................ sst_filled.nc
  final_file............................. sst_final.nc
  subset_file............................ sst_subset.nc
  trend_params_file...................... trend_params.nc
  standardise_params_file................ std_params.nc
  resume_after_stage..................... 0
  chunks................................. {'time': 366, 'lat': 720, 'lon': 720}
  dask_scheduler......................... threads
  dask_n_workers......................... 8
  dask_threads_per_worker................ 2
  dask_memory_limit...................... 12GB
  dask_num_threads....................... 16

-- Model Training --

  model_name............................. sfno
  input_history.......................... 10
  output_channels........................ 1
  embed_dim.............................. 80
  num_layers............................. 5
  num_modes_lat.......................... 32
  num_modes_lon.......................... 64
  mlp_ratio.............................. 2.5
  activation............................. gelu
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norm_type.............................. instance_norm
dropout_rate........................... 0.05
use_residual........................... True
spectral_bias.......................... True
add_grid_features...................... True
add_seasonal_encoding.................. False
forecast_horizon....................... 90
initial_rollout_steps.................. 1
max_rollout_steps...................... 8
rollout_epoch_start.................... 10
rollout_epoch_end...................... 60
rollout_loss_decay..................... 0.92
train_years............................ (1982, 2015)
val_years.............................. (2016, 2019)
test_years............................. (2020, 2024)
learning_rate.......................... 0.0002
min_learning_rate...................... 1e-06
batch_size............................. 16
num_epochs............................. 120
weight_decay........................... 1e-05
optimizer.............................. adamw
scheduler.............................. cosine
scheduler_patience..................... 10
scheduler_factor....................... 0.5
warmup_epochs.......................... 5
gradient_clip_norm..................... 1
loss_function.......................... huber
early_stopping_patience................ 20
early_stopping_metric.................. val_loss
input_data_path........................ data/processed_128x256/sst_final.nc
checkpoint_dir......................... checkpoints_128x256/
best_model_path........................ checkpoints_128x256/best_model.pt
training_log_path...................... results_128x256/training_log.csv
random_seed............................ 42
num_workers............................ 4
distributed............................ True
dist_backend........................... nccl
find_unused_parameters................. False
sync_batchnorm......................... False
master_port............................ 29500

-- Model Inference --

model_path............................. checkpoints_128x256/best_model.pt
input_data_path........................ data/processed_128x256/sst_final.nc
run_validation......................... True
run_test............................... True
forecast_lead_steps.................... 90
chain_stride_steps..................... 90
output_dir............................. results_128x256/
val_predictions_file................... predictions_val.nc
test_predictions_file.................. predictions_test.nc

-- Concat Chains --
```

```
  predictions_dir..... /scratch/um09/xl5758/DHW-project/results_128x256
  predictions_file_template..... predictions_{split}.nc
  std_params_path..... data/processed_128x256/std_params.nc
  trend_params_path..... data/processed_128x256/trend_params.nc
  output_dir..... /scratch/um09/xl5758/DHW-project/results_128x256
  actual_file_template..... sst_actual_{split}_continuous.nc
  predicted_file_template..... sst_predicted_{split}_chains.nc
  splits..... ['test', 'val']
  lon_correction_threshold..... 0.6

-- Calc DHW --

  mmm_path..... /home/272/xl5758/ct5km_climatology_v3.1.nc
  results_dir..... /scratch/um09/xl5758/DHW-project/results_128x256
  predicted_file_template..... sst_predicted_{split}_chains.nc
  actual_file_template..... sst_actual_{split}_continuous.nc
  dhw_output_dir..... /scratch/um09/xl5758/DHW-project/dhw_output_128x256
  output_file_template..... dhw_products_{split}.nc
  run_test..... True
  run_validation..... True
  window_days..... 84
  hotspot_threshold..... 1

-- DHW Plots --

  dhw_dir..... /scratch/um09/xl5758/DHW-project/dhw_output_128x256
  output_pdf..... dhw_analysis_combined.pdf
  output_csv..... dhw_metrics_combined.csv
  spatial_lead_steps..... [1, 7, 14, 30, 60, 90]
  n_sample_chains..... 6
  plot_init_dates..... ['2016-05-20', '2016-10-15', '2017-02-10', '2017-07-09', '20
  dhw_vmax..... 20
  dhw_diff_vmax..... 8
  tropic_lat..... 30
  dhw_cmap..... ['#CCe6F0', '#441C94', '#6F47BF', '#815CCC', '#A383E6', '#ED
  error_cmap..... ['#053061', '#2166AC', '#92C5DE', '#F7F7F7', '#F4A582', '#D6
  rmse_cmap..... YlOrRd
  bias_cmap..... RdBu_r
  bleaching_thresholds..... [(0, 4, 'No Stress', '#27AE60'), (4, 8, 'Warning', '#F39C12'
  gbr_lat_bounds..... (-25.0, -10.0)
  gbr_lon_bounds..... (142.0, 157.0)
  reef_boxes..... [{'name': 'Great Barrier Reef', 'lat': (-25.0, -10.0), 'lon'
```

DHW Analysis - Summary

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DEGREE HEATING WEEKS (DHW) - COMBINED ANALYSIS
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METHODOLOGY  (NOAA Coral Reef Watch v3.1)
  SSTA      = SST - DC              (DC = daily climatology)
  HotSpot   = max(SST - MMM, 0)     (MMM = Maximum Monthly Mean)
  DHW_i     = (1/7) x sum(HotSpot_j) for j=i-83..i, HotSpot_j >= 1.0C

SLIDING WINDOW CONSTRUCTION  (always 84 days ending on valid date)
  Lead L:  (84-L) actual + L predicted SST days
  Lead 1:  83 actual +  1 predicted
  Lead 30: 54 actual + 30 predicted
  Lead 84:  0 actual + 84 predicted
  Lead 90: previous 84 predicted days (days 7-90)

VERIFICATION
  Actual DHW  : 84 observed days ending on the forecast valid date
  Persistence : observed DHW on the day before lead 1, held constant
  Sample      : chains defined at every lead step (same chains for
                model and persistence at all leads)

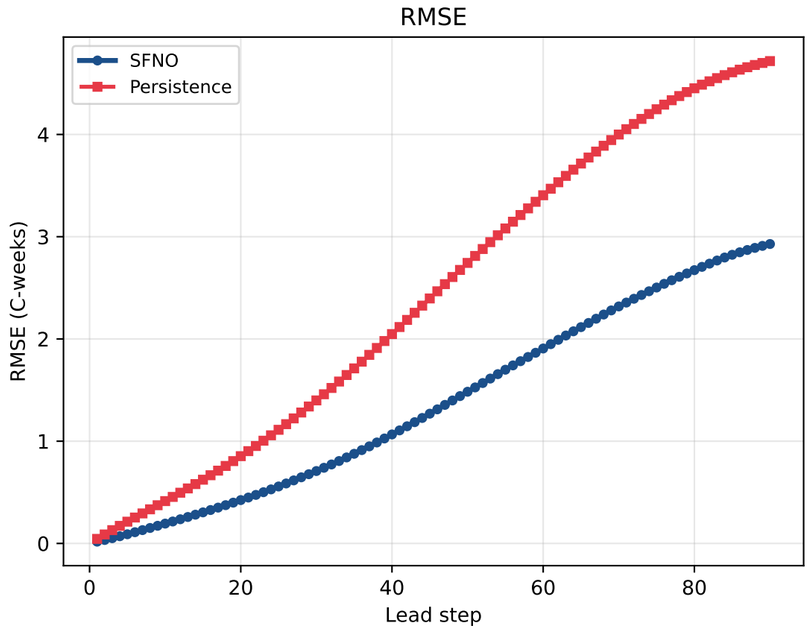
BLEACHING ALERT THRESHOLDS
  DHW  0-4  : No Stress
  DHW  4-8  : Warning
  DHW  8-12 : Alert Level 1
  DHW 12-16 : Alert Level 2
  DHW 16-20 : Alert Level 3
  DHW 20->= : Alert Level 4-5

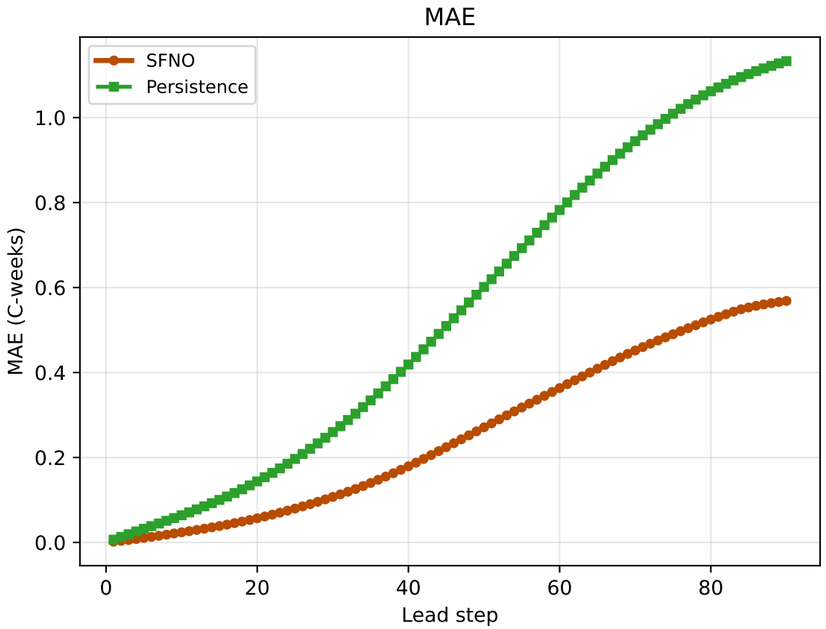
DATASETS
  Test          19 chains x 90 leads x 128 lat x 256 lon
  Val           16 chains x 90 leads x 128 lat x 256 lon
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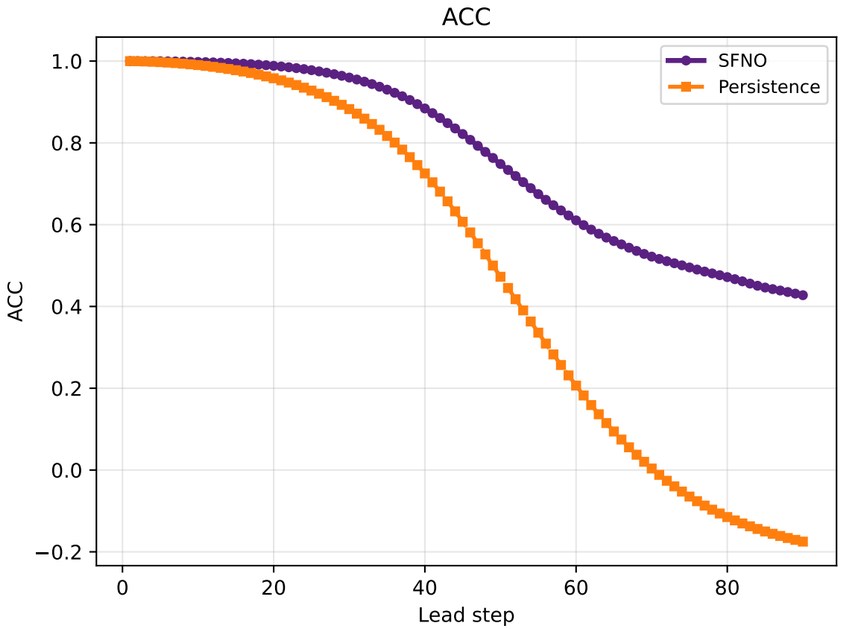
DHW Metrics by Lead Step - Test (18 chains scored at every lead)

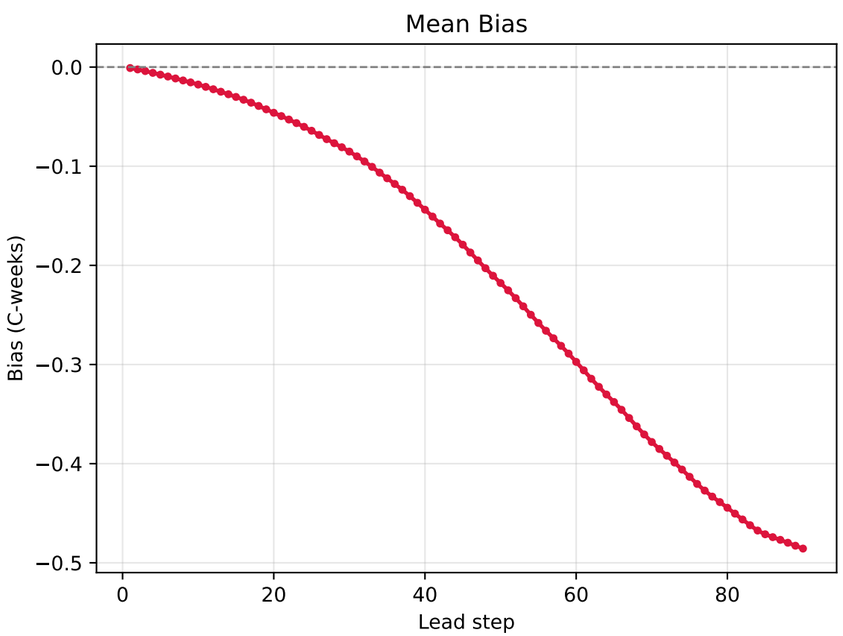
Lead Step	SFNO RMSE	Persist RMSE	Skill %	SFNO MAE	Persist MAE	SFNO Bias	SFNO ACC	Persist ACC	SFNO R²
1	0.0177	0.0446	60.3%	0.0019	0.0066	-0.0010	1.0000	0.9999	1.0000
2	0.0339	0.0878	61.4%	0.0038	0.0130	-0.0024	0.9999	0.9996	0.9999
3	0.0515	0.1301	60.4%	0.0059	0.0194	-0.0040	0.9999	0.9990	0.9997
4	0.0704	0.1717	59.0%	0.0082	0.0258	-0.0058	0.9997	0.9983	0.9995
5	0.0897	0.2128	57.8%	0.0106	0.0321	-0.0077	0.9996	0.9974	0.9992
6	0.1099	0.2534	56.6%	0.0131	0.0383	-0.0095	0.9993	0.9964	0.9989
7	0.1303	0.2936	55.6%	0.0156	0.0446	-0.0115	0.9991	0.9951	0.9984
8	0.1513	0.3337	54.7%	0.0183	0.0509	-0.0135	0.9987	0.9937	0.9978
9	0.1726	0.3737	53.8%	0.0211	0.0573	-0.0154	0.9983	0.9920	0.9971
10	0.1940	0.4140	53.2%	0.0239	0.0639	-0.0176	0.9979	0.9902	0.9964
11	0.2147	0.4549	52.8%	0.0266	0.0707	-0.0200	0.9974	0.9882	0.9955
12	0.2365	0.4961	52.3%	0.0295	0.0778	-0.0224	0.9968	0.9859	0.9945
13	0.2587	0.5380	51.9%	0.0324	0.0850	-0.0249	0.9961	0.9834	0.9934
14	0.2811	0.5805	51.6%	0.0355	0.0924	-0.0274	0.9954	0.9807	0.9921
15	0.3039	0.6234	51.3%	0.0387	0.1001	-0.0301	0.9945	0.9777	0.9907
16	0.3268	0.6671	51.0%	0.0420	0.1080	-0.0329	0.9936	0.9744	0.9891
17	0.3504	0.7119	50.8%	0.0455	0.1164	-0.0359	0.9925	0.9709	0.9874
18	0.3745	0.7579	50.6%	0.0491	0.1251	-0.0392	0.9913	0.9669	0.9854
19	0.3990	0.8052	50.5%	0.0529	0.1343	-0.0426	0.9899	0.9627	0.9832
20	0.4241	0.8536	50.3%	0.0568	0.1437	-0.0461	0.9884	0.9580	0.9807
21	0.4489	0.9027	50.3%	0.0610	0.1535	-0.0494	0.9867	0.9529	0.9781
22	0.4746	0.9530	50.2%	0.0654	0.1637	-0.0529	0.9849	0.9474	0.9751
23	0.5011	1.0048	50.1%	0.0700	0.1743	-0.0565	0.9828	0.9414	0.9718
24	0.5287	1.0576	50.0%	0.0748	0.1853	-0.0603	0.9804	0.9349	0.9681
25	0.5570	1.1114	49.9%	0.0798	0.1966	-0.0642	0.9778	0.9278	0.9640
26	0.5865	1.1666	49.7%	0.0848	0.2083	-0.0685	0.9748	0.9201	0.9595
27	0.6167	1.2230	49.6%	0.0901	0.2205	-0.0727	0.9715	0.9118	0.9544
28	0.6467	1.2805	49.5%	0.0956	0.2332	-0.0768	0.9680	0.9028	0.9490
29	0.6772	1.3394	49.4%	0.1014	0.2463	-0.0809	0.9640	0.8930	0.9431
30	0.7081	1.3987	49.4%	0.1072	0.2598	-0.0852	0.9597	0.8826	0.9367
31	0.7400	1.4591	49.3%	0.1132	0.2737	-0.0901	0.9549	0.8713	0.9297
32	0.7724	1.5206	49.2%	0.1194	0.2881	-0.0952	0.9497	0.8592	0.9220
33	0.8061	1.5834	49.1%	0.1258	0.3031	-0.1006	0.9438	0.8461	0.9136
34	0.8411	1.6473	48.9%	0.1327	0.3185	-0.1065	0.9373	0.8320	0.9044
35	0.8768	1.7121	48.8%	0.1399	0.3345	-0.1122	0.9302	0.8169	0.8944
36	0.9128	1.7778	48.7%	0.1474	0.3508	-0.1179	0.9224	0.8007	0.8837
37	0.9494	1.8444	48.5%	0.1551	0.3674	-0.1237	0.9140	0.7835	0.8723
38	0.9874	1.9117	48.3%	0.1629	0.3844	-0.1302	0.9048	0.7651	0.8599
39	1.0265	1.9795	48.1%	0.1710	0.4015	-0.1369	0.8948	0.7456	0.8466
40	1.0659	2.0477	47.9%	0.1792	0.4189	-0.1439	0.8841	0.7252	0.8326
41	1.1057	2.1165	47.8%	0.1877	0.4365	-0.1508	0.8727	0.7035	0.8180
42	1.1460	2.1861	47.6%	0.1966	0.4545	-0.1578	0.8607	0.6808	0.8028
43	1.1861	2.2558	47.4%	0.2056	0.4725	-0.1646	0.8483	0.6571	0.7873
44	1.2272	2.3259	47.2%	0.2149	0.4908	-0.1717	0.8353	0.6325	0.7713
45	1.2691	2.3960	47.0%	0.2242	0.5093	-0.1792	0.8216	0.6070	0.7547
46	1.3115	2.4661	46.8%	0.2335	0.5277	-0.1870	0.8074	0.5810	0.7378
47	1.3550	2.5360	46.6%	0.2428	0.5463	-0.1950	0.7927	0.5544	0.7205
48	1.3983	2.6057	46.3%	0.2521	0.5647	-0.2029	0.7779	0.5273	0.7031
49	1.4411	2.6749	46.1%	0.2615	0.5830	-0.2105	0.7631	0.5000	0.6861
50	1.4837	2.7437	45.9%	0.2709	0.6013	-0.2178	0.7484	0.4726	0.6692
51	1.5259	2.8119	45.7%	0.2803	0.6195	-0.2252	0.7338	0.4452	0.6527
52	1.5689	2.8796	45.5%	0.2897	0.6378	-0.2330	0.7190	0.4177	0.6361
53	1.6124	2.9468	45.3%	0.2991	0.6561	-0.2413	0.7041	0.3904	0.6196
54	1.6558	3.0138	45.1%	0.3084	0.6744	-0.2498	0.6892	0.3631	0.6032
55	1.6990	3.0805	44.8%	0.3176	0.6926	-0.2581	0.6747	0.3360	0.5873
56	1.7413	3.1466	44.7%	0.3268	0.7108	-0.2660	0.6607	0.3092	0.5721
57	1.7825	3.2122	44.5%	0.3359	0.7289	-0.2735	0.6475	0.2827	0.5577
58	1.8236	3.2770	44.4%	0.3451	0.7469	-0.2812	0.6348	0.2568	0.5438
59	1.8648	3.3414	44.2%	0.3542	0.7647	-0.2890	0.6225	0.2314	0.5304
60	1.9066	3.4054	44.0%	0.3633	0.7824	-0.2974	0.6105	0.2065	0.5173
61	1.9491	3.4692	43.8%	0.3725	0.8002	-0.3059	0.5989	0.1823	0.5046
62	1.9915	3.5325	43.6%	0.3817	0.8178	-0.3144	0.5879	0.1589	0.4925
63	2.0336	3.5947	43.4%	0.3908	0.8350	-0.3225	0.5778	0.1364	0.4812
64	2.0748	3.6556	43.2%	0.3999	0.8518	-0.3302	0.5684	0.1149	0.4707
65	2.1155	3.7155	43.1%	0.4089	0.8682	-0.3378	0.5600	0.0943	0.4609
66	2.1564	3.7745	42.9%	0.4179	0.8842	-0.3457	0.5518	0.0746	0.4515
67	2.1977	3.8325	42.7%	0.4267	0.8999	-0.3539	0.5436	0.0556	0.4421
68	2.2388	3.8894	42.4%	0.4353	0.9152	-0.3624	0.5358	0.0375	0.4331
69	2.2791	3.9451	42.2%	0.4438	0.9300	-0.3705	0.5285	0.0202	0.4246
70	2.3181	3.9993	42.0%	0.4519	0.9444	-0.3782	0.5219	0.0037	0.4169
71	2.3558	4.0518	41.9%	0.4599	0.9582	-0.3852	0.5162	-0.0117	0.4100
72	2.3931	4.1026	41.7%	0.4677	0.9716	-0.3920	0.5107	-0.0262	0.4034
73	2.4299	4.1521	41.5%	0.4753	0.9845	-0.3989	0.5056	-0.0399	0.3971
74	2.4663	4.2000	41.3%	0.4827	0.9969	-0.4060	0.5006	-0.0527	0.3910
75	2.5029	4.2466	41.1%	0.4901	1.0090	-0.4133	0.4954	-0.0648	0.3849
76	2.5390	4.2915	40.8%	0.4972	1.0206	-0.4204	0.4902	-0.0762	0.3789
77	2.5741	4.3346	40.6%	0.5043	1.0318	-0.4272	0.4853	-0.0869	0.3731
78	2.6079	4.3756	40.4%	0.5112	1.0423	-0.4332	0.4808	-0.0969	0.3675
79	2.6405	4.4146	40.2%	0.5180	1.0523	-0.4388	0.4765	-0.1062	0.3621
80	2.6729	4.4516	40.0%	0.5246	1.0618	-0.4445	0.4718	-0.1149	0.3564
81	2.7050	4.4868	39.7%	0.5310	1.0709	-0.4504	0.4667	-0.1231	0.3505
82	2.7364	4.5199	39.5%	0.5371	1.0795	-0.4563	0.4614	-0.1306	0.3445
83	2.7671	4.5510	39.2%	0.5431	1.0876	-0.4621	0.4559	-0.1375	0.3385
84	2.7964	4.5803	38.9%	0.5488	1.0953	-0.4675	0.4506	-0.1441	0.3327
85	2.8231	4.6080	38.7%	0.5530	1.1025	-0.4712	0.4462	-0.1502	0.3277
86	2.8476	4.6338	38.5%	0.5569	1.1093	-0.4741	0.4424	-0.1558	0.3233
87	2.8699	4.6576	38.4%	0.5603	1.1158	-0.4769	0.4389	-0.1612	0.3194
88	2.8906	4.6794	38.2%	0.5632	1.1218	-0.4798	0.4353	-0.1662	0.3155
89	2.9102	4.6993	38.1%	0.5660	1.1275	-0.4828	0.4314	-0.1708	0.3116
90	2.9285	4.7176	37.9%	0.5685	1.1330	-0.4857	0.4275	-0.1750	0.3079

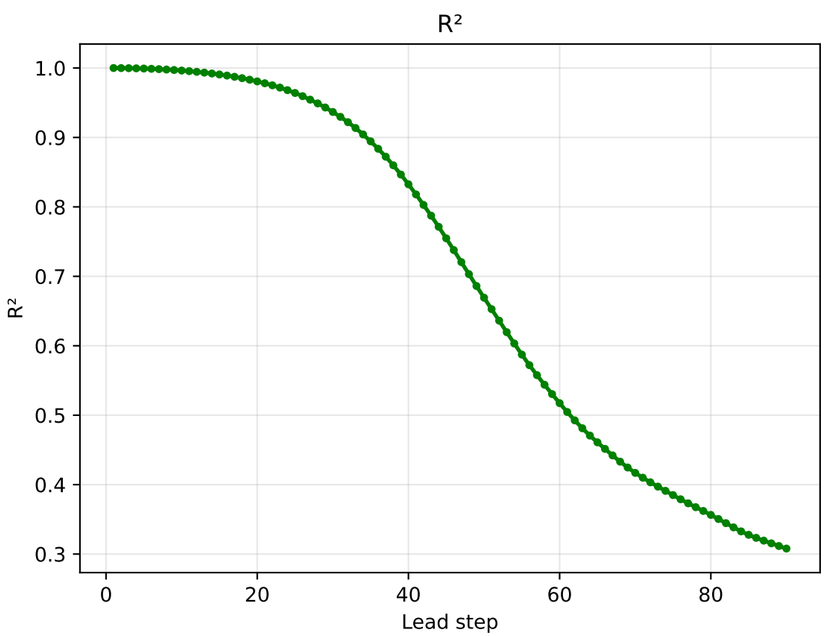
DHW Forecast Skill vs Lead Step - Test

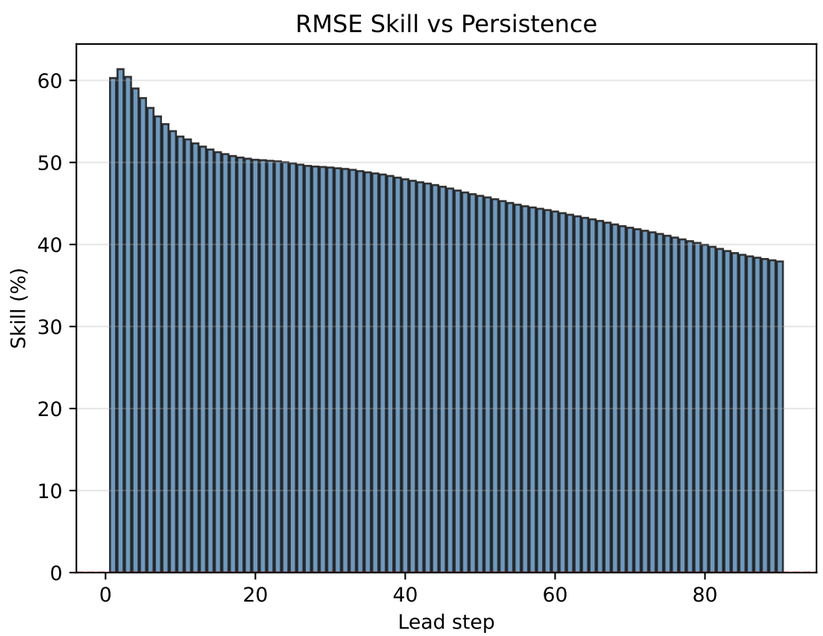


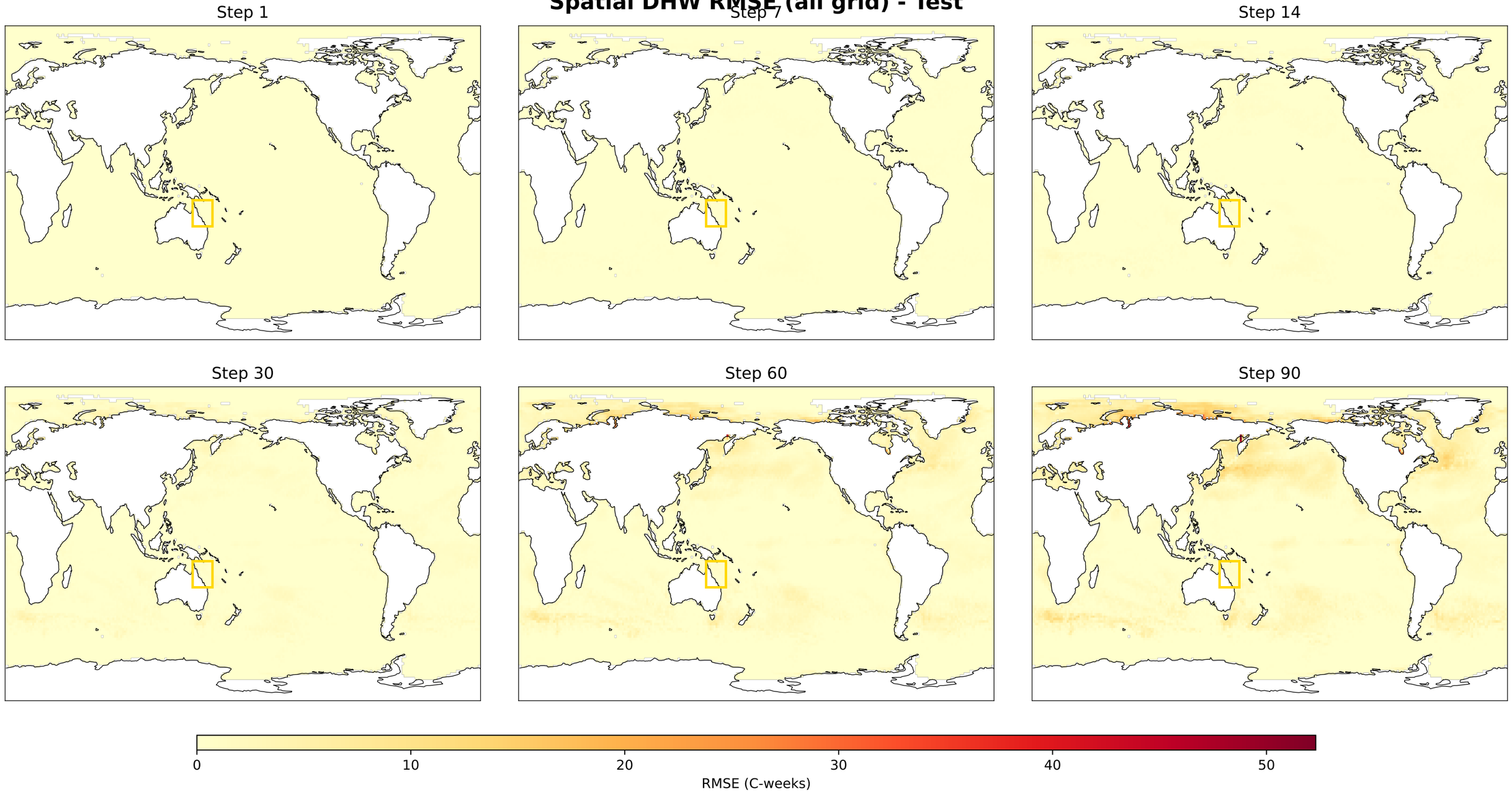


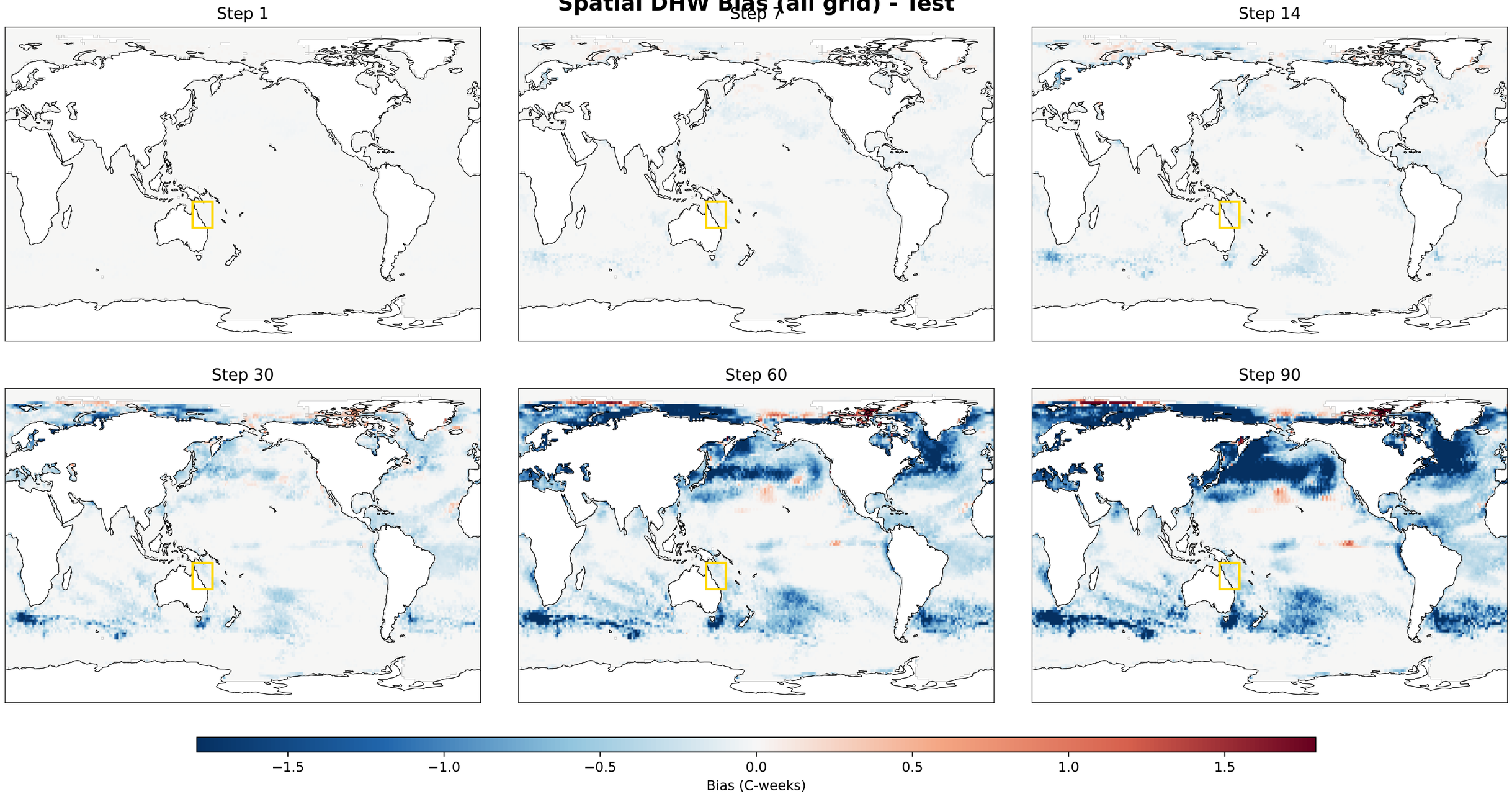


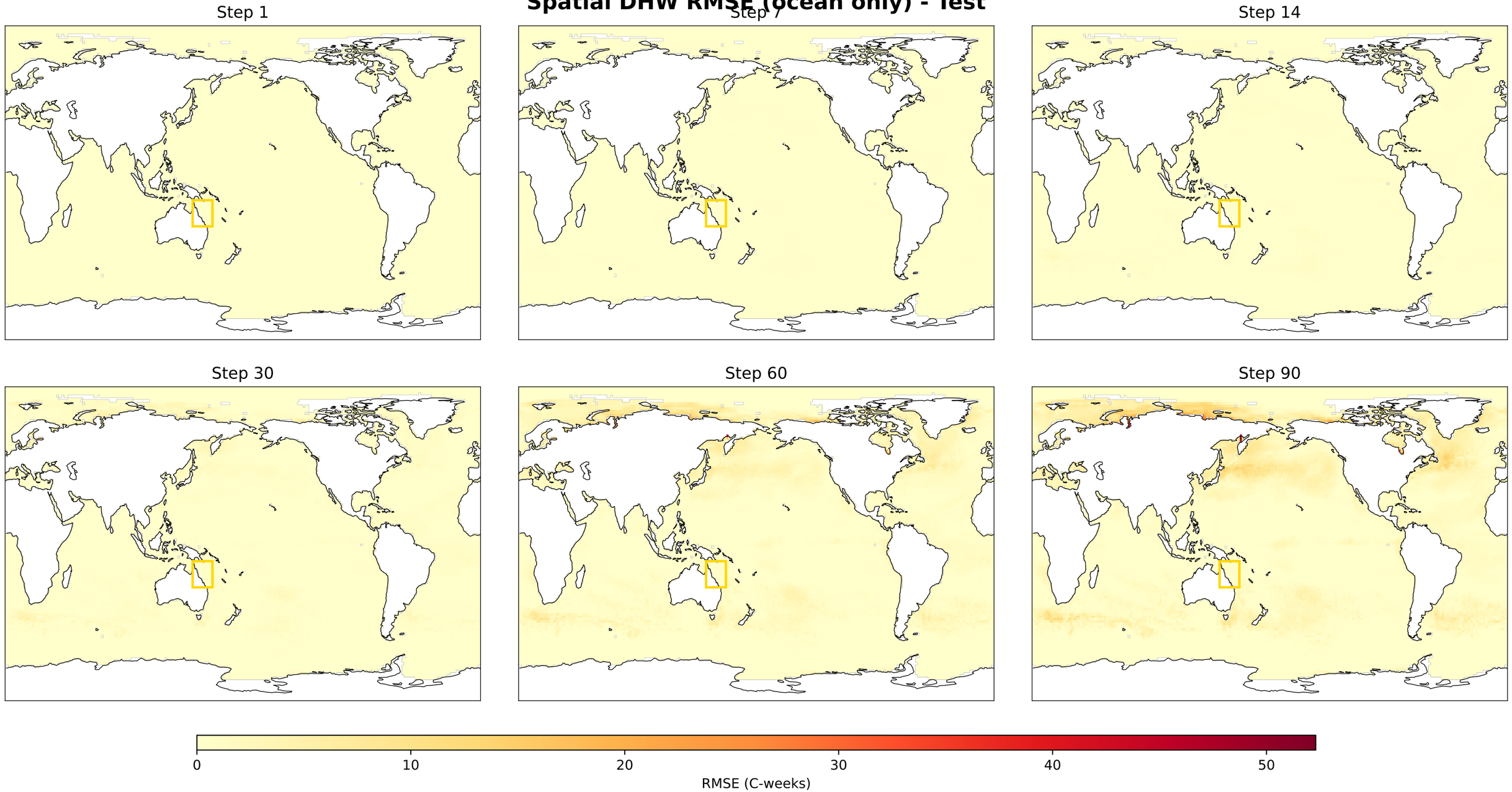


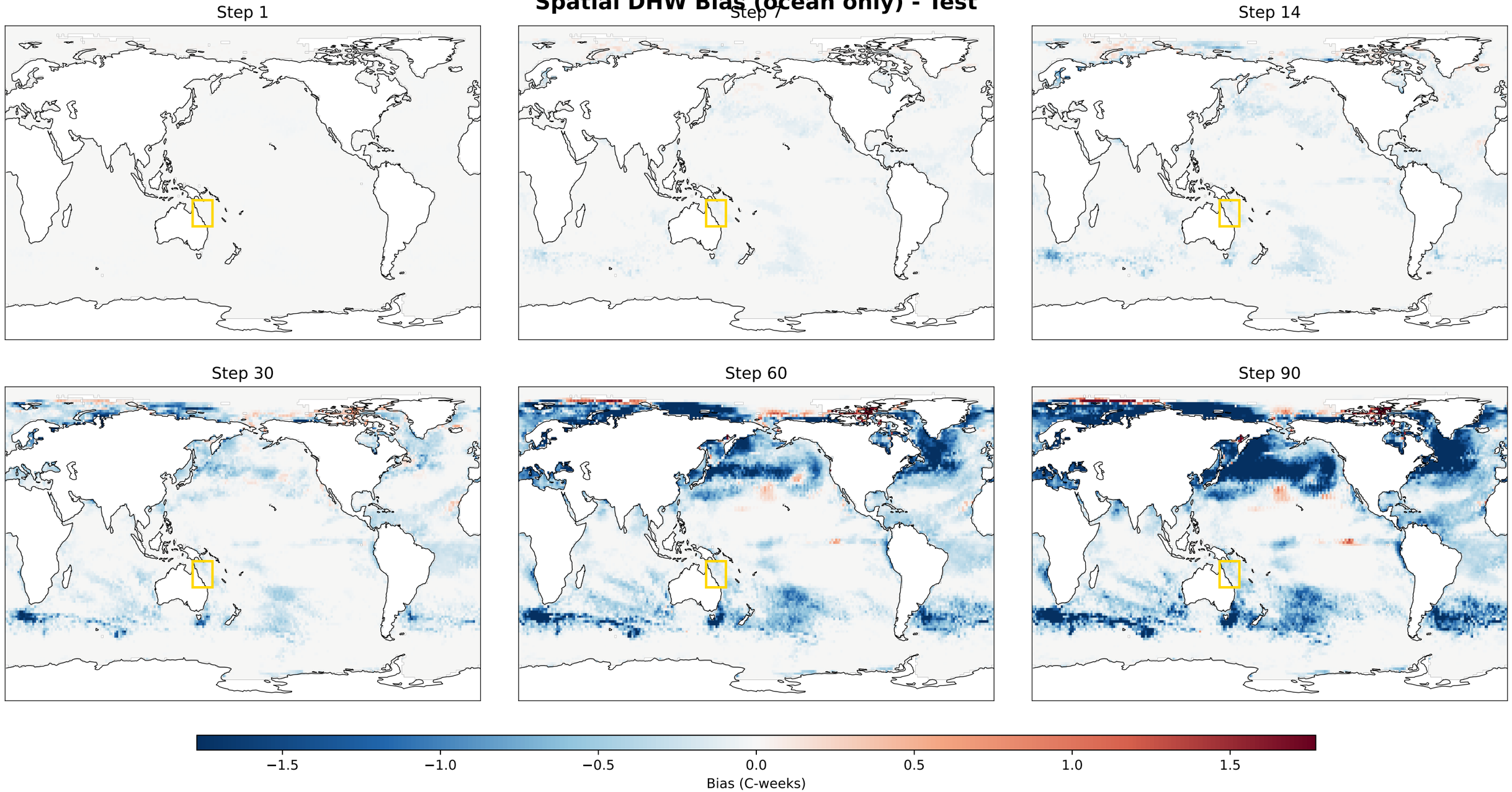





Spatial DHW RMSE (all grid) - Test
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0
10
20
30
40
50
RMSE (C-weeks)

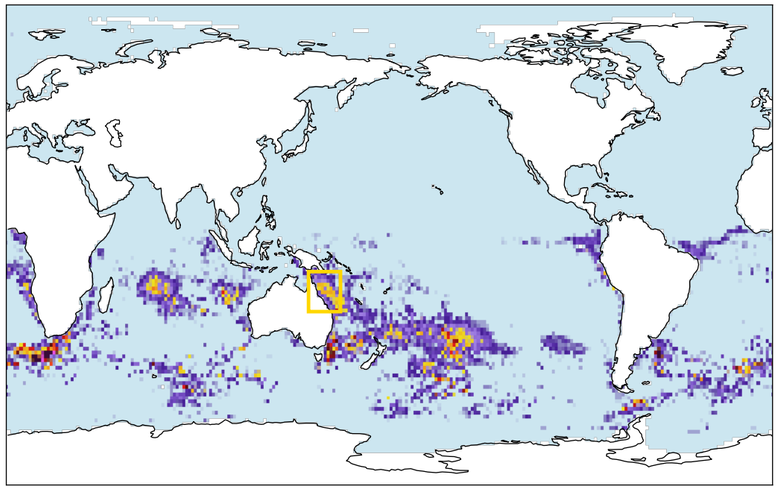

Spatial DHW Bias (all grid) - Test
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
Bias (C-weeks)


Spatial DHW RMSE (ocean only) - Test
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0
10
20
30
40
50
RMSE (C-weeks)

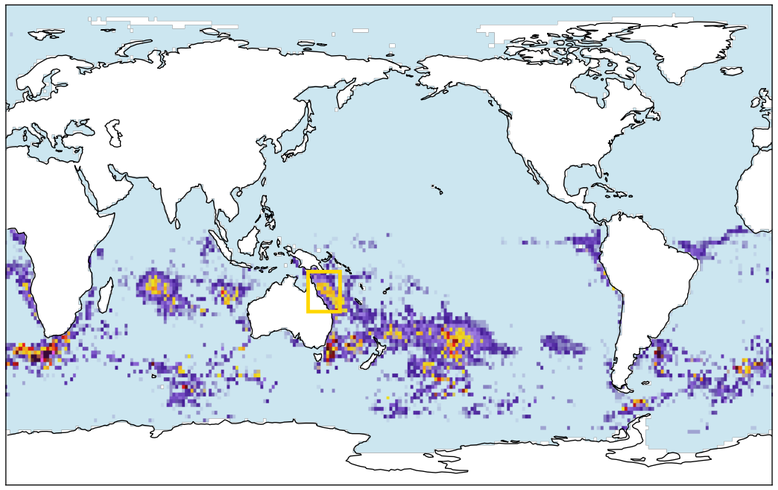

Spatial DHW Bias (ocean only) - Test
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
Bias (C-weeks)

DHW Chain init: 2020-04-10 - Test (C-weeks)

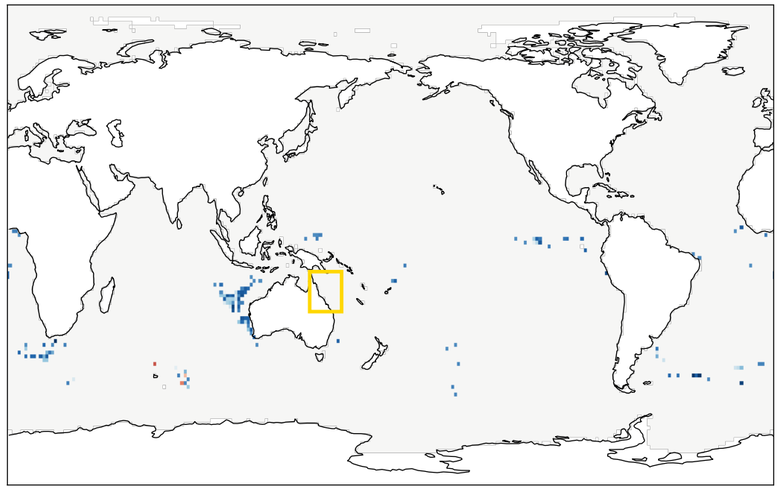
Actual DHW - Step 1



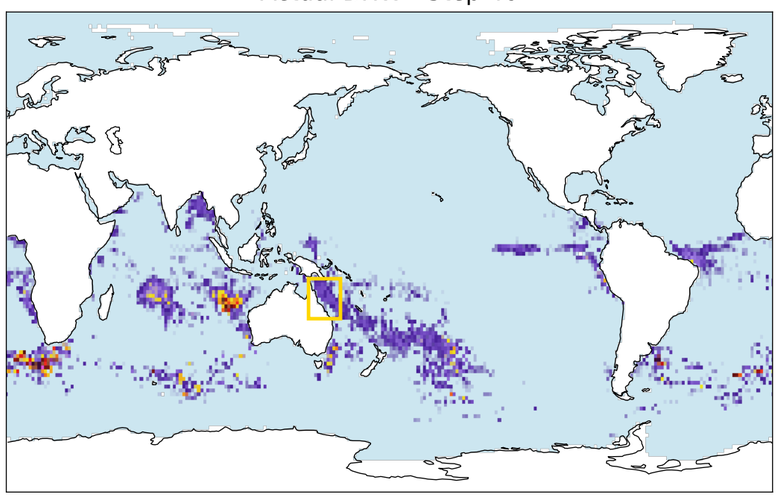
Predicted DHW - Step 1



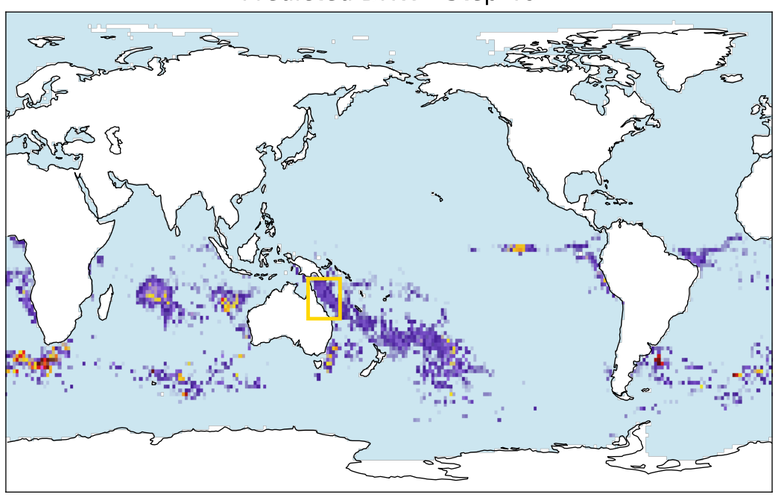
Error (Pred - Actual) - Step 1 [scale ±0.28]



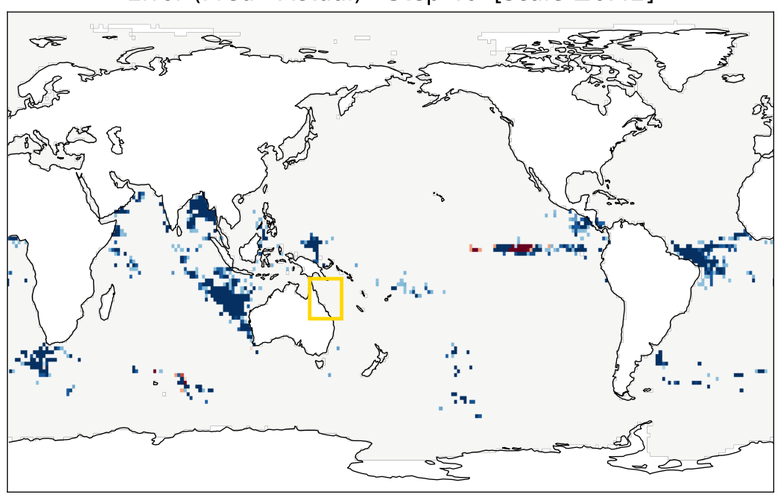
Actual DHW - Step 46



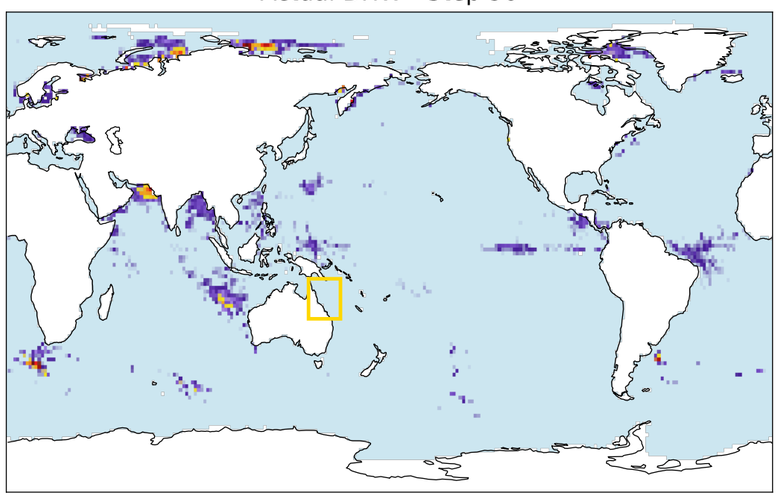
Predicted DHW - Step 46



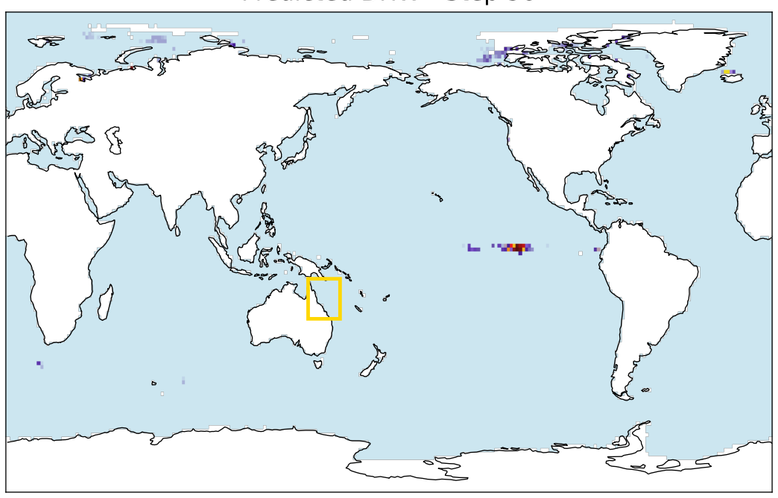
Error (Pred - Actual) - Step 46 [scale ±0.42]



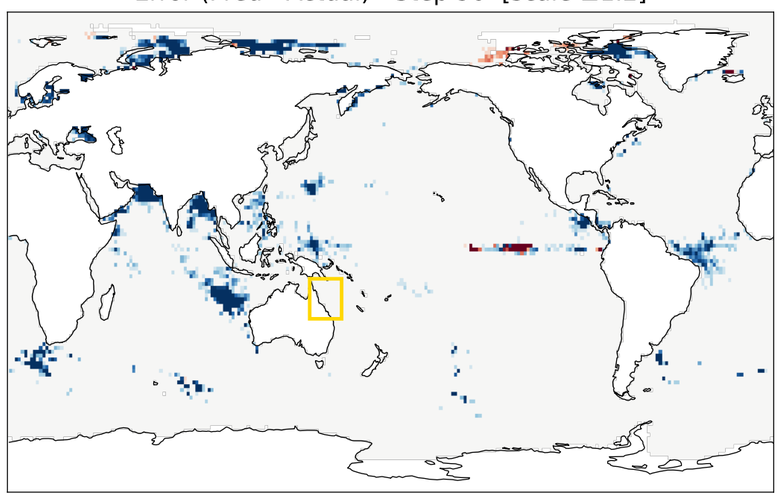
Actual DHW - Step 90

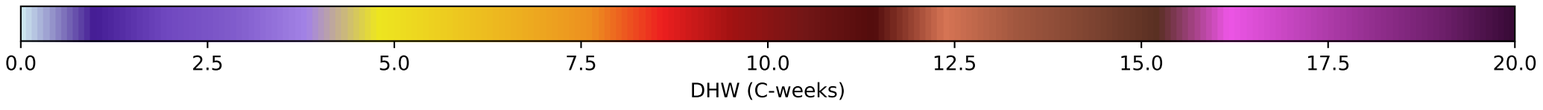


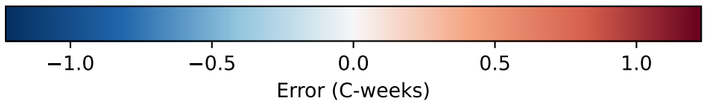
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±1.2]

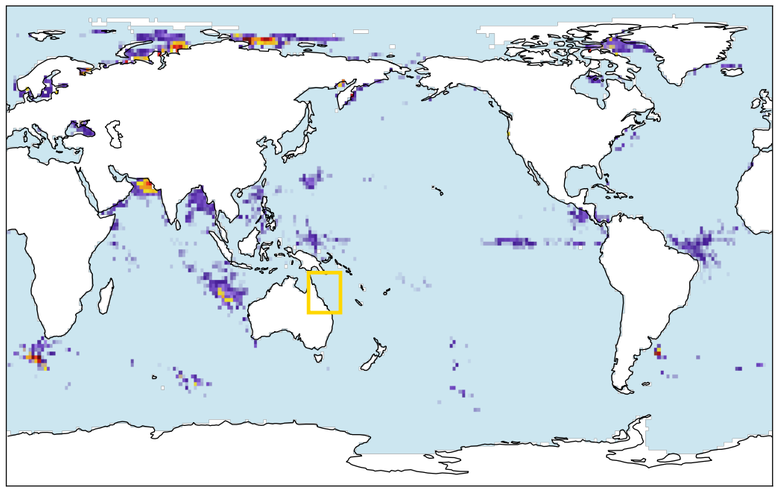




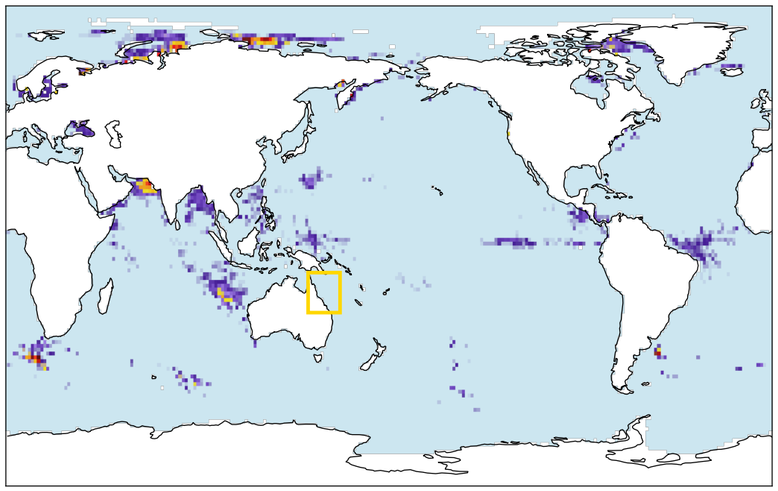


DHW Chain init: 2020-07-09 - Test (C-weeks)

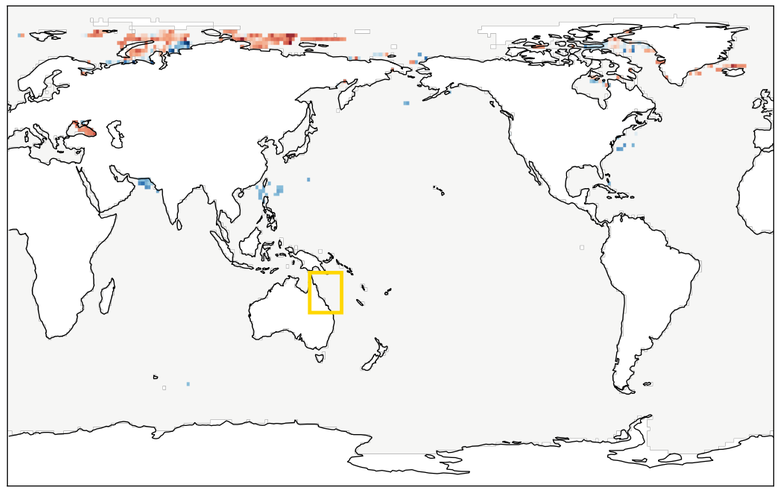
Actual DHW - Step 1



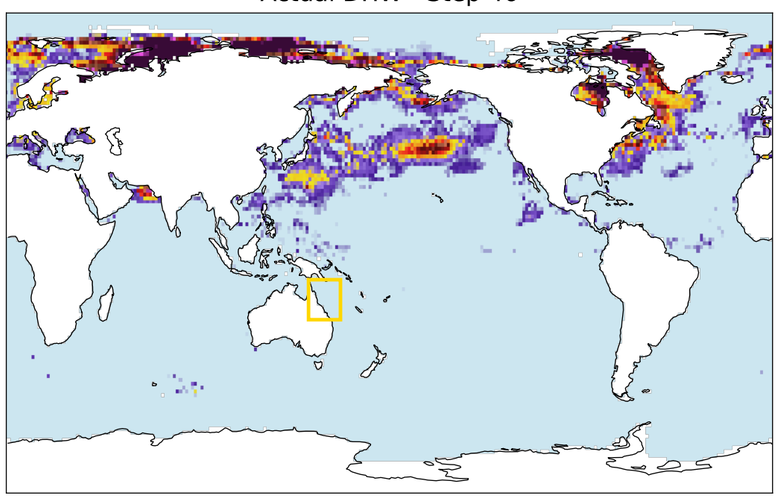
Predicted DHW - Step 1



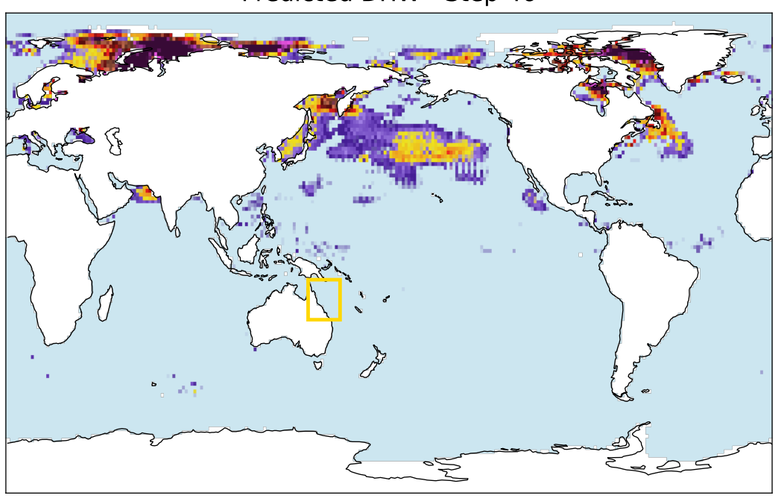
Error (Pred - Actual) - Step 1 [scale ±0.4]



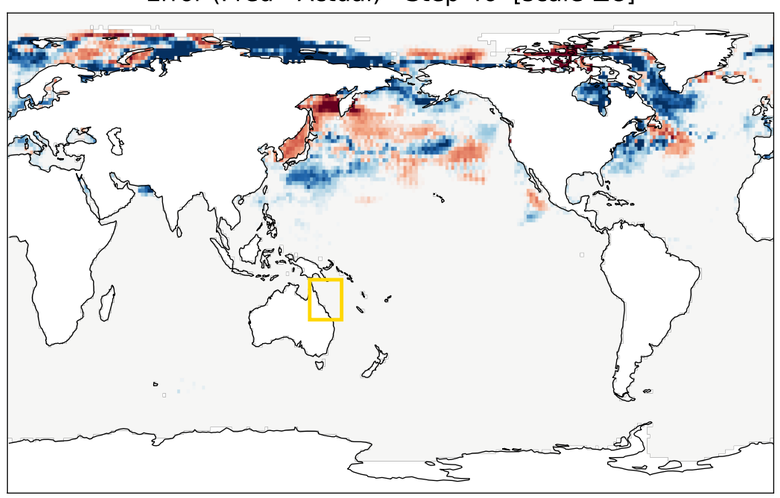
Actual DHW - Step 46



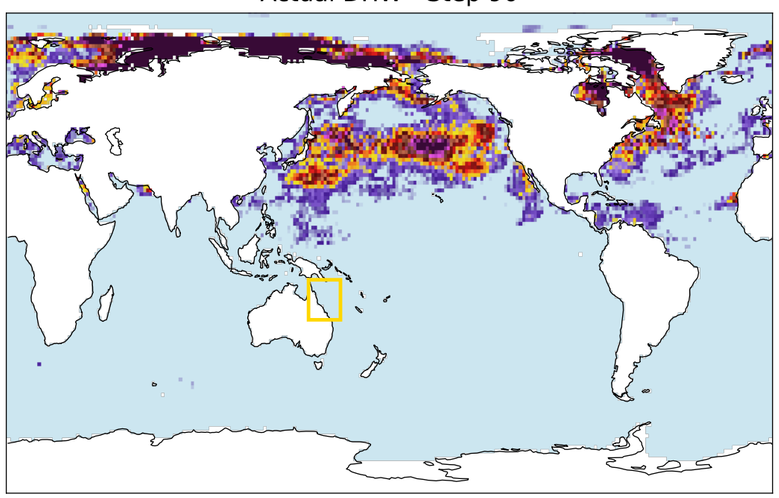
Predicted DHW - Step 46



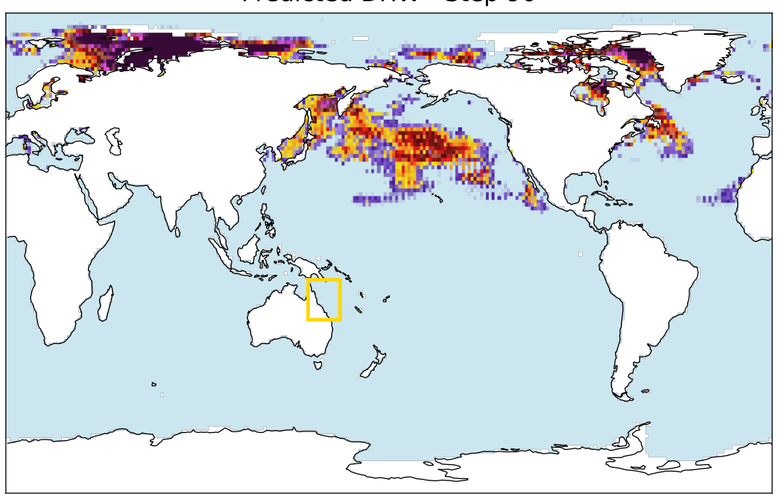
Error (Pred - Actual) - Step 46 [scale ±8]



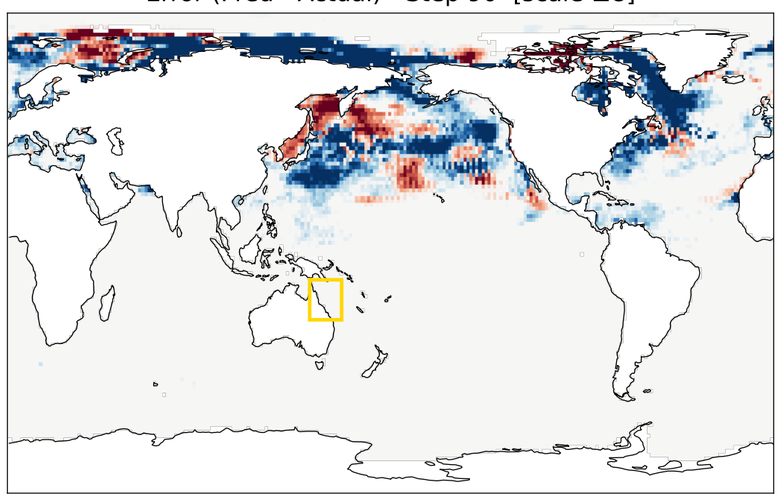
Actual DHW - Step 90

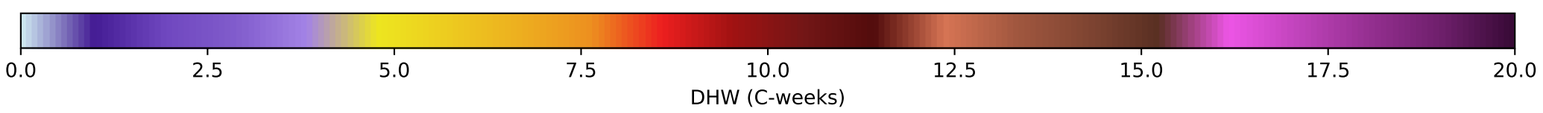


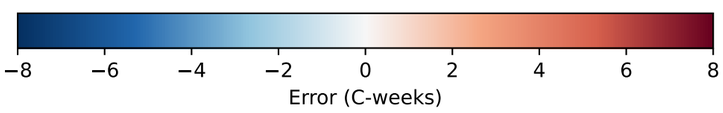
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

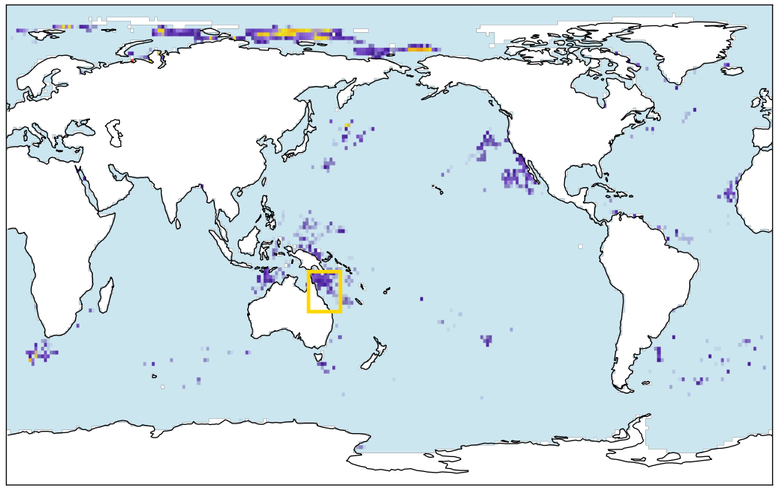




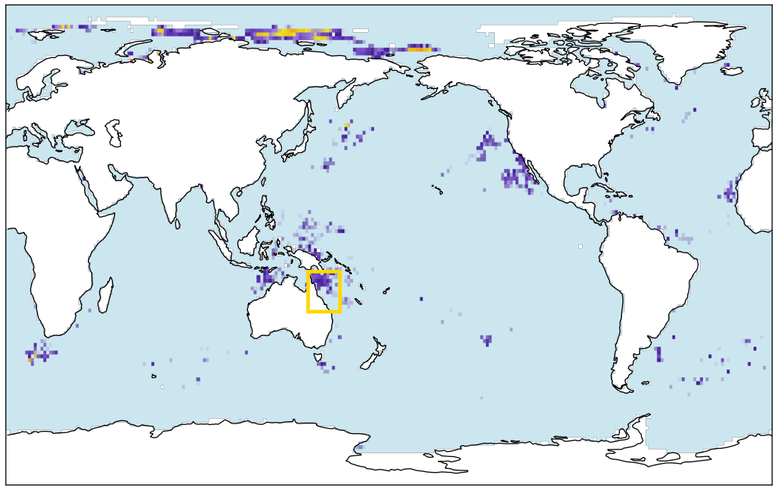


DHW Chain init: 2021-01-06 - Test (C-weeks)

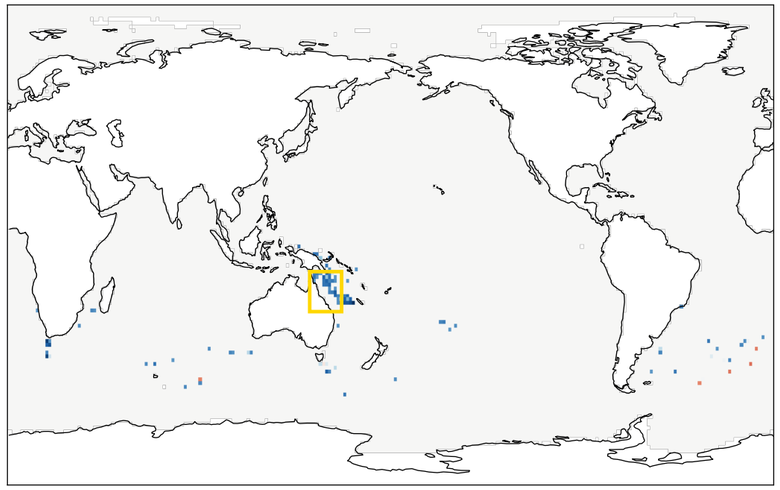
Actual DHW - Step 1



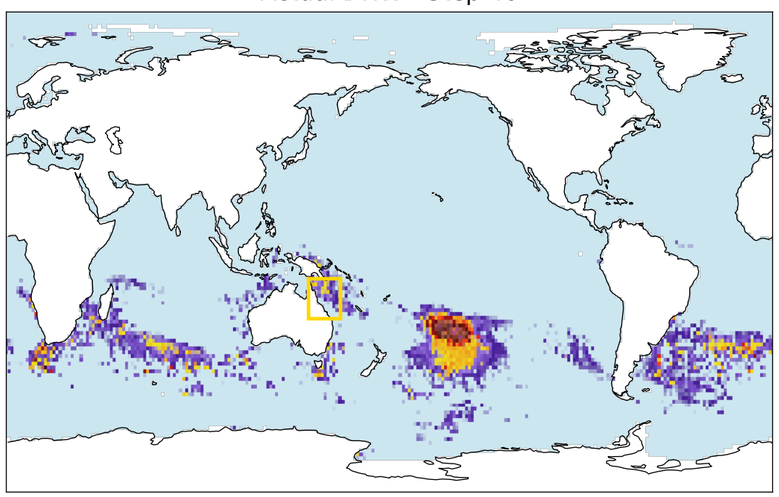
Predicted DHW - Step 1



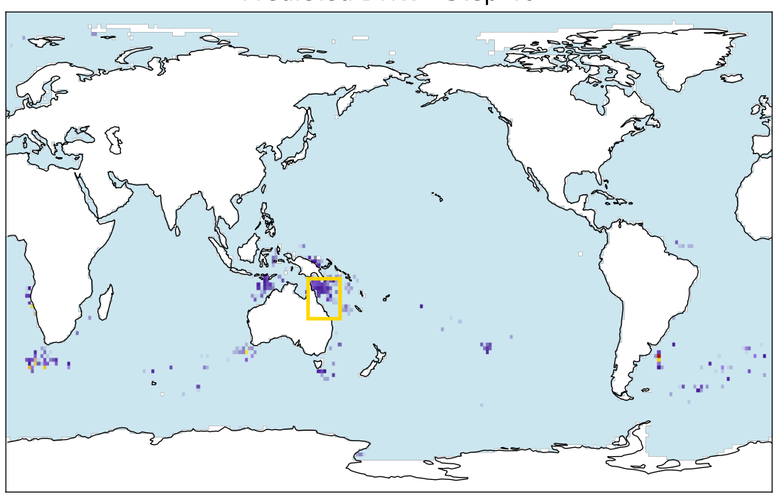
Error (Pred - Actual) - Step 1 [scale ±0.29]



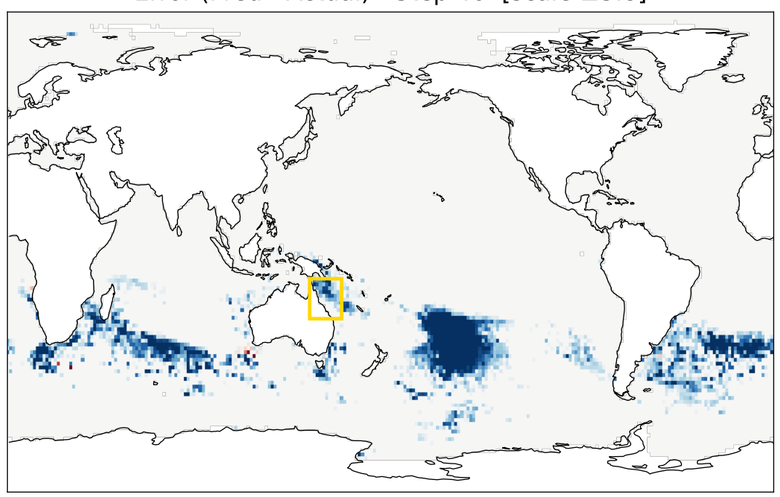
Actual DHW - Step 46



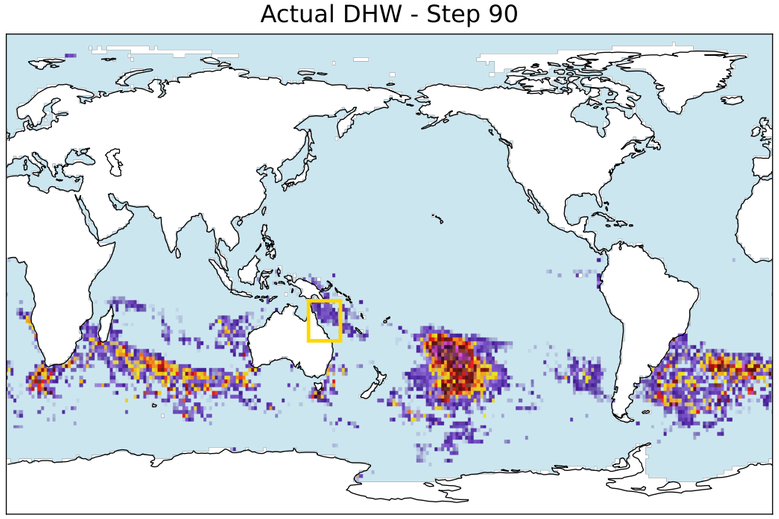
Predicted DHW - Step 46



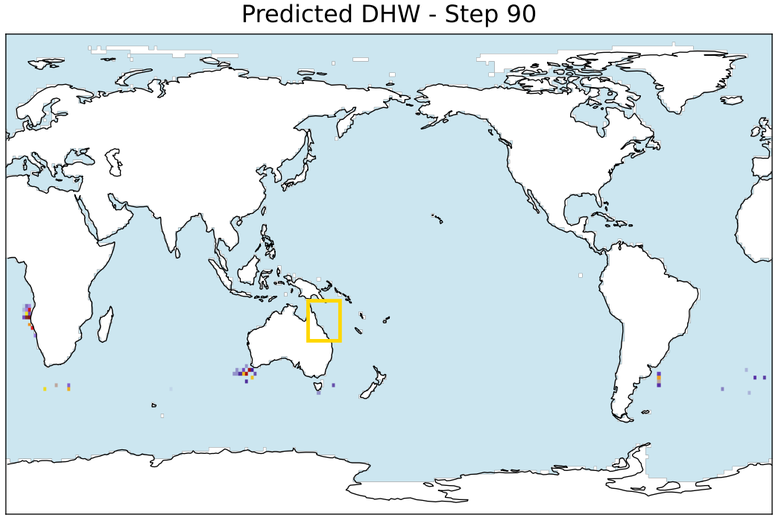
Error (Pred - Actual) - Step 46 [scale ±3.8]



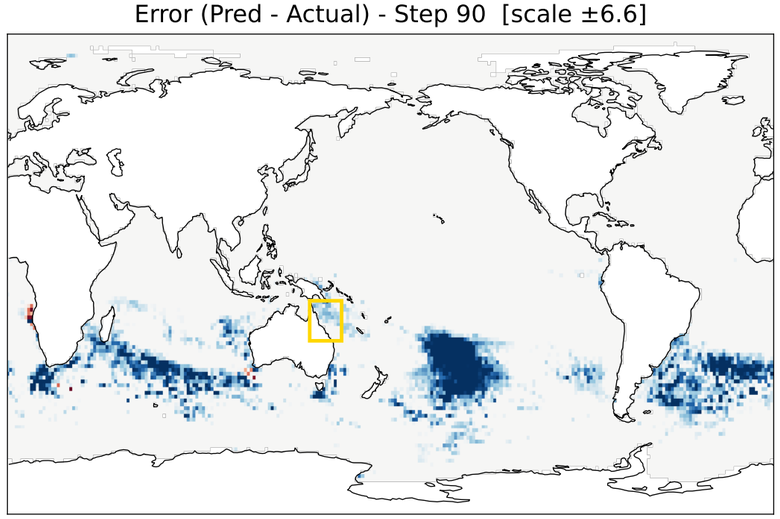
Actual DHW - Step 90

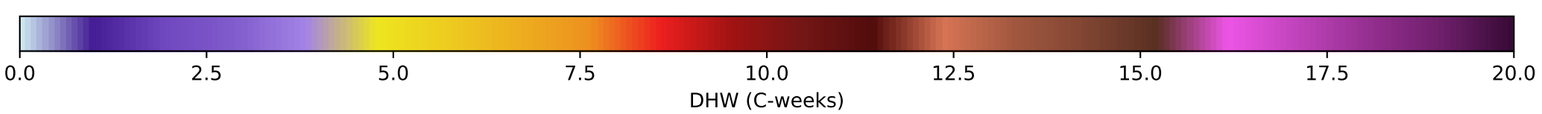


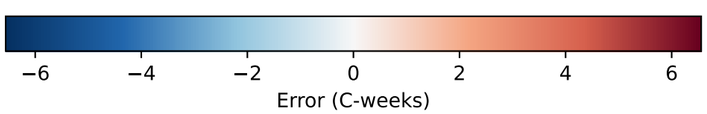
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±6.6]

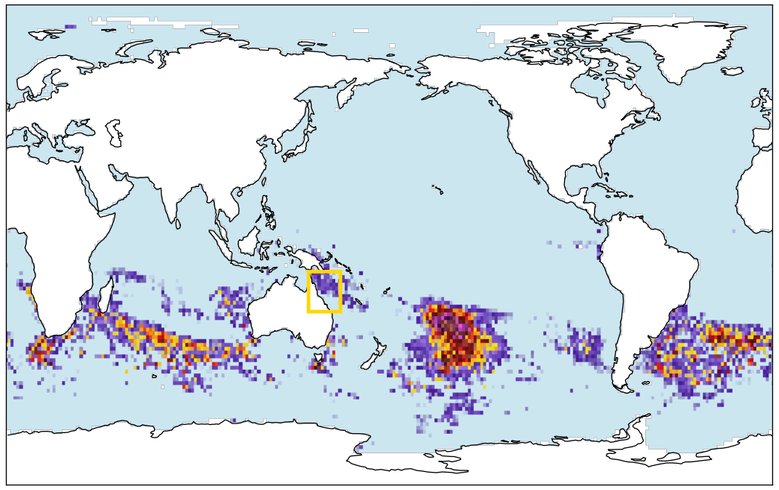




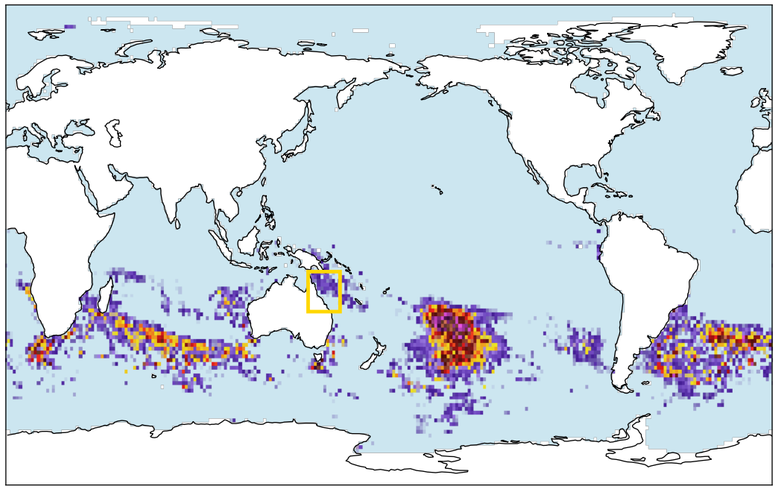


DHW Chain init: 2021-04-06 - Test (C-weeks)

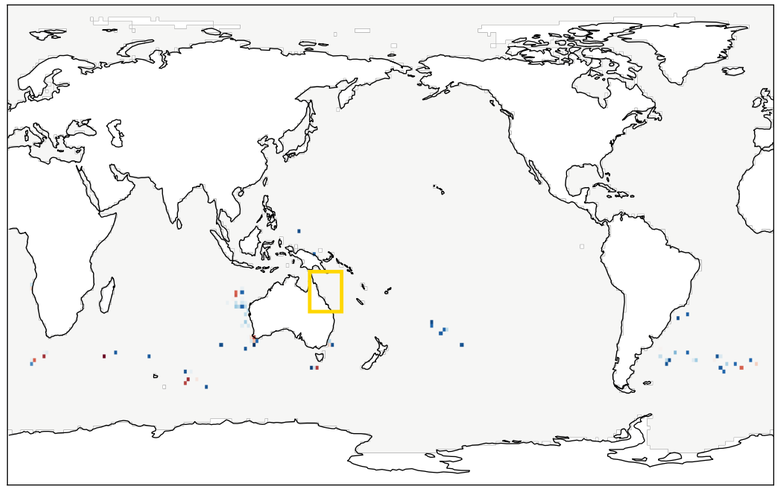
Actual DHW - Step 1



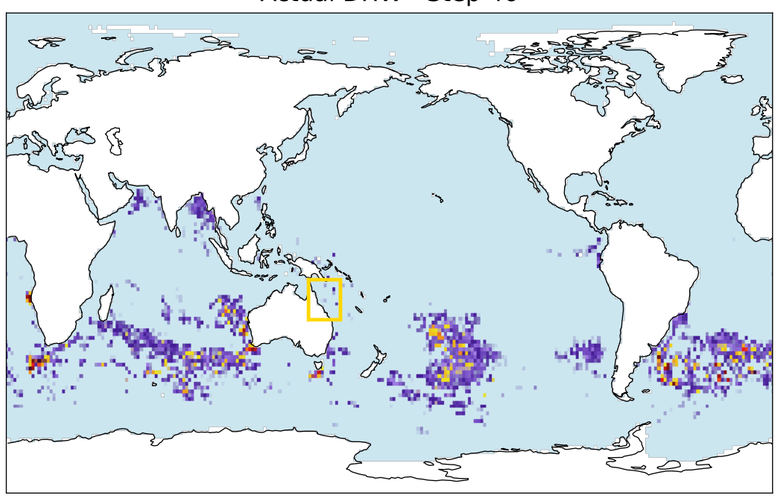
Predicted DHW - Step 1



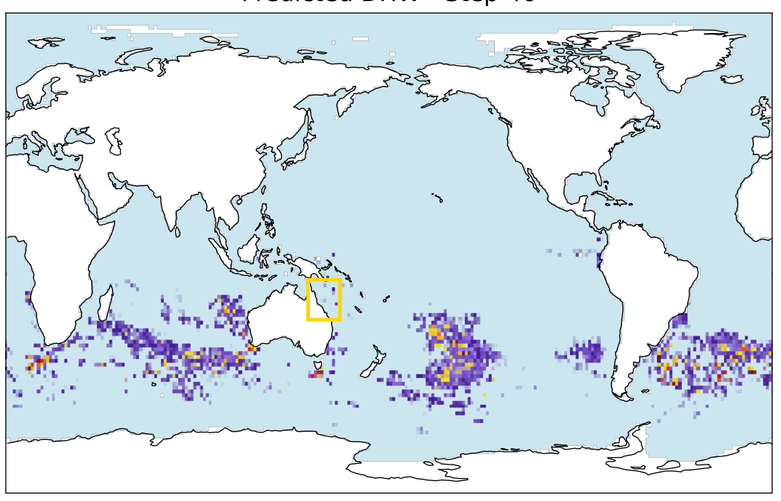
Error (Pred - Actual) - Step 1 [scale ±0.22]



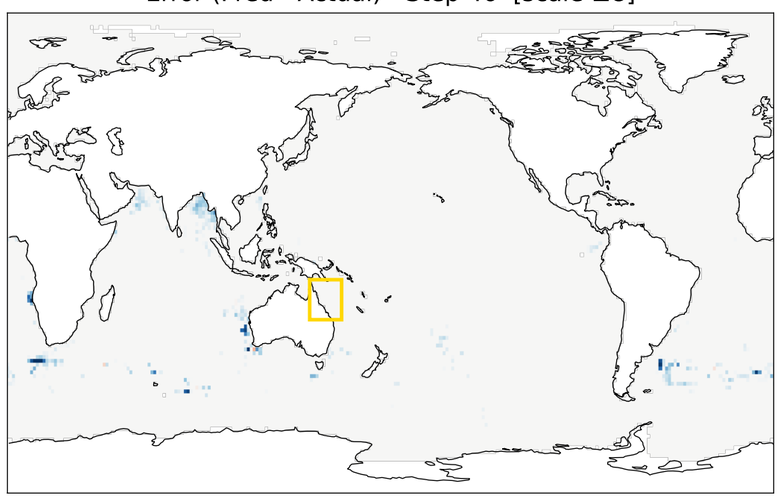
Actual DHW - Step 46



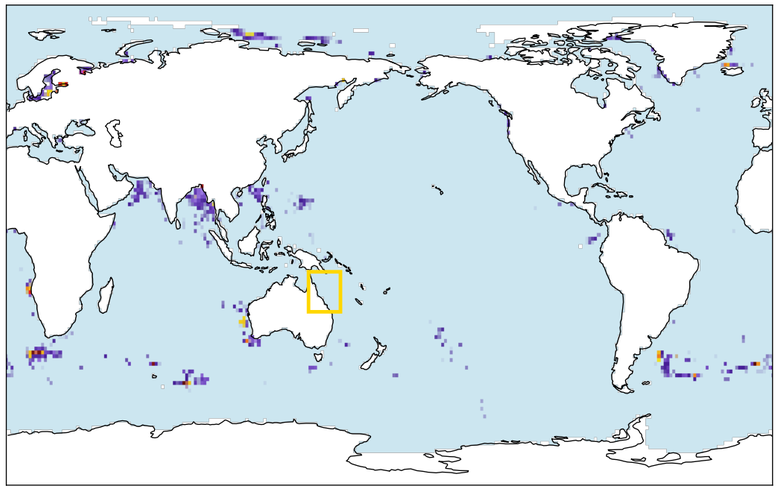
Predicted DHW - Step 46



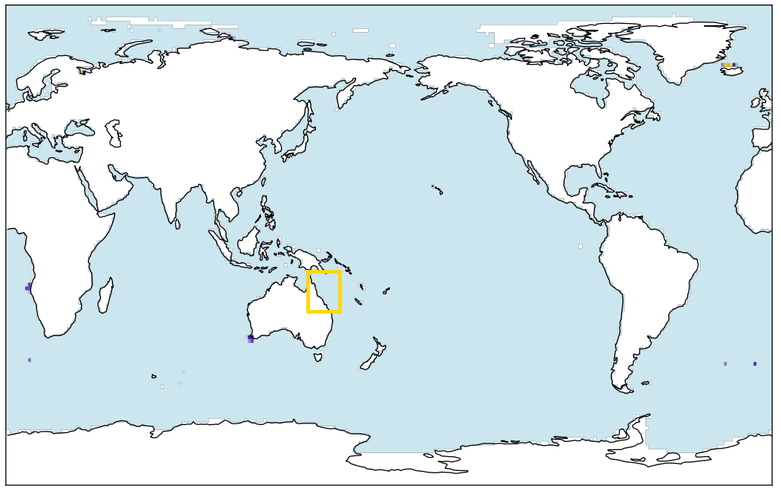
Error (Pred - Actual) - Step 46 [scale ±8]



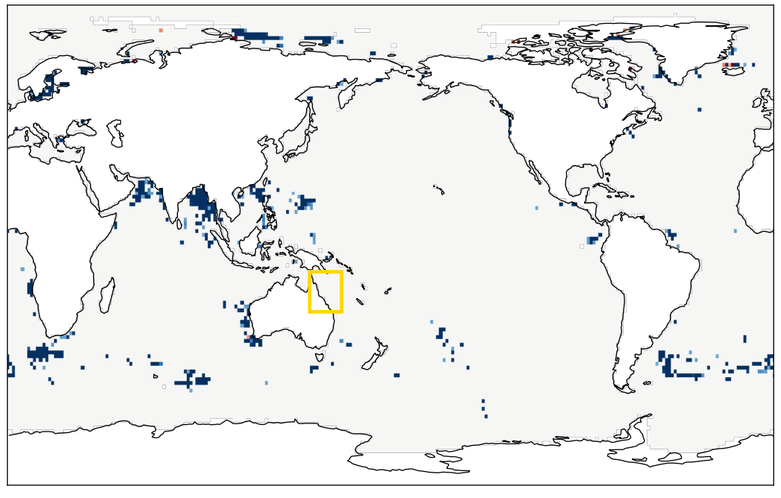
Actual DHW - Step 90

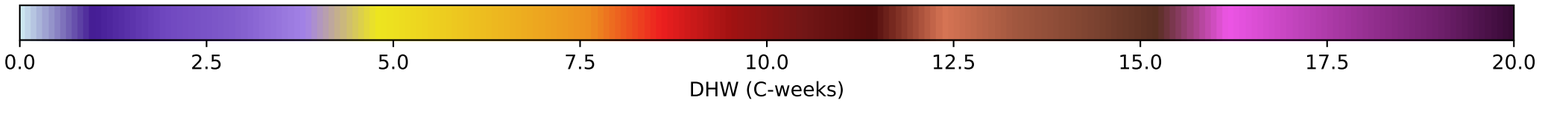


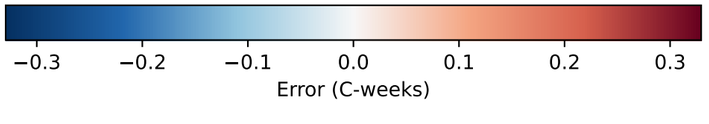
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±0.33]

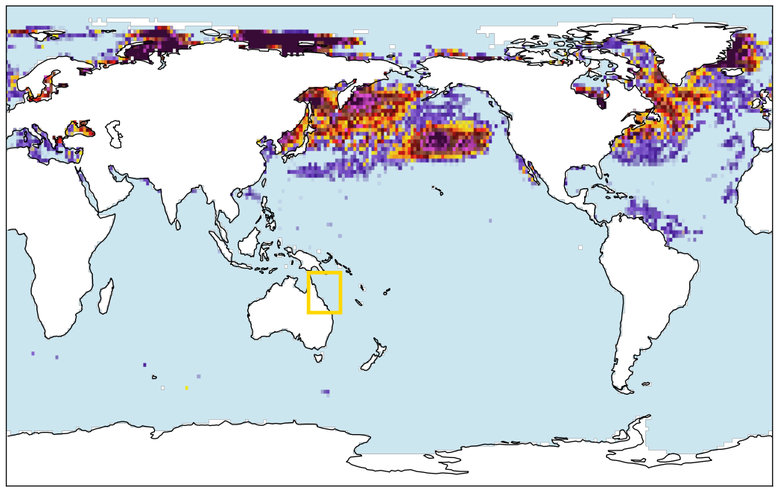




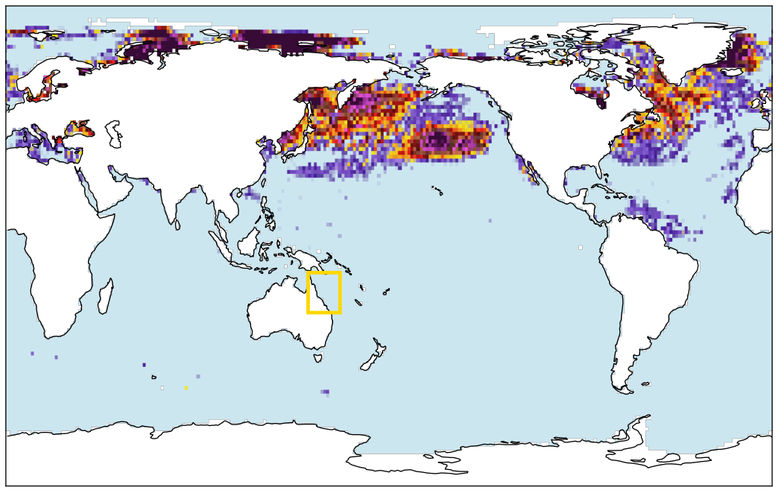


DHW Chain init: 2021-10-03 - Test (C-weeks)

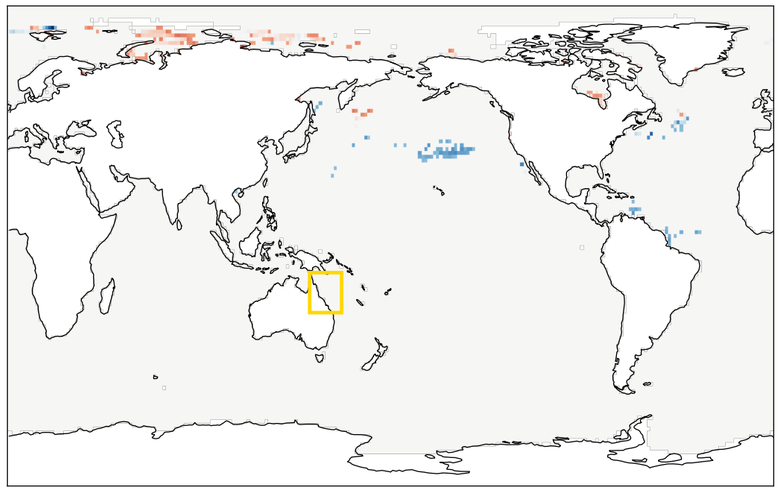
Actual DHW - Step 1



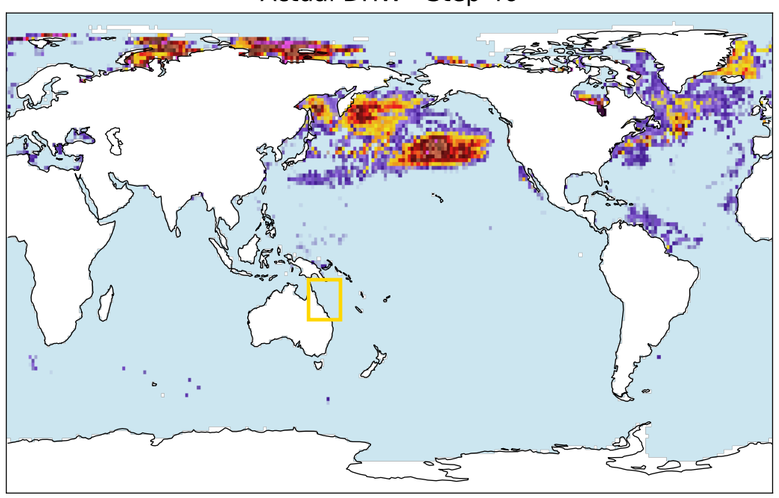
Predicted DHW - Step 1



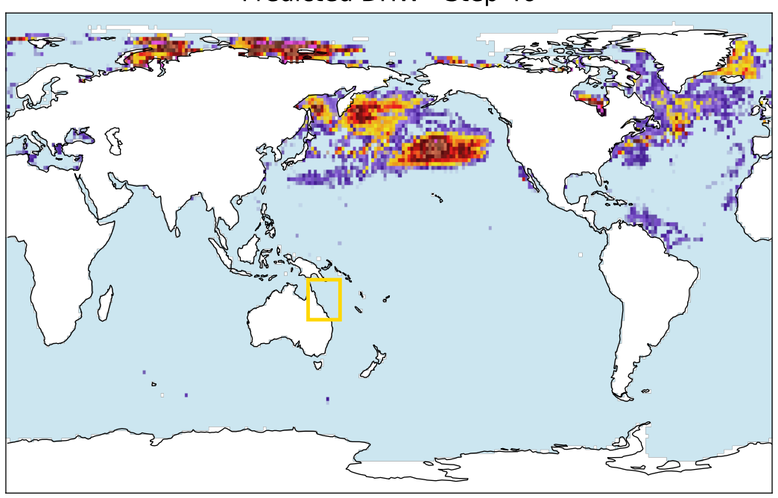
Error (Pred - Actual) - Step 1 [scale ±0.4]



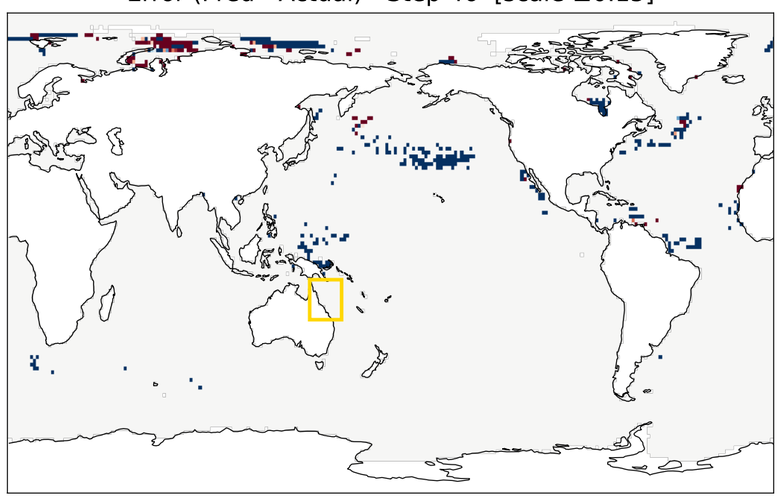
Actual DHW - Step 46



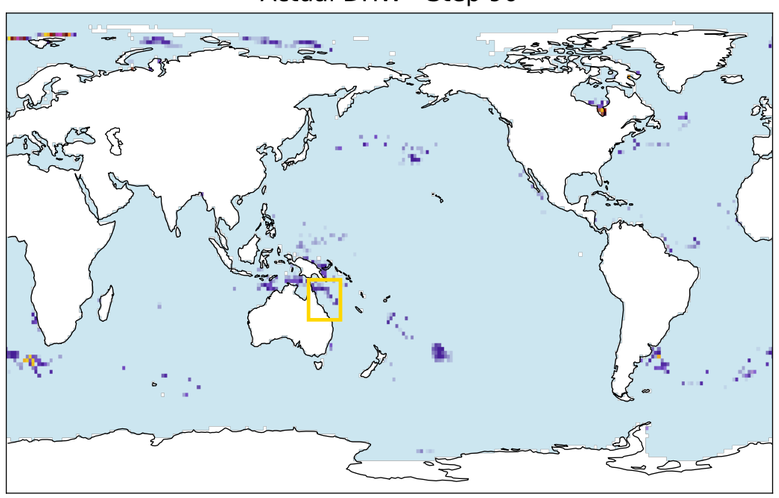
Predicted DHW - Step 46



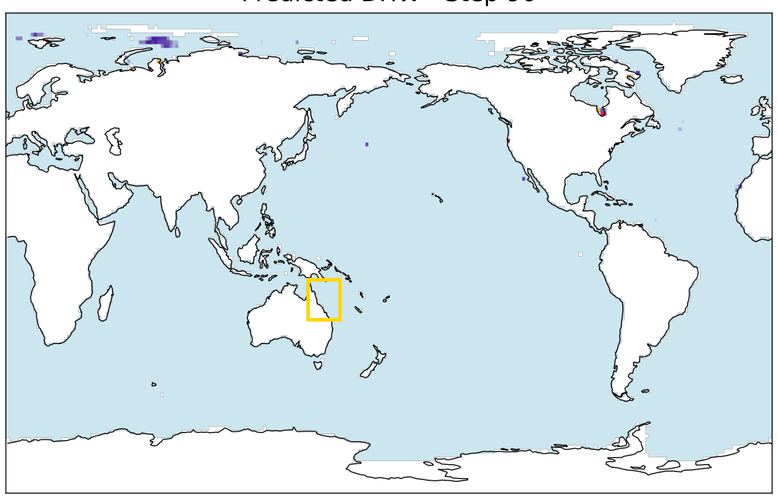
Error (Pred - Actual) - Step 46 [scale ±0.15]



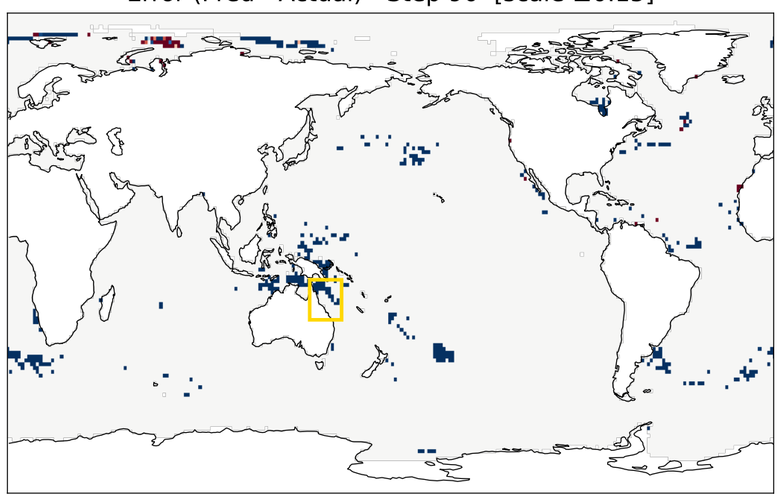
Actual DHW - Step 90

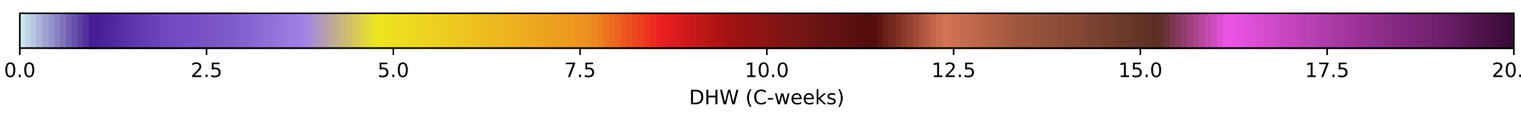


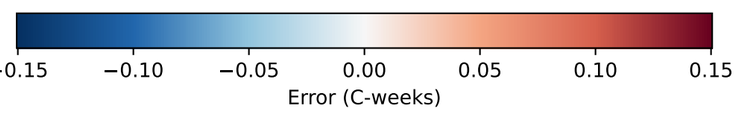
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±0.15]

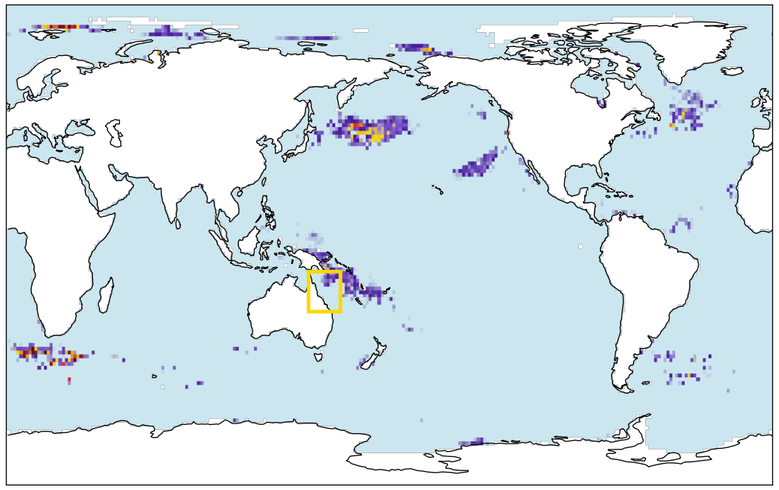




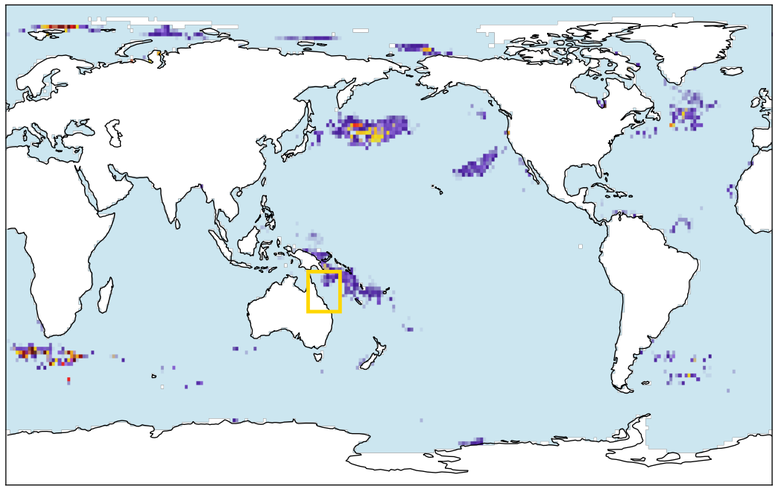


DHW Chain init: 2022-12-27 - Test (C-weeks)

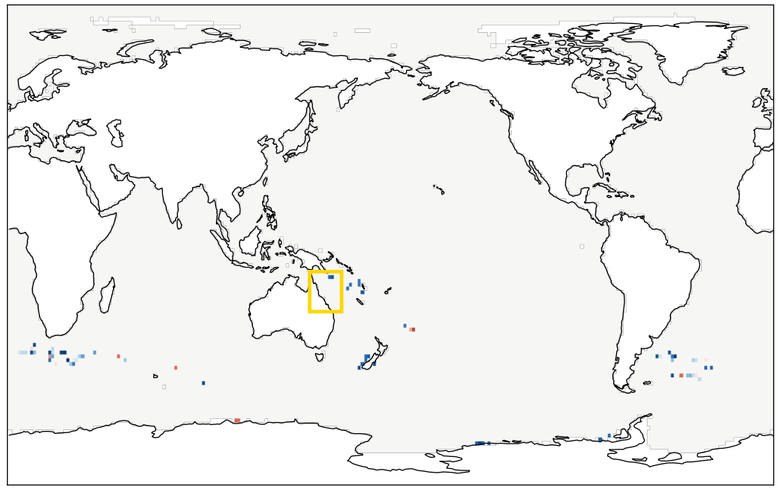
Actual DHW - Step 1



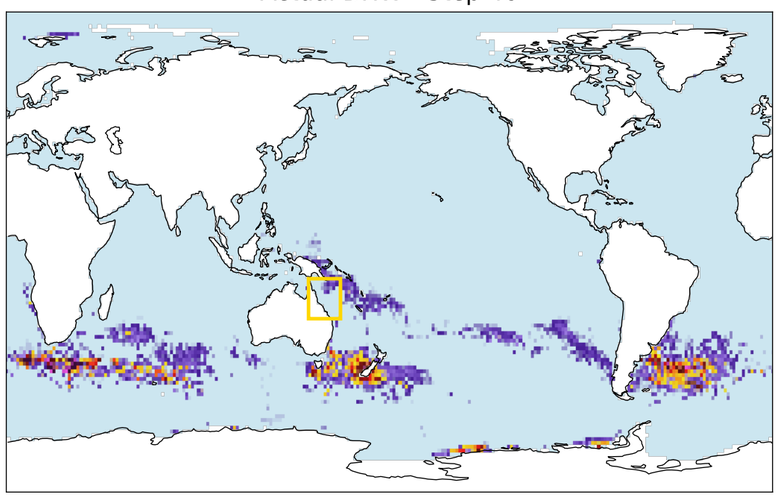
Predicted DHW - Step 1



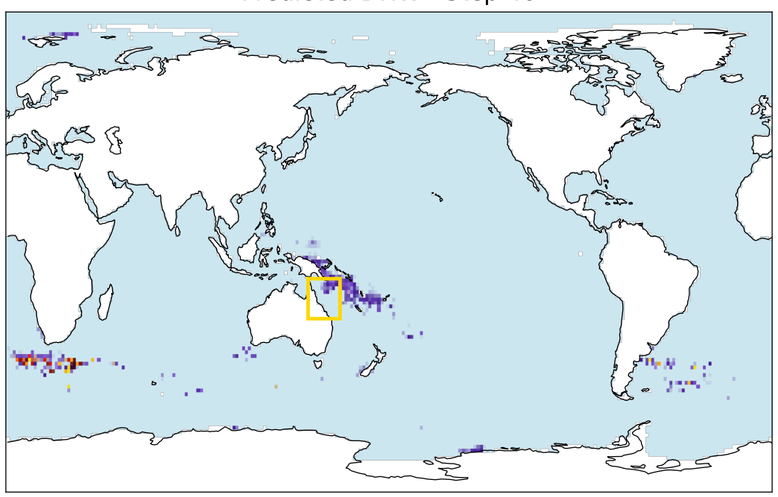
Error (Pred - Actual) - Step 1 [scale ±0.24]



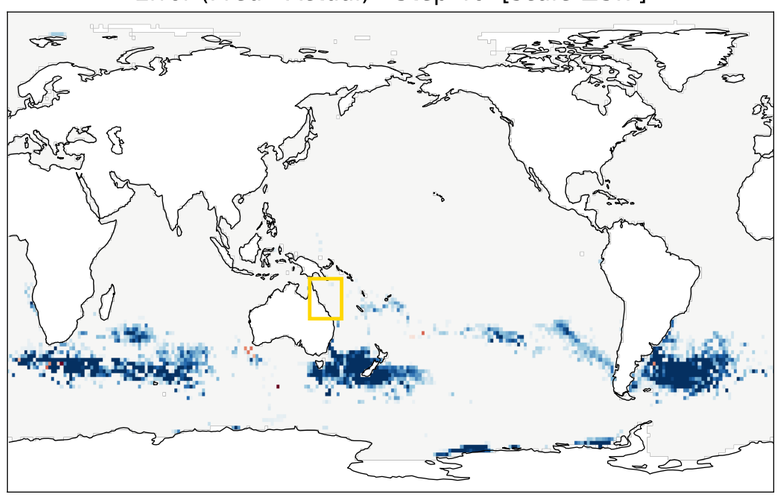
Actual DHW - Step 46



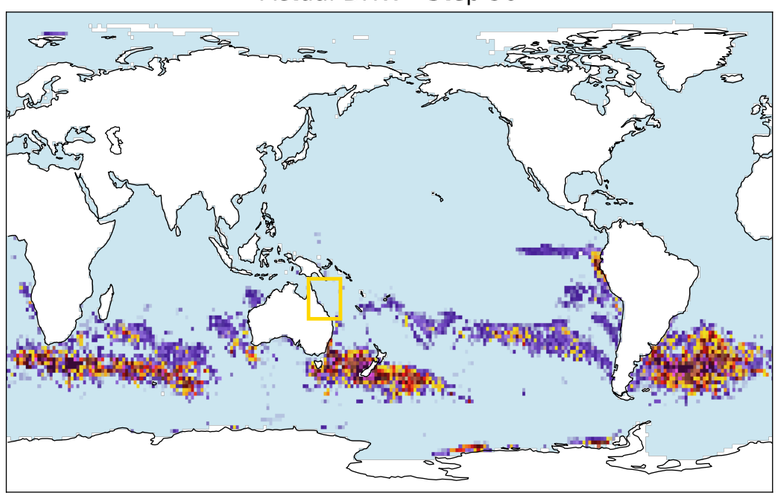
Predicted DHW - Step 46



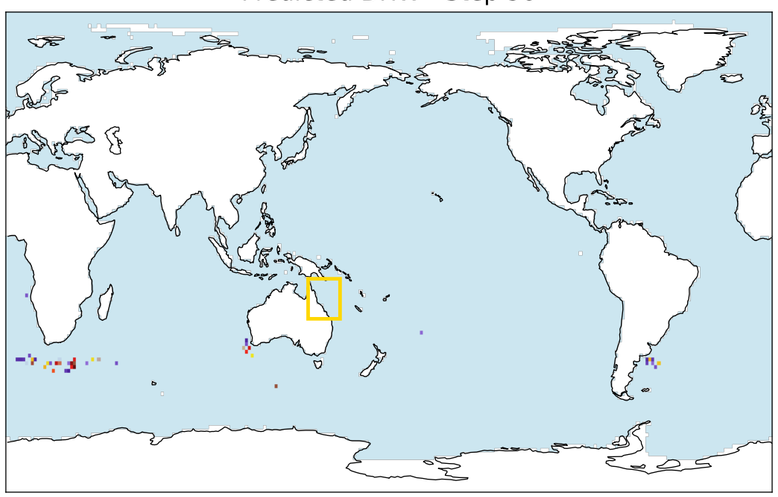
Error (Pred - Actual) - Step 46 [scale ±3.7]



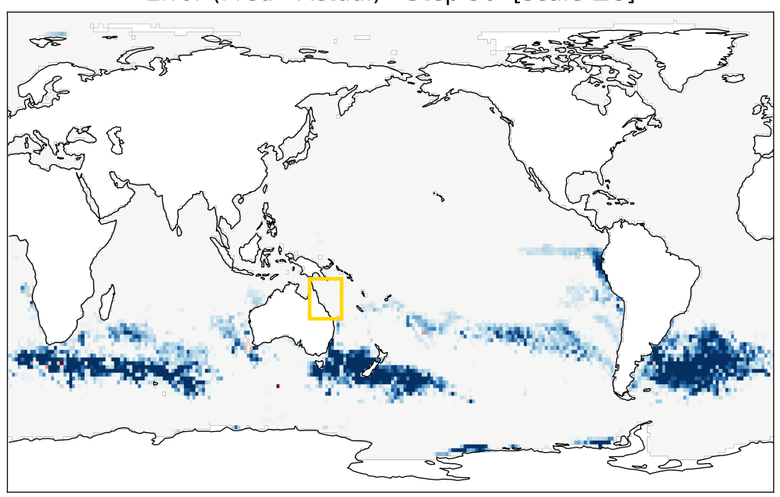
Actual DHW - Step 90

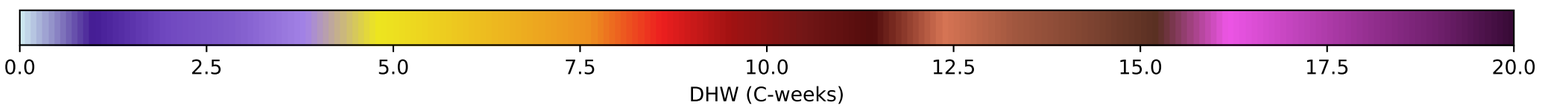


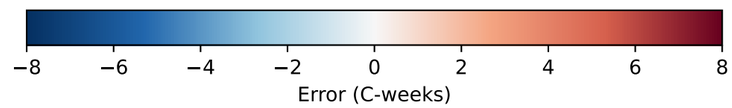
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

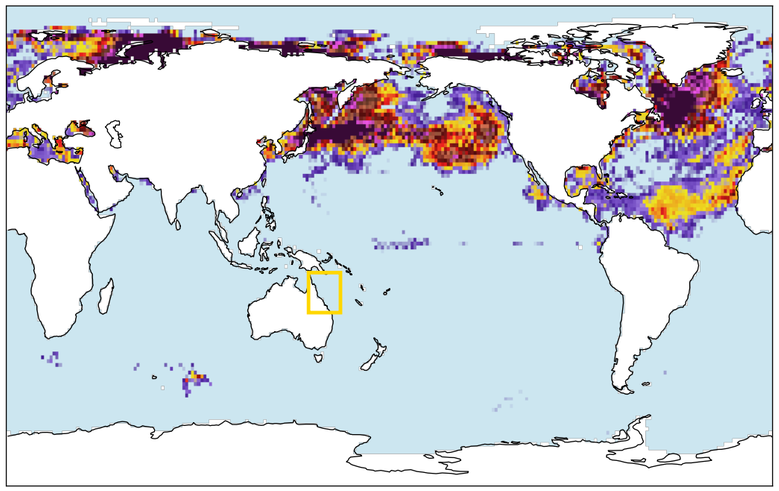




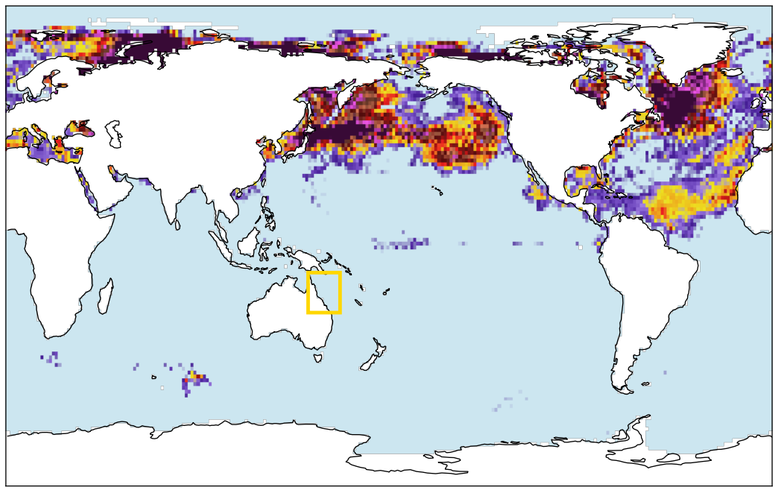


DHW Chain init: 2023-09-23 - Test (C-weeks)

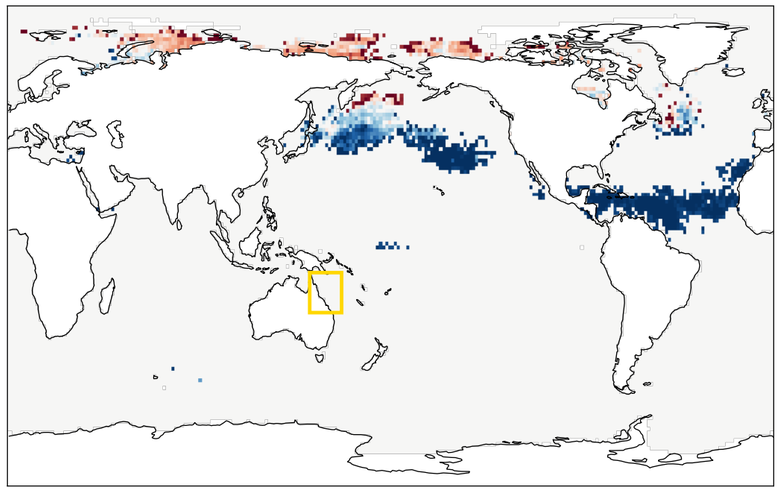
Actual DHW - Step 1



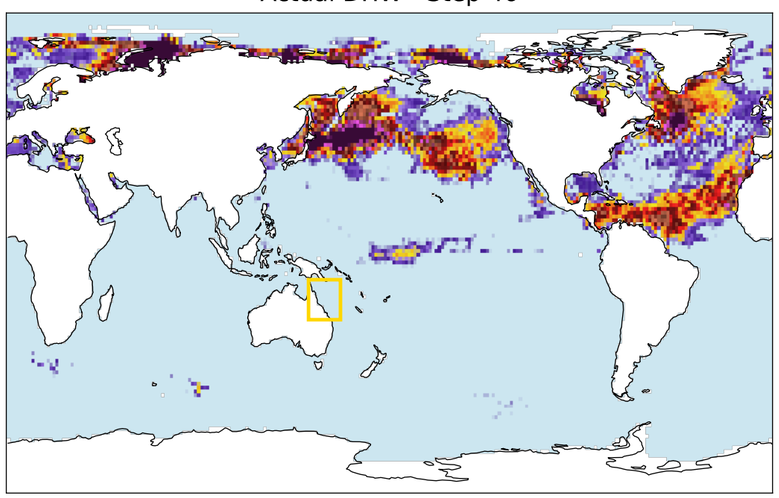
Predicted DHW - Step 1



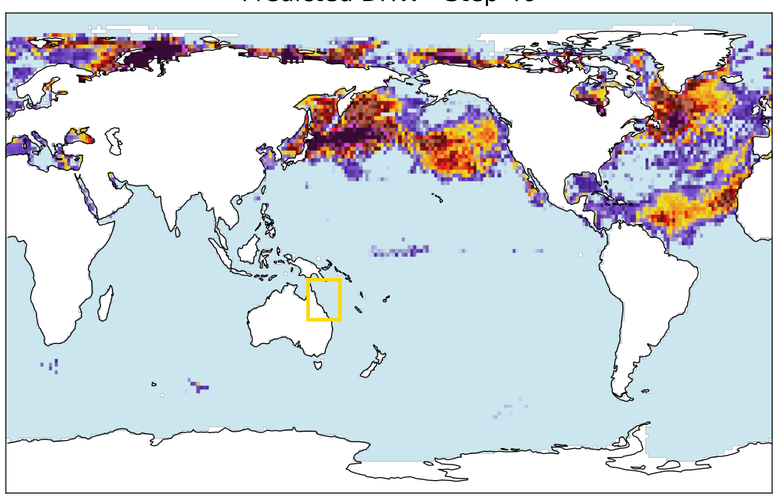
Error (Pred - Actual) - Step 1 [scale ±0.17]



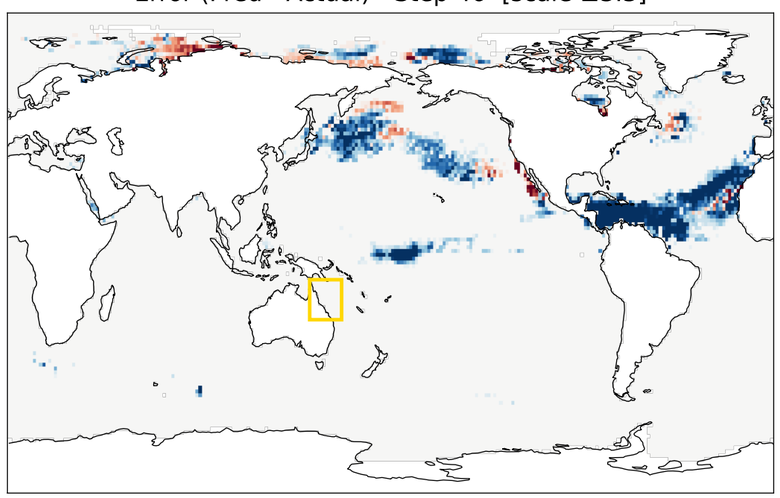
Actual DHW - Step 46



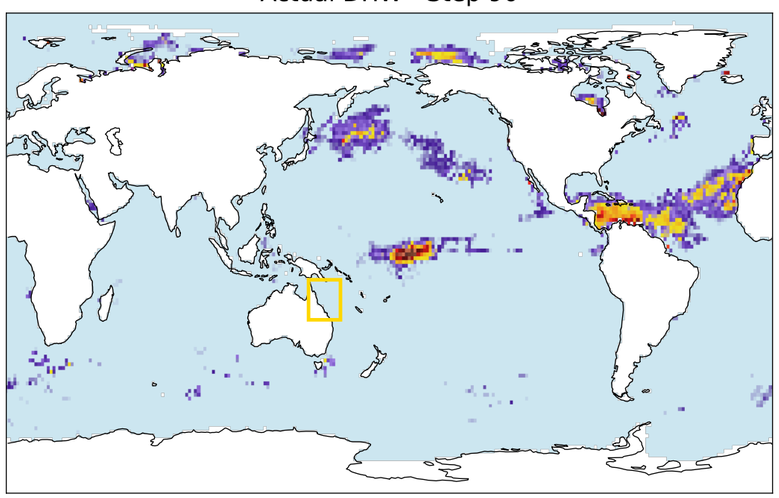
Predicted DHW - Step 46



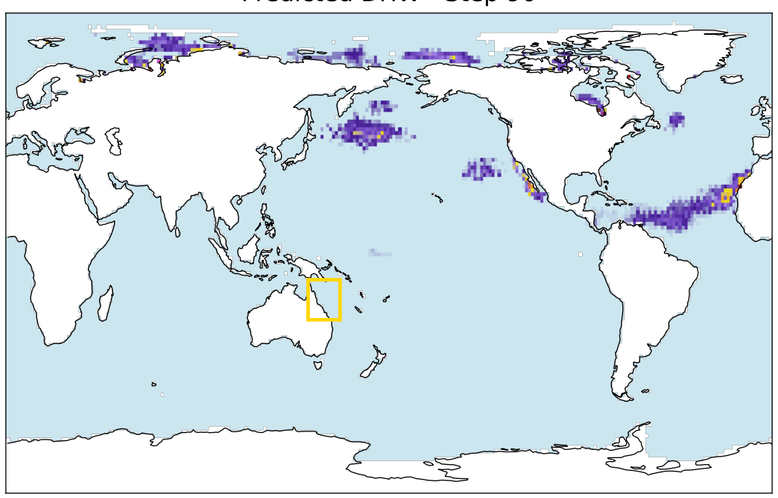
Error (Pred - Actual) - Step 46 [scale ±3.3]



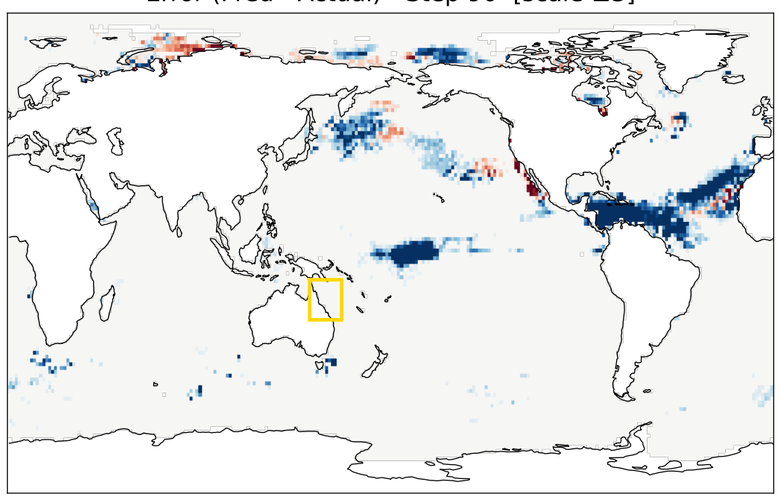
Actual DHW - Step 90

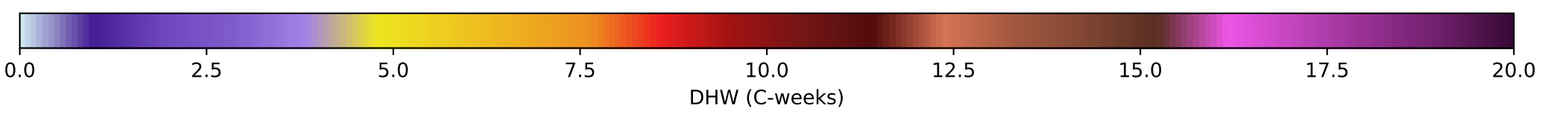


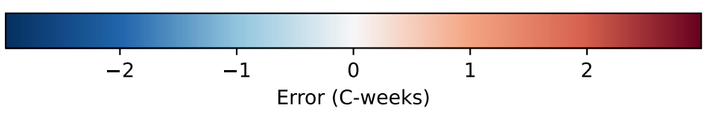
Predicted DHW - Step 90



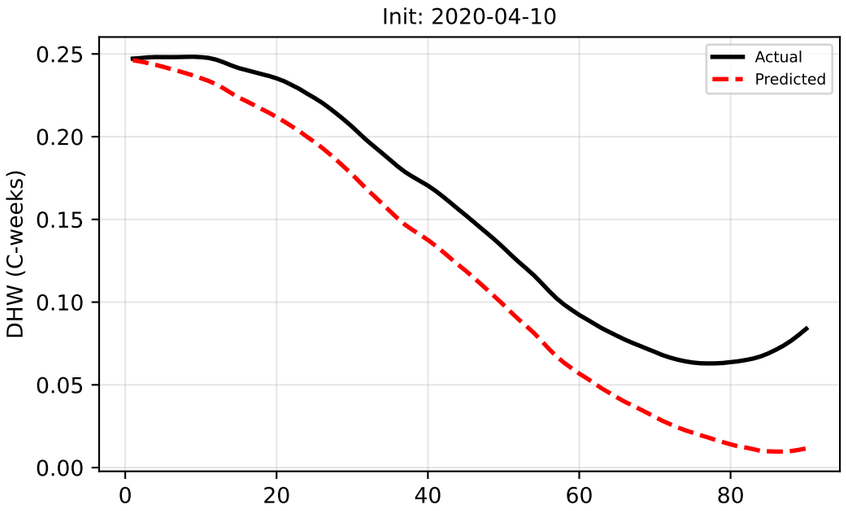
Error (Pred - Actual) - Step 90 [scale ±3]

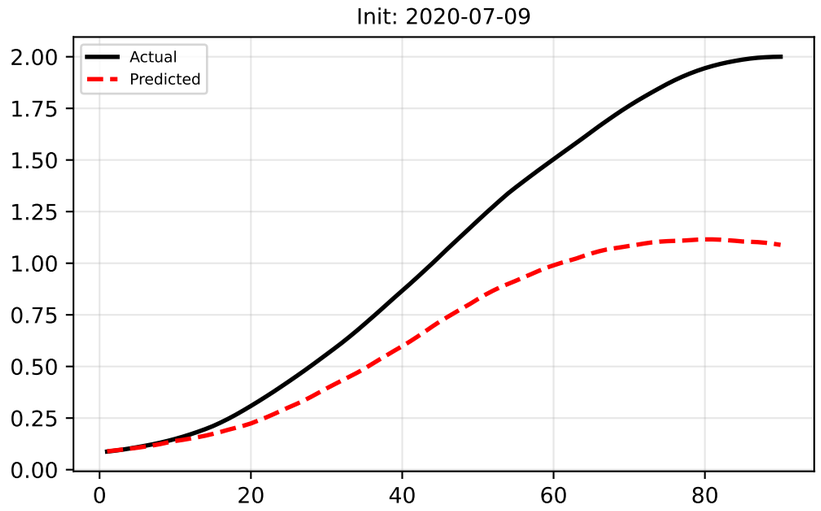


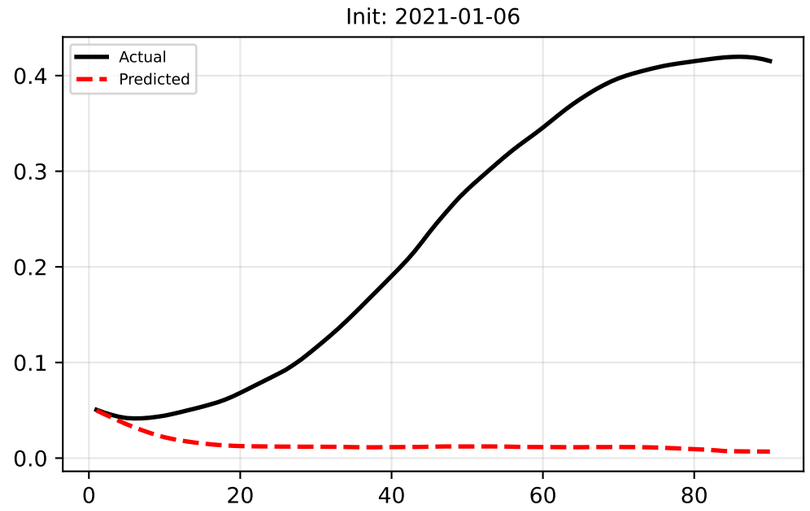


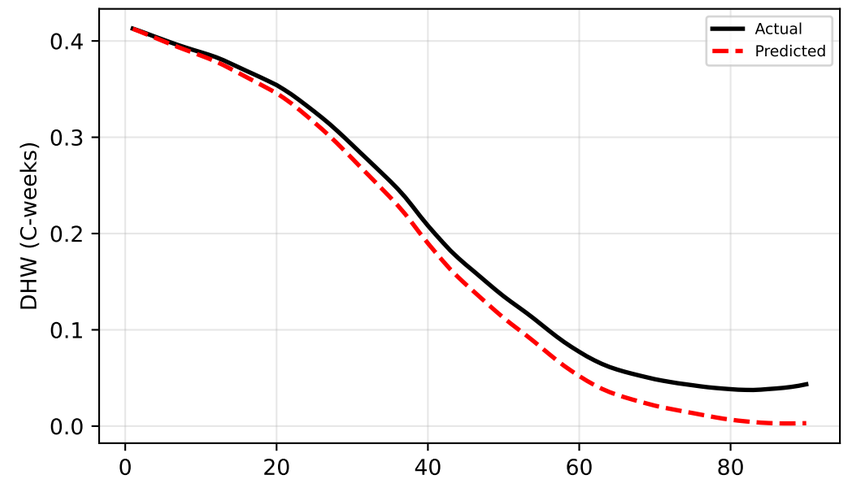


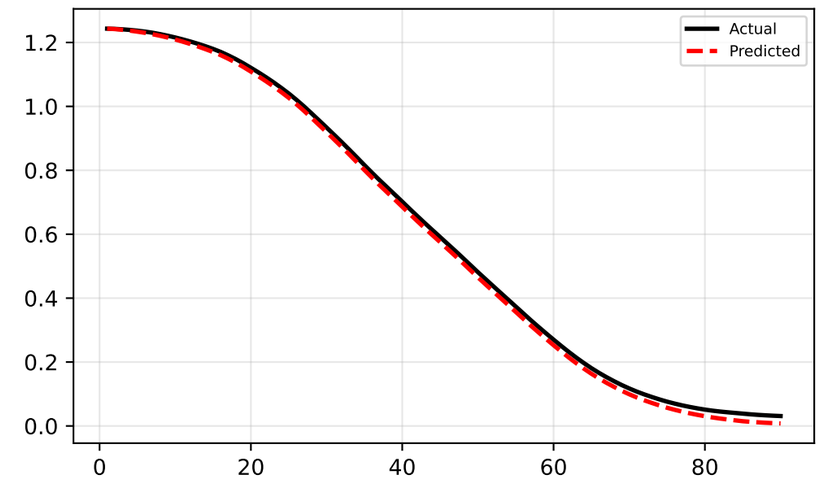
Global-Mean DHW - Test (C-weeks)

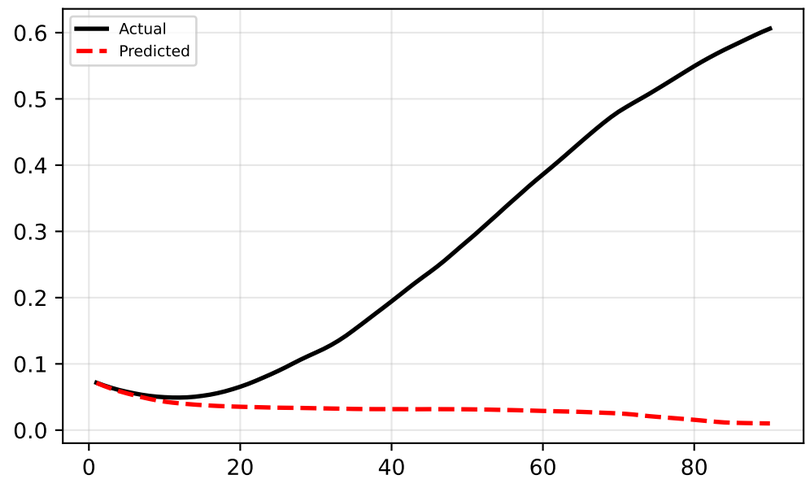


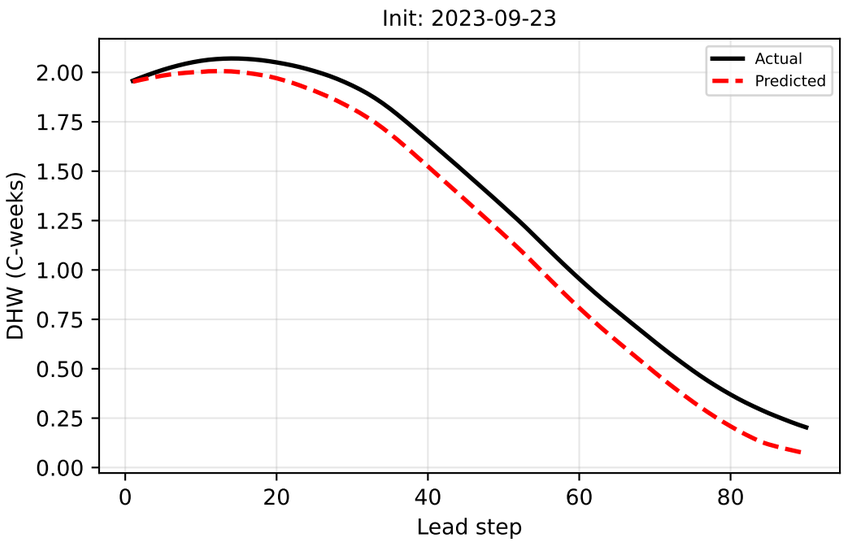




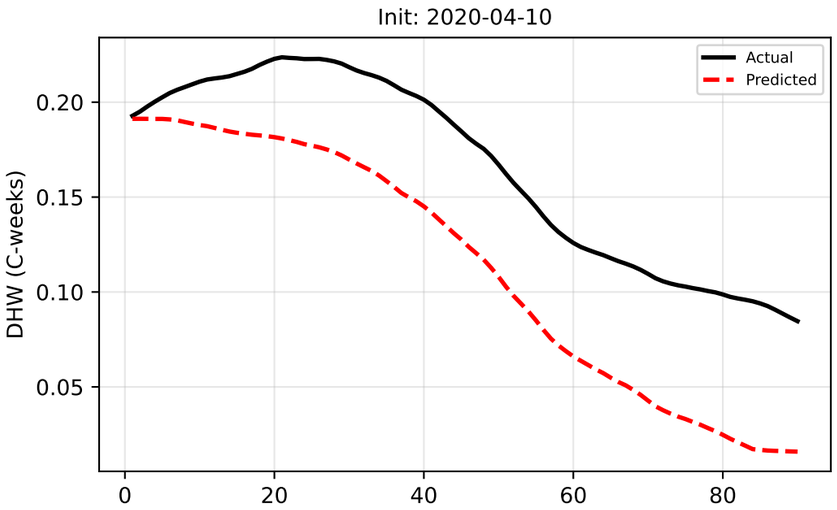


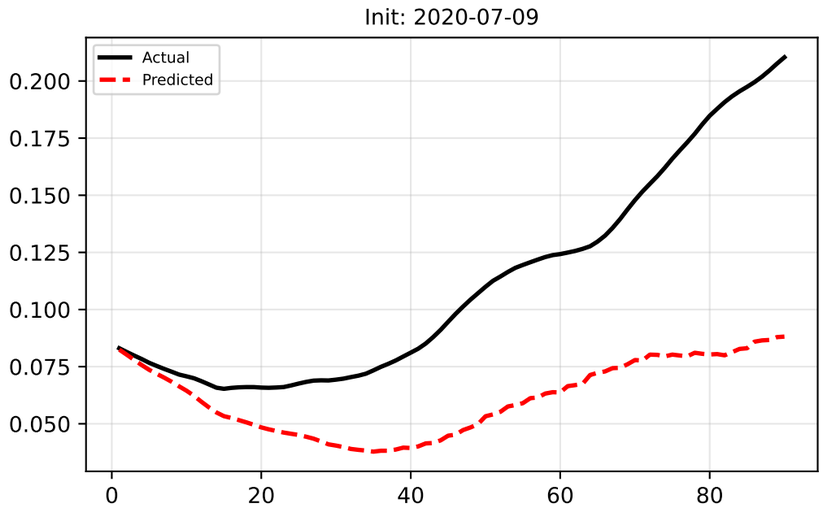


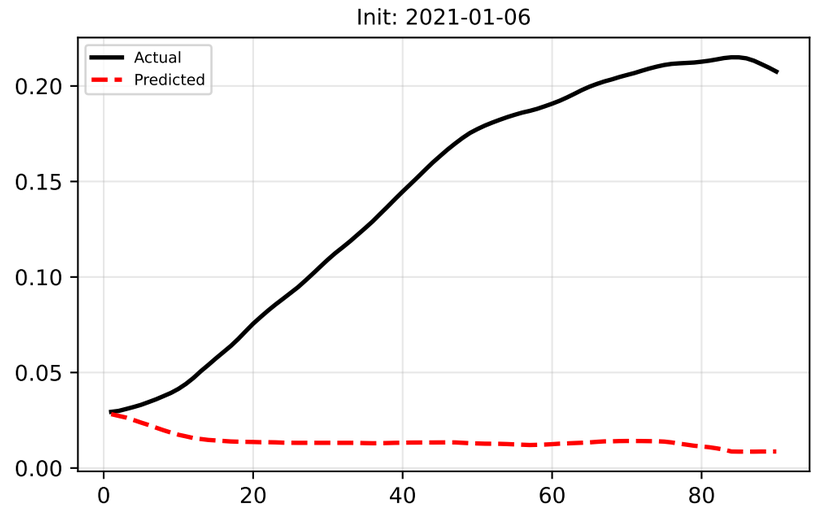


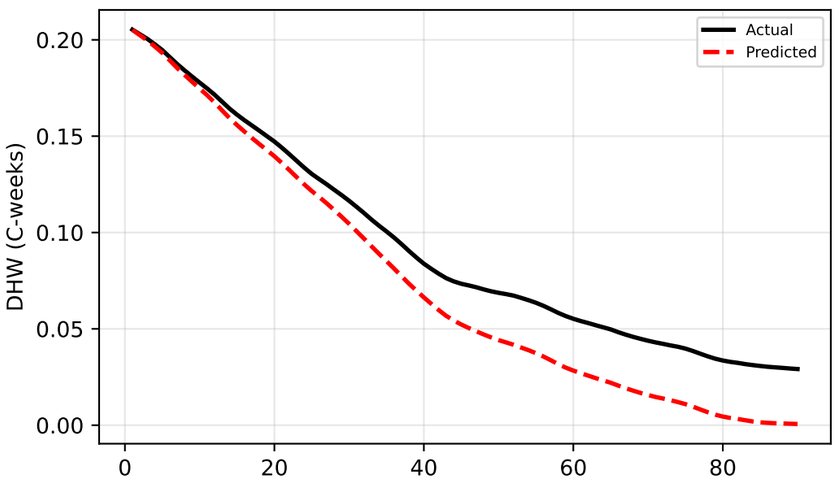


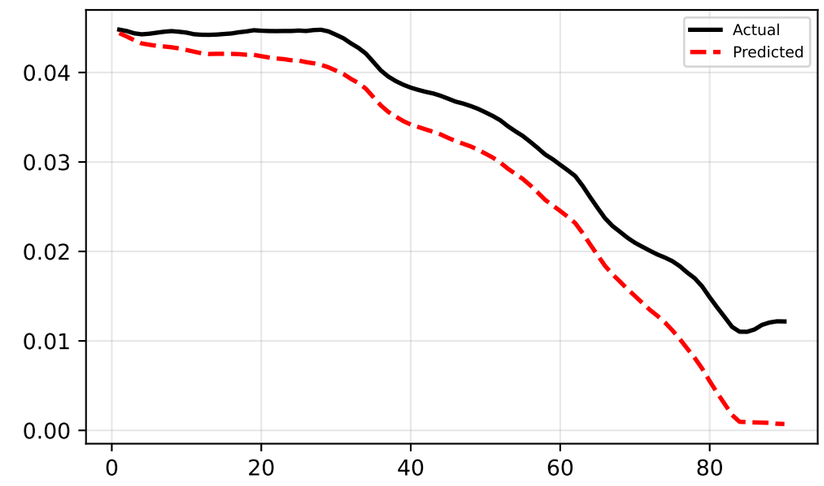
Tropical-Mean DHW (|lat|<=30) - Test (C-weeks)

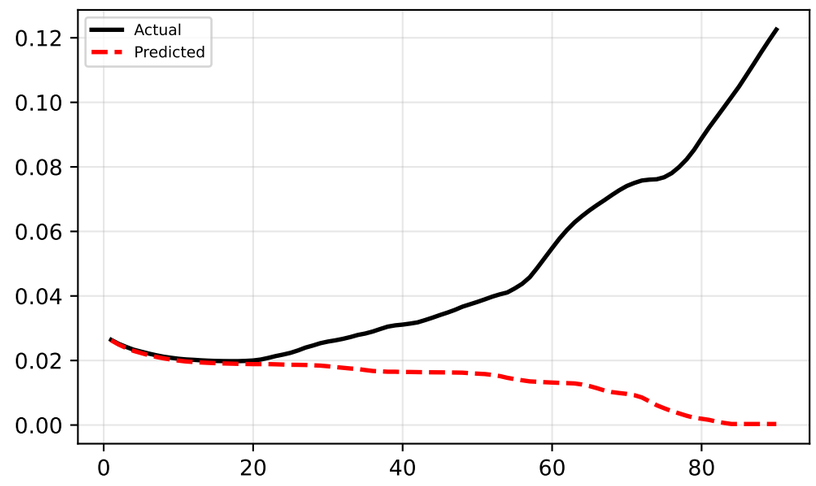


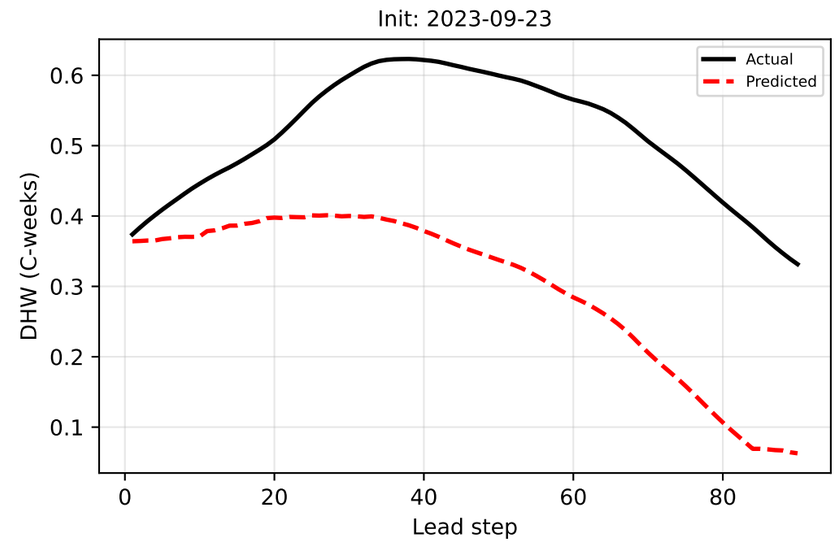




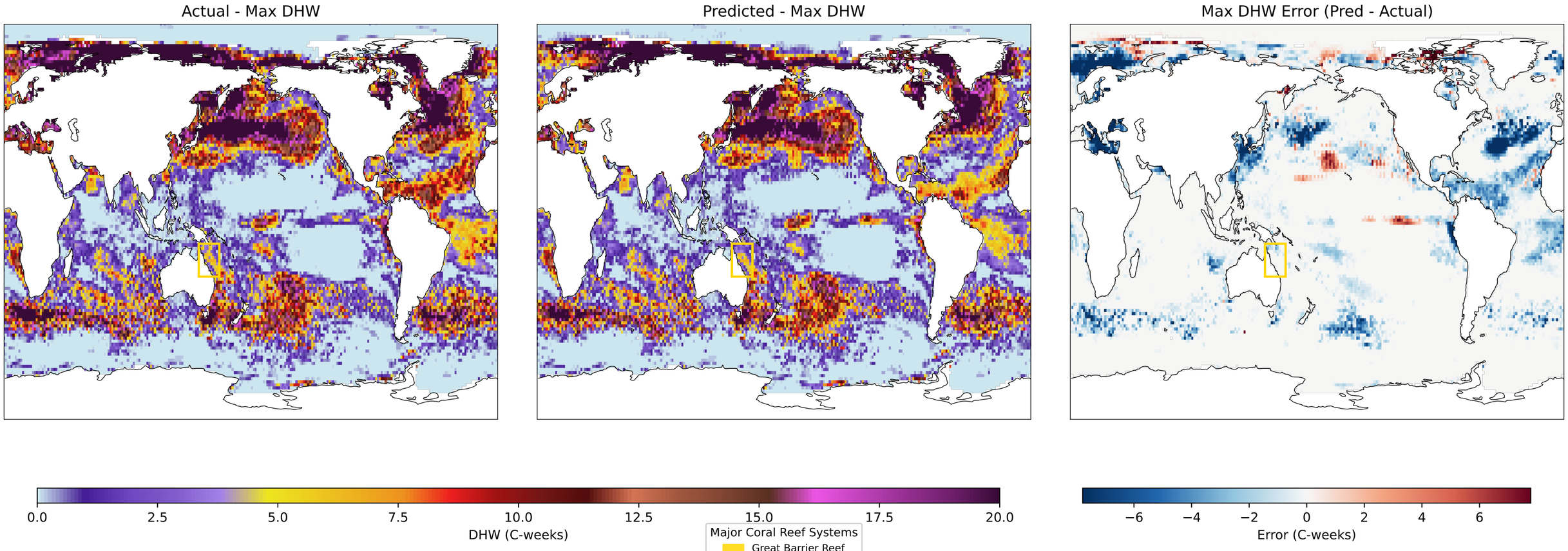


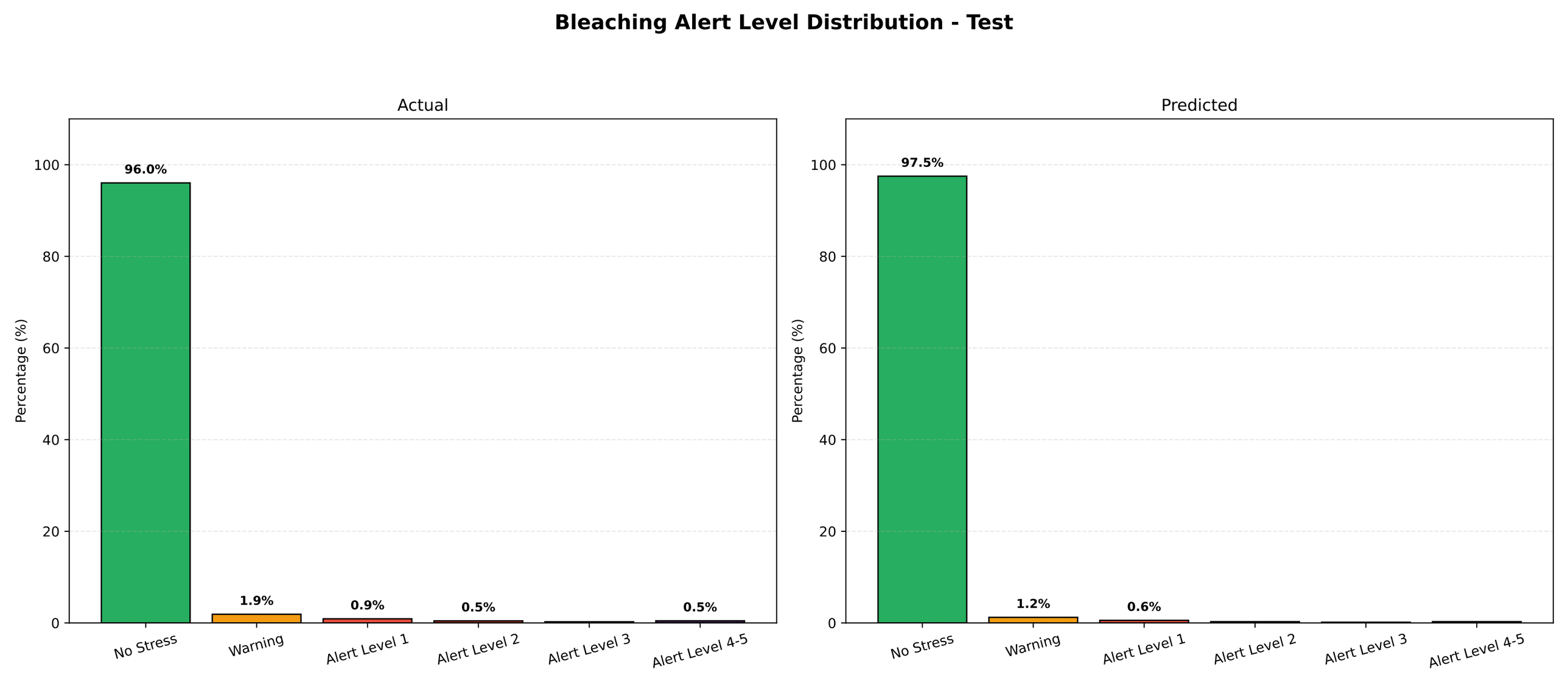






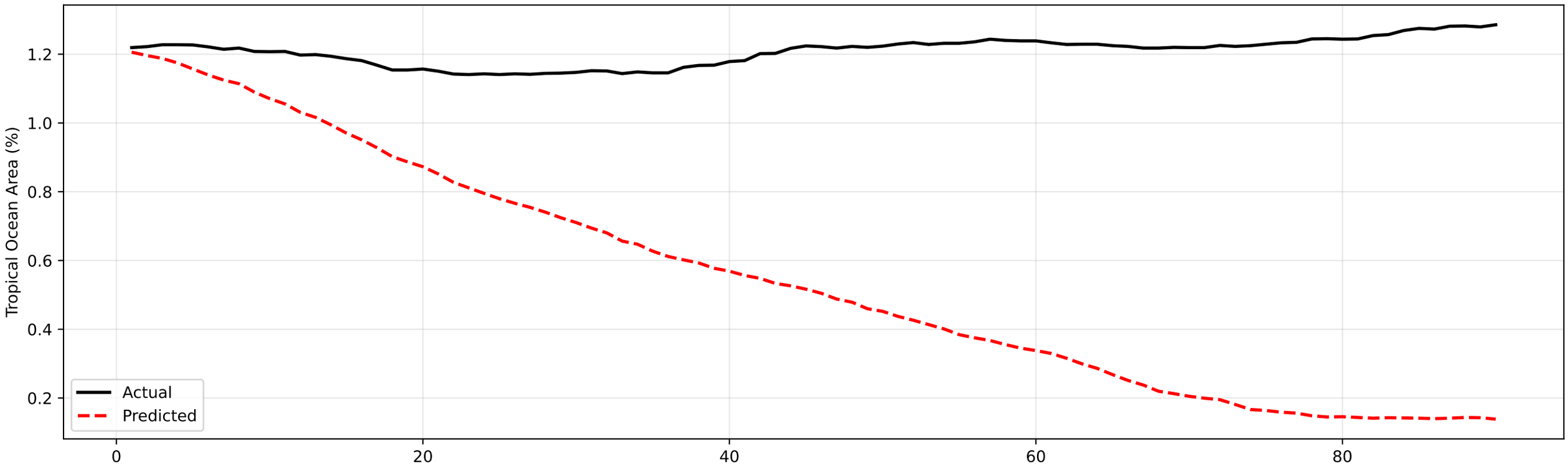
Peak DHW (max over all chains & leads) - Test (C-weeks)



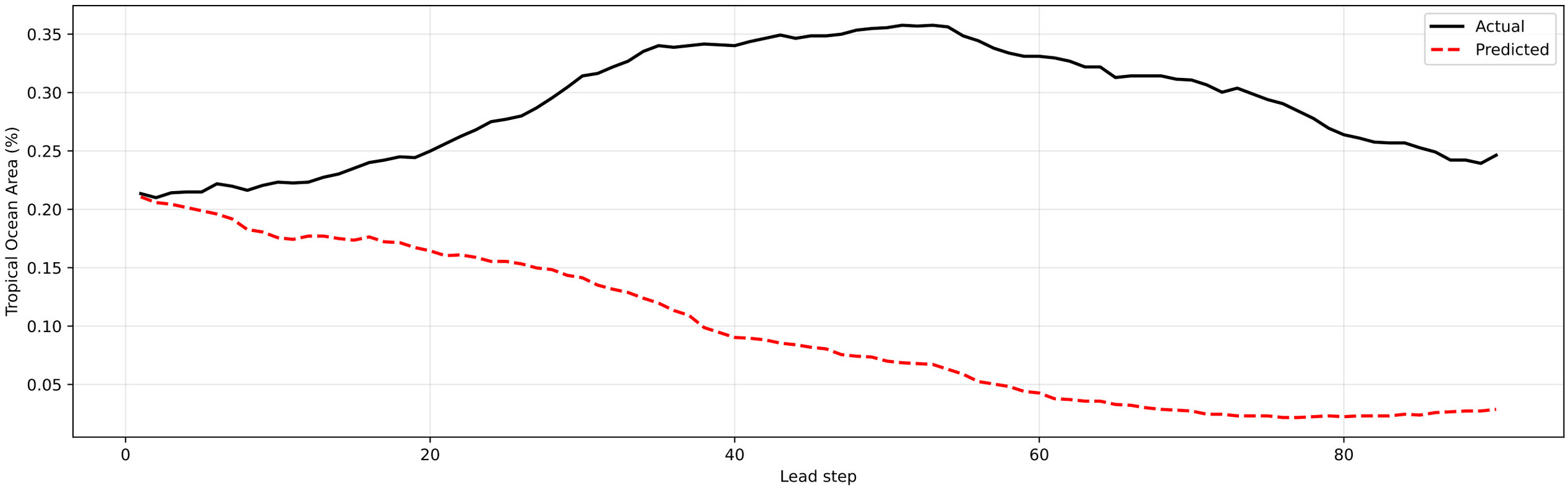

Bleaching Alert Level Distribution - Test
Actual
Predicted
Percentage (%)
0
20
40
60
80
100
96.0%
1.9%
0.9%
0.5%
0.5%
97.5%
1.2%
0.6%
No Stress
Warning
Alert Level 1
Alert Level 2
Alert Level 3
Alert Level 4-5

Tropical Ocean Area Under Bleaching Stress - Test

DHW >= 4 C-weeks (Alert Level 1 DHW threshold)



DHW >= 8 C-weeks (Alert Level 2 DHW threshold)



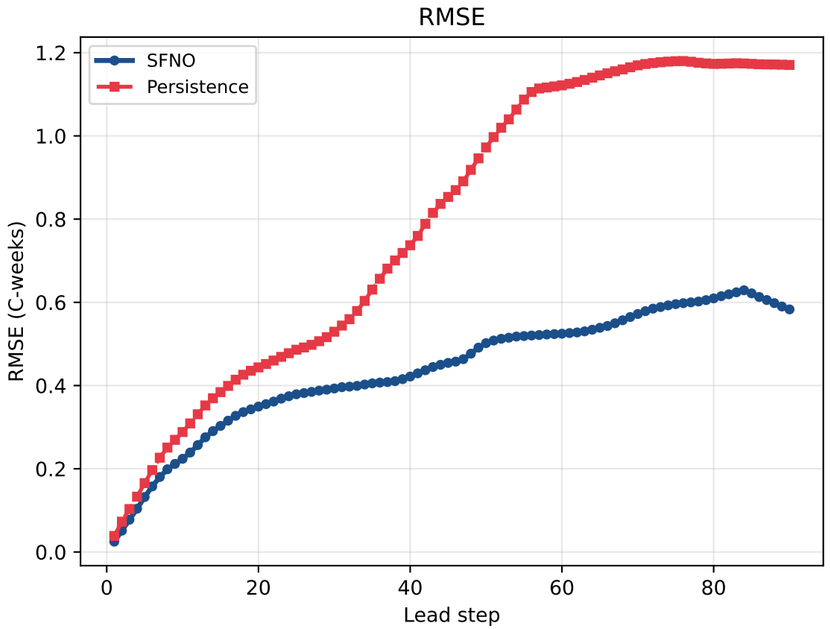
Great Barrier Reef Analysis - Test

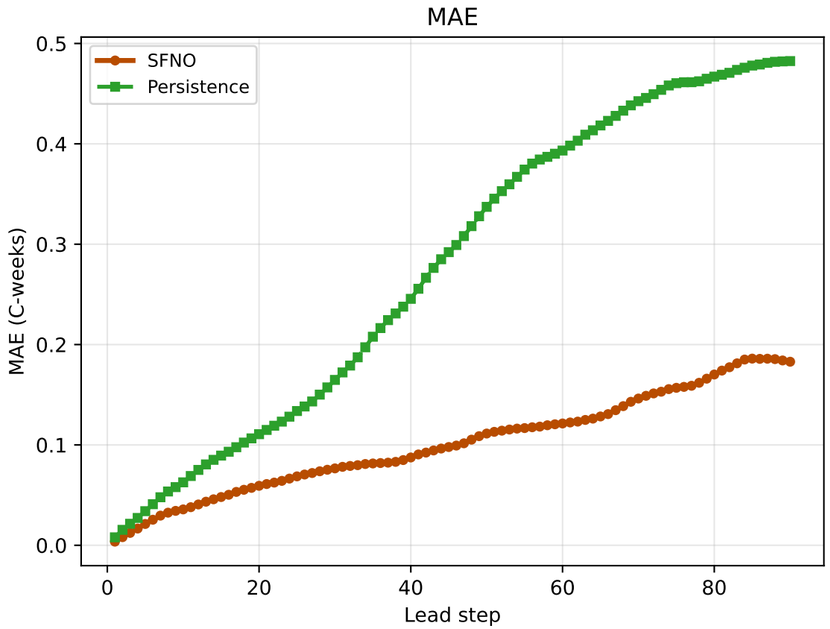
```
                    GREAT BARRIER REEF - TEST
Lat -25.0 deg to -10.0 deg  |  Lon 142.0 deg to 157.0 deg  (0-360)
                    Grid: 11 lat x 11 lon points
```

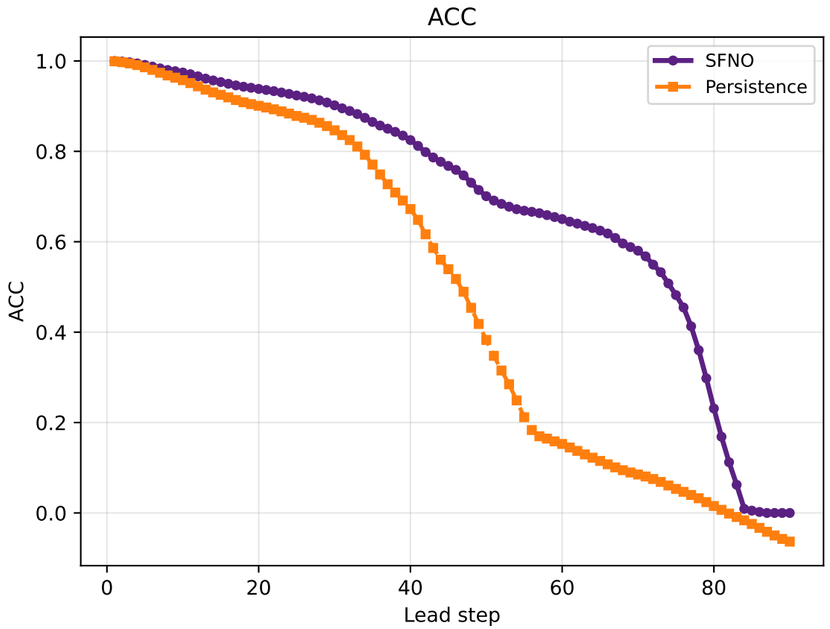
DHW Metrics by Lead Step - Test - GBR (18 chains scored at every lead)

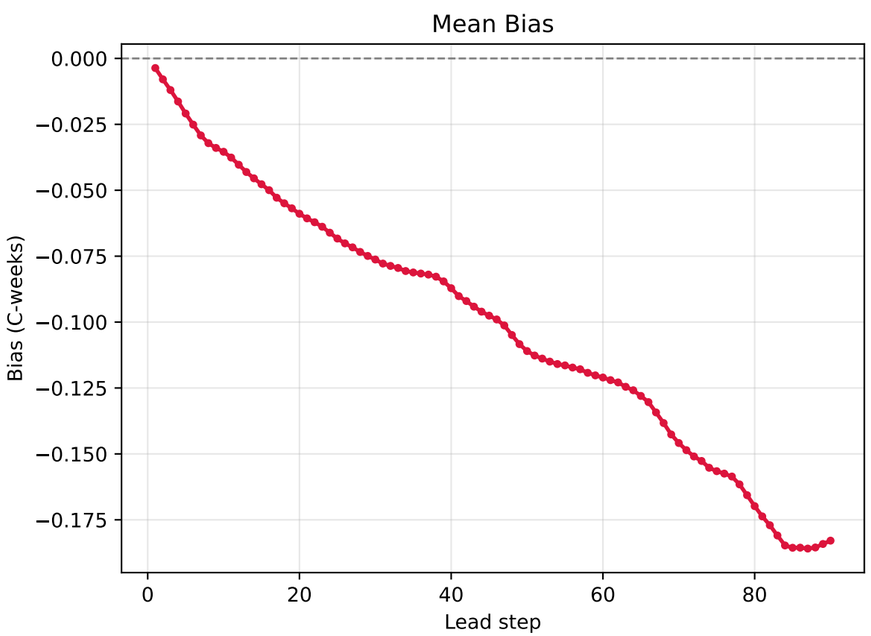
Lead Step	SFNO RMSE	Persist RMSE	Skill %	SFNO MAE	Persist MAE	SFNO Bias	SFNO ACC	Persist ACC	SFNO R²
1	0.0249	0.0386	35.4%	0.0036	0.0079	-0.0036	0.9997	0.9992	0.9994
2	0.0512	0.0729	29.7%	0.0081	0.0154	-0.0079	0.9987	0.9973	0.9974
3	0.0776	0.1032	24.8%	0.0123	0.0214	-0.0120	0.9969	0.9945	0.9940
4	0.1040	0.1329	21.8%	0.0167	0.0272	-0.0163	0.9945	0.9909	0.9894
5	0.1321	0.1655	20.2%	0.0212	0.0340	-0.0209	0.9911	0.9860	0.9829
6	0.1576	0.1968	19.9%	0.0255	0.0409	-0.0251	0.9874	0.9802	0.9758
7	0.1803	0.2266	20.5%	0.0295	0.0479	-0.0292	0.9836	0.9738	0.9685
8	0.1988	0.2509	20.8%	0.0325	0.0536	-0.0321	0.9800	0.9679	0.9617
9	0.2117	0.2696	21.5%	0.0343	0.0580	-0.0339	0.9773	0.9630	0.9566
10	0.2240	0.2883	22.3%	0.0358	0.0627	-0.0354	0.9744	0.9575	0.9511
11	0.2390	0.3090	22.6%	0.0380	0.0690	-0.0376	0.9706	0.9510	0.9440
12	0.2569	0.3310	22.4%	0.0407	0.0751	-0.0403	0.9660	0.9438	0.9353
13	0.2757	0.3523	21.7%	0.0434	0.0806	-0.0431	0.9609	0.9365	0.9259
14	0.2905	0.3694	21.4%	0.0458	0.0851	-0.0455	0.9569	0.9305	0.9183
15	0.3030	0.3842	21.1%	0.0481	0.0894	-0.0477	0.9533	0.9250	0.9116
16	0.3156	0.3991	20.9%	0.0503	0.0932	-0.0500	0.9495	0.9194	0.9046
17	0.3273	0.4140	20.9%	0.0532	0.0977	-0.0528	0.9459	0.9135	0.8980
18	0.3362	0.4260	21.1%	0.0553	0.1023	-0.0550	0.9430	0.9086	0.8926
19	0.3429	0.4355	21.3%	0.0572	0.1065	-0.0569	0.9406	0.9044	0.8882
20	0.3494	0.4437	21.2%	0.0593	0.1107	-0.0589	0.9381	0.9007	0.8837
21	0.3554	0.4518	21.3%	0.0610	0.1149	-0.0607	0.9358	0.8970	0.8795
22	0.3615	0.4602	21.4%	0.0625	0.1191	-0.0621	0.9333	0.8929	0.8750
23	0.3685	0.4690	21.4%	0.0642	0.1232	-0.0638	0.9302	0.8885	0.8694
24	0.3743	0.4777	21.6%	0.0664	0.1281	-0.0661	0.9271	0.8836	0.8640
25	0.3791	0.4861	22.0%	0.0687	0.1338	-0.0683	0.9236	0.8783	0.8579
26	0.3819	0.4916	22.3%	0.0705	0.1383	-0.0702	0.9207	0.8743	0.8529
27	0.3847	0.4979	22.7%	0.0720	0.1433	-0.0717	0.9172	0.8695	0.8468
28	0.3878	0.5066	23.4%	0.0737	0.1501	-0.0734	0.9129	0.8632	0.8393
29	0.3901	0.5163	24.4%	0.0753	0.1573	-0.0749	0.9080	0.8559	0.8309
30	0.3930	0.5294	25.8%	0.0766	0.1648	-0.0763	0.9018	0.8465	0.8204
31	0.3961	0.5442	27.2%	0.0782	0.1722	-0.0778	0.8952	0.8360	0.8093
32	0.3975	0.5596	29.0%	0.0790	0.1791	-0.0787	0.8892	0.8249	0.7992
33	0.3992	0.5792	31.1%	0.0798	0.1873	-0.0795	0.8825	0.8106	0.7880
34	0.4025	0.6035	33.3%	0.0810	0.1973	-0.0806	0.8742	0.7924	0.7742
35	0.4053	0.6307	35.7%	0.0815	0.2078	-0.0812	0.8650	0.7708	0.7594
36	0.4071	0.6566	38.0%	0.0819	0.2164	-0.0816	0.8570	0.7489	0.7466
37	0.4084	0.6811	40.0%	0.0823	0.2244	-0.0820	0.8501	0.7272	0.7356
38	0.4106	0.7006	41.4%	0.0831	0.2310	-0.0828	0.8431	0.7089	0.7246
39	0.4154	0.7187	42.2%	0.0849	0.2378	-0.0846	0.8349	0.6912	0.7118
40	0.4216	0.7372	42.8%	0.0875	0.2456	-0.0871	0.8249	0.6723	0.6962
41	0.4293	0.7596	43.5%	0.0905	0.2556	-0.0902	0.8121	0.6484	0.6765
42	0.4368	0.7886	44.6%	0.0924	0.2666	-0.0920	0.7984	0.6164	0.6551
43	0.4446	0.8150	45.4%	0.0945	0.2764	-0.0942	0.7864	0.5862	0.6363
44	0.4499	0.8367	46.2%	0.0964	0.2850	-0.0960	0.7769	0.5603	0.6211
45	0.4544	0.8533	46.7%	0.0979	0.2921	-0.0976	0.7677	0.5393	0.6064
46	0.4576	0.8698	47.4%	0.0994	0.2992	-0.0990	0.7590	0.5176	0.5926
47	0.4635	0.8909	48.0%	0.1016	0.3081	-0.1013	0.7469	0.4893	0.5738
48	0.4766	0.9184	48.1%	0.1053	0.3180	-0.1049	0.7307	0.4538	0.5492
49	0.4912	0.9460	48.1%	0.1087	0.3279	-0.1084	0.7145	0.4181	0.5248
50	0.5019	0.9723	48.4%	0.1113	0.3372	-0.1110	0.7008	0.3828	0.5041
51	0.5082	0.9973	49.0%	0.1130	0.3453	-0.1127	0.6911	0.3476	0.4892
52	0.5124	1.0196	49.7%	0.1142	0.3529	-0.1138	0.6837	0.3152	0.4773
53	0.5151	1.0399	50.5%	0.1153	0.3597	-0.1150	0.6776	0.2847	0.4674
54	0.5178	1.0634	51.3%	0.1163	0.3671	-0.1159	0.6720	0.2490	0.4581
55	0.5190	1.0873	52.3%	0.1168	0.3743	-0.1164	0.6686	0.2118	0.4518
56	0.5203	1.1055	52.9%	0.1176	0.3803	-0.1172	0.6663	0.1835	0.4473
57	0.5215	1.1139	53.2%	0.1182	0.3844	-0.1179	0.6632	0.1696	0.4425
58	0.5227	1.1164	53.2%	0.1196	0.3870	-0.1192	0.6590	0.1644	0.4370
59	0.5237	1.1190	53.2%	0.1206	0.3902	-0.1202	0.6547	0.1585	0.4309
60	0.5246	1.1215	53.2%	0.1214	0.3933	-0.1210	0.6502	0.1526	0.4243
61	0.5262	1.1253	53.2%	0.1224	0.3982	-0.1220	0.6444	0.1448	0.4155
62	0.5277	1.1295	53.3%	0.1232	0.4031	-0.1229	0.6401	0.1373	0.4086
63	0.5303	1.1345	53.3%	0.1249	0.4091	-0.1245	0.6351	0.1295	0.4020
64	0.5343	1.1400	53.1%	0.1262	0.4135	-0.1259	0.6304	0.1223	0.3958
65	0.5388	1.1455	53.0%	0.1284	0.4183	-0.1280	0.6246	0.1151	0.3888
66	0.5432	1.1506	52.8%	0.1307	0.4229	-0.1304	0.6182	0.1077	0.3809
67	0.5497	1.1553	52.4%	0.1346	0.4278	-0.1343	0.6083	0.1009	0.3700
68	0.5570	1.1599	52.0%	0.1386	0.4330	-0.1383	0.5962	0.0946	0.3581
69	0.5647	1.1649	51.5%	0.1430	0.4384	-0.1426	0.5882	0.0895	0.3501
70	0.5720	1.1696	51.1%	0.1462	0.4425	-0.1459	0.5802	0.0850	0.3420
71	0.5788	1.1728	50.7%	0.1489	0.4457	-0.1486	0.5677	0.0807	0.3306
72	0.5849	1.1751	50.2%	0.1513	0.4495	-0.1510	0.5493	0.0751	0.3132
73	0.5887	1.1771	50.0%	0.1530	0.4538	-0.1527	0.5326	0.0686	0.2969
74	0.5928	1.1786	49.7%	0.1556	0.4582	-0.1553	0.5077	0.0608	0.2744
75	0.5957	1.1796	49.5%	0.1569	0.4604	-0.1566	0.4820	0.0533	0.2524
76	0.5980	1.1798	49.3%	0.1578	0.4614	-0.1575	0.4546	0.0467	0.2306
77	0.6000	1.1783	49.1%	0.1589	0.4614	-0.1586	0.4129	0.0398	0.1992
78	0.6020	1.1759	48.8%	0.1619	0.4623	-0.1616	0.3601	0.0324	0.1616
79	0.6053	1.1743	48.5%	0.1660	0.4649	-0.1657	0.2980	0.0239	0.1190
80	0.6095	1.1731	48.0%	0.1702	0.4670	-0.1698	0.2310	0.0154	0.0741
81	0.6148	1.1734	47.6%	0.1741	0.4688	-0.1737	0.1686	0.0069	0.0319
82	0.6194	1.1740	47.2%	0.1774	0.4708	-0.1771	0.1122	-0.0013	-0.0078
83	0.6241	1.1746	46.9%	0.1813	0.4737	-0.1809	0.0623	-0.0089	-0.0447
84	0.6290	1.1741	46.4%	0.1851	0.4758	-0.1847	0.0090	-0.0162	-0.0874
85	0.6216	1.1732	47.0%	0.1860	0.4780	-0.1856	0.0052	-0.0243	-0.0936
86	0.6129	1.1722	47.7%	0.1857	0.4791	-0.1856	0.0022	-0.0330	-0.0989
87	0.6056	1.1718	48.3%	0.1859	0.4807	-0.1859	-0.0000	-0.0417	-0.1040
88	0.5982	1.1717	48.9%	0.1855	0.4817	-0.1855	-0.0000	-0.0498	-0.1064
89	0.5900	1.1712	49.6%	0.1842	0.4821	-0.1842	-0.0000	-0.0575	-0.1079
90	0.5830	1.1706	50.2%	0.1829	0.4825	-0.1829	0.0000	-0.0634	-0.1091

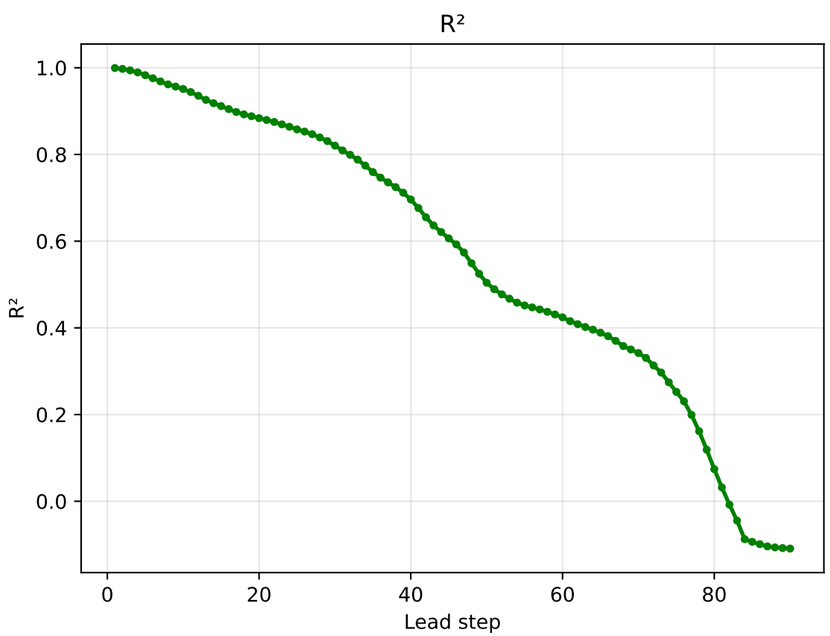
DHW Forecast Skill vs Lead Step - Test - GBR

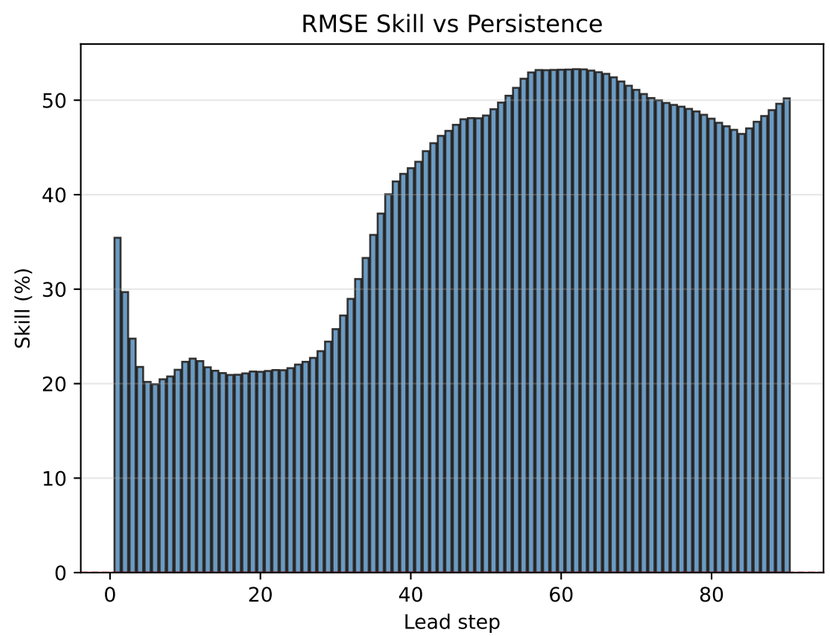


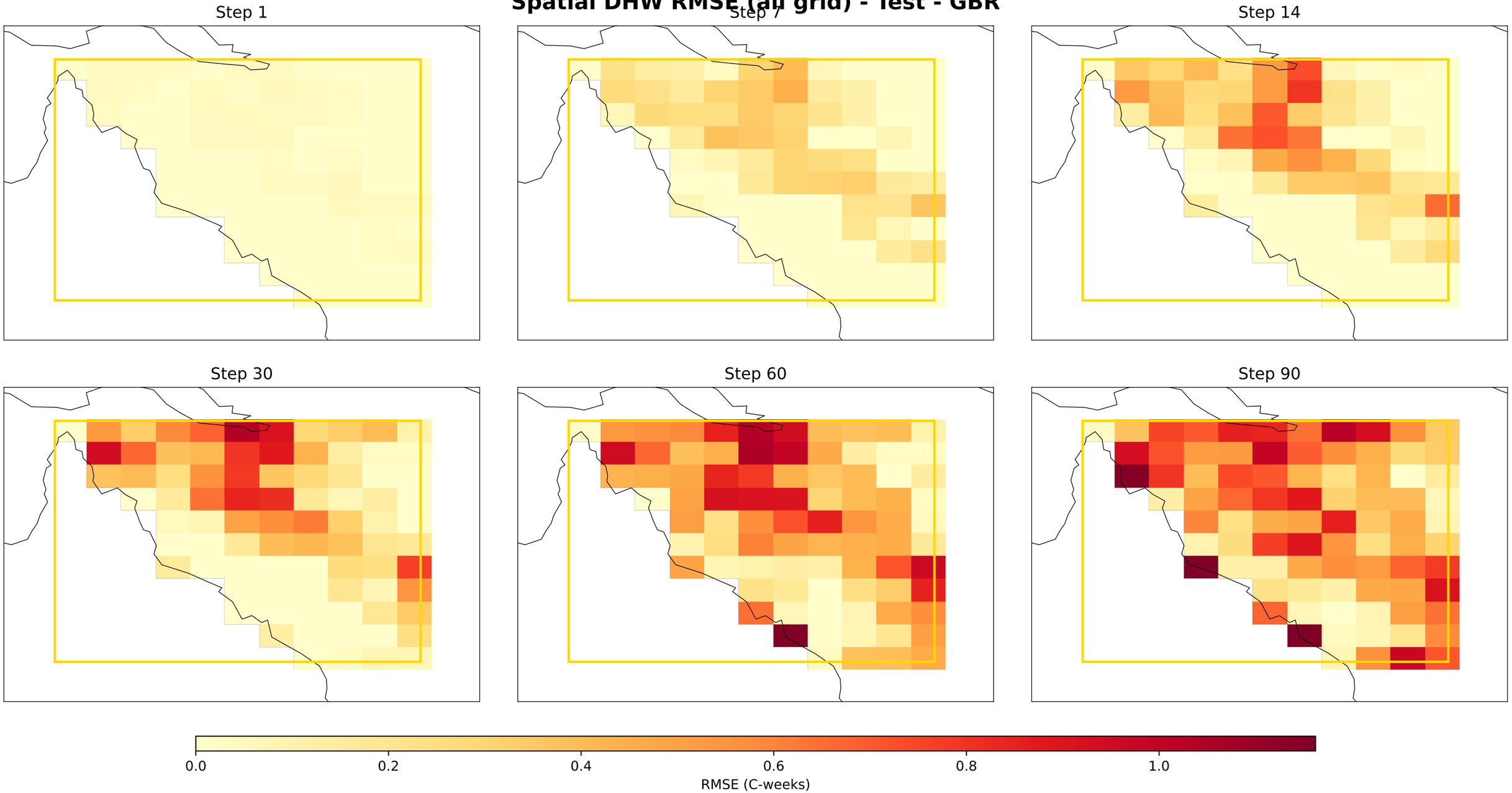


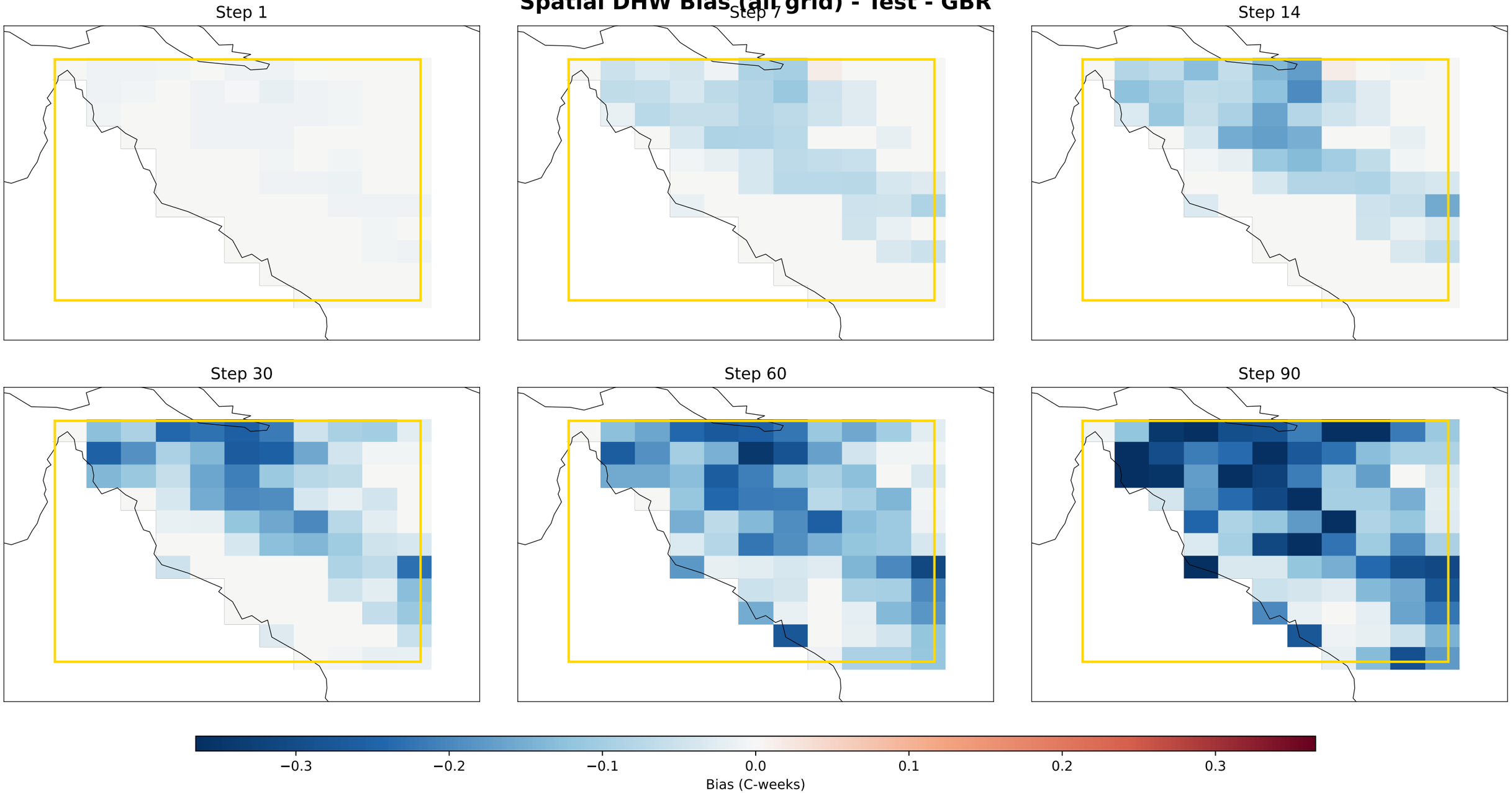


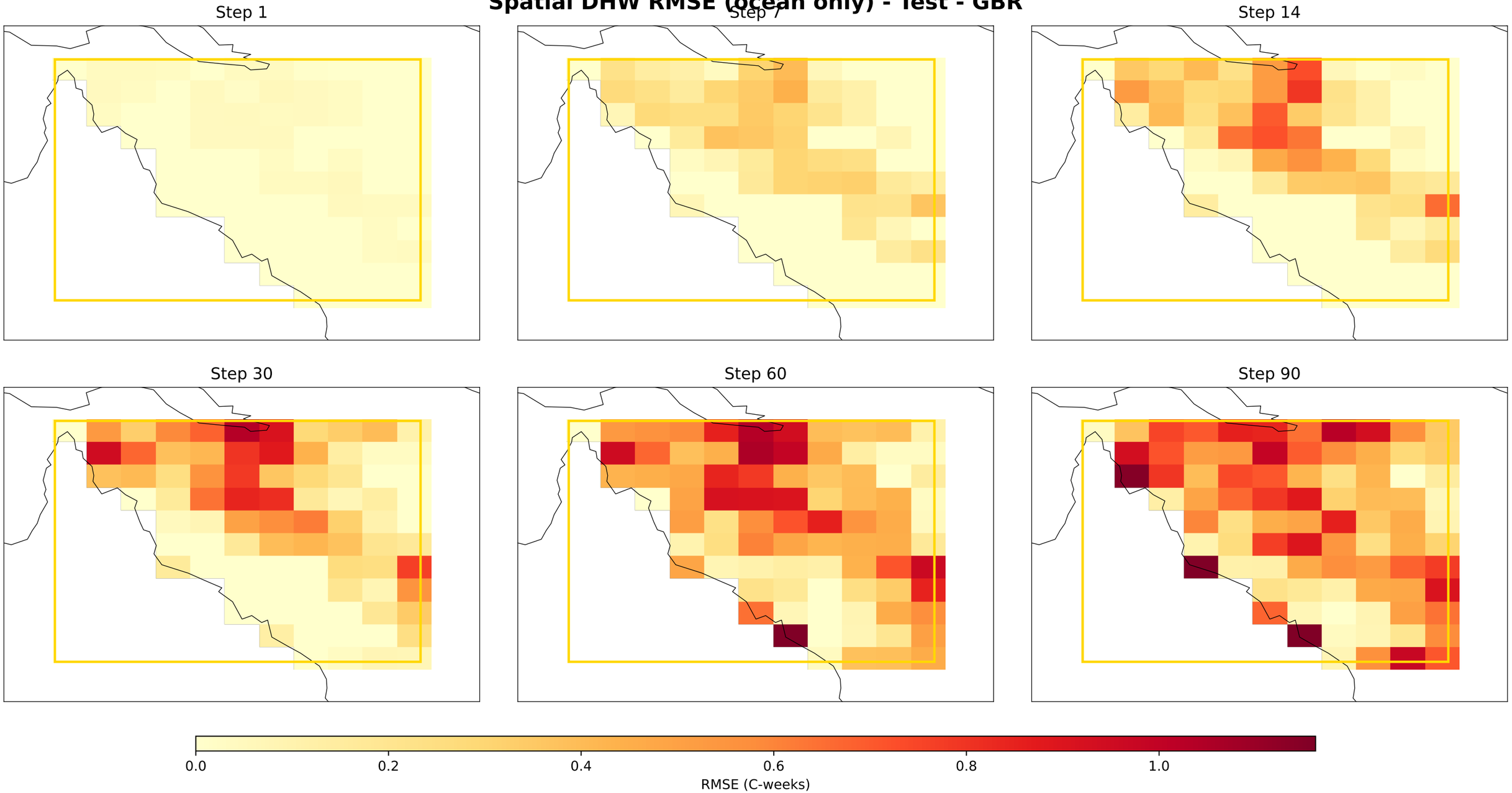


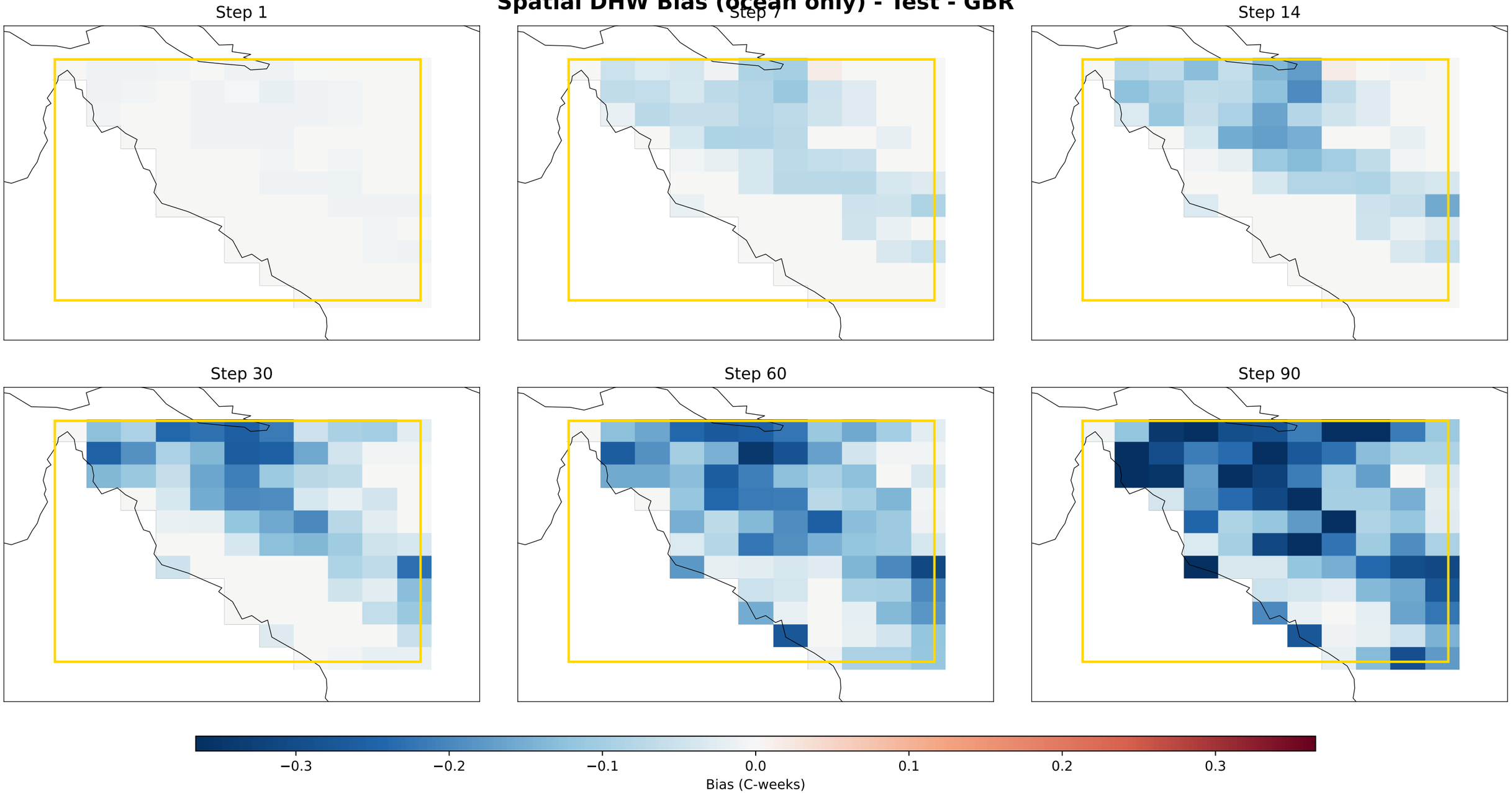





Spatial DHW RMSE (all grid) - Test - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0.0
0.2
0.4
0.6
0.8
1.0
RMSE (C-weeks)

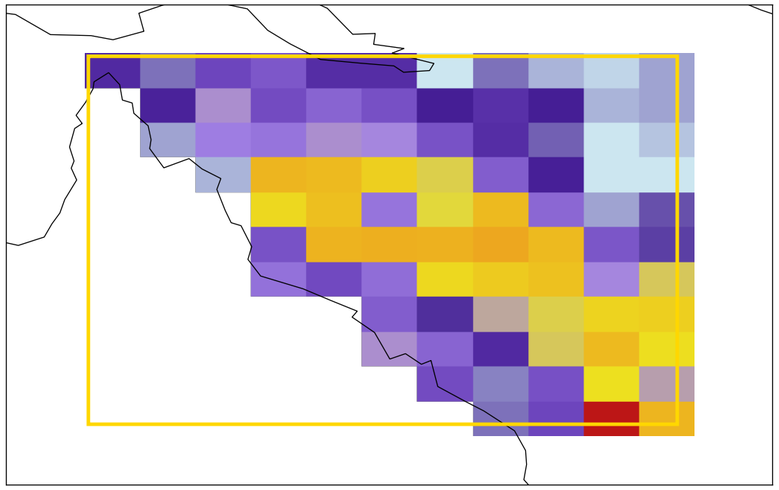

Spatial DHW Bias (all grid) - Test - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3
Bias (C-weeks)


Spatial DHW RMSE (ocean only) - Test - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0.0
0.2
0.4
0.6
0.8
1.0
RMSE (C-weeks)

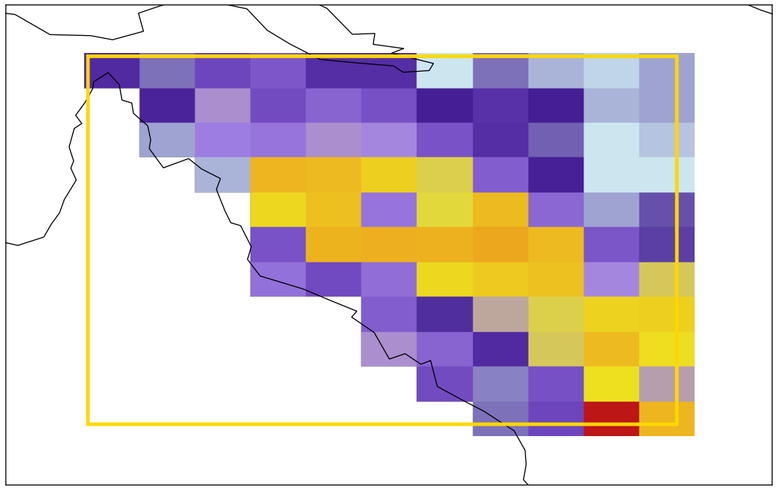

Spatial DHW Bias (ocean only) - Test - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3
Bias (C-weeks)

DHW Chain init: 2020-04-10 - Test - GBR (C-weeks)

Actual DHW - Step 1



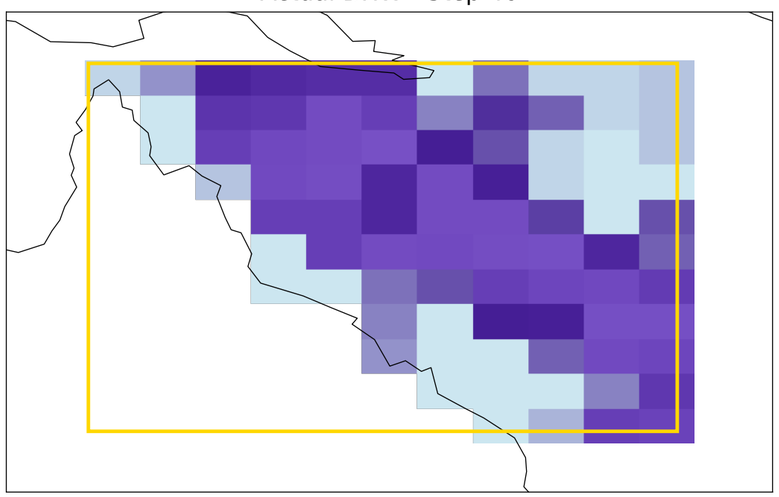
Predicted DHW - Step 1



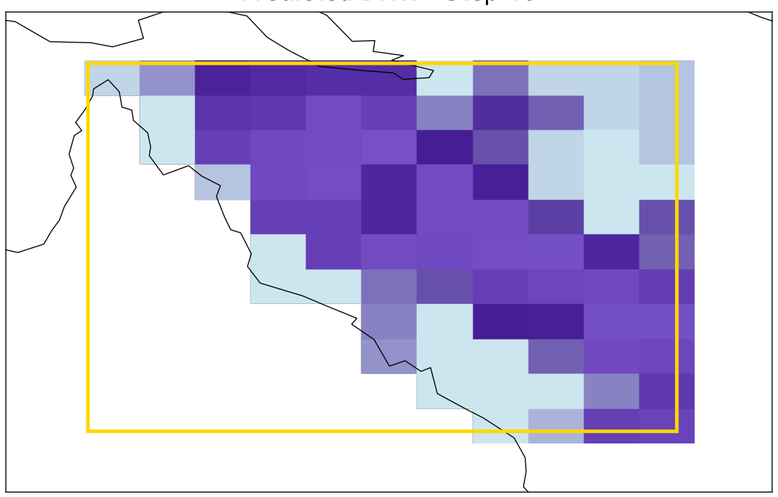
Error (Pred - Actual) - Step 1 [scale ±8]



Actual DHW - Step 46



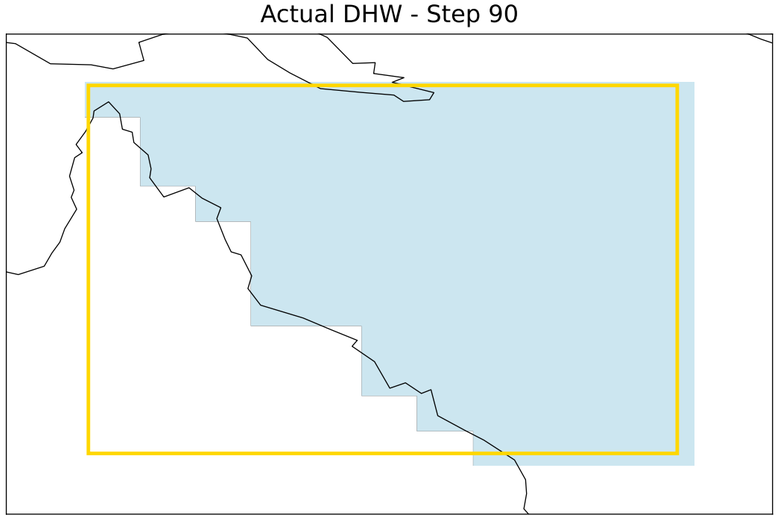
Predicted DHW - Step 46



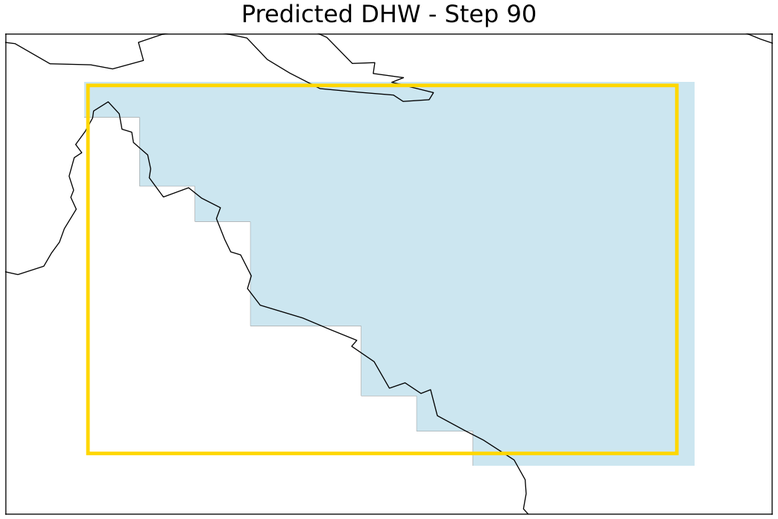
Error (Pred - Actual) - Step 46 [scale ±8]



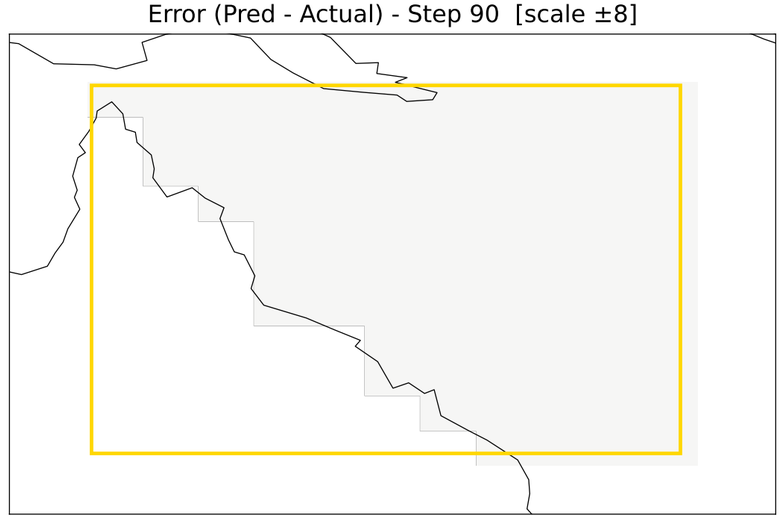
Actual DHW - Step 90

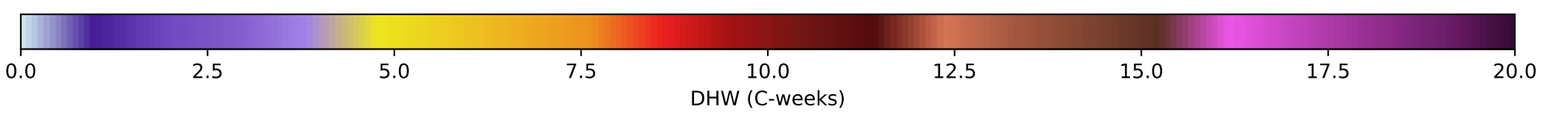


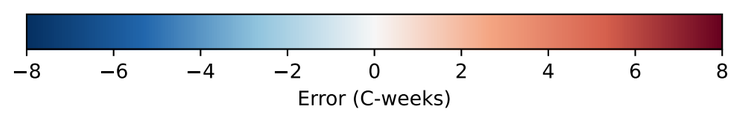
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

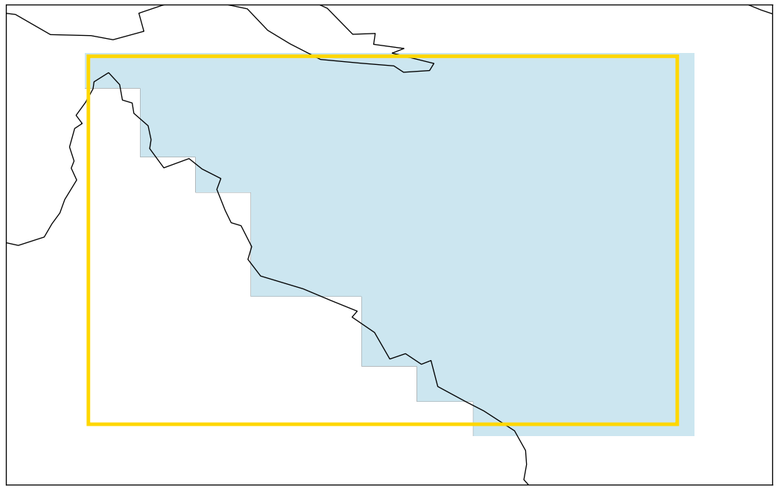




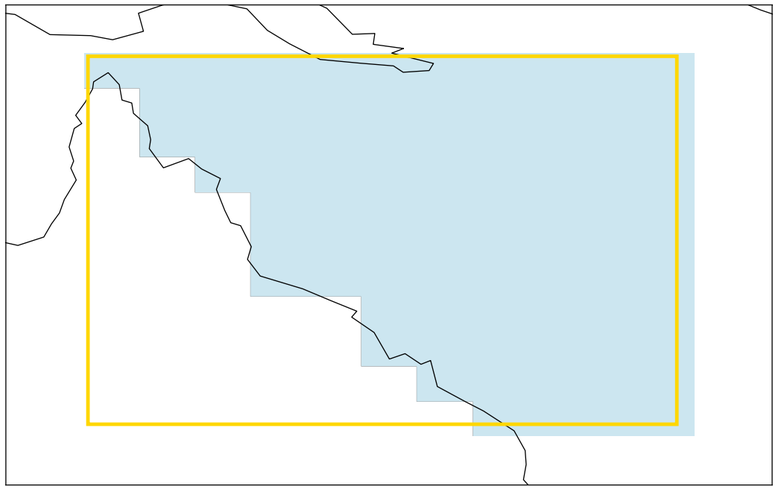


DHW Chain init: 2020-07-09 - Test - GBR (C-weeks)

Actual DHW - Step 1



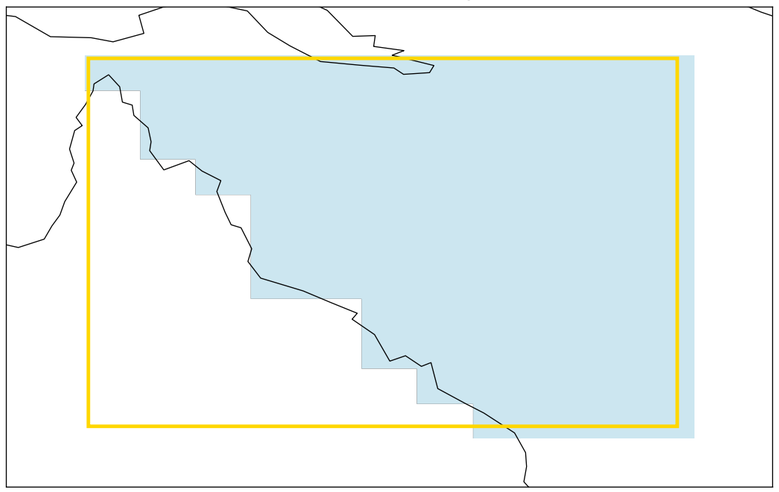
Predicted DHW - Step 1



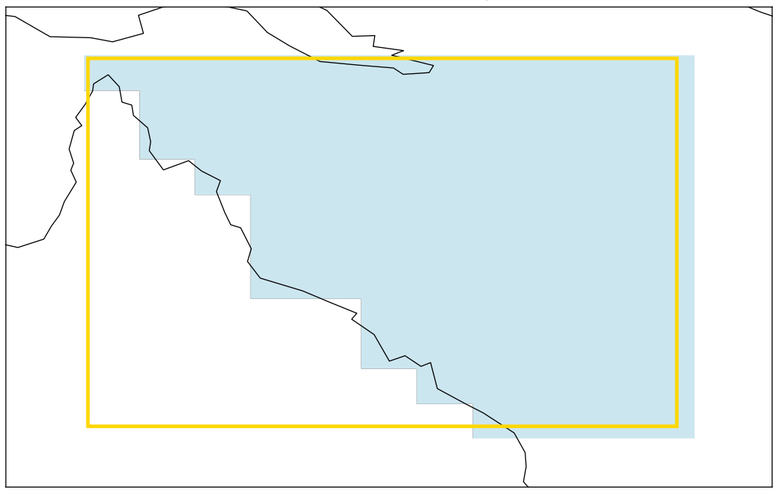
Error (Pred - Actual) - Step 1 [scale ±8]



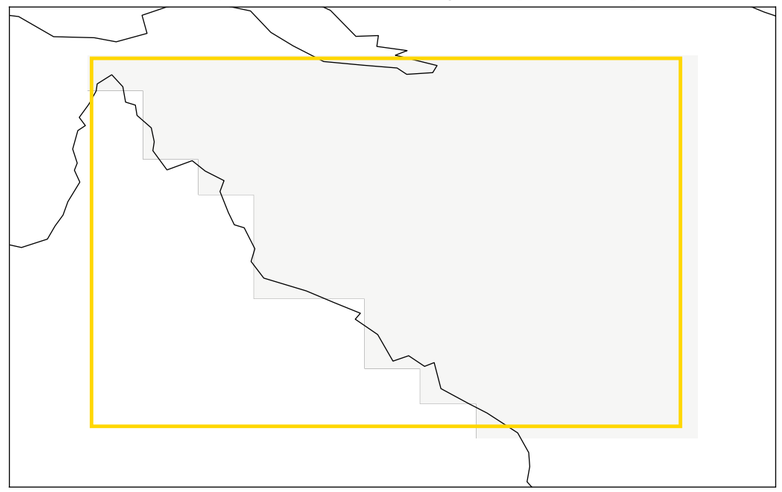
Actual DHW - Step 46



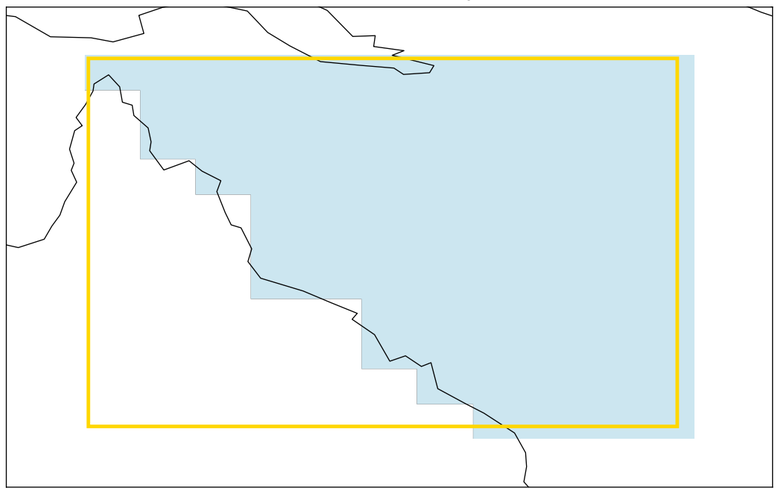
Predicted DHW - Step 46



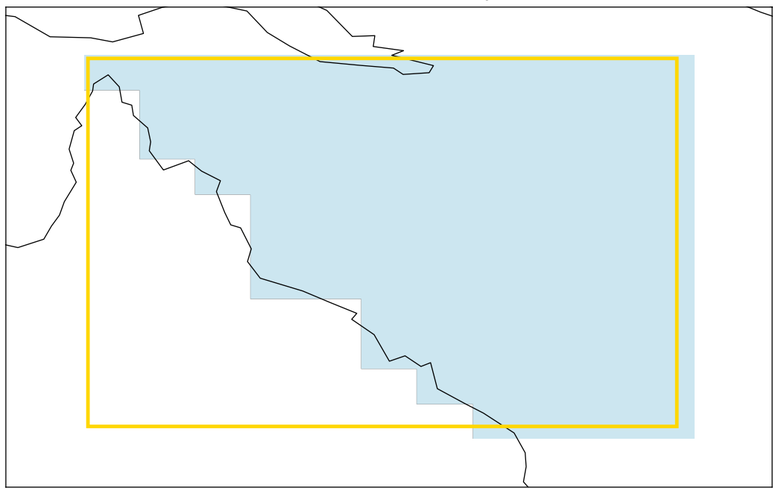
Error (Pred - Actual) - Step 46 [scale ±8]



Actual DHW - Step 90

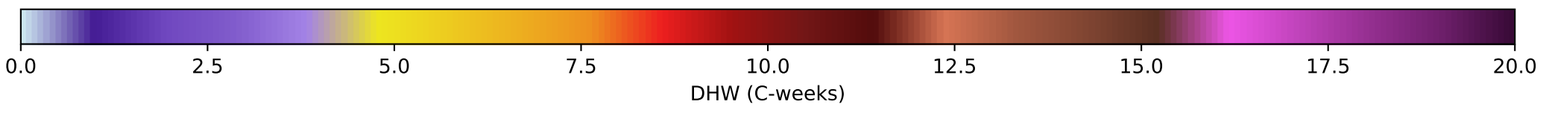


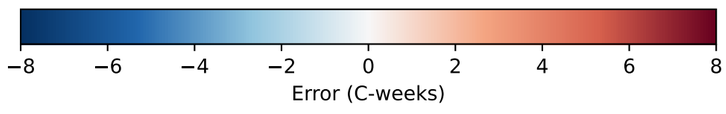
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

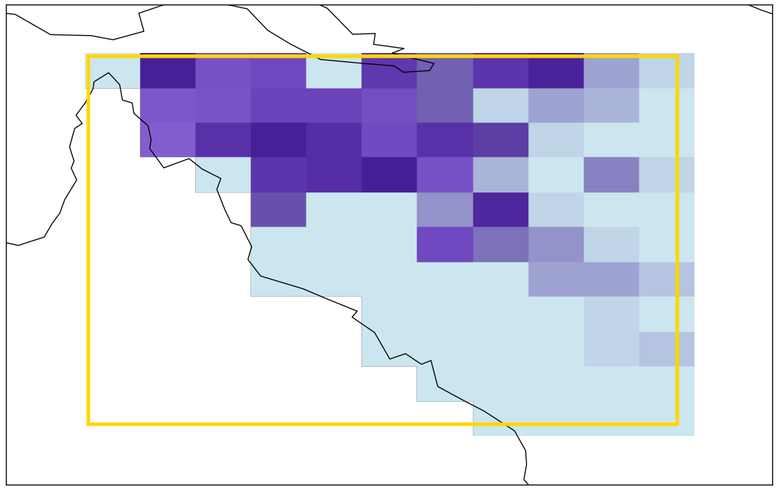




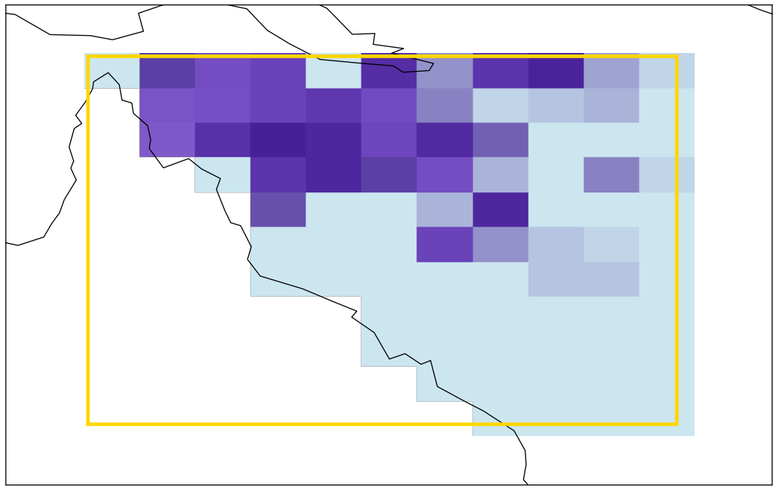


DHW Chain init: 2021-01-06 - Test - GBR (C-weeks)

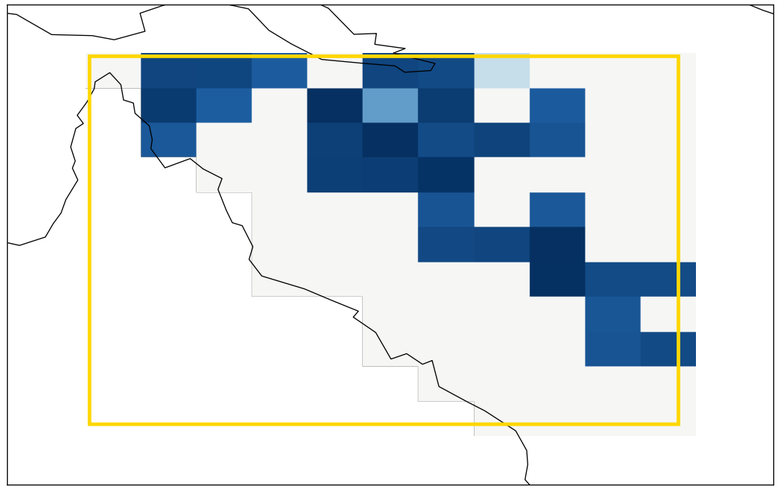
Actual DHW - Step 1



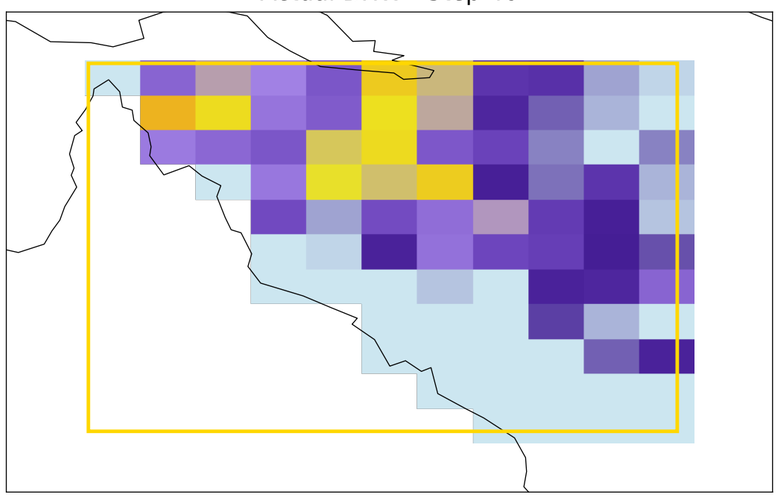
Predicted DHW - Step 1



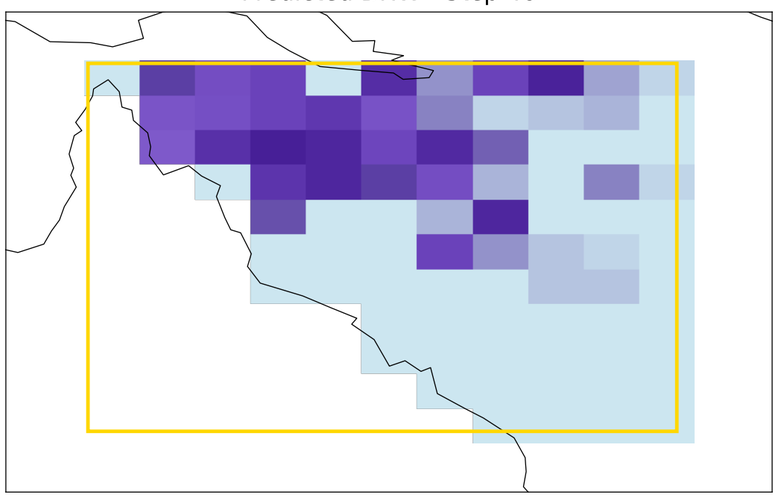
Error (Pred - Actual) - Step 1 [scale ±0.2]



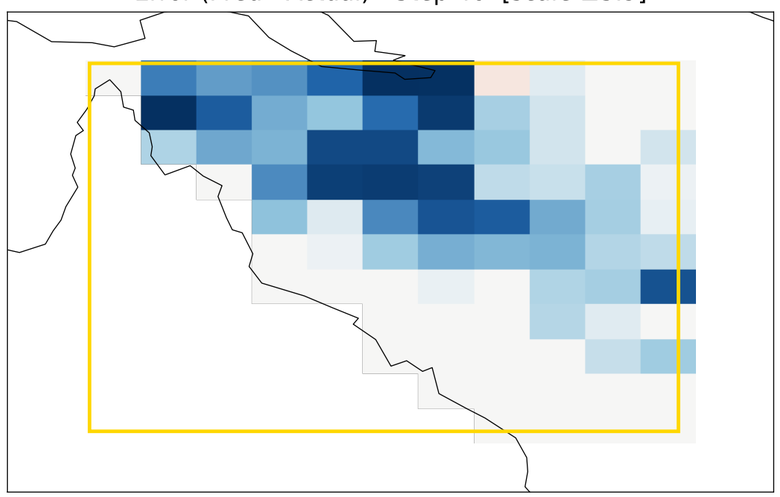
Actual DHW - Step 46



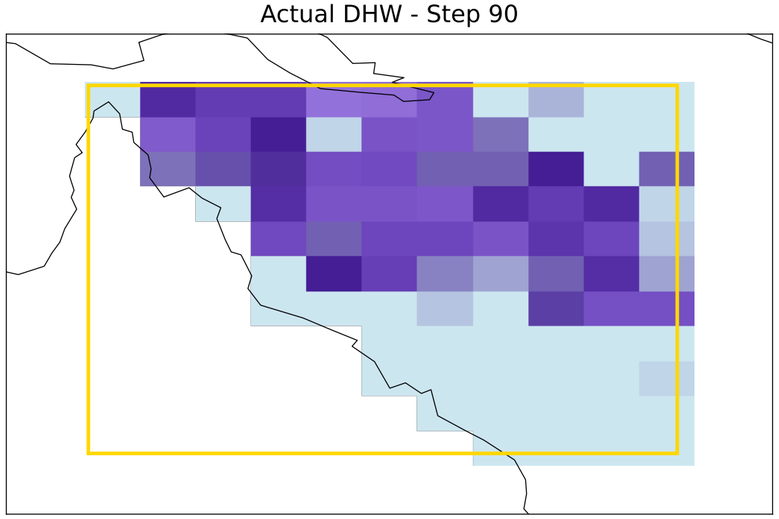
Predicted DHW - Step 46



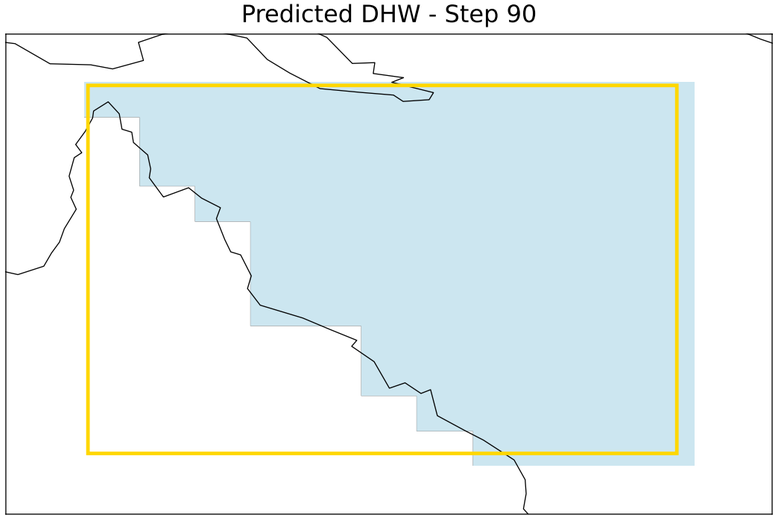
Error (Pred - Actual) - Step 46 [scale ±3.9]



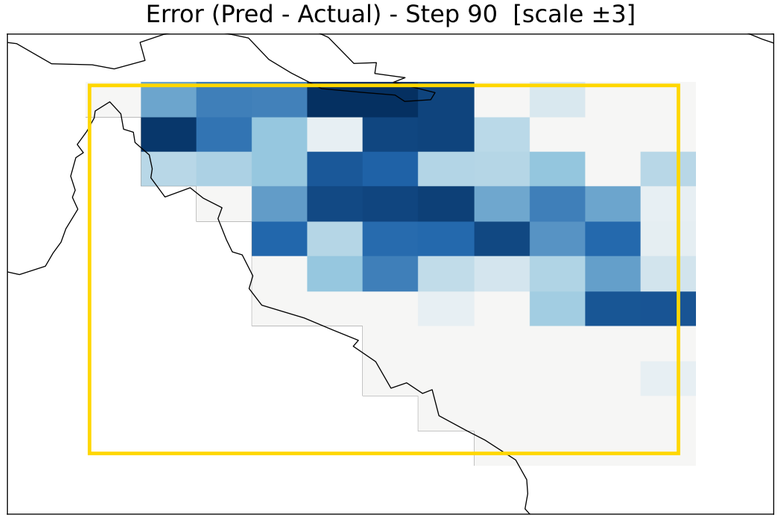
Actual DHW - Step 90

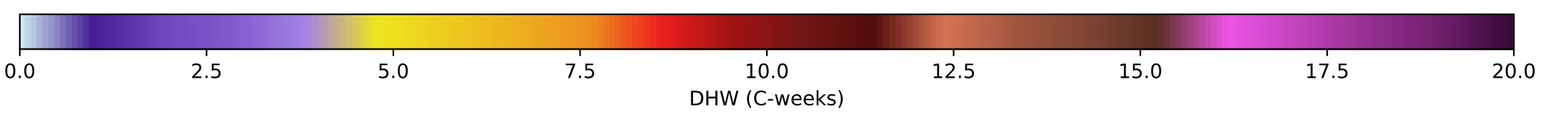


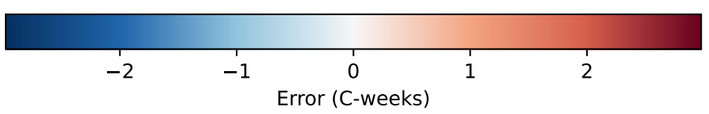
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±3]

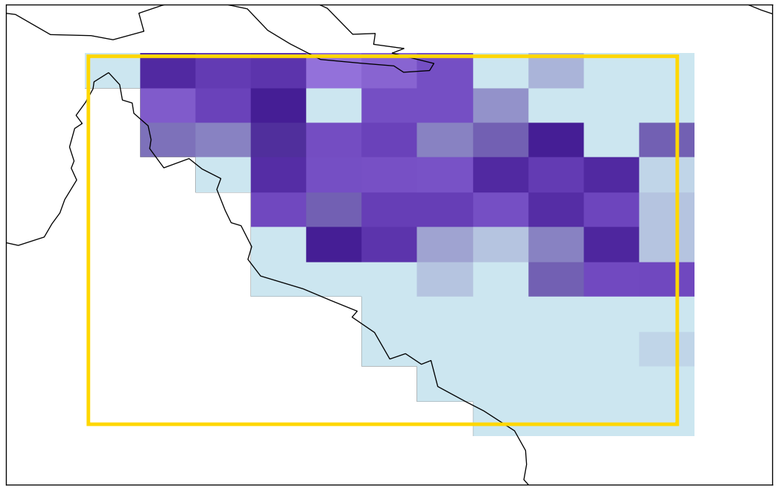




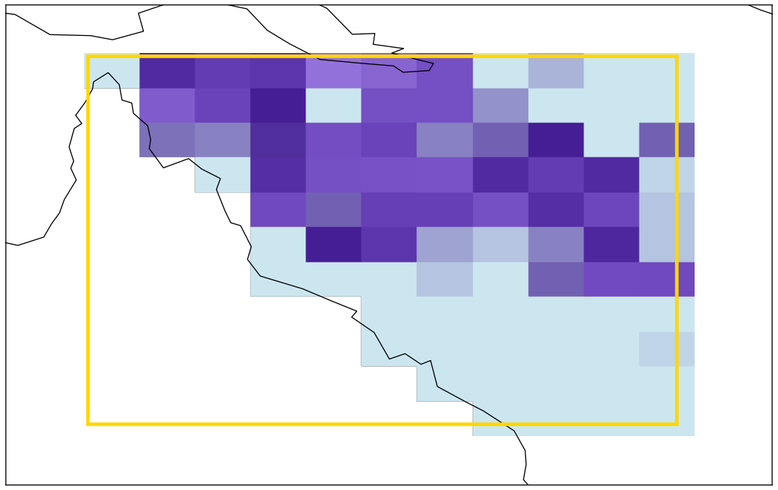


DHW Chain init: 2021-04-06 - Test - GBR (C-weeks)

Actual DHW - Step 1



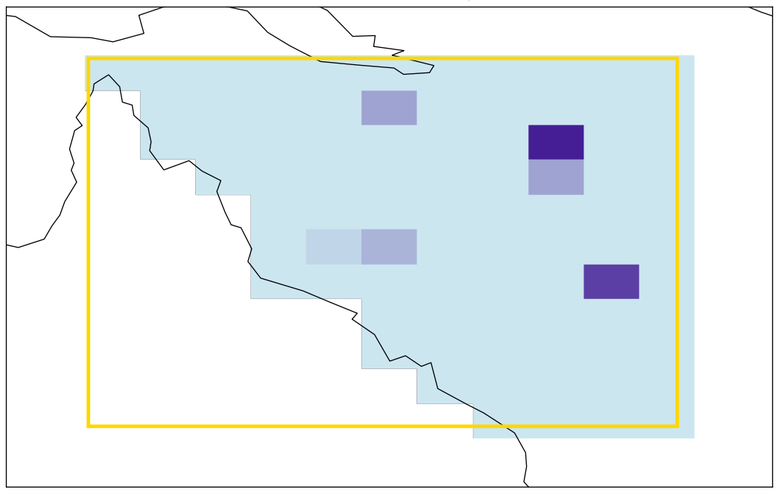
Predicted DHW - Step 1



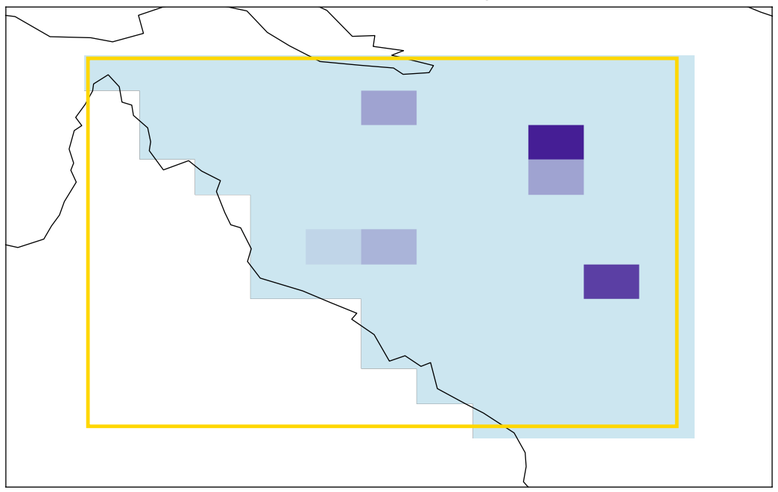
Error (Pred - Actual) - Step 1 [scale ±8]



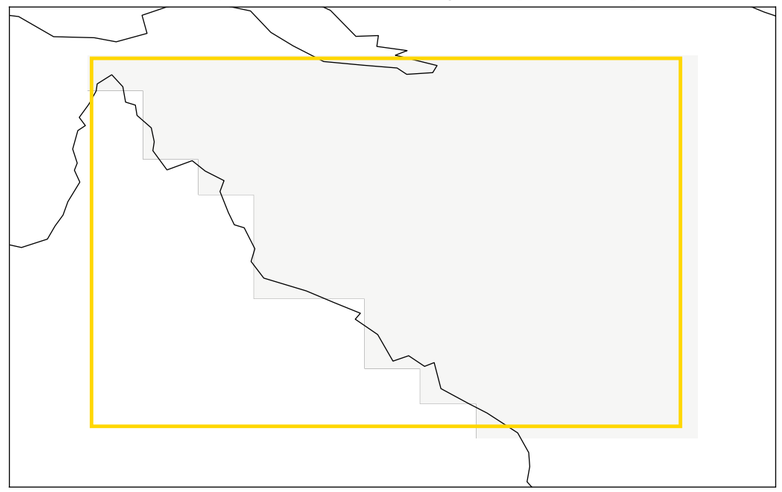
Actual DHW - Step 46



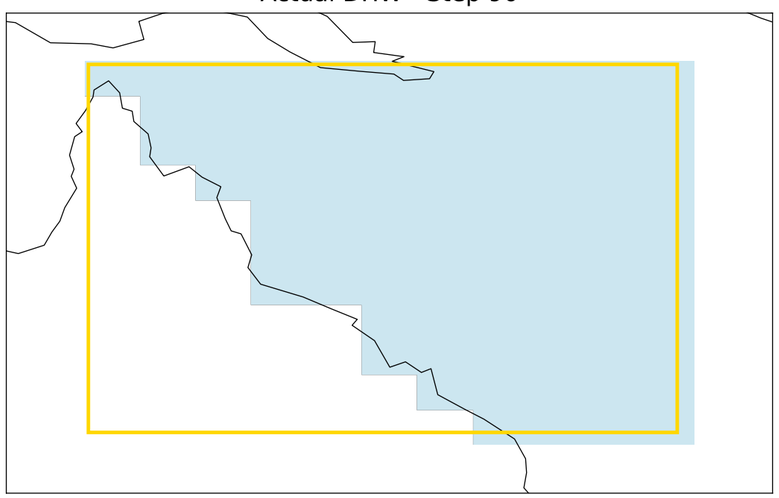
Predicted DHW - Step 46



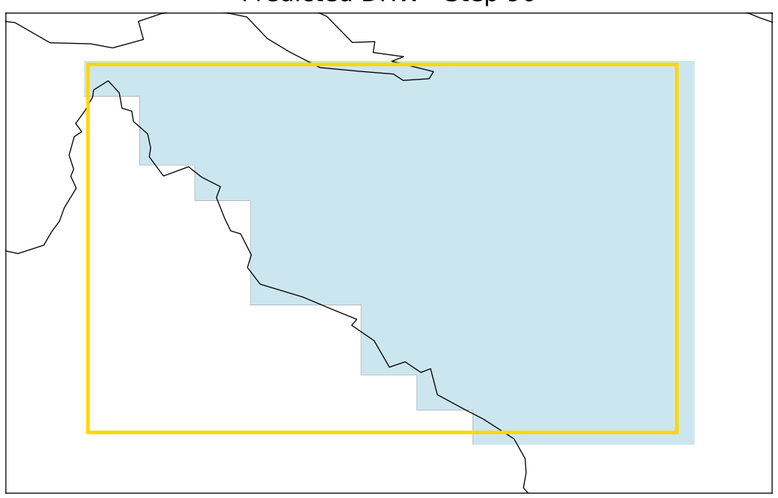
Error (Pred - Actual) - Step 46 [scale ±8]



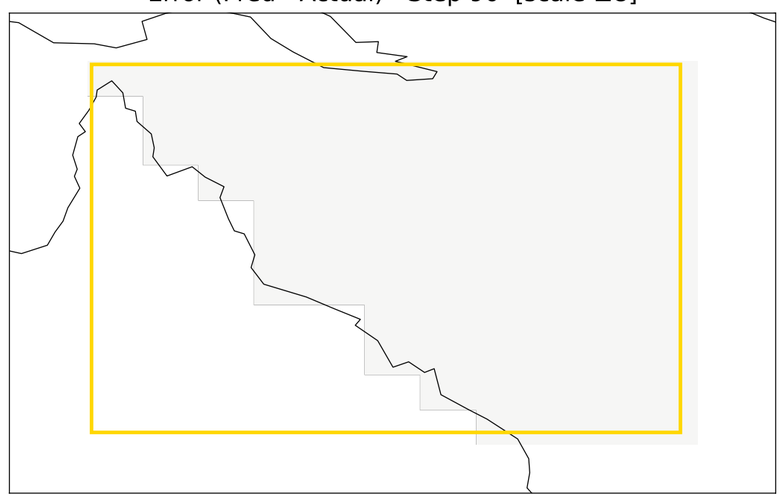
Actual DHW - Step 90

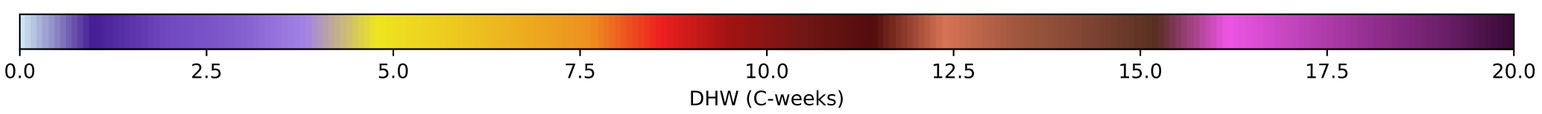


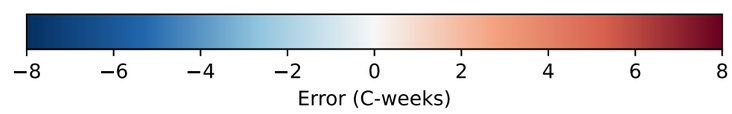
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

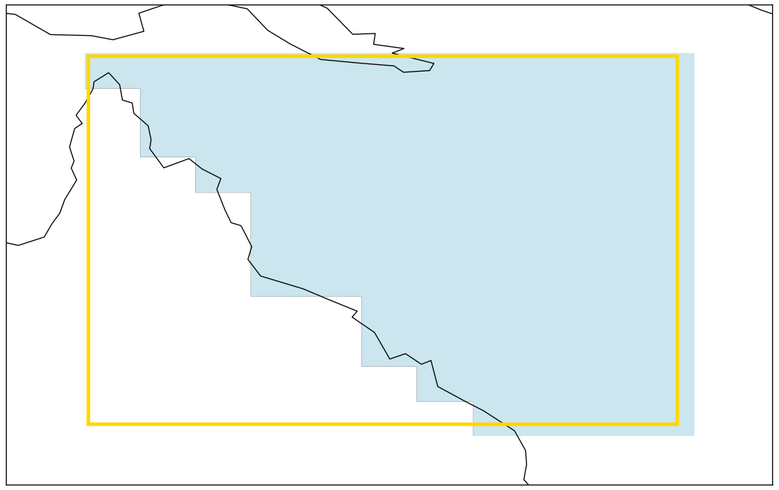




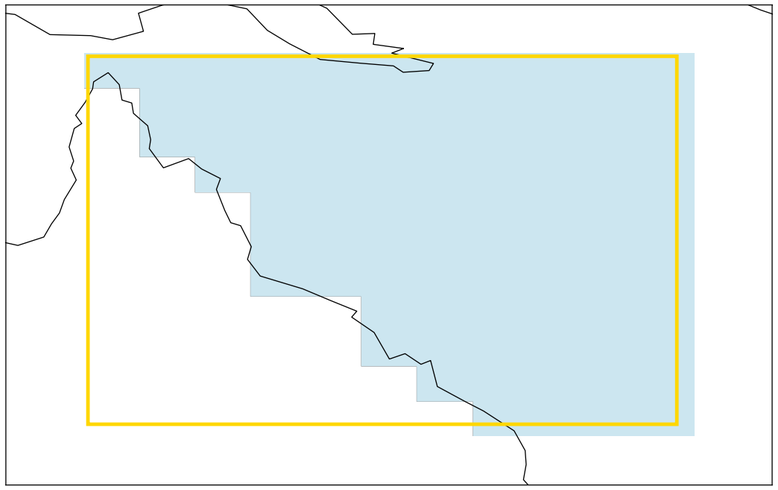


DHW Chain init: 2021-10-03 - Test - GBR (C-weeks)

Actual DHW - Step 1



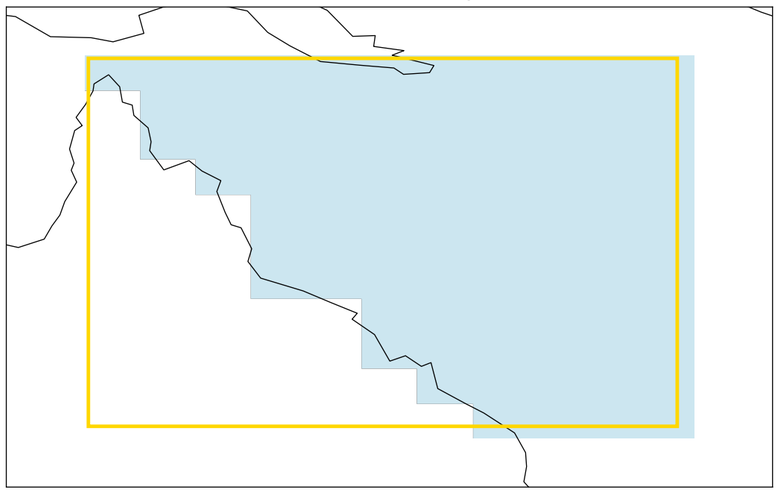
Predicted DHW - Step 1



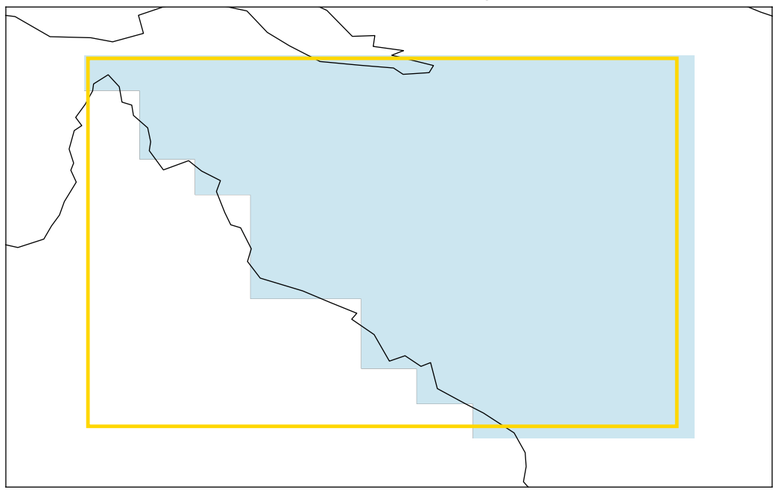
Error (Pred - Actual) - Step 1 [scale ±8]



Actual DHW - Step 46



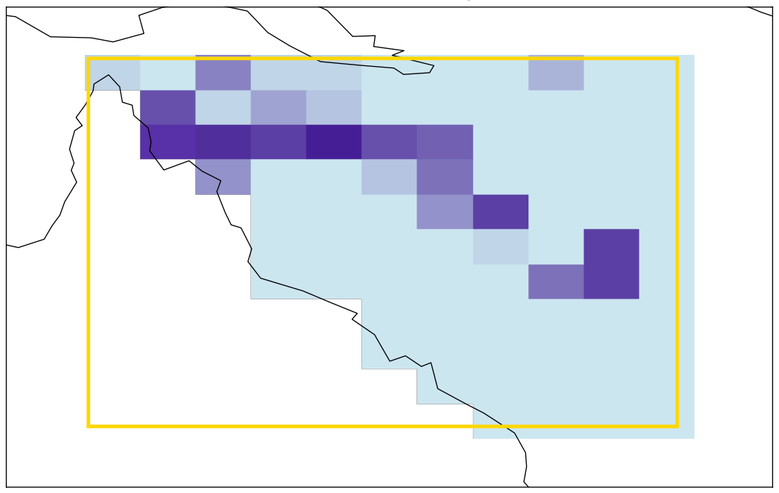
Predicted DHW - Step 46



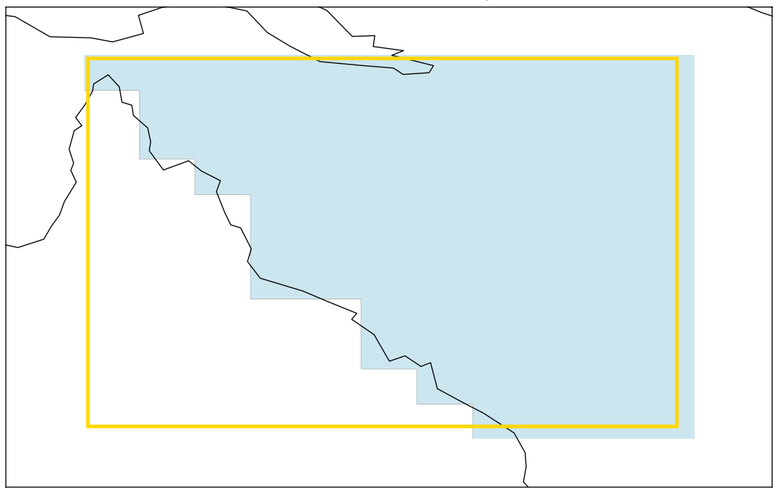
Error (Pred - Actual) - Step 46 [scale ±8]



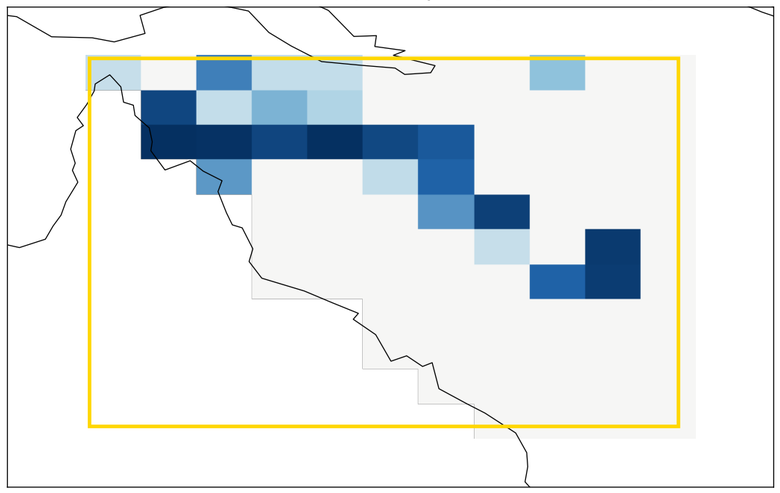
Actual DHW - Step 90

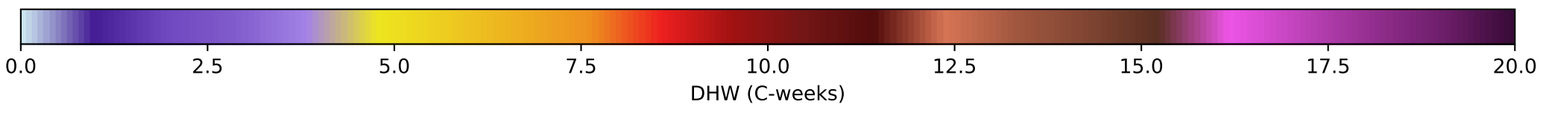


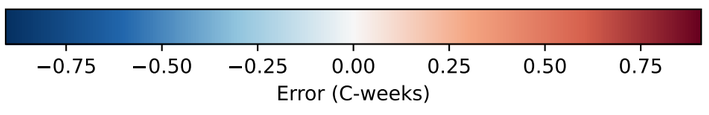
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±0.91]

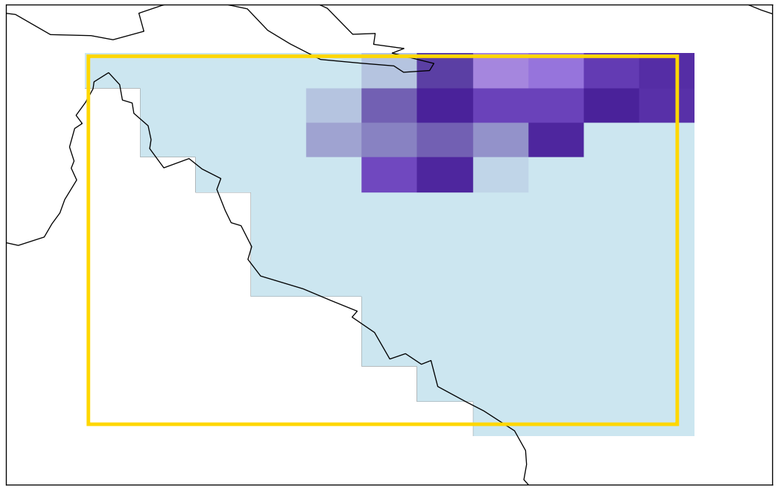




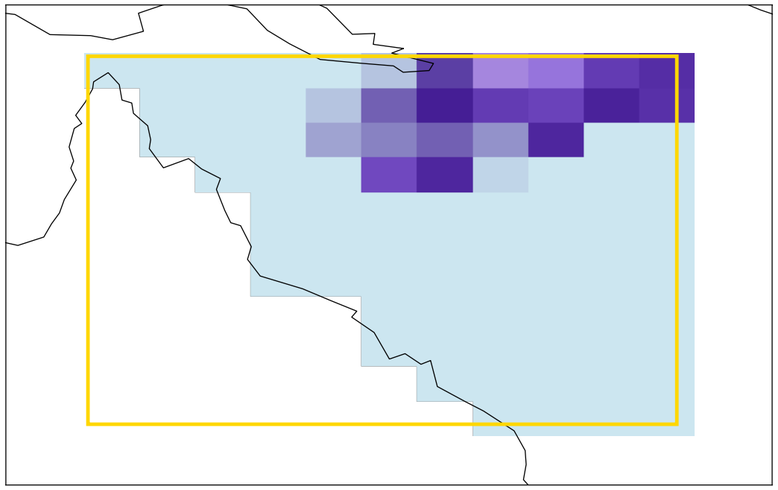


DHW Chain init: 2022-12-27 - Test - GBR (C-weeks)

Actual DHW - Step 1



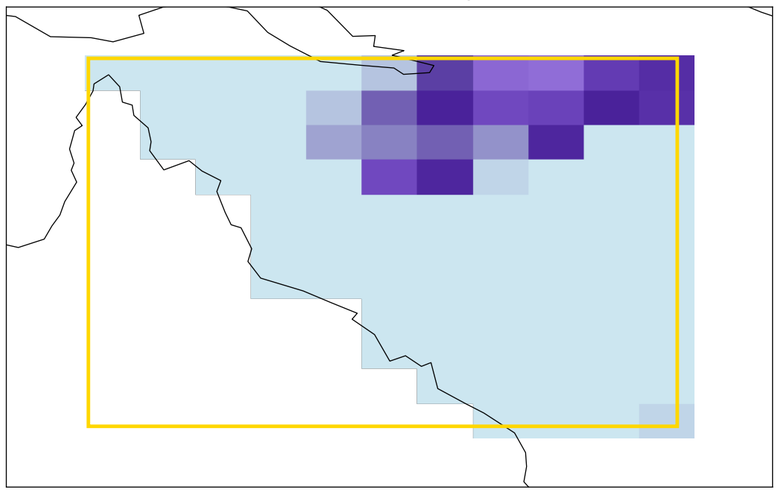
Predicted DHW - Step 1



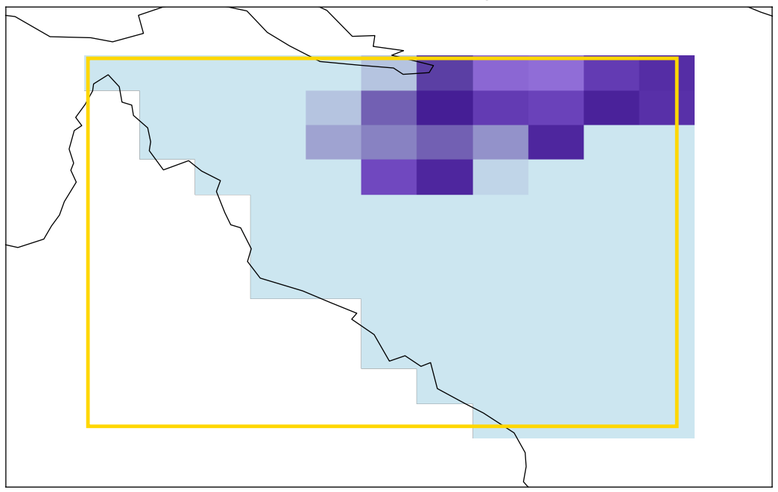
Error (Pred - Actual) - Step 1 [scale ±0.051]



Actual DHW - Step 46



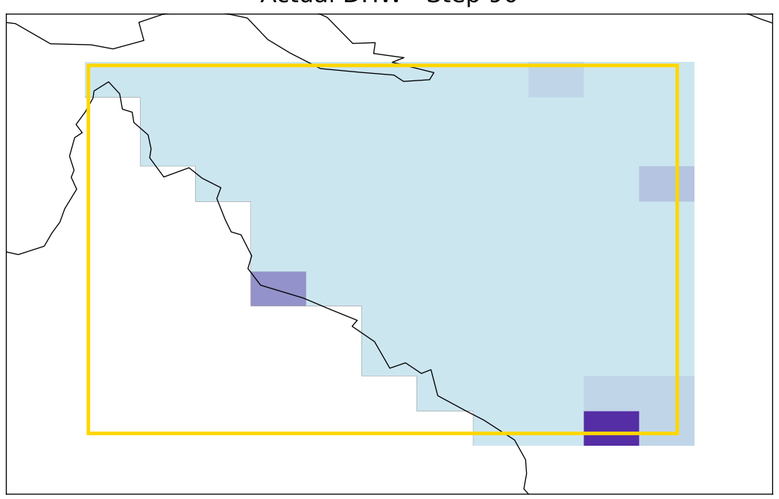
Predicted DHW - Step 46



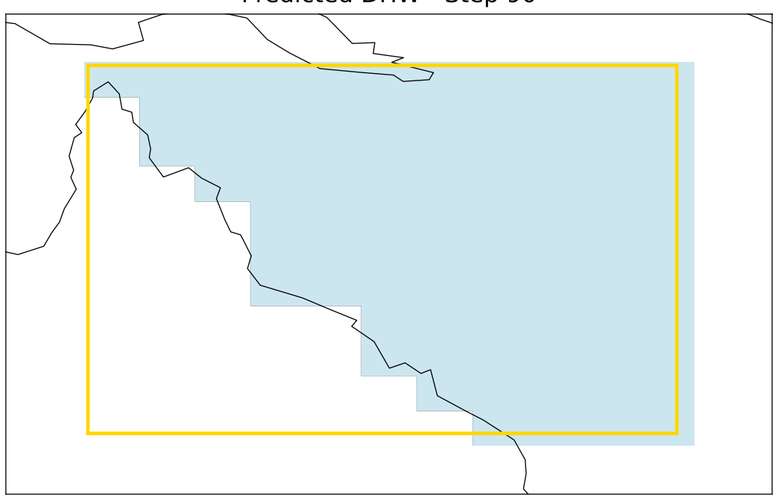
Error (Pred - Actual) - Step 46 [scale ±0.15]



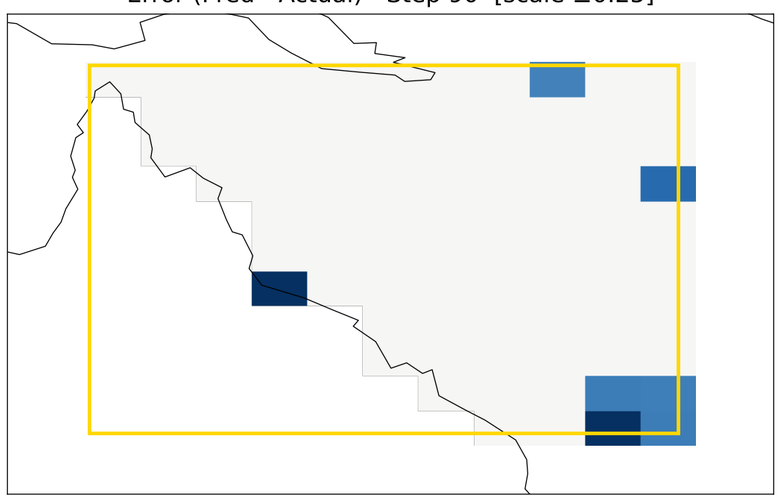
Actual DHW - Step 90

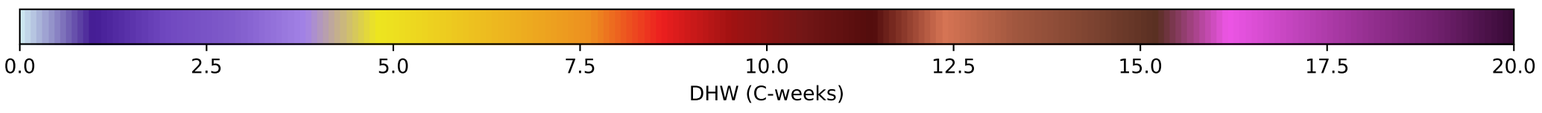


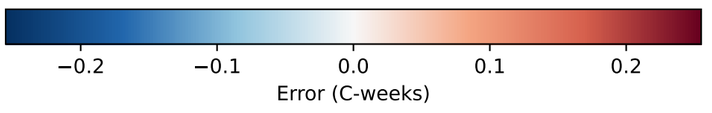
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±0.25]

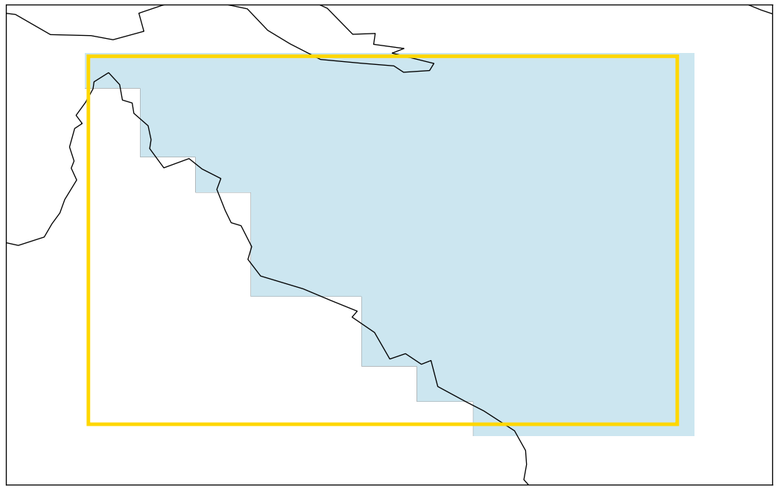




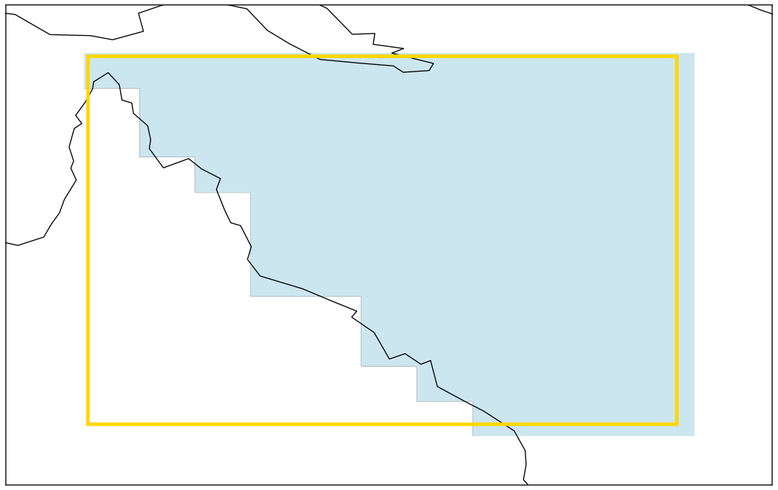


DHW Chain init: 2023-09-23 - Test - GBR (C-weeks)

Actual DHW - Step 1



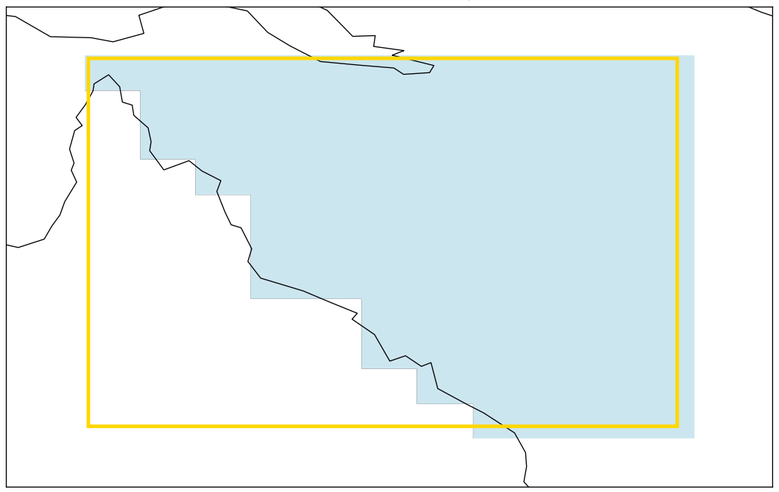
Predicted DHW - Step 1



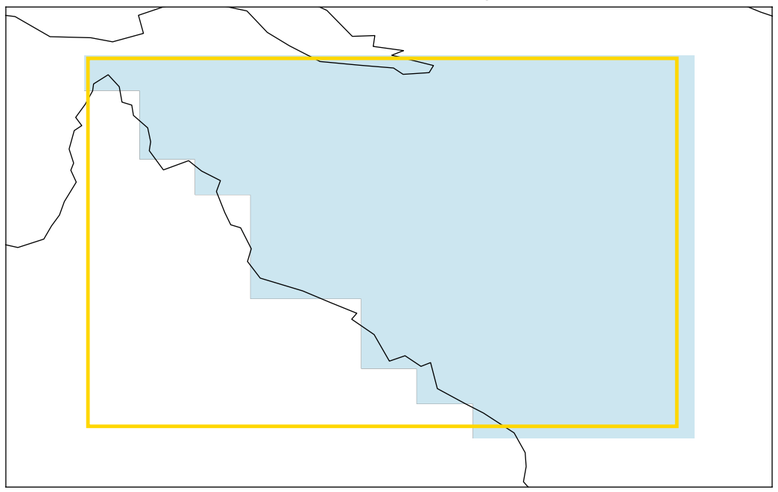
Error (Pred - Actual) - Step 1 [scale ±8]



Actual DHW - Step 46



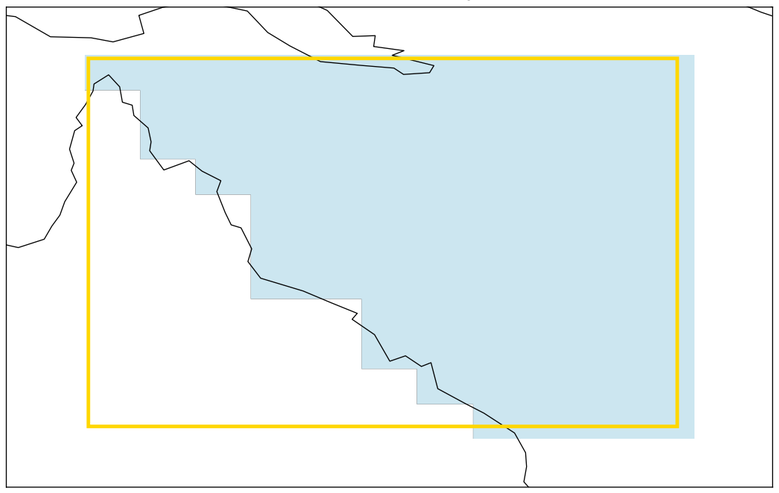
Predicted DHW - Step 46



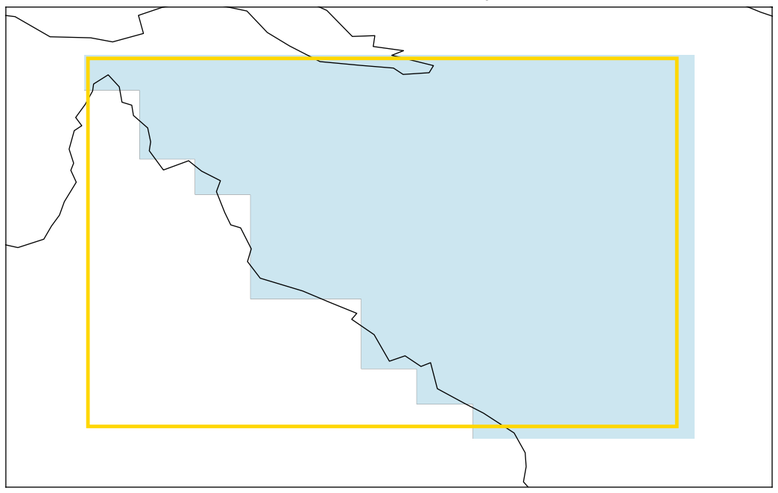
Error (Pred - Actual) - Step 46 [scale ±8]



Actual DHW - Step 90

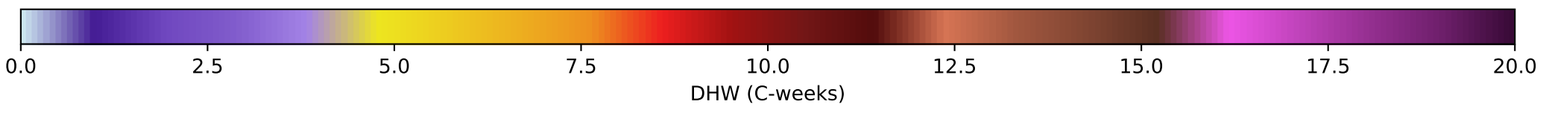


Predicted DHW - Step 90



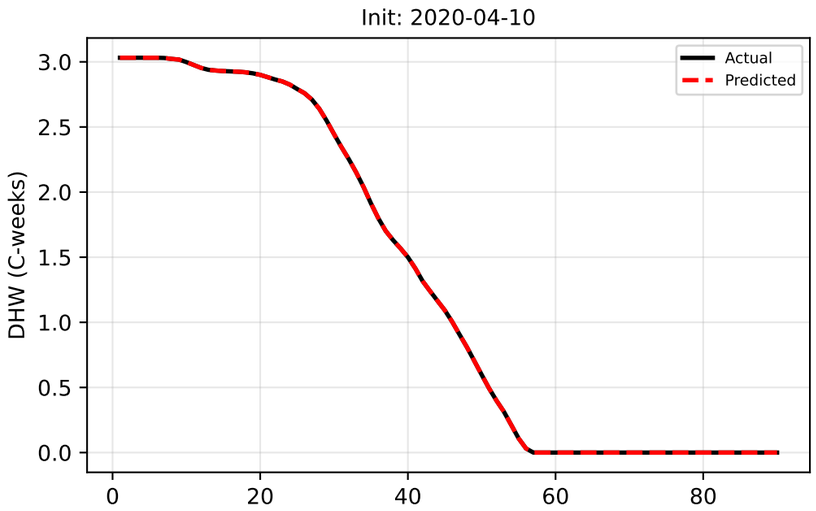
Error (Pred - Actual) - Step 90 [scale ±8]

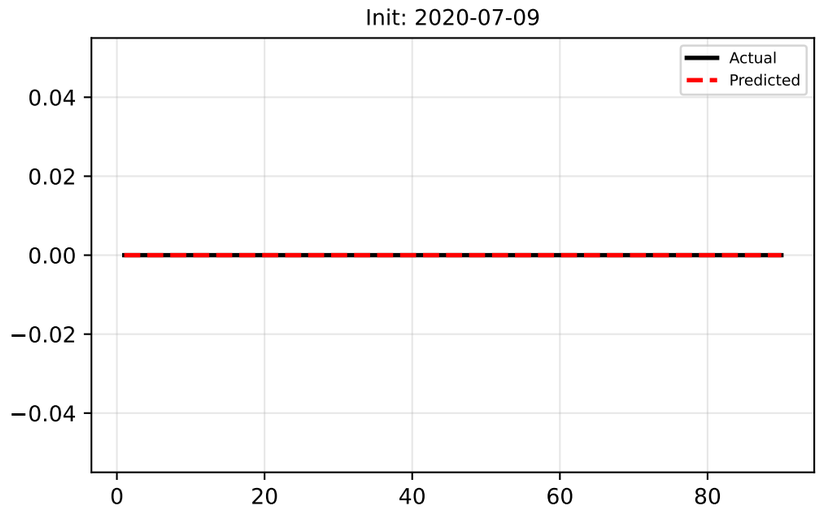


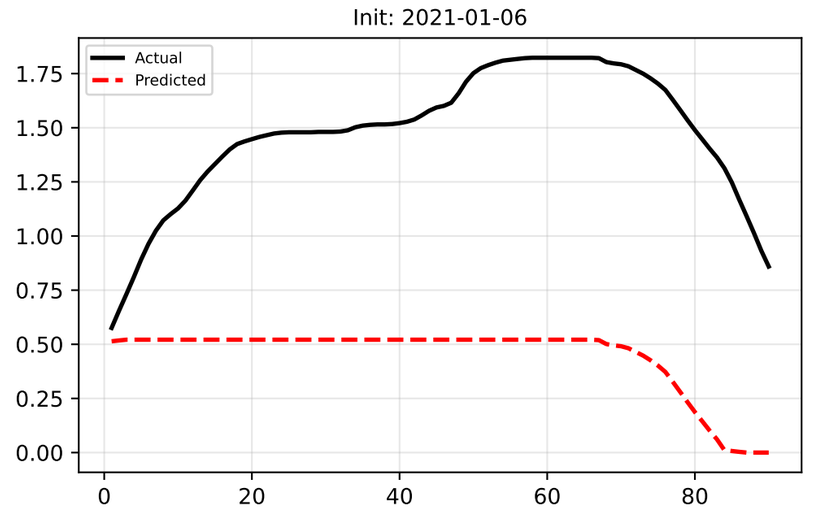


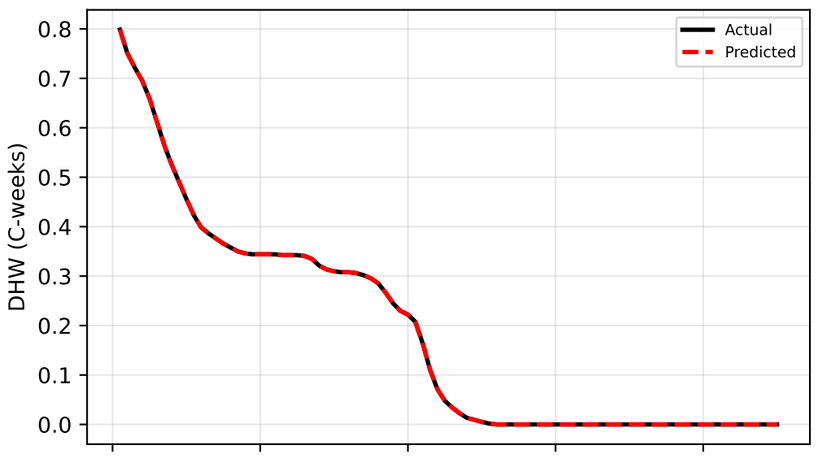


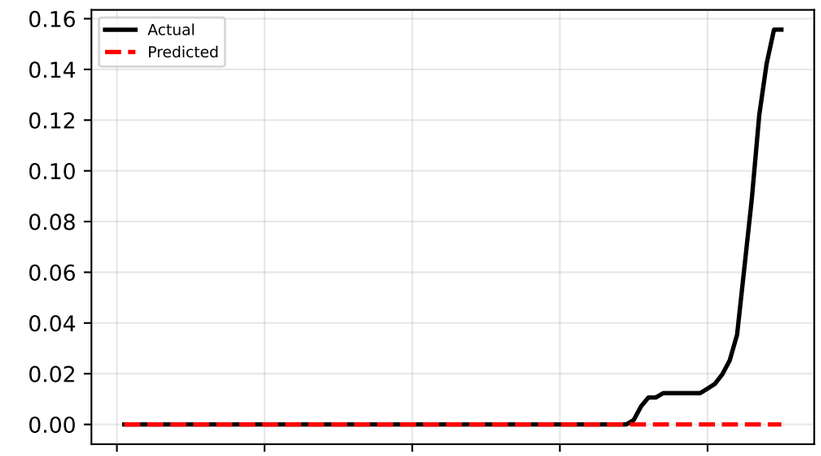
Global-Mean DHW - Test - GBR (C-weeks)

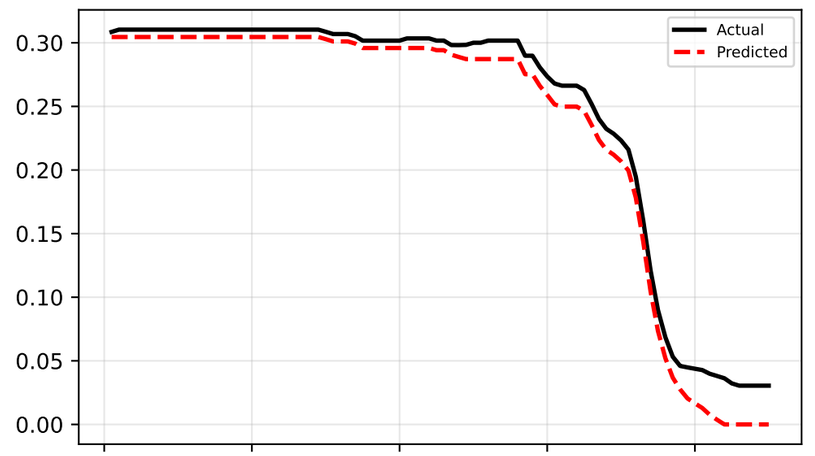


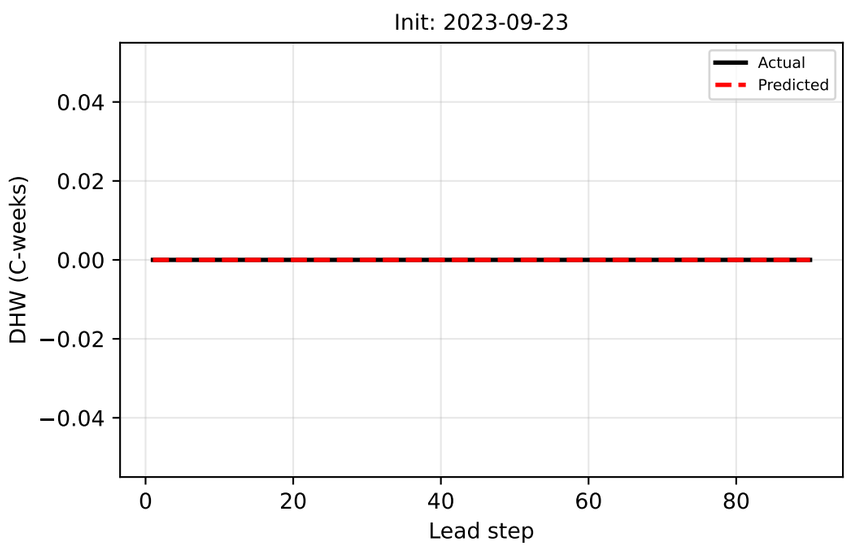


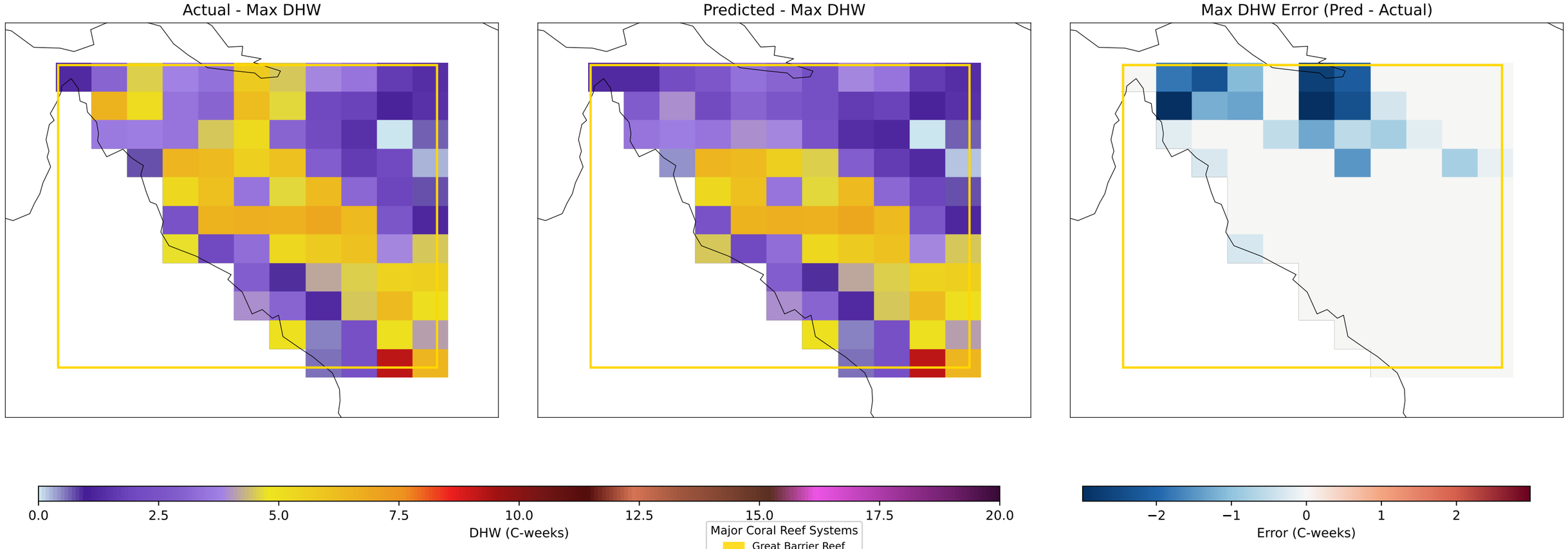


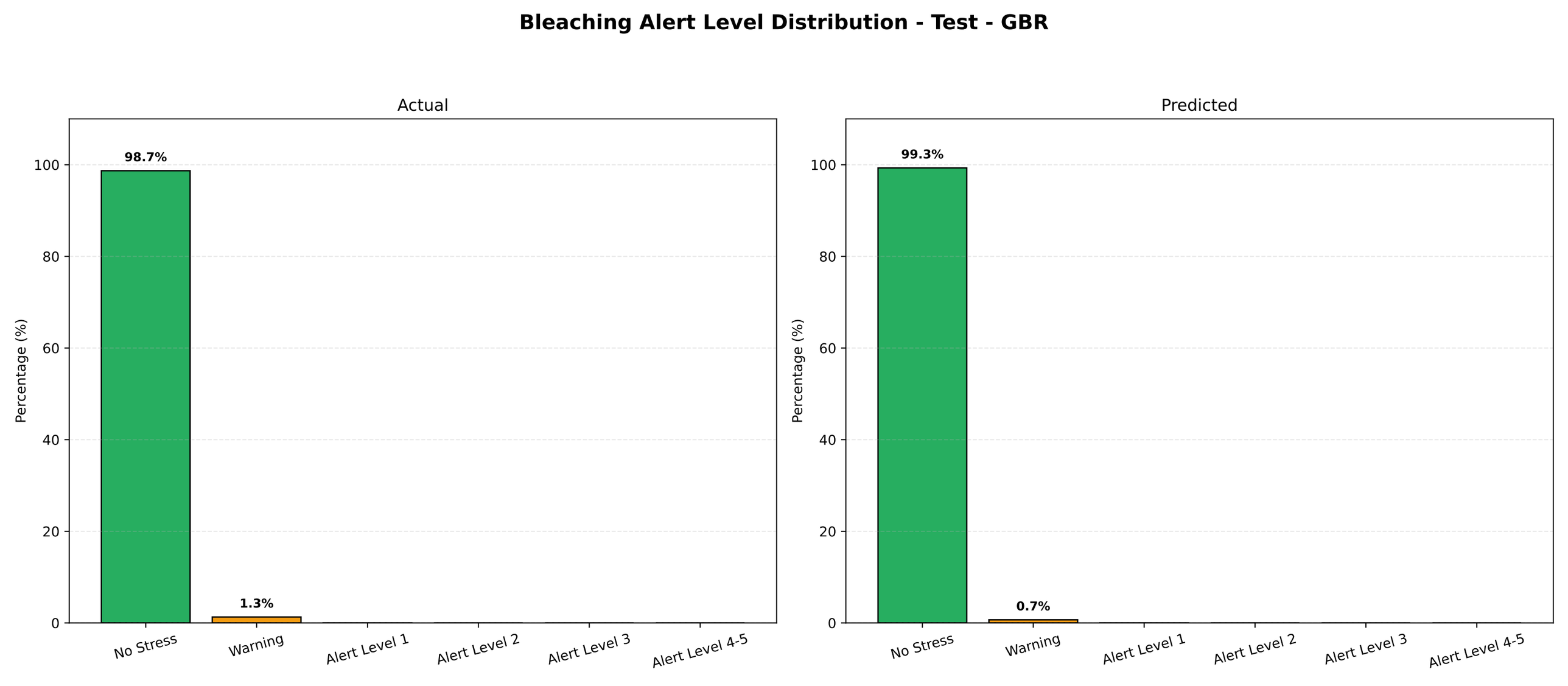






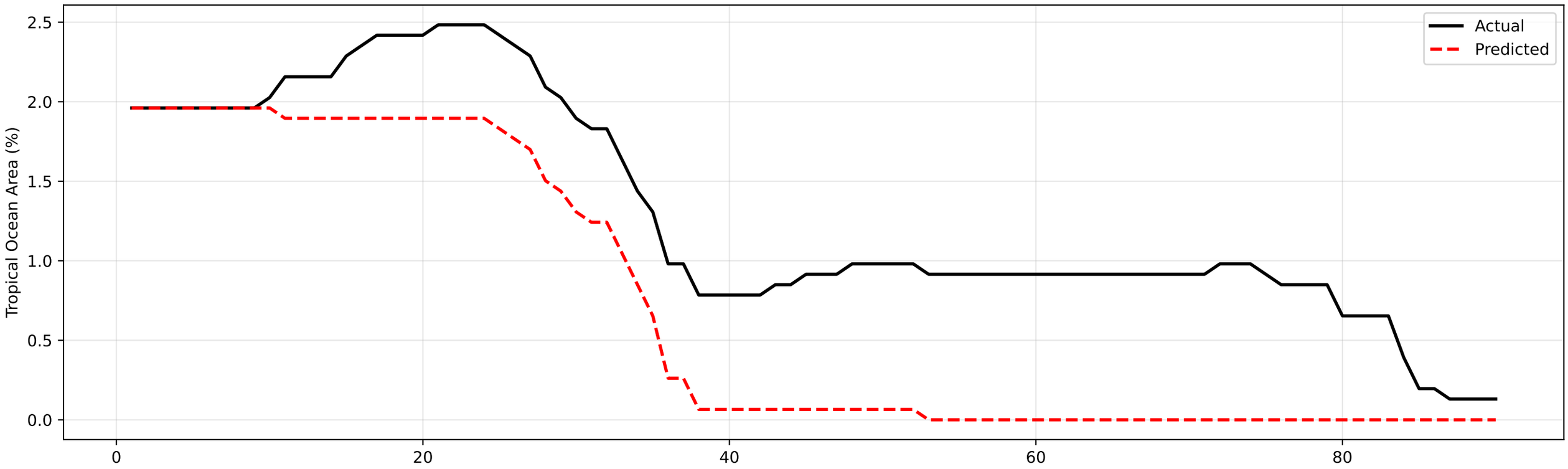



Peak DHW (max over all chains & leads) - Test - GBR (C-weeks)
Actual - Max DHW
Predicted - Max DHW
Max DHW Error (Pred - Actual)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
DHW (C-weeks)
Major Coral Reef Systems
Great Barrier Reef
−2
−1
0
1
2
Error (C-weeks)

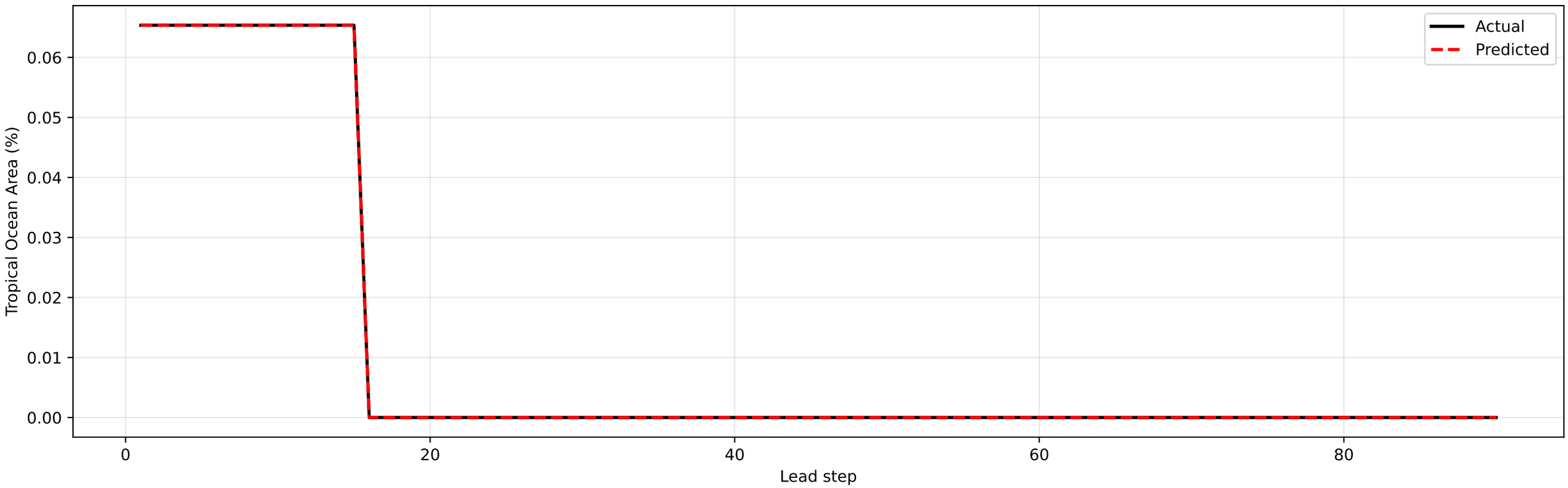

Bleaching Alert Level Distribution - Test - GBR
Actual
Predicted
Percentage (%)
0
20
40
60
80
100
98.7%
1.3%
99.3%
0.7%
No Stress
Warning
Alert Level 1
Alert Level 2
Alert Level 3
Alert Level 4-5

Tropical Ocean Area Under Bleaching Stress - Test - GBR

DHW >= 4 C-weeks (Alert Level 1 DHW threshold)



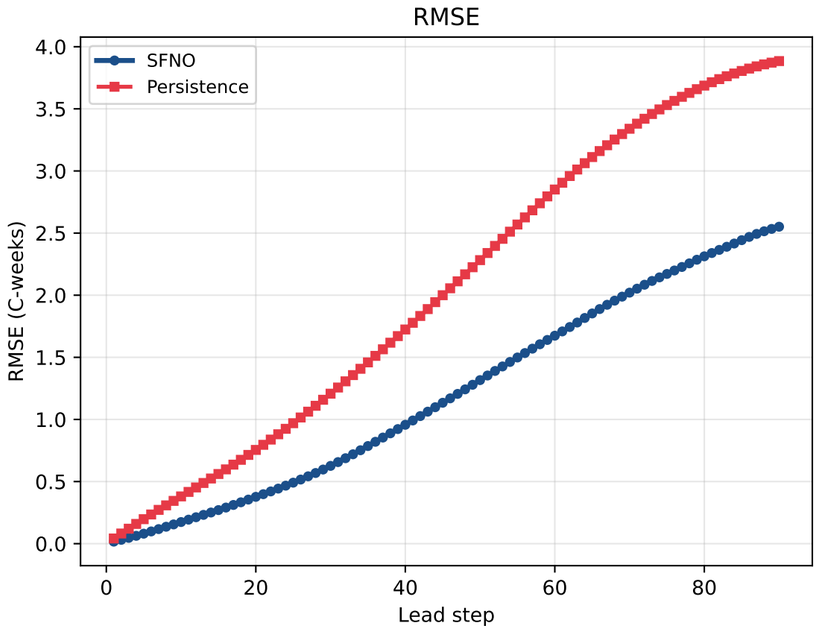
DHW >= 8 C-weeks (Alert Level 2 DHW threshold)

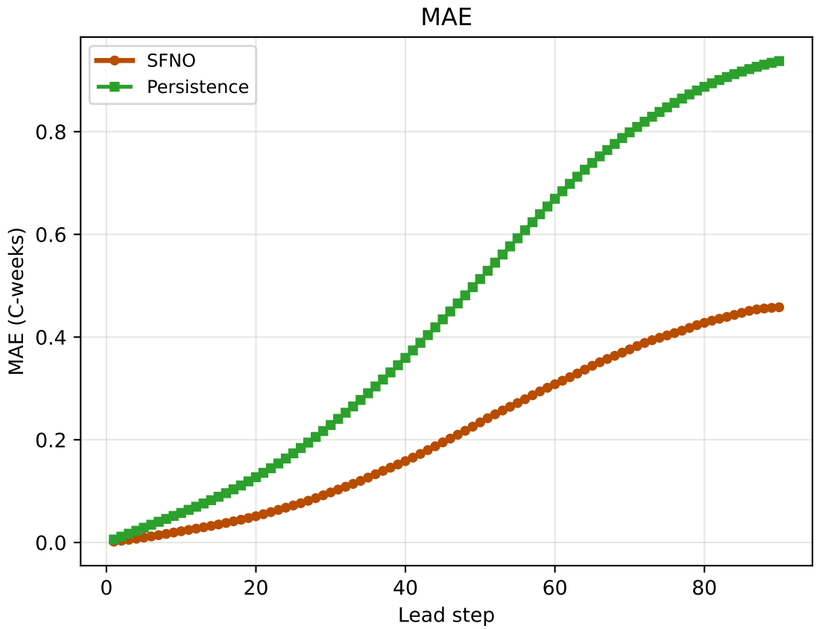


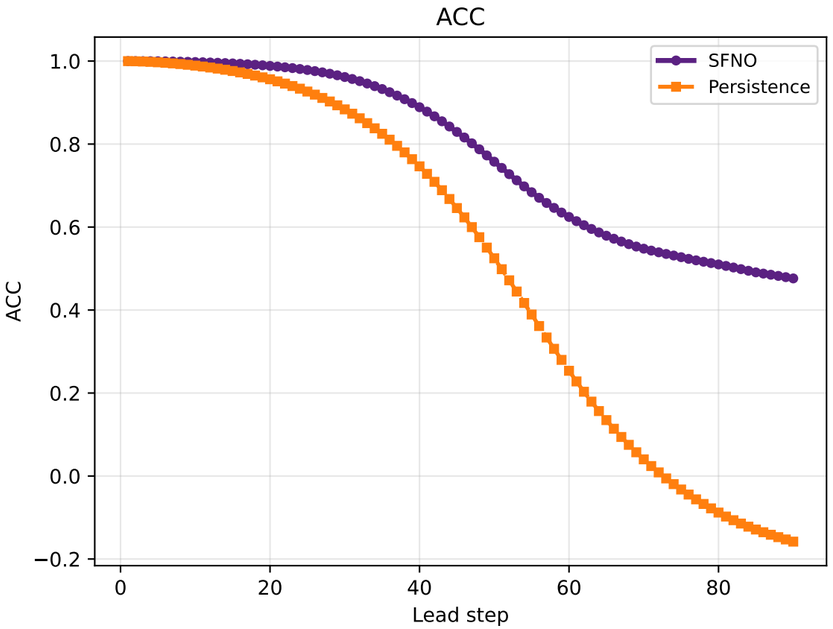
DHW Metrics by Lead Step - Val (15 chains scored at every lead)

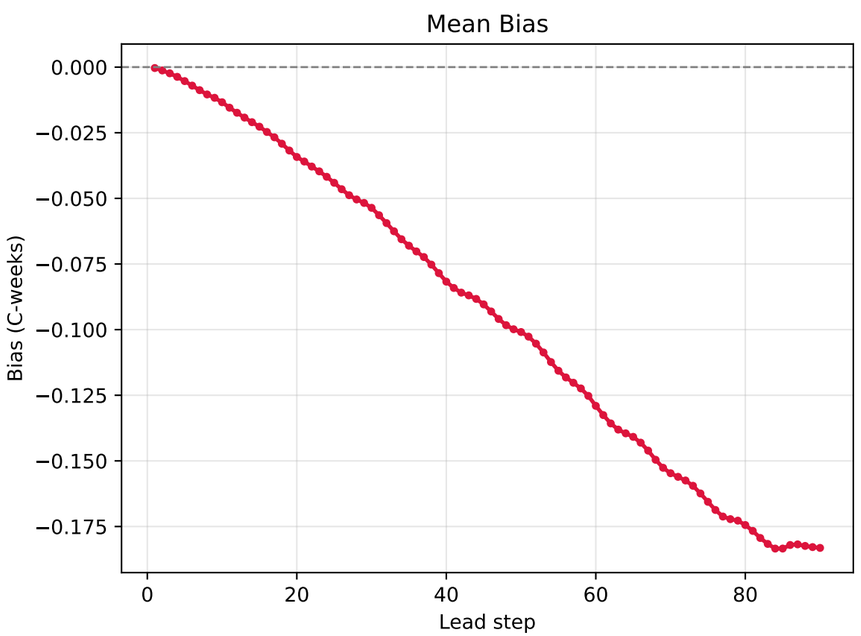
Lead Step	SFNO RMSE	Persist RMSE	Skill %	SFNO MAE	Persist MAE	SFNO Bias	SFNO ACC	Persist ACC	SFNO R²
1	0.0165	0.0418	60.6%	0.0017	0.0057	-0.0004	1.0000	0.9999	1.0000
2	0.0311	0.0818	61.9%	0.0034	0.0114	-0.0013	0.9999	0.9995	0.9999
3	0.0468	0.1205	61.1%	0.0053	0.0171	-0.0024	0.9998	0.9989	0.9997
4	0.0632	0.1589	60.2%	0.0073	0.0228	-0.0037	0.9997	0.9981	0.9995
5	0.0804	0.1972	59.2%	0.0095	0.0287	-0.0053	0.9995	0.9971	0.9992
6	0.0985	0.2350	58.1%	0.0118	0.0346	-0.0070	0.9993	0.9959	0.9988
7	0.1173	0.2720	56.9%	0.0143	0.0403	-0.0088	0.9990	0.9944	0.9983
8	0.1363	0.3084	55.8%	0.0167	0.0460	-0.0104	0.9986	0.9928	0.9977
9	0.1549	0.3444	55.0%	0.0194	0.0517	-0.0117	0.9982	0.9911	0.9970
10	0.1741	0.3804	54.2%	0.0219	0.0576	-0.0134	0.9977	0.9891	0.9962
11	0.1933	0.4166	53.6%	0.0244	0.0636	-0.0155	0.9972	0.9869	0.9954
12	0.2127	0.4526	53.0%	0.0270	0.0697	-0.0174	0.9966	0.9845	0.9943
13	0.2318	0.4885	52.5%	0.0296	0.0760	-0.0192	0.9959	0.9819	0.9932
14	0.2509	0.5245	52.2%	0.0322	0.0824	-0.0210	0.9952	0.9791	0.9920
15	0.2704	0.5608	51.8%	0.0350	0.0890	-0.0227	0.9943	0.9760	0.9907
16	0.2906	0.5981	51.4%	0.0380	0.0961	-0.0247	0.9934	0.9727	0.9892
17	0.3116	0.6362	51.0%	0.0411	0.1034	-0.0267	0.9923	0.9691	0.9874
18	0.3330	0.6749	50.7%	0.0443	0.1111	-0.0292	0.9911	0.9651	0.9855
19	0.3549	0.7145	50.3%	0.0476	0.1189	-0.0318	0.9898	0.9608	0.9834
20	0.3769	0.7547	50.1%	0.0511	0.1271	-0.0342	0.9883	0.9562	0.9811
21	0.3983	0.7957	49.9%	0.0551	0.1356	-0.0360	0.9868	0.9512	0.9787
22	0.4206	0.8377	49.8%	0.0592	0.1446	-0.0379	0.9850	0.9457	0.9759
23	0.4433	0.8806	49.7%	0.0635	0.1539	-0.0397	0.9831	0.9399	0.9730
24	0.4670	0.9245	49.5%	0.0678	0.1636	-0.0418	0.9809	0.9335	0.9696
25	0.4912	0.9697	49.3%	0.0721	0.1736	-0.0441	0.9785	0.9266	0.9659
26	0.5161	1.0157	49.2%	0.0765	0.1839	-0.0465	0.9758	0.9191	0.9619
27	0.5423	1.0625	49.0%	0.0812	0.1945	-0.0488	0.9728	0.9112	0.9573
28	0.5688	1.1099	48.7%	0.0863	0.2055	-0.0504	0.9695	0.9026	0.9524
29	0.5967	1.1584	48.5%	0.0919	0.2170	-0.0517	0.9658	0.8934	0.9469
30	0.6261	1.2074	48.1%	0.0975	0.2288	-0.0536	0.9616	0.8836	0.9407
31	0.6567	1.2569	47.8%	0.1030	0.2407	-0.0564	0.9568	0.8732	0.9338
32	0.6879	1.3067	47.4%	0.1085	0.2527	-0.0594	0.9516	0.8621	0.9263
33	0.7195	1.3569	47.0%	0.1141	0.2650	-0.0625	0.9459	0.8504	0.9182
34	0.7521	1.4076	46.6%	0.1200	0.2776	-0.0656	0.9396	0.8380	0.9093
35	0.7859	1.4593	46.1%	0.1264	0.2906	-0.0680	0.9326	0.8247	0.8996
36	0.8201	1.5117	45.8%	0.1329	0.3038	-0.0702	0.9250	0.8106	0.8890
37	0.8542	1.5646	45.4%	0.1394	0.3173	-0.0724	0.9168	0.7957	0.8779
38	0.8882	1.6177	45.1%	0.1457	0.3312	-0.0752	0.9081	0.7800	0.8660
39	0.9224	1.6708	44.8%	0.1520	0.3452	-0.0785	0.8988	0.7636	0.8535
40	0.9569	1.7241	44.5%	0.1584	0.3594	-0.0817	0.8888	0.7464	0.8403
41	0.9916	1.7780	44.2%	0.1652	0.3740	-0.0841	0.8782	0.7282	0.8263
42	1.0270	1.8327	44.0%	0.1724	0.3888	-0.0859	0.8669	0.7090	0.8114
43	1.0625	1.8883	43.7%	0.1799	0.4038	-0.0870	0.8549	0.6888	0.7959
44	1.0984	1.9443	43.5%	0.1874	0.4191	-0.0883	0.8424	0.6676	0.7796
45	1.1342	1.9999	43.3%	0.1949	0.4345	-0.0904	0.8293	0.6458	0.7628
46	1.1698	2.0555	43.1%	0.2023	0.4499	-0.0931	0.8158	0.6233	0.7455
47	1.2057	2.1115	42.9%	0.2098	0.4656	-0.0959	0.8018	0.5999	0.7277
48	1.2421	2.1682	42.7%	0.2175	0.4813	-0.0983	0.7874	0.5756	0.7091
49	1.2791	2.2251	42.5%	0.2255	0.4971	-0.0999	0.7726	0.5505	0.6901
50	1.3166	2.2823	42.3%	0.2339	0.5130	-0.1009	0.7576	0.5248	0.6705
51	1.3536	2.3396	42.1%	0.2420	0.5290	-0.1027	0.7425	0.4984	0.6508
52	1.3903	2.3967	42.0%	0.2496	0.5449	-0.1053	0.7275	0.4717	0.6310
53	1.4266	2.4537	41.9%	0.2570	0.5607	-0.1087	0.7126	0.4446	0.6116
54	1.4627	2.5113	41.8%	0.2643	0.5766	-0.1124	0.6980	0.4170	0.5924
55	1.4986	2.5689	41.7%	0.2715	0.5923	-0.1157	0.6840	0.3893	0.5736
56	1.5344	2.6262	41.6%	0.2790	0.6080	-0.1182	0.6705	0.3615	0.5553
57	1.5698	2.6833	41.5%	0.2867	0.6237	-0.1202	0.6580	0.3339	0.5378
58	1.6051	2.7398	41.4%	0.2942	0.6392	-0.1224	0.6462	0.3066	0.5209
59	1.6398	2.7954	41.3%	0.3013	0.6543	-0.1252	0.6350	0.2799	0.5048
60	1.6740	2.8507	41.3%	0.3081	0.6692	-0.1290	0.6244	0.2536	0.4897
61	1.7085	2.9054	41.2%	0.3149	0.6839	-0.1326	0.6142	0.2280	0.4751
62	1.7439	2.9593	41.1%	0.3219	0.6983	-0.1357	0.6045	0.2031	0.4606
63	1.7801	3.0117	40.9%	0.3290	0.7123	-0.1381	0.5954	0.1793	0.4460
64	1.8167	3.0626	40.7%	0.3366	0.7259	-0.1395	0.5870	0.1565	0.4315
65	1.8531	3.1120	40.5%	0.3440	0.7391	-0.1408	0.5791	0.1348	0.4172
66	1.8885	3.1601	40.2%	0.3509	0.7519	-0.1431	0.5719	0.1140	0.4040
67	1.9229	3.2071	40.0%	0.3573	0.7641	-0.1461	0.5651	0.0941	0.3918
68	1.9561	3.2530	39.9%	0.3635	0.7761	-0.1496	0.5588	0.0752	0.3806
69	1.9884	3.2971	39.7%	0.3697	0.7876	-0.1526	0.5531	0.0572	0.3700
70	2.0207	3.3397	39.5%	0.3760	0.7986	-0.1547	0.5479	0.0402	0.3594
71	2.0526	3.3806	39.3%	0.3825	0.8092	-0.1561	0.5433	0.0240	0.3489
72	2.0843	3.4203	39.1%	0.3885	0.8194	-0.1575	0.5391	0.0088	0.3388
73	2.1147	3.4585	38.9%	0.3939	0.8291	-0.1595	0.5352	-0.0057	0.3293
74	2.1435	3.4954	38.7%	0.3987	0.8384	-0.1624	0.5314	-0.0195	0.3209
75	2.1712	3.5305	38.5%	0.4032	0.8474	-0.1656	0.5274	-0.0325	0.3128
76	2.1989	3.5641	38.3%	0.4078	0.8560	-0.1687	0.5234	-0.0446	0.3046
77	2.2271	3.5969	38.1%	0.4125	0.8644	-0.1712	0.5198	-0.0562	0.2962
78	2.2564	3.6284	37.8%	0.4178	0.8724	-0.1722	0.5164	-0.0673	0.2868
79	2.2854	3.6583	37.5%	0.4232	0.8801	-0.1728	0.5132	-0.0779	0.2771
80	2.3130	3.6871	37.3%	0.4277	0.8873	-0.1744	0.5099	-0.0880	0.2678
81	2.3395	3.7142	37.0%	0.4319	0.8942	-0.1767	0.5063	-0.0976	0.2587
82	2.3647	3.7392	36.8%	0.4356	0.9006	-0.1794	0.5025	-0.1065	0.2499
83	2.3898	3.7626	36.5%	0.4392	0.9064	-0.1816	0.4986	-0.1145	0.2408
84	2.4159	3.7845	36.2%	0.4431	0.9119	-0.1834	0.4946	-0.1220	0.2309
85	2.4430	3.8050	35.8%	0.4471	0.9171	-0.1834	0.4908	-0.1289	0.2201
86	2.4691	3.8241	35.4%	0.4511	0.9218	-0.1820	0.4879	-0.1354	0.2093
87	2.4934	3.8419	35.1%	0.4539	0.9263	-0.1818	0.4851	-0.1415	0.1993
88	2.5145	3.8579	34.8%	0.4556	0.9304	-0.1824	0.4823	-0.1473	0.1902
89	2.5336	3.8719	34.6%	0.4569	0.9341	-0.1828	0.4793	-0.1528	0.1813
90	2.5513	3.8845	34.3%	0.4580	0.9374	-0.1831	0.4762	-0.1581	0.1724

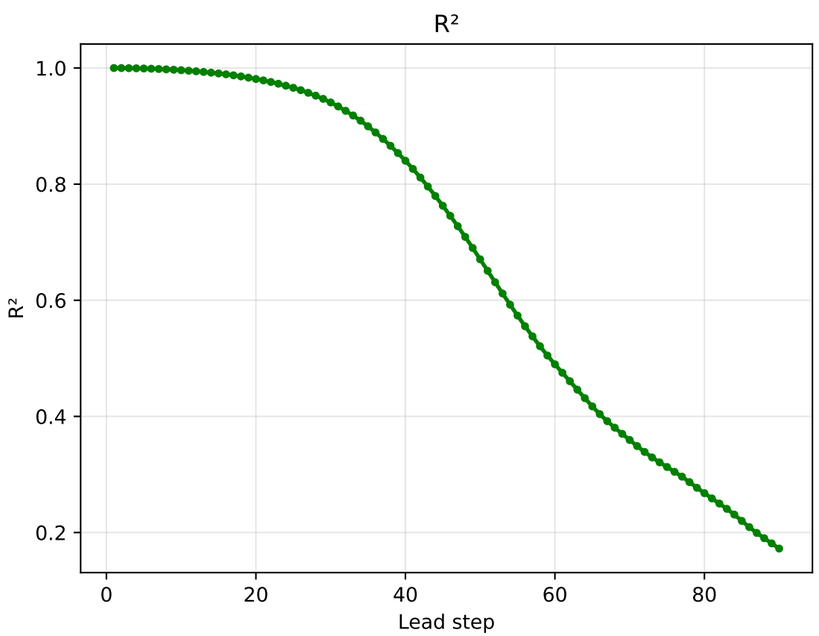
DHW Forecast Skill vs Lead Step - Val

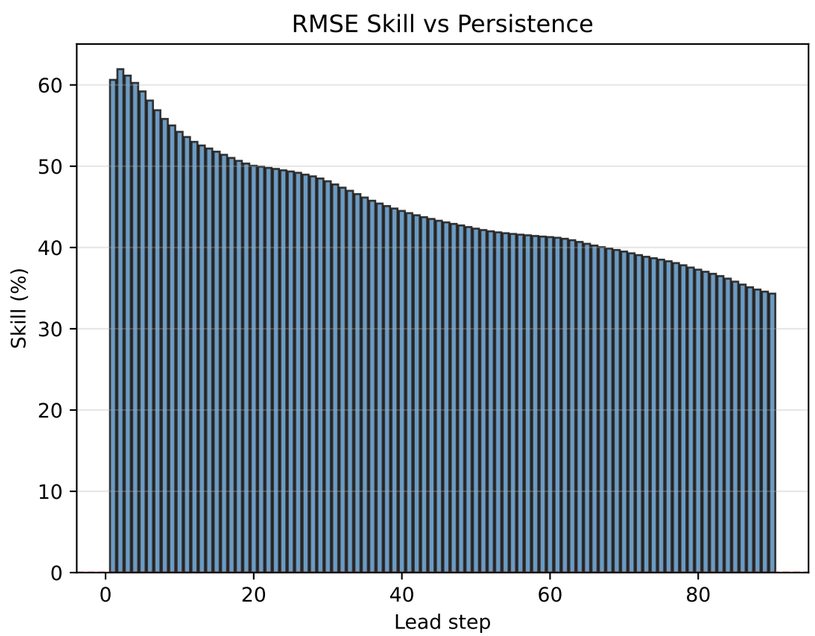


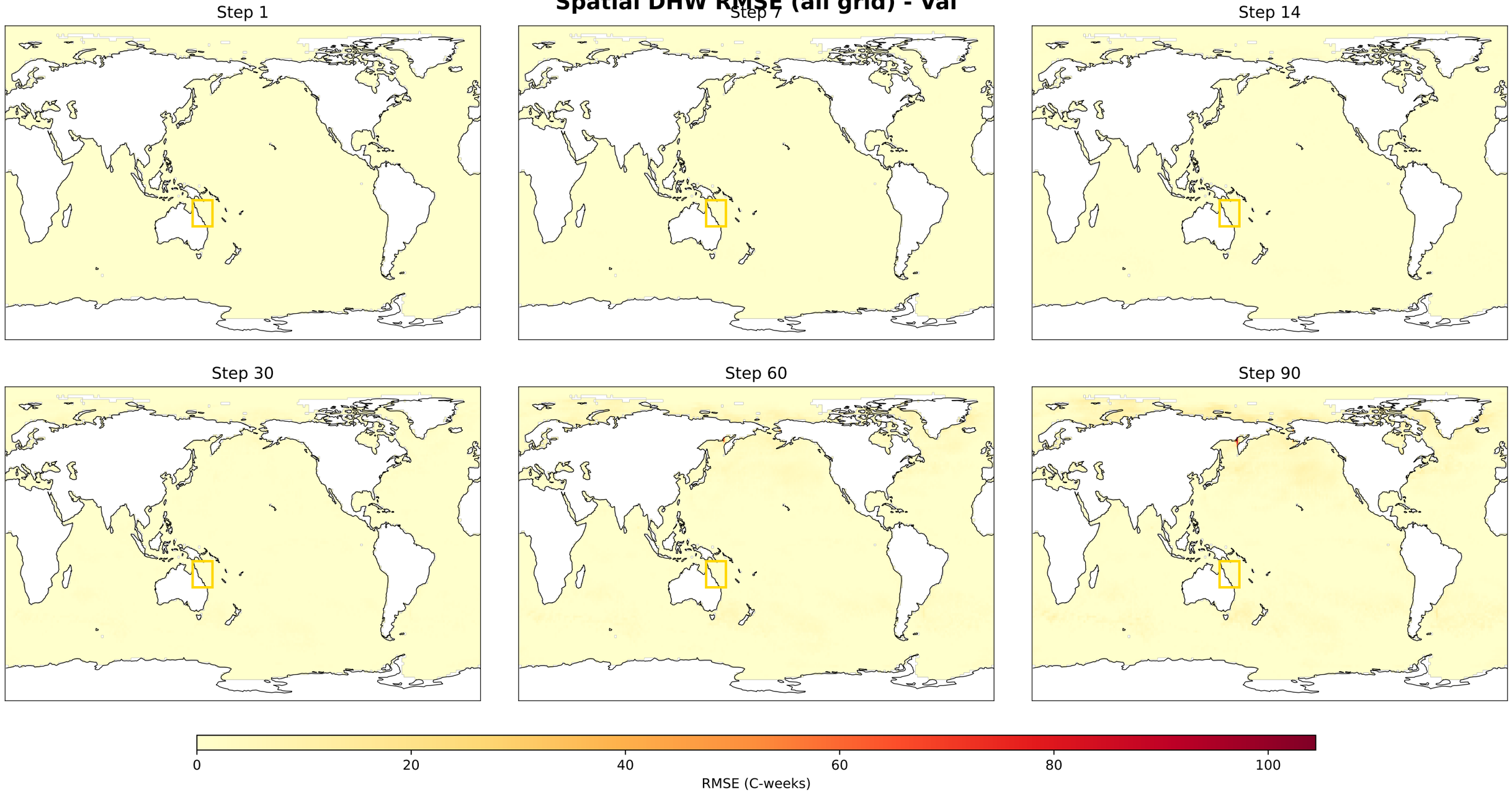


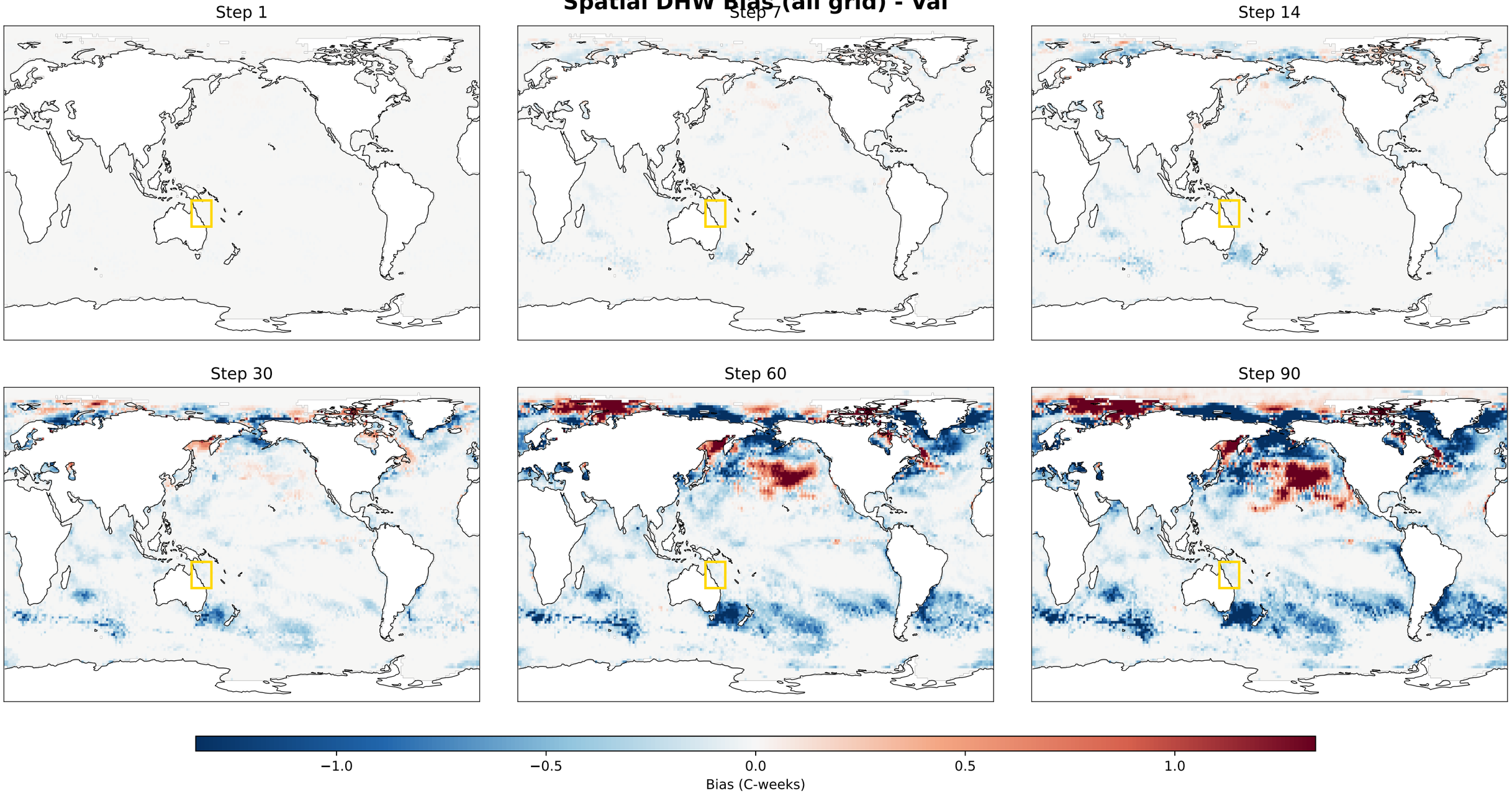


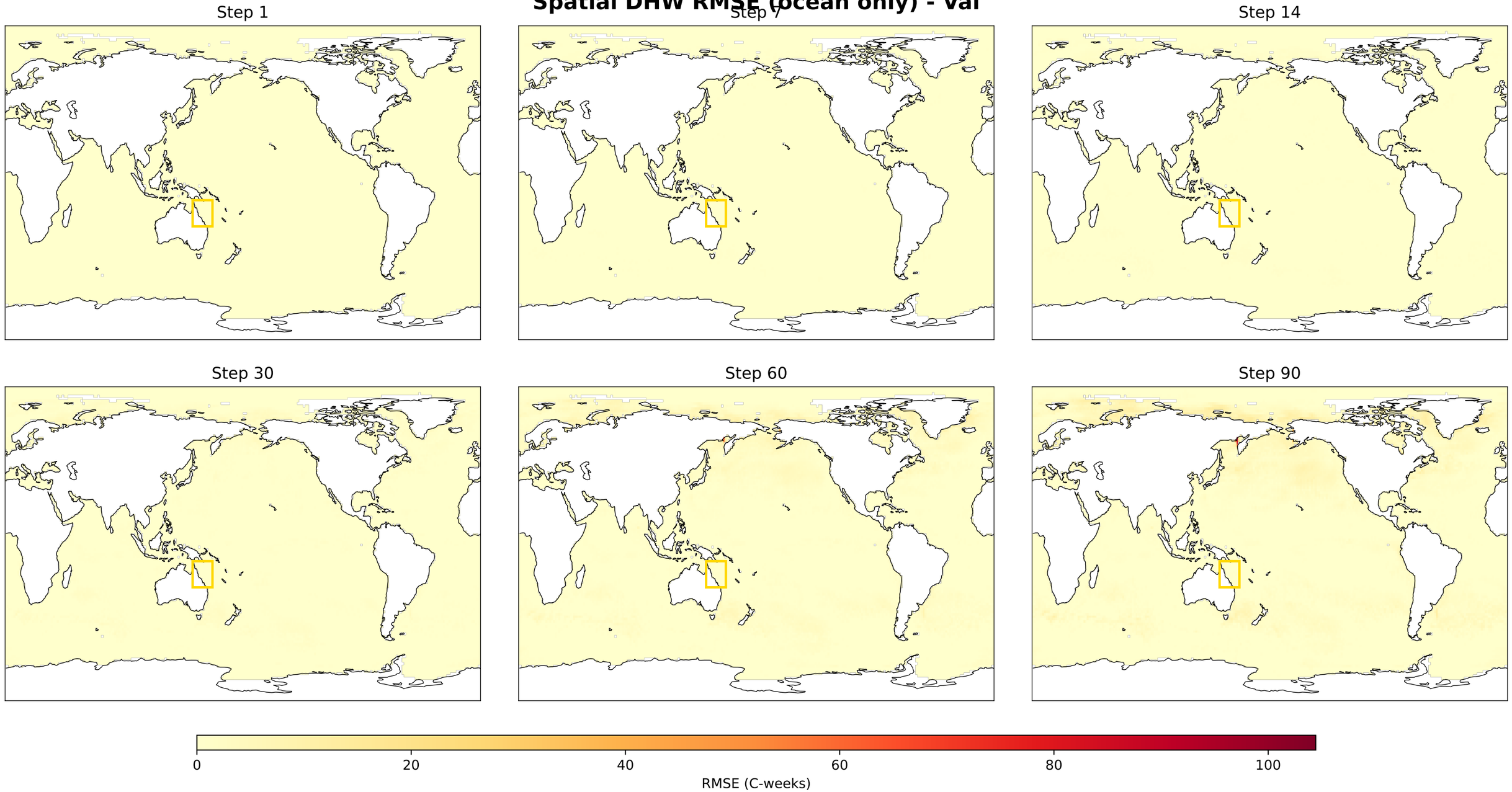


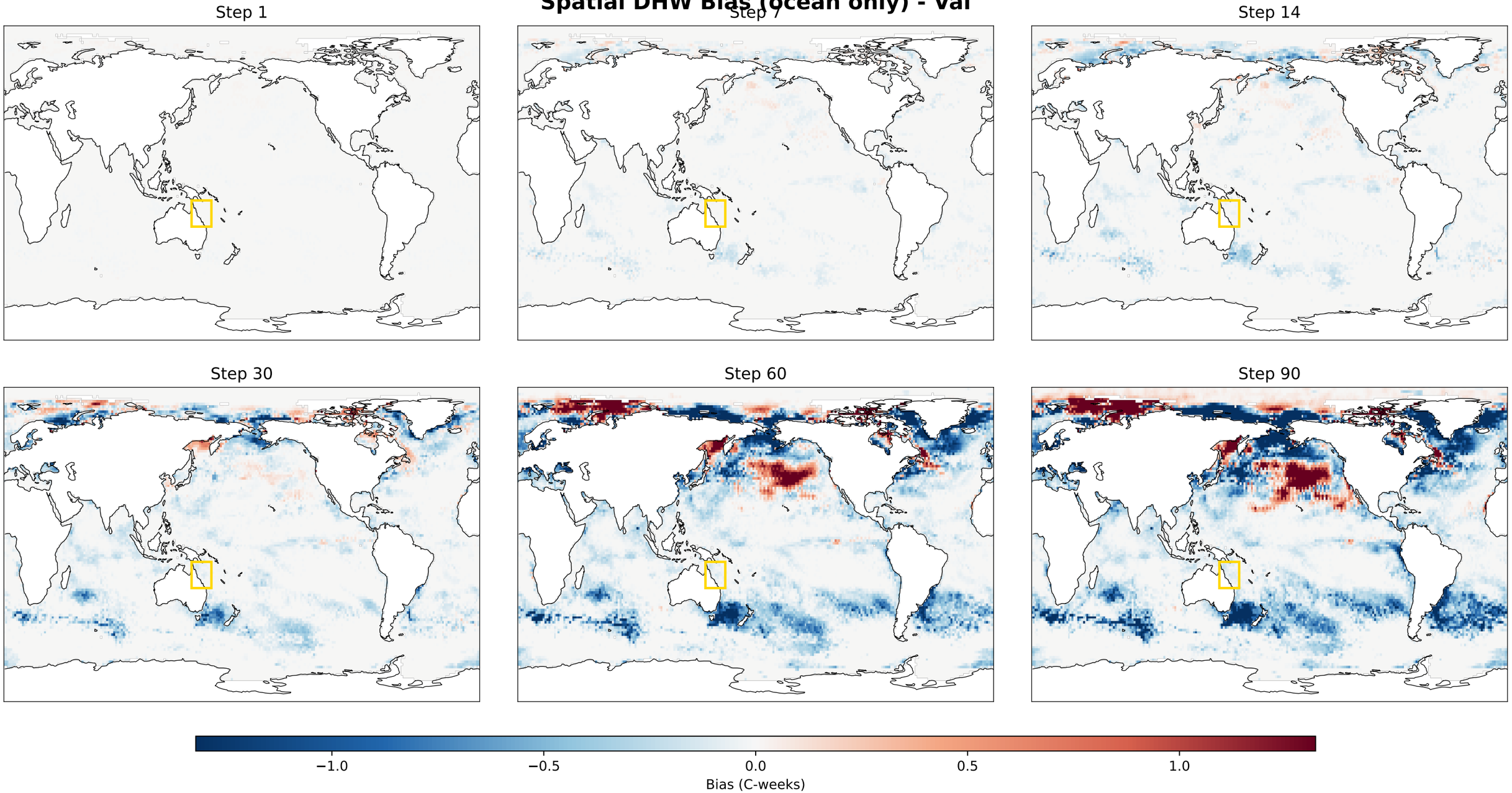





Spatial DHW RMSE (all grid) - Val
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0
20
40
60
80
100
RMSE (C-weeks)

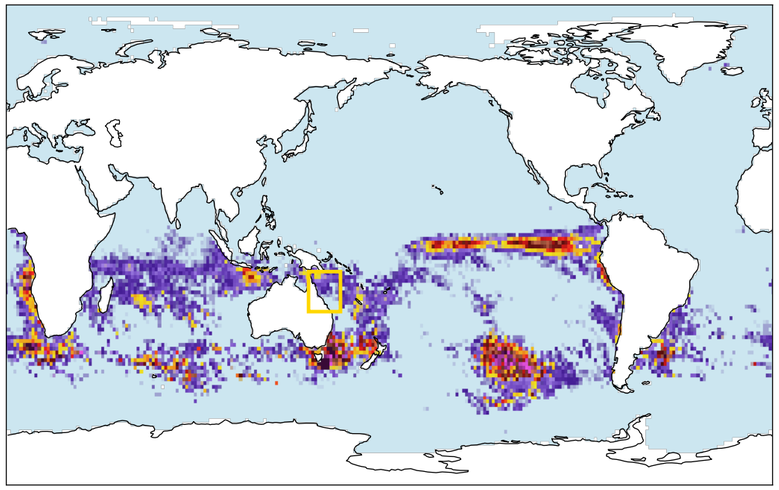

Spatial DHW Bias (all grid) - Val
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−1.0
−0.5
0.0
0.5
1.0
Bias (C-weeks)


Spatial DHW RMSE (ocean only) - Val
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0
20
40
60
80
100
RMSE (C-weeks)

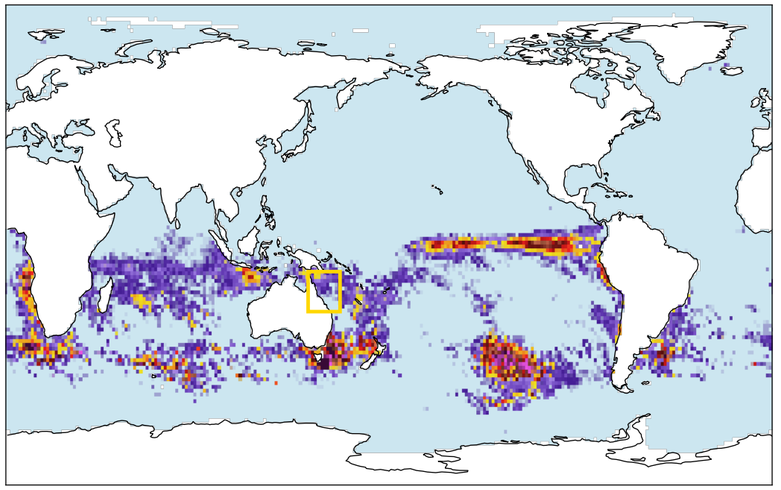

Spatial DHW Bias (ocean only) - Val
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−1.0
−0.5
0.0
0.5
1.0
Bias (C-weeks)

DHW Chain init: 2016-04-10 - Val (C-weeks)

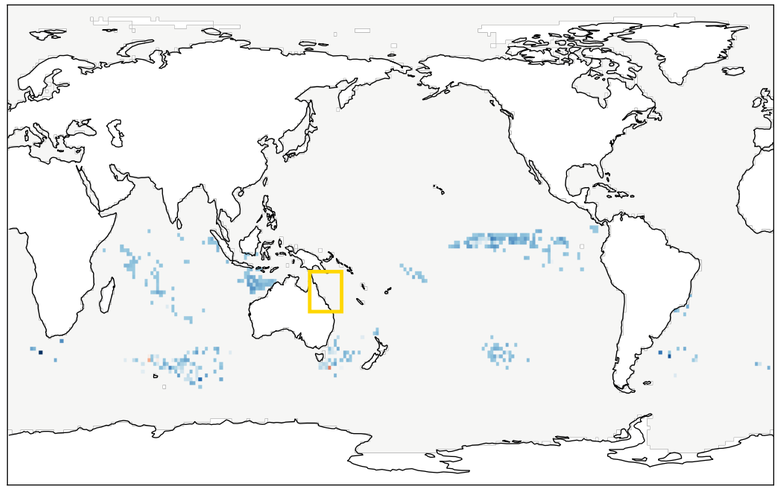
Actual DHW - Step 1



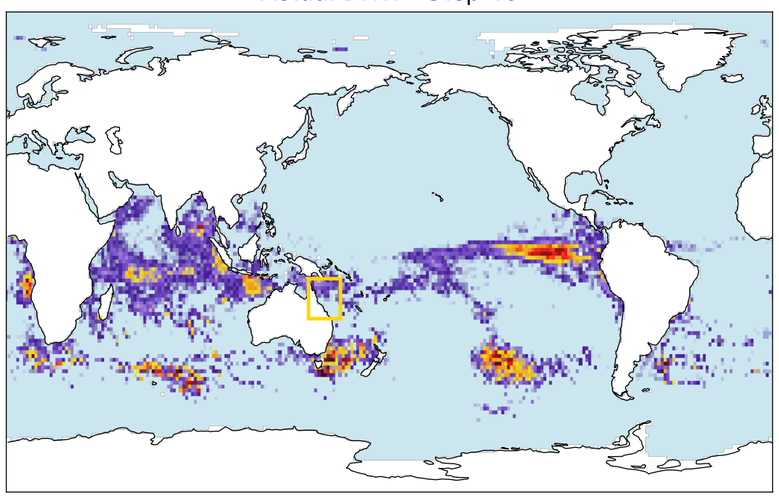
Predicted DHW - Step 1



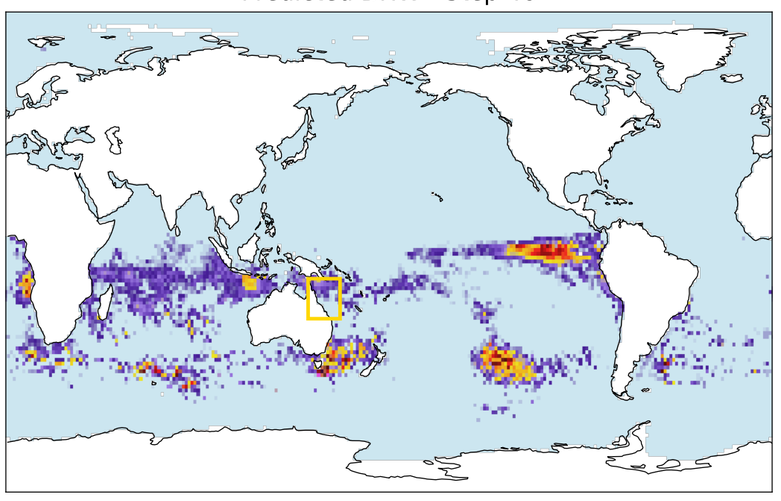
Error (Pred - Actual) - Step 1 [scale ±0.48]



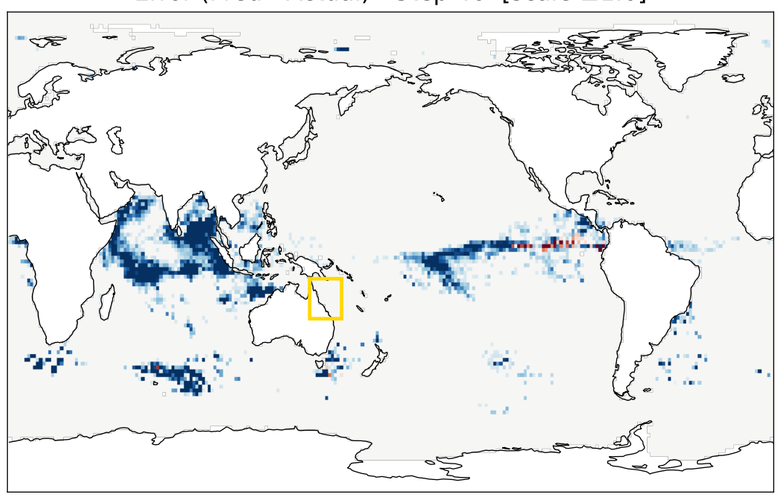
Actual DHW - Step 46



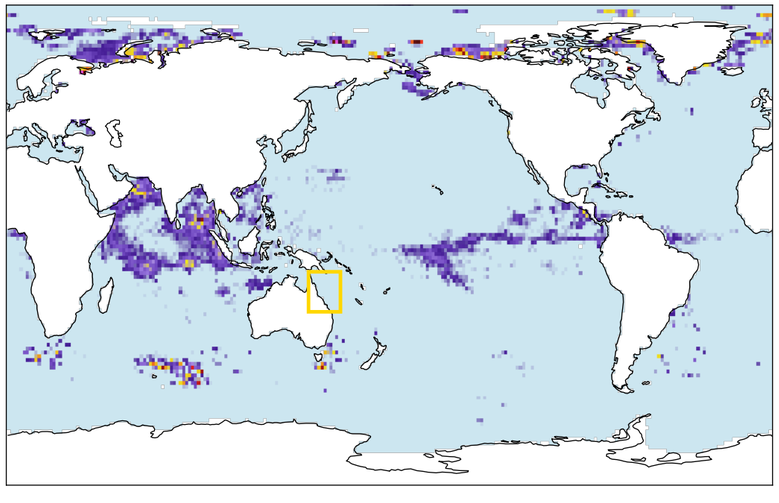
Predicted DHW - Step 46



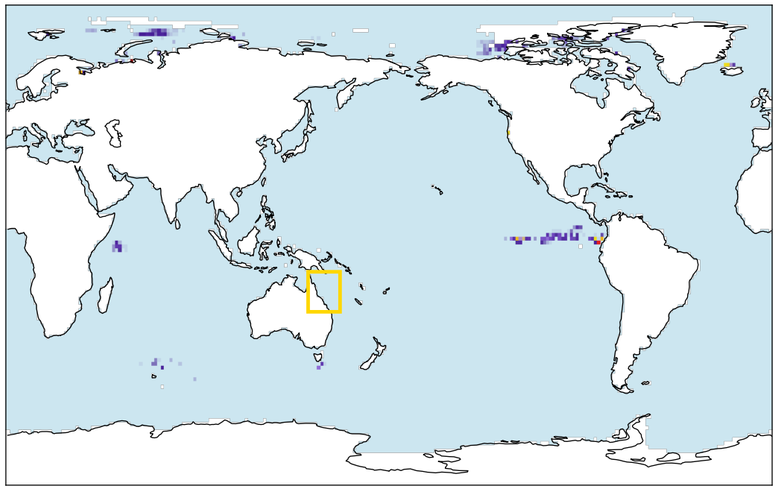
Error (Pred - Actual) - Step 46 [scale ±1.6]



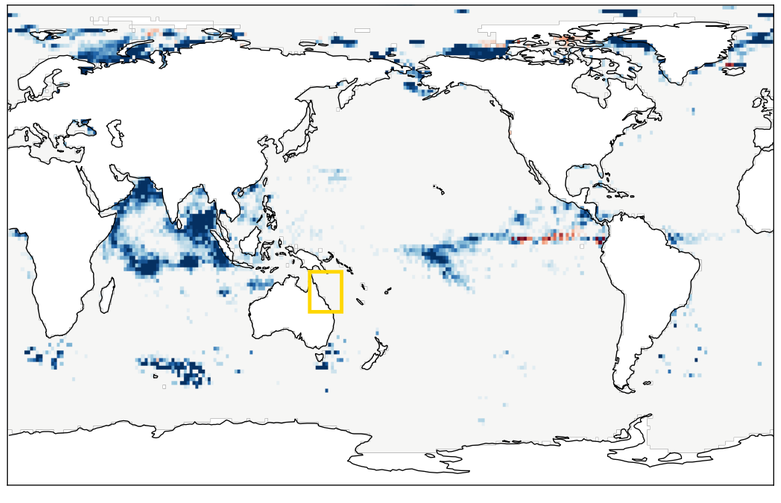
Actual DHW - Step 90

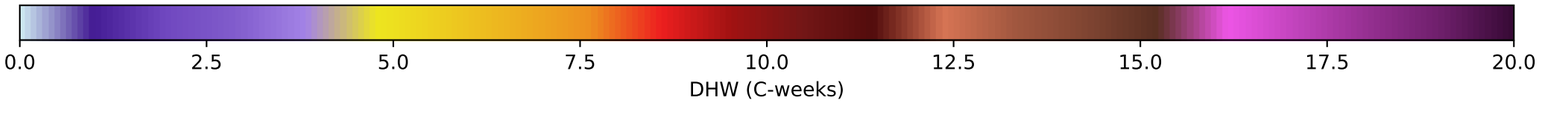


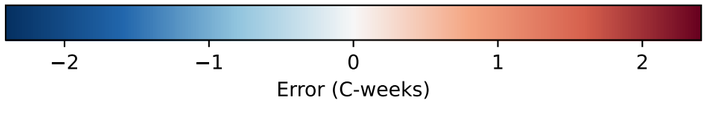
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±2.4]

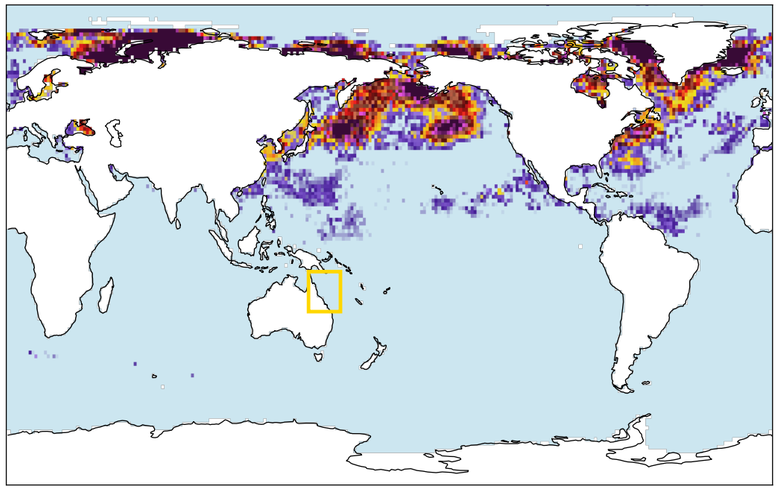




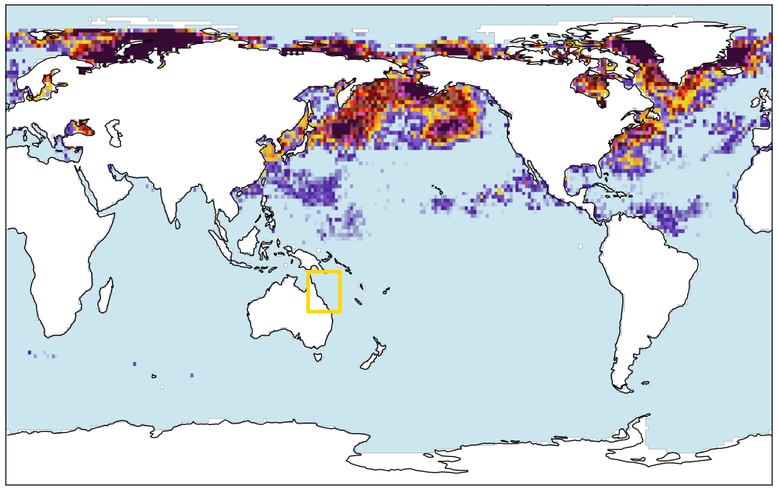


DHW Chain init: 2016-10-07 - Val (C-weeks)

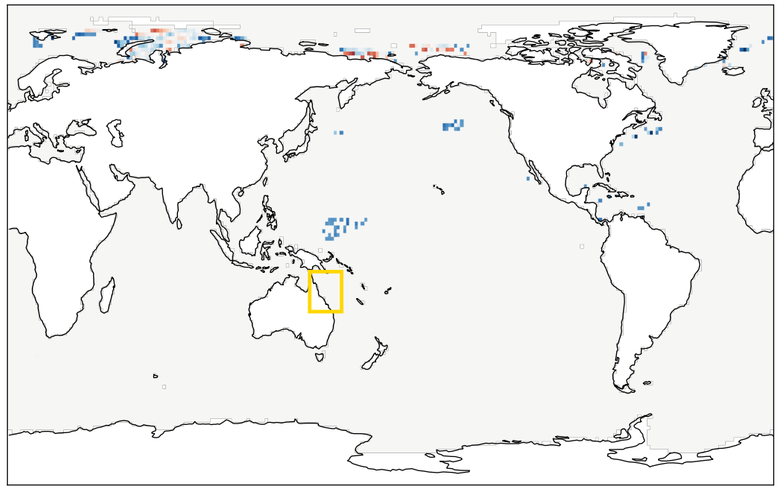
Actual DHW - Step 1



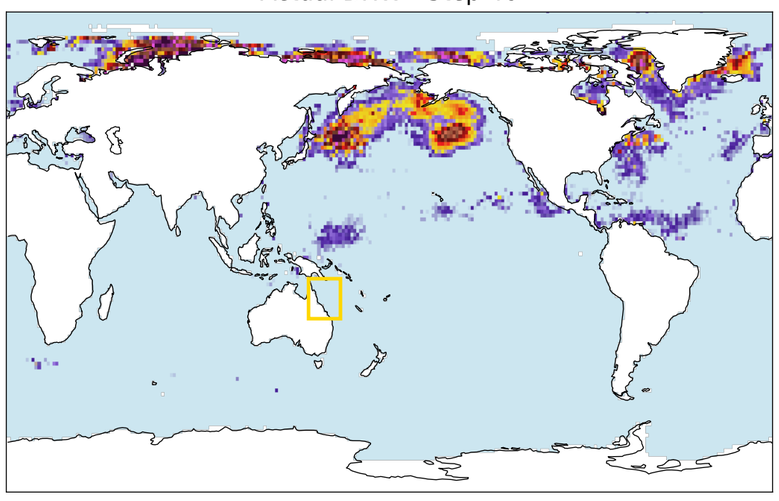
Predicted DHW - Step 1



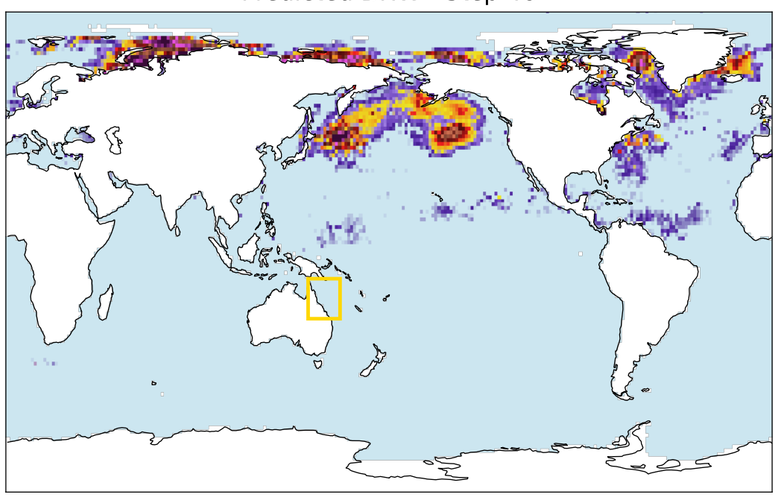
Error (Pred - Actual) - Step 1 [scale ±0.3]



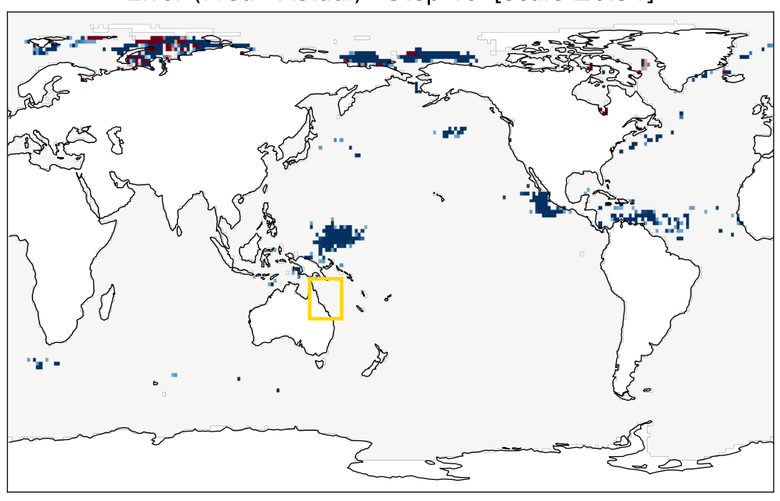
Actual DHW - Step 46



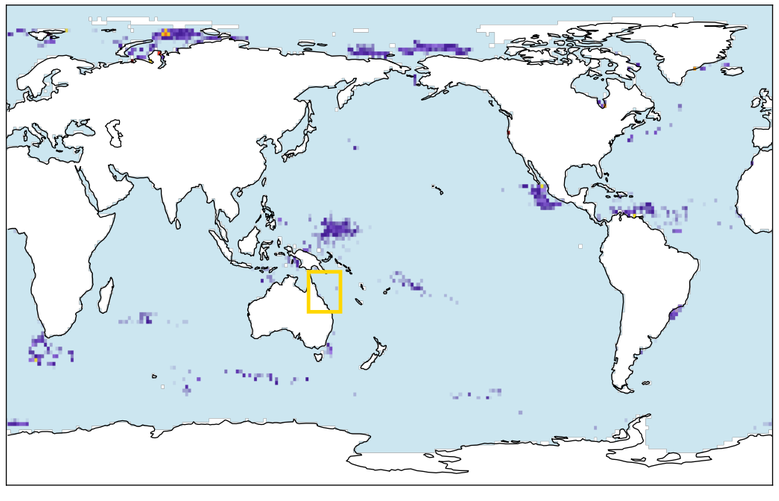
Predicted DHW - Step 46



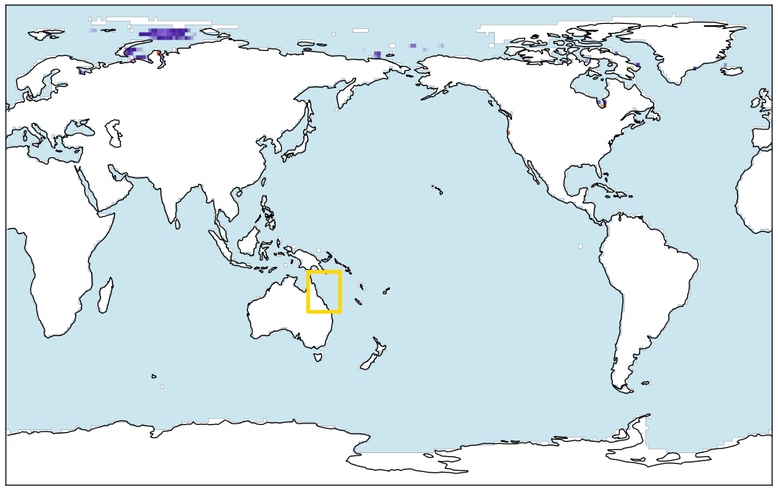
Error (Pred - Actual) - Step 46 [scale ±0.34]



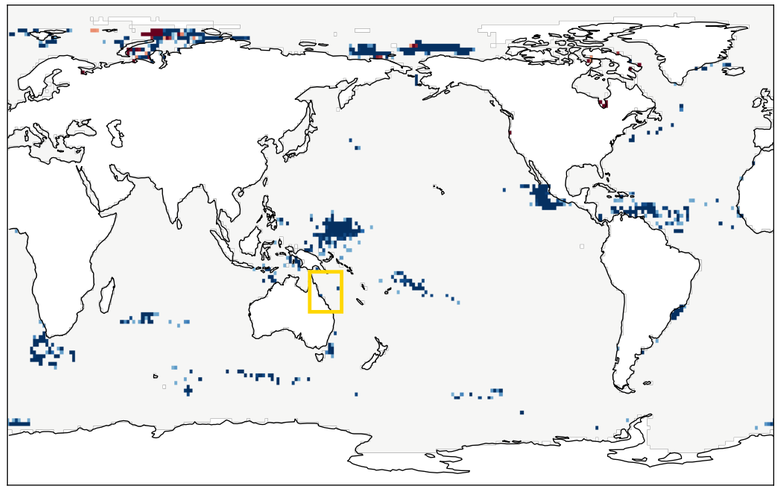
Actual DHW - Step 90

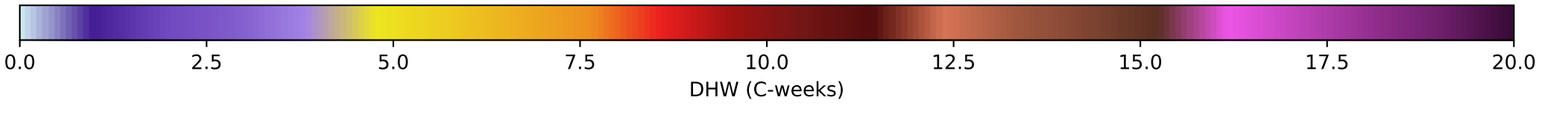


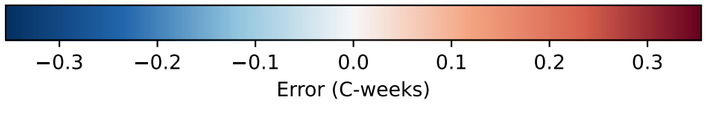
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±0.35]

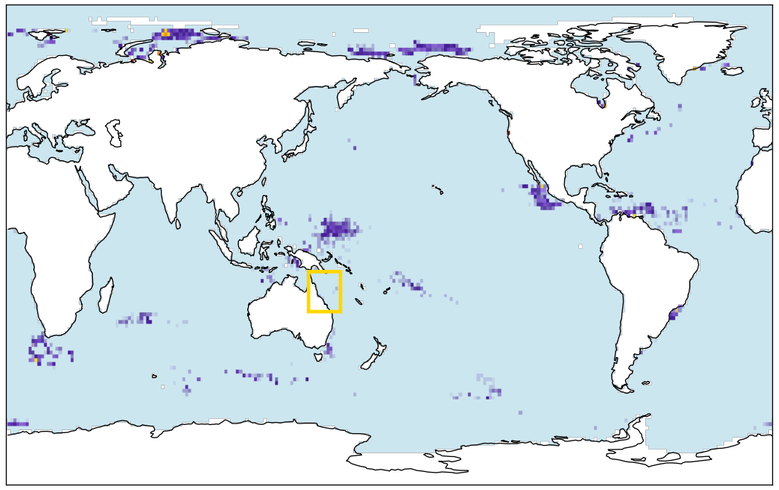




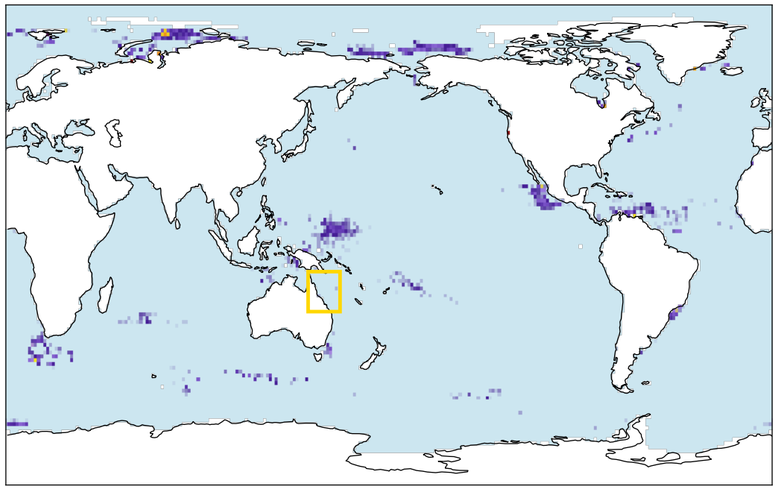


DHW Chain init: 2017-01-05 - Val (C-weeks)

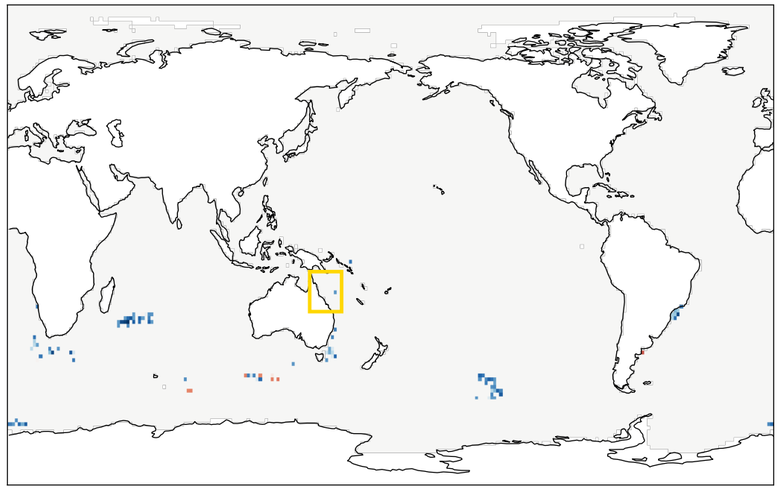
Actual DHW - Step 1



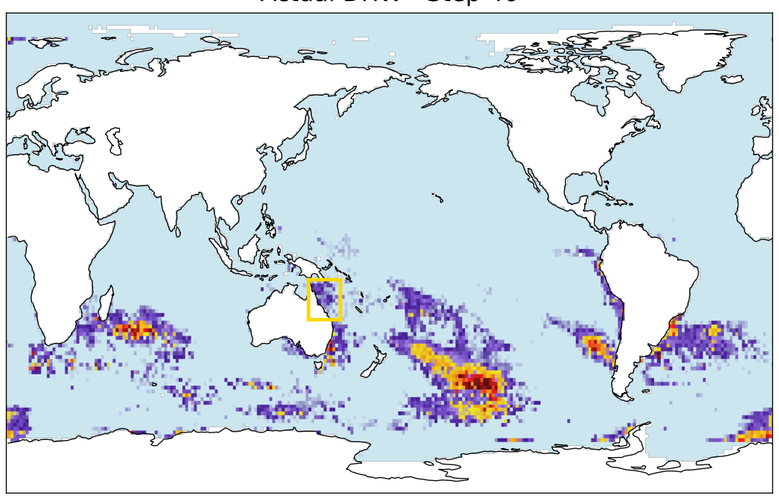
Predicted DHW - Step 1



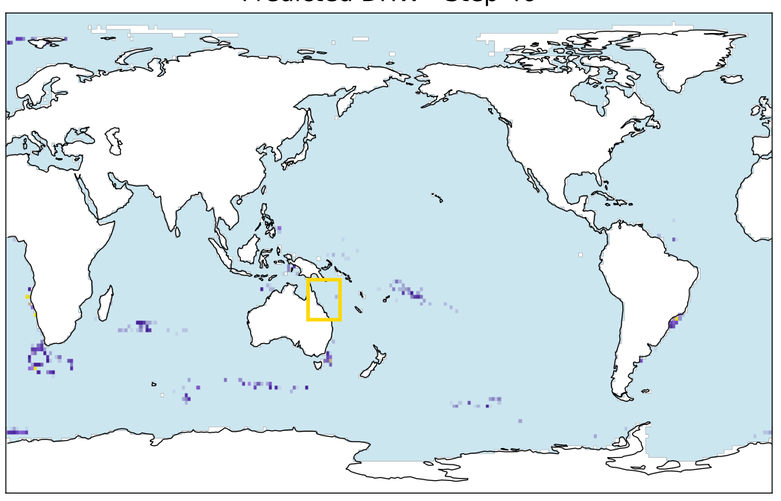
Error (Pred - Actual) - Step 1 [scale ±0.31]



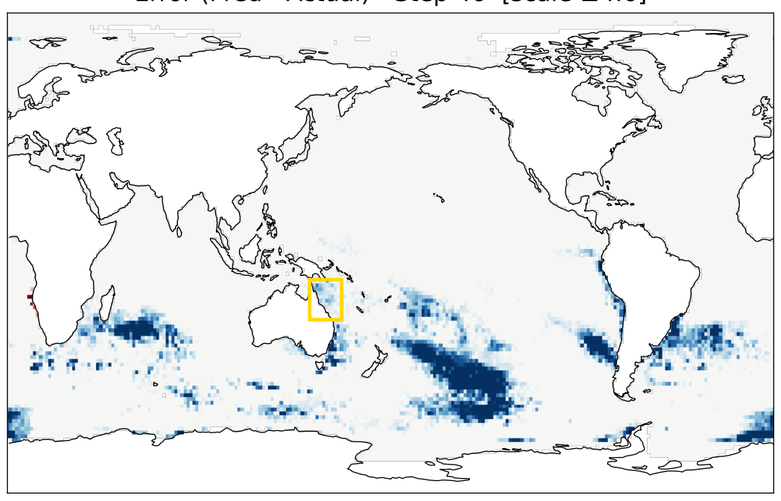
Actual DHW - Step 46



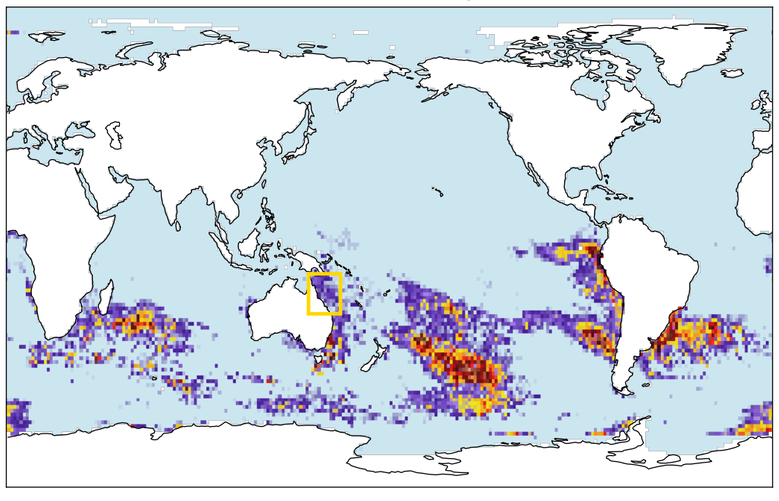
Predicted DHW - Step 46



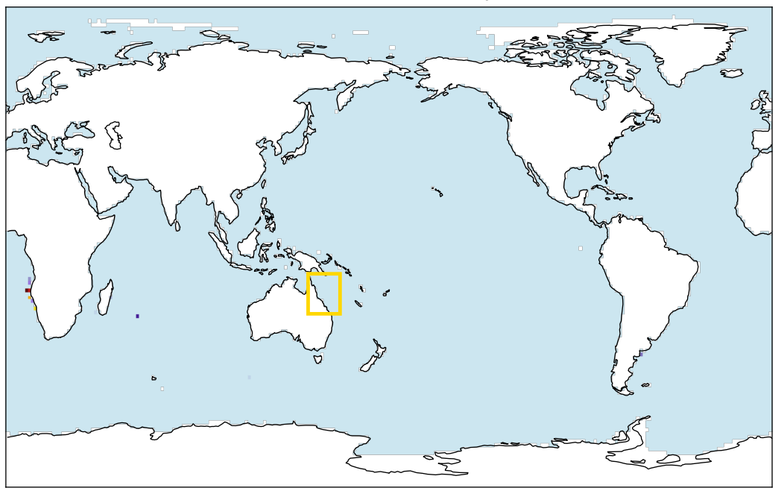
Error (Pred - Actual) - Step 46 [scale ±4.6]



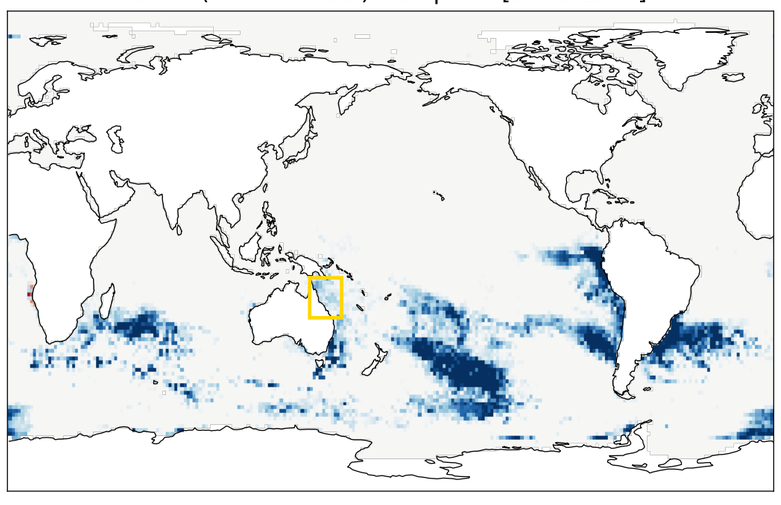
Actual DHW - Step 90

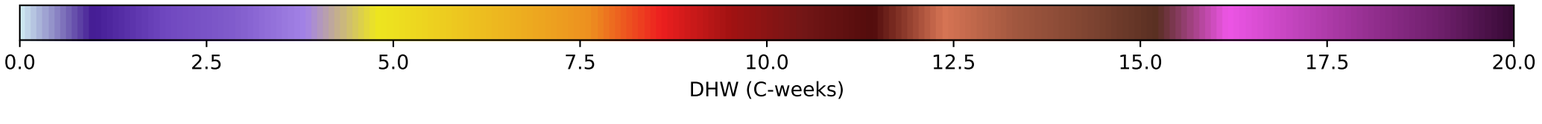


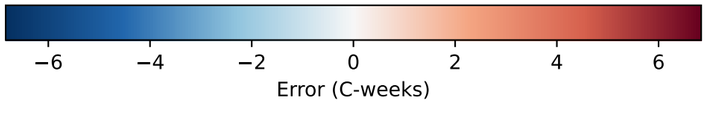
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±6.8]

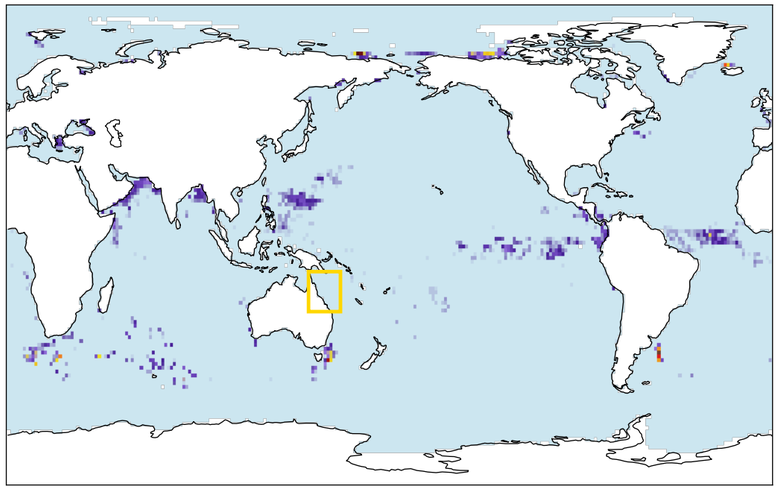




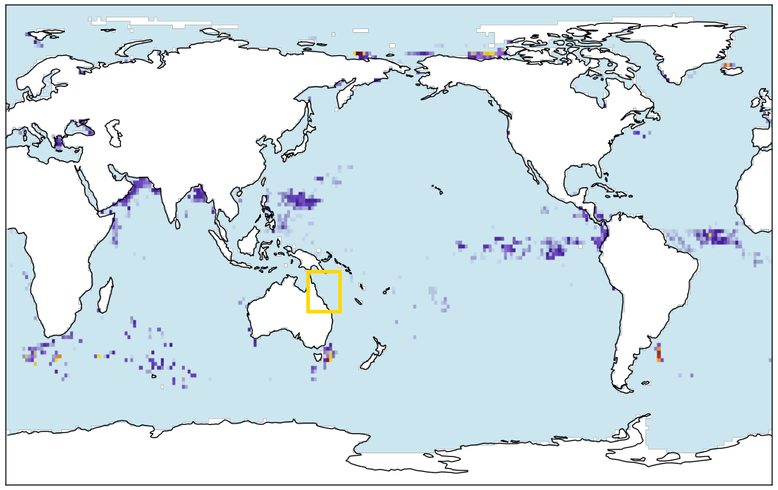


DHW Chain init: 2017-07-04 - Val (C-weeks)

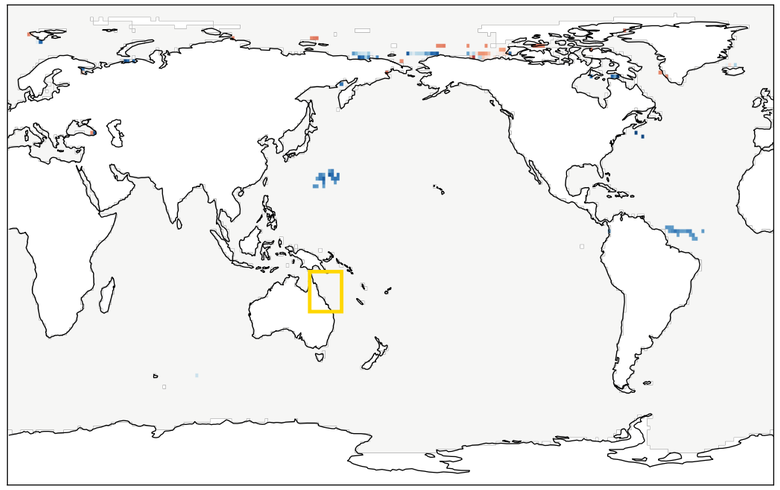
Actual DHW - Step 1



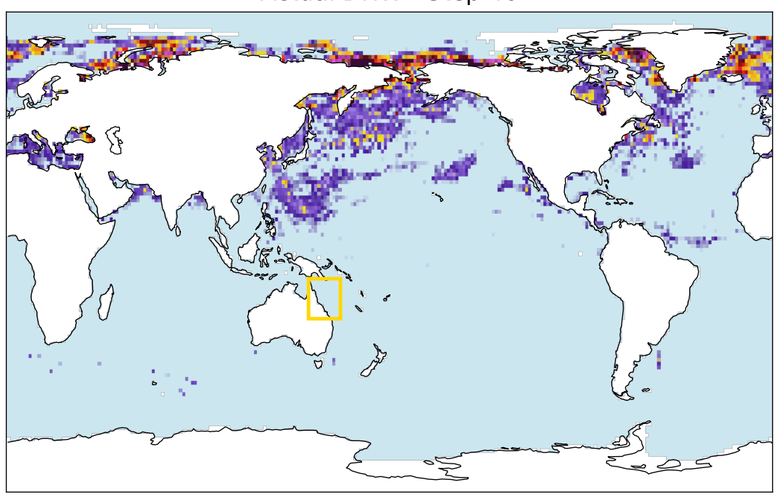
Predicted DHW - Step 1



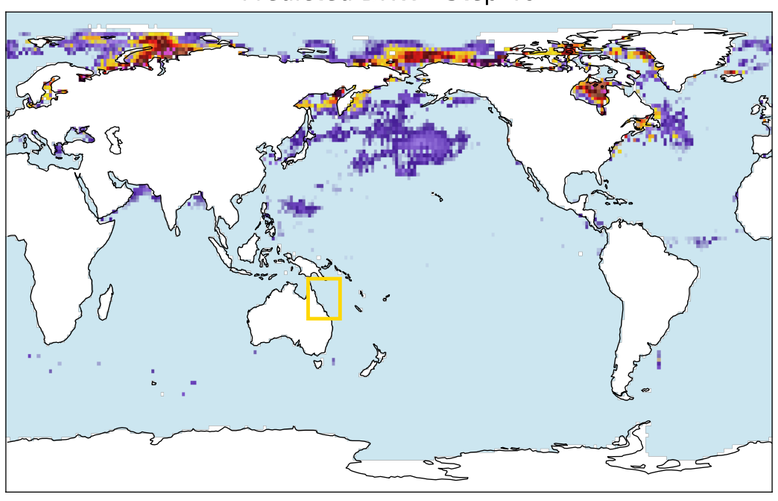
Error (Pred - Actual) - Step 1 [scale ±0.32]



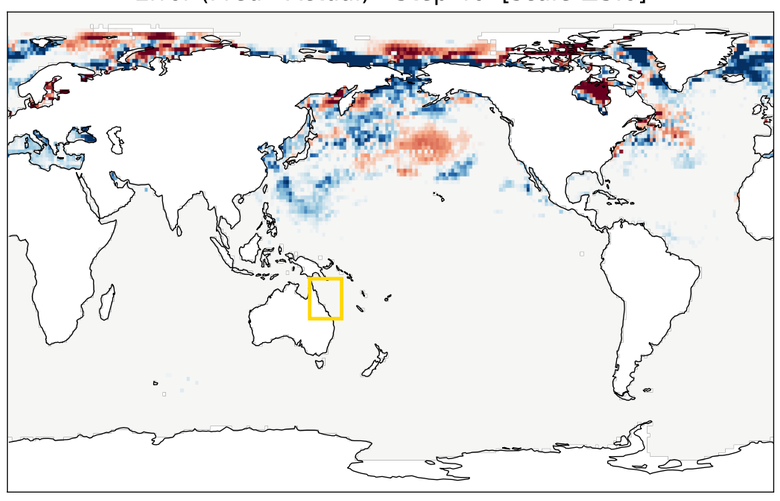
Actual DHW - Step 46



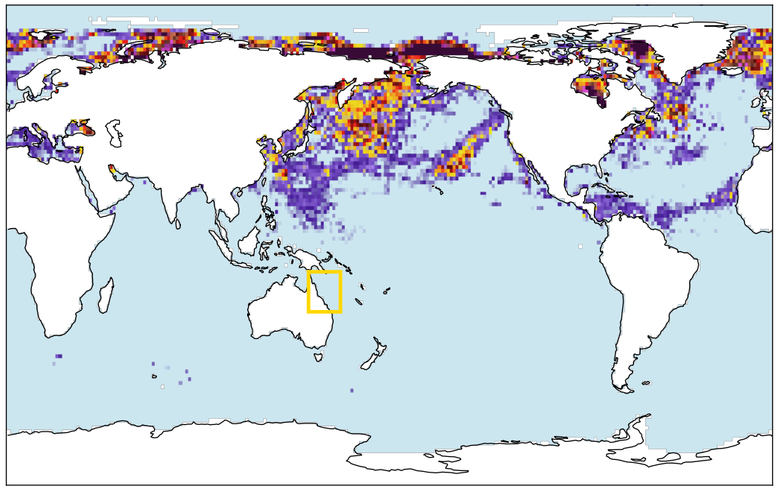
Predicted DHW - Step 46



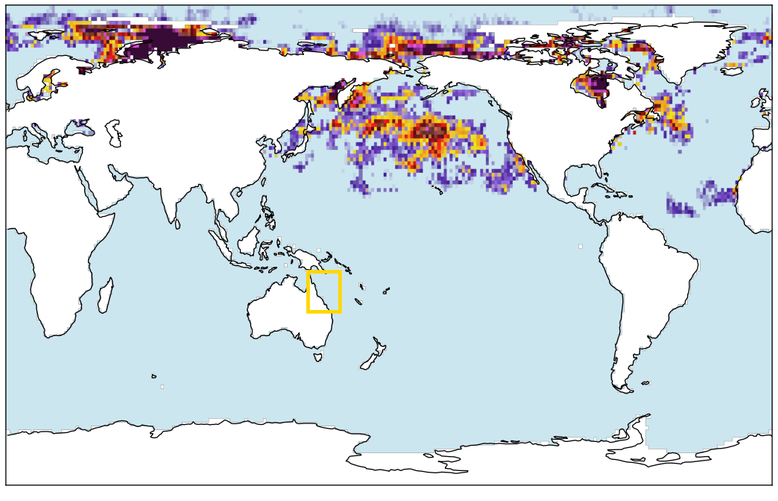
Error (Pred - Actual) - Step 46 [scale ±5.6]



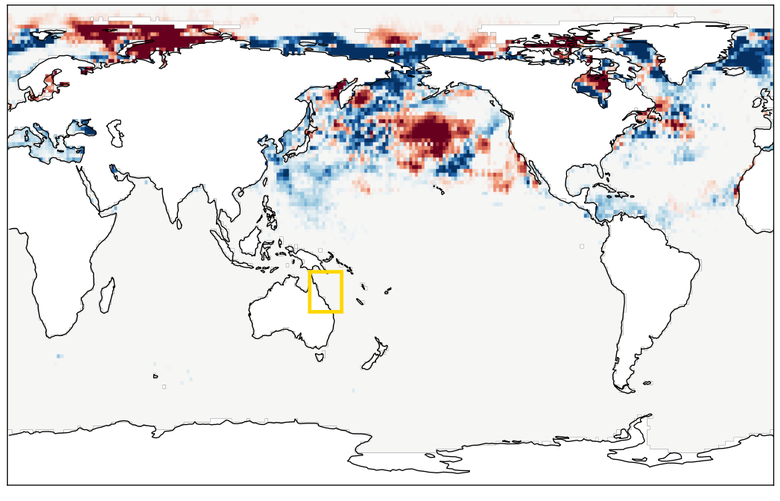
Actual DHW - Step 90

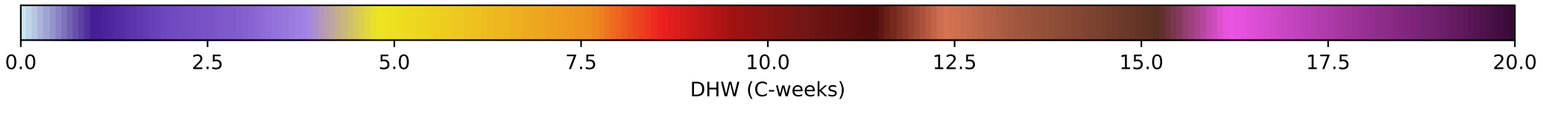


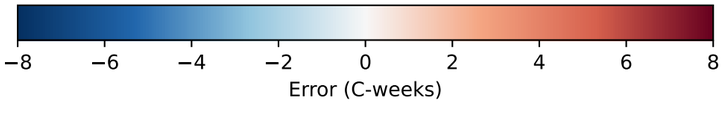
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

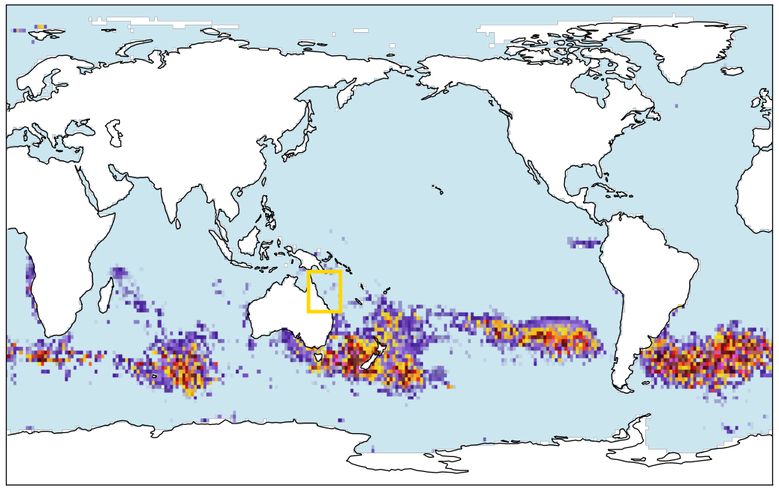




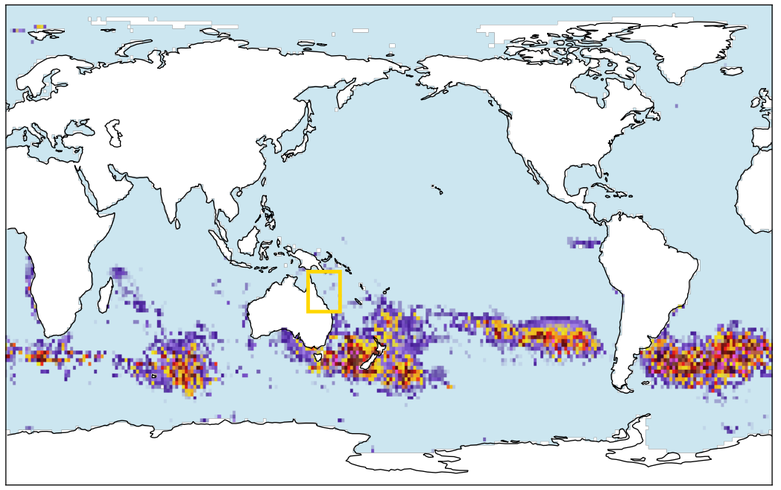


DHW Chain init: 2018-03-31 - Val (C-weeks)

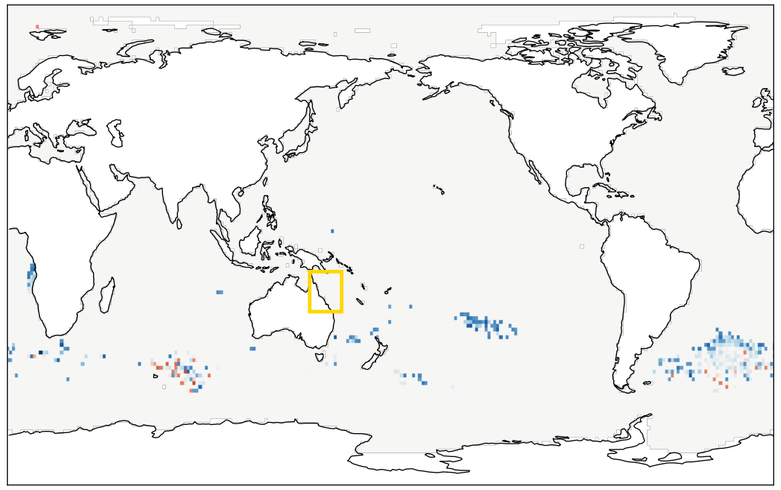
Actual DHW - Step 1



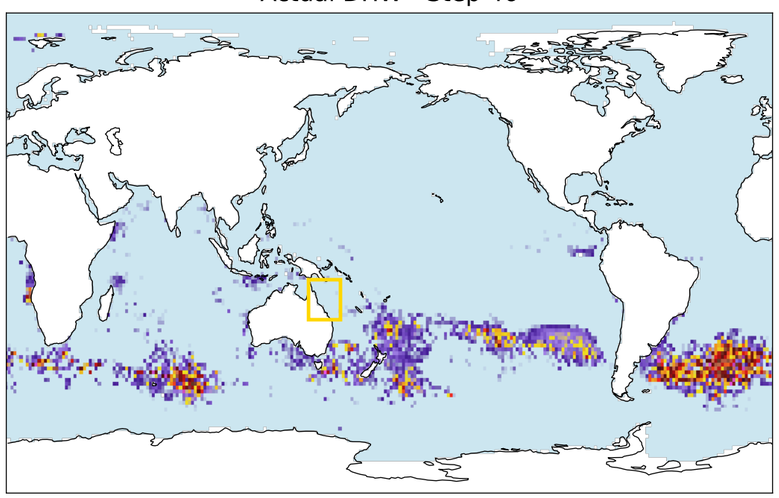
Predicted DHW - Step 1



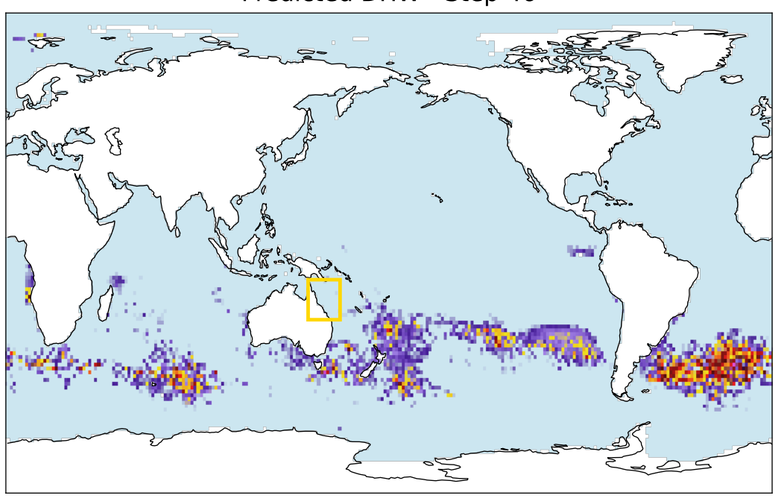
Error (Pred - Actual) - Step 1 [scale ±0.31]



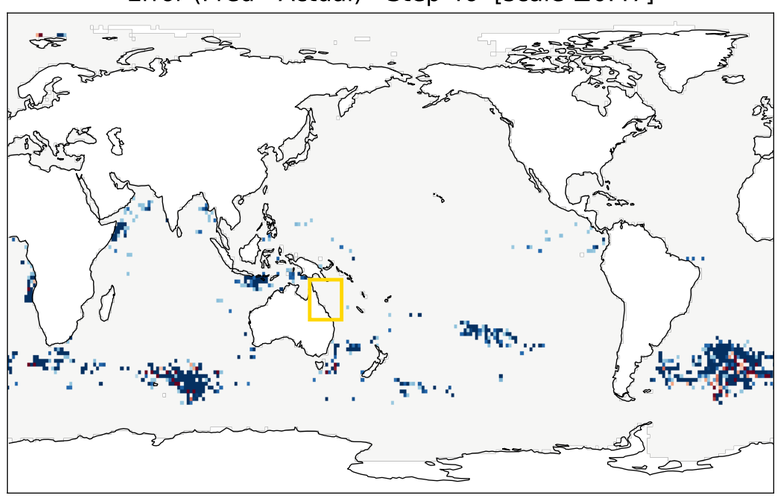
Actual DHW - Step 46



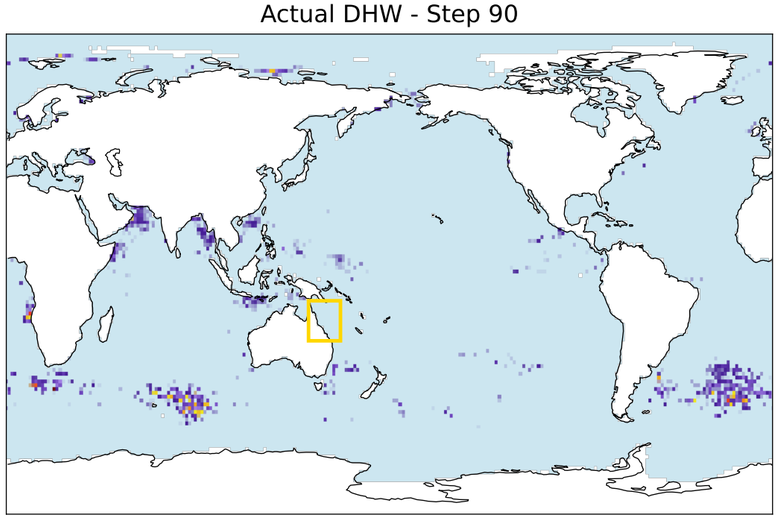
Predicted DHW - Step 46



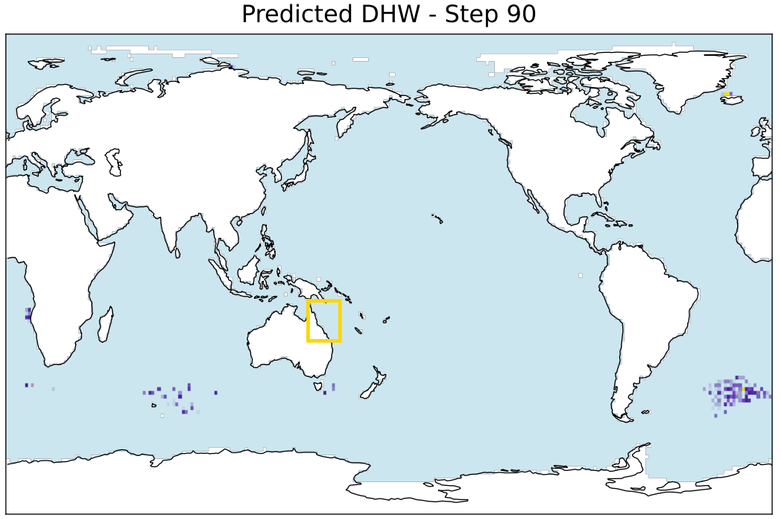
Error (Pred - Actual) - Step 46 [scale ±0.47]



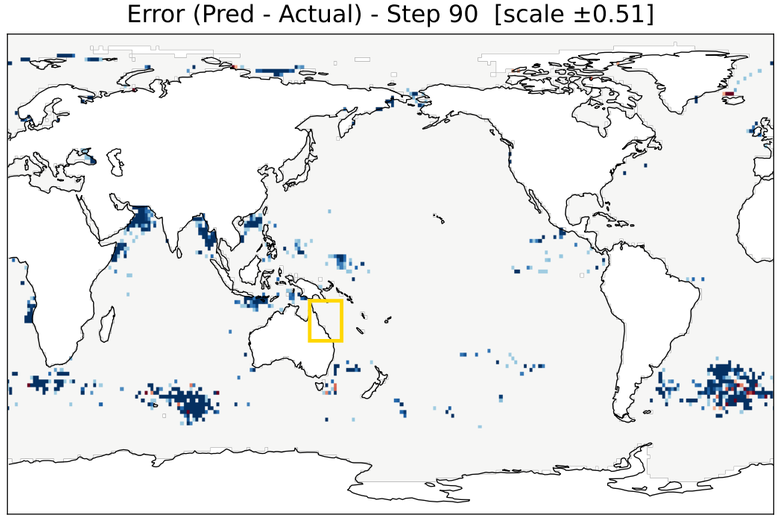
Actual DHW - Step 90

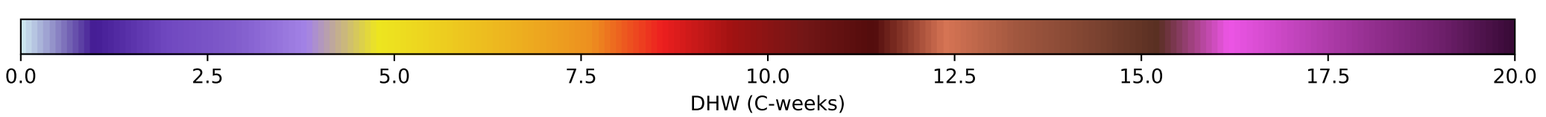


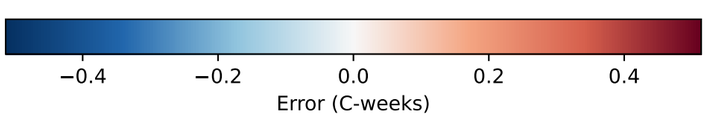
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±0.51]

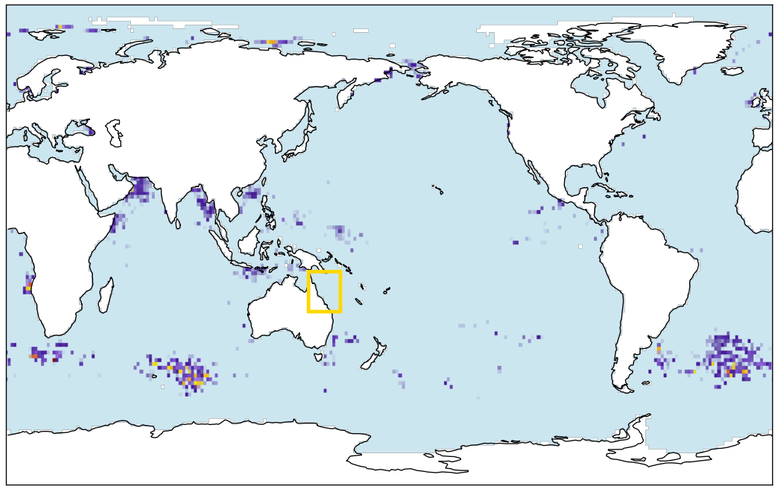




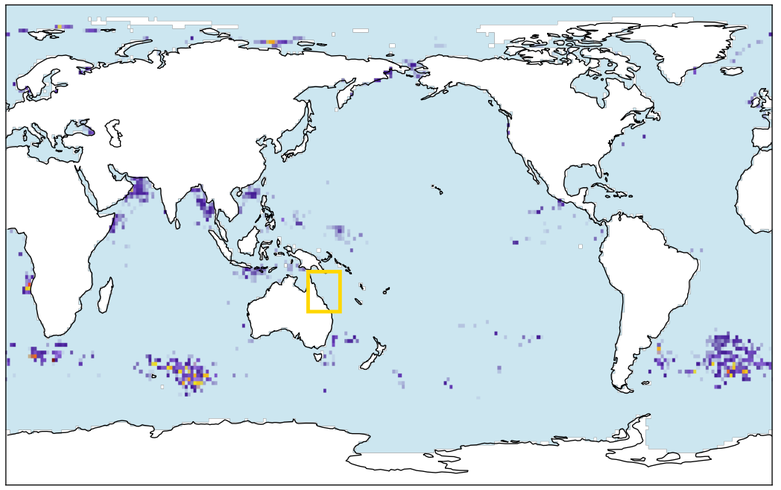


DHW Chain init: 2018-06-29 - Val (C-weeks)

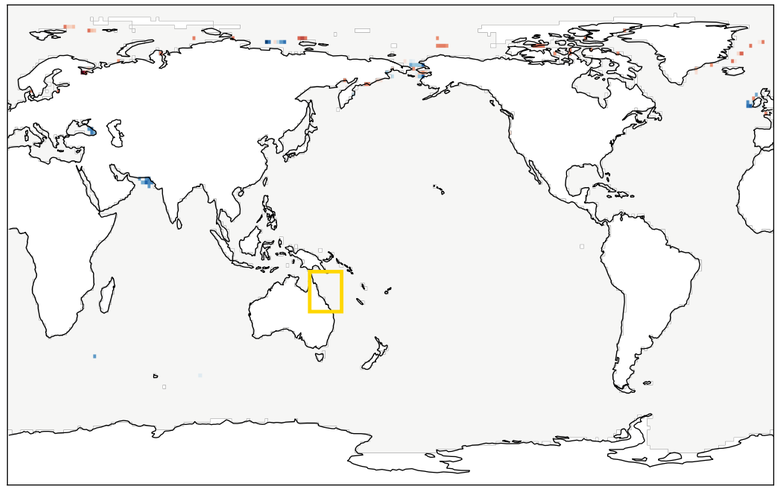
Actual DHW - Step 1



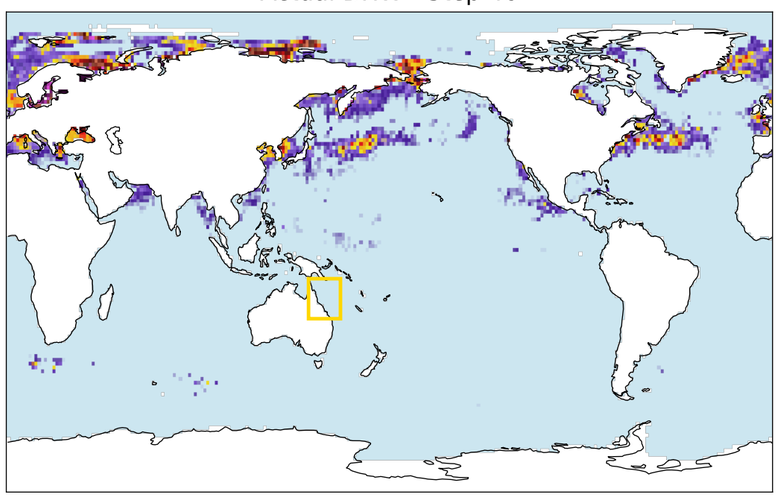
Predicted DHW - Step 1



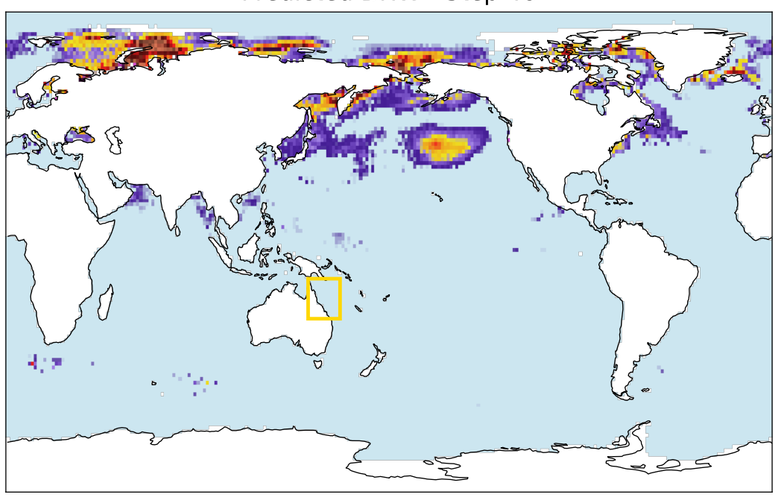
Error (Pred - Actual) - Step 1 [scale ±0.35]



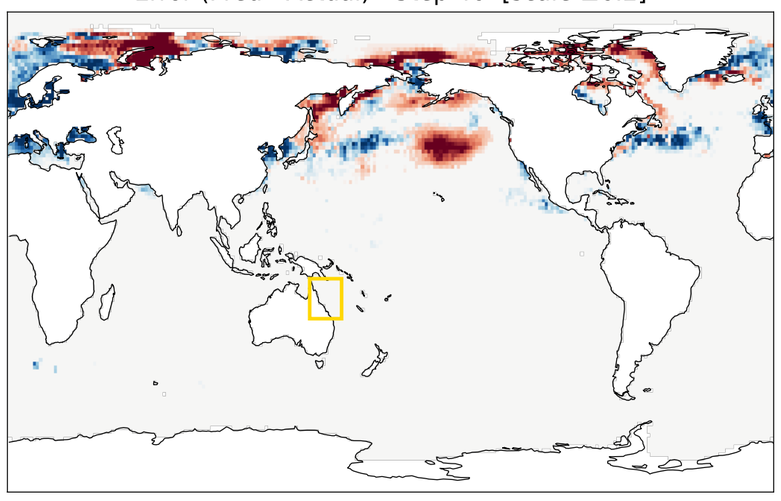
Actual DHW - Step 46



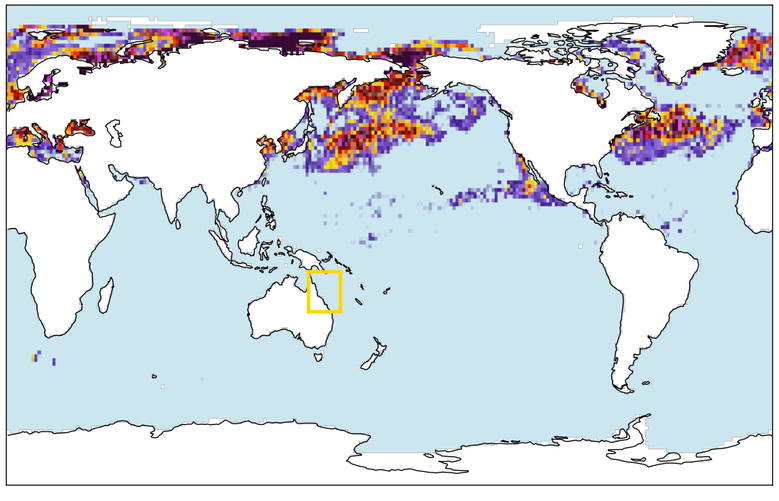
Predicted DHW - Step 46



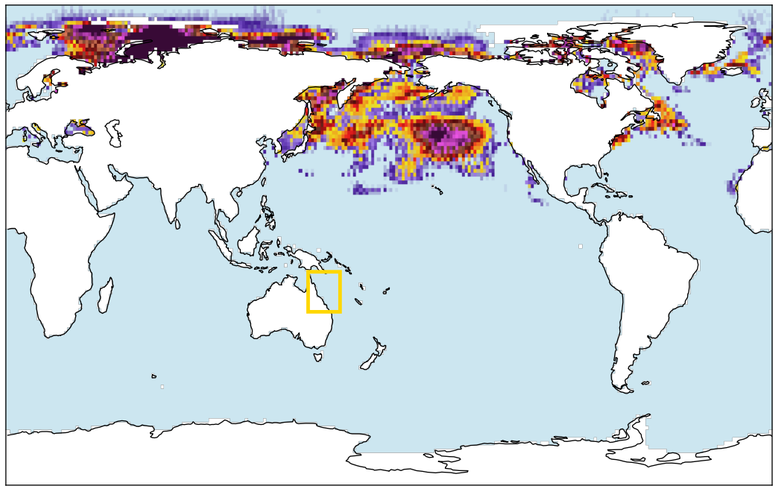
Error (Pred - Actual) - Step 46 [scale ±6.2]



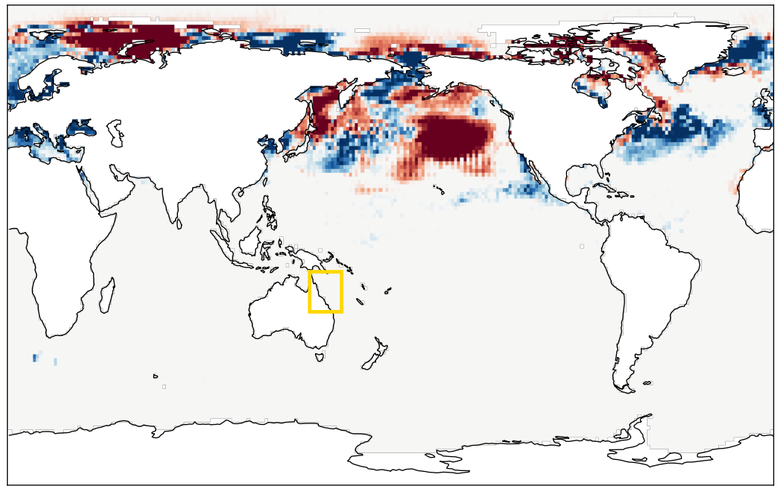
Actual DHW - Step 90

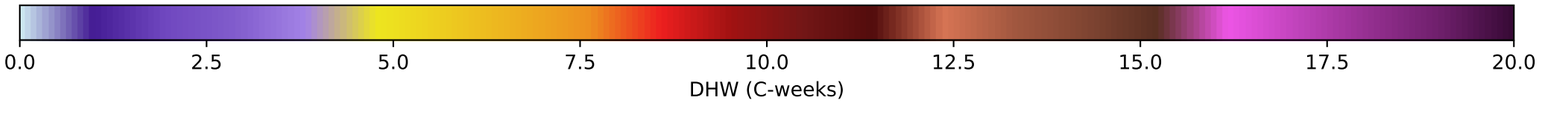


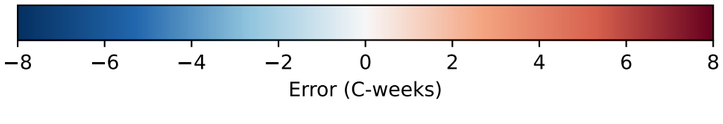
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

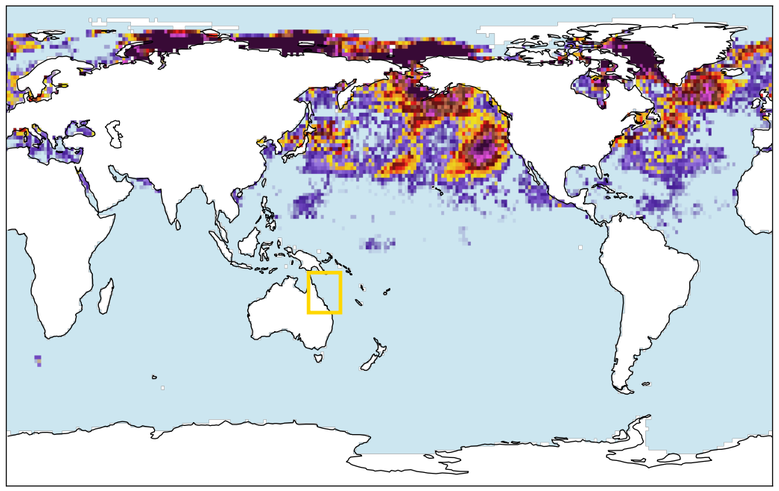




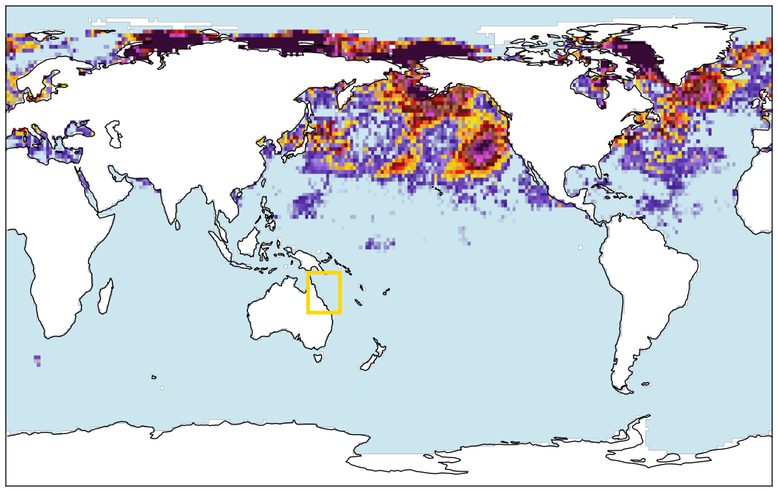


DHW Chain init: 2019-09-22 - Val (C-weeks)

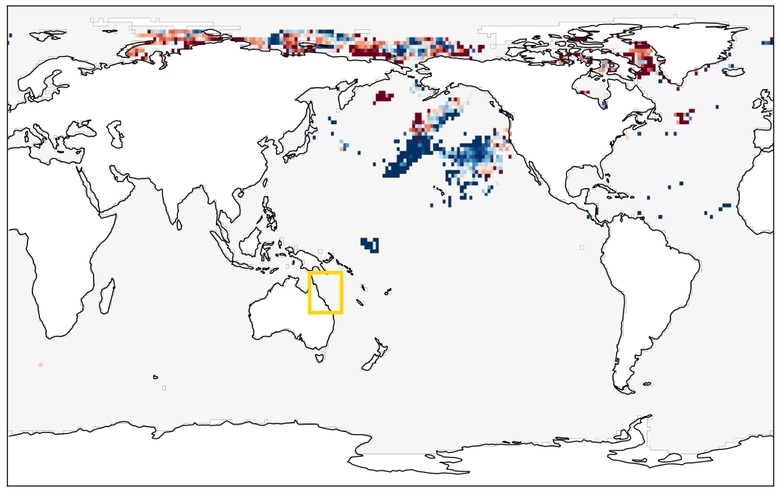
Actual DHW - Step 1



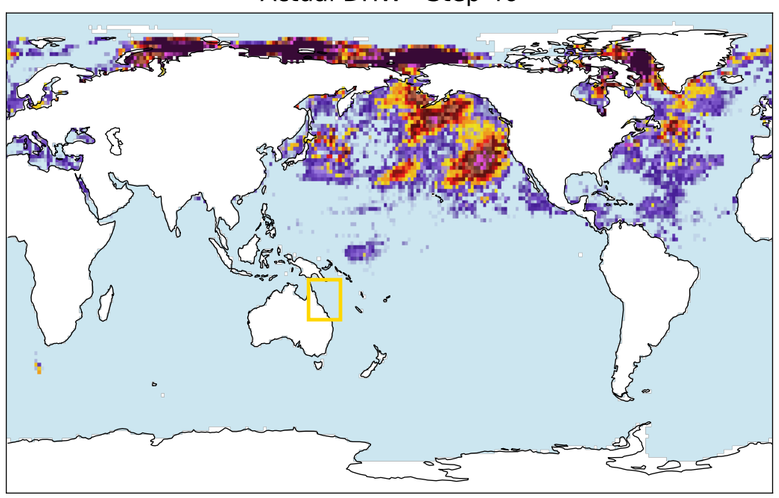
Predicted DHW - Step 1



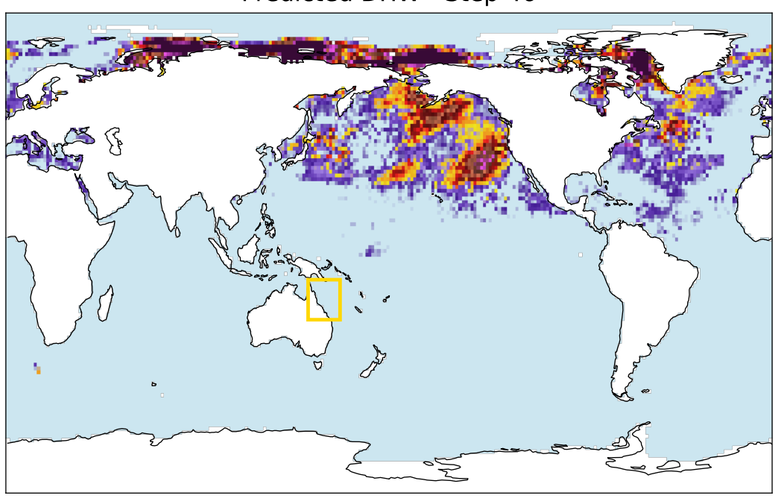
Error (Pred - Actual) - Step 1 [scale ±0.07]



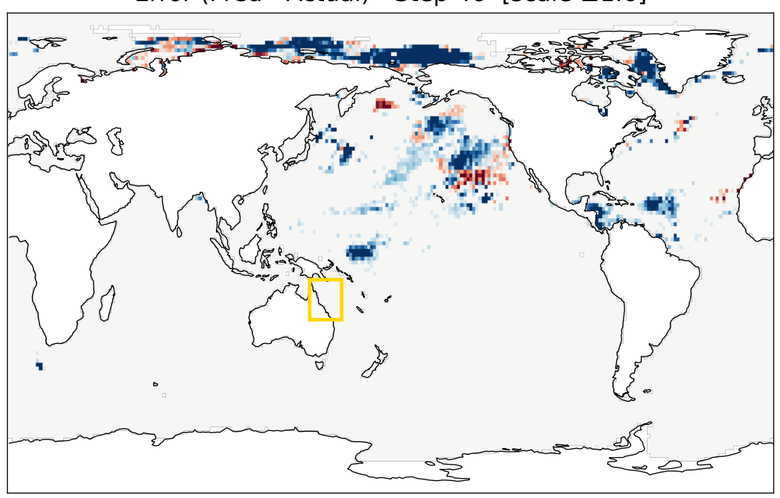
Actual DHW - Step 46



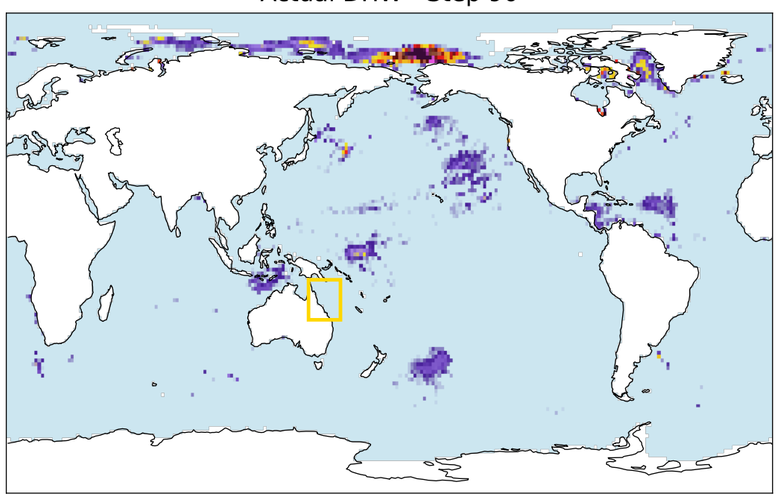
Predicted DHW - Step 46



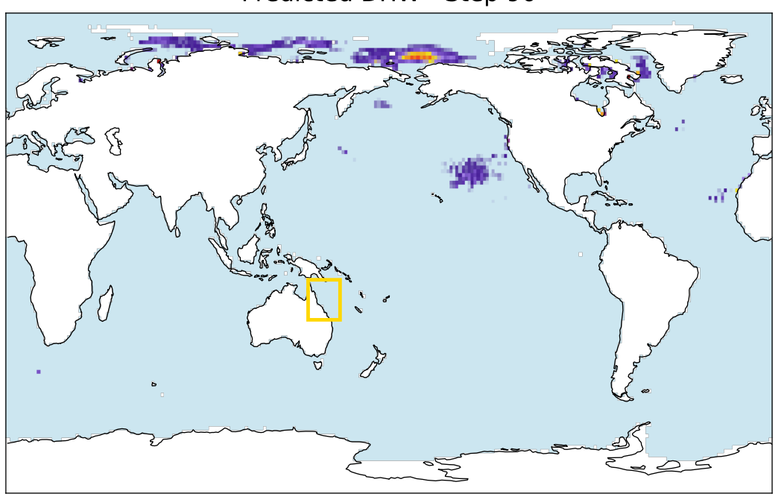
Error (Pred - Actual) - Step 46 [scale ±1.6]



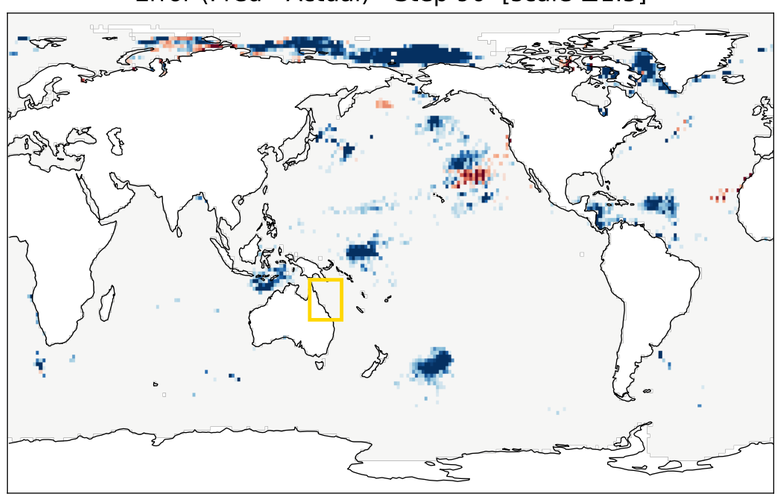
Actual DHW - Step 90

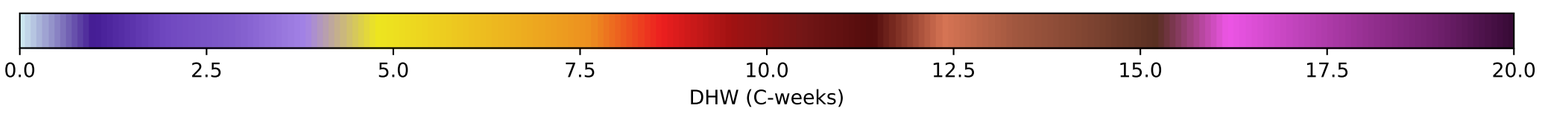


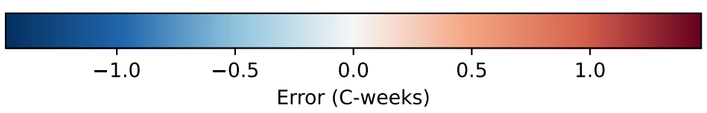
Predicted DHW - Step 90



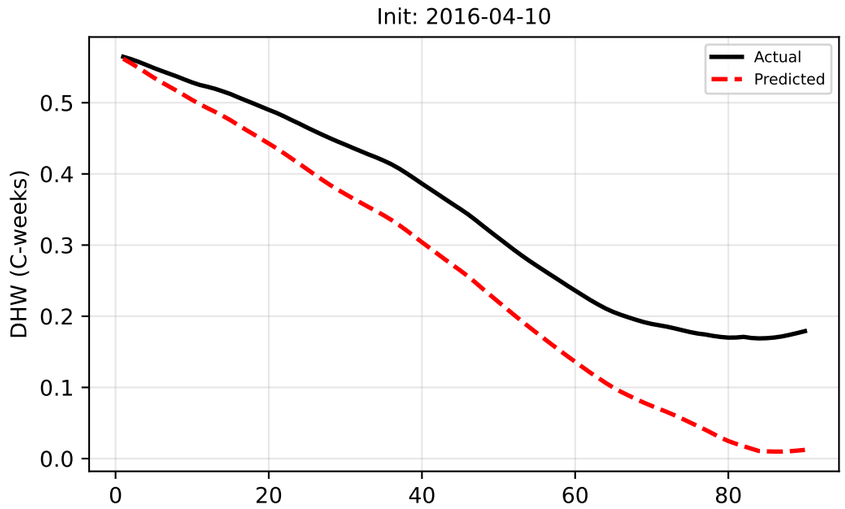
Error (Pred - Actual) - Step 90 [scale ±1.5]

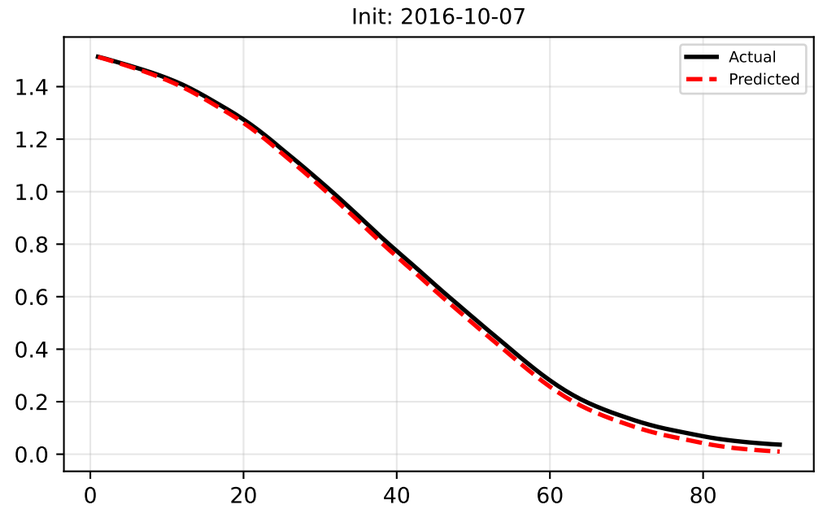


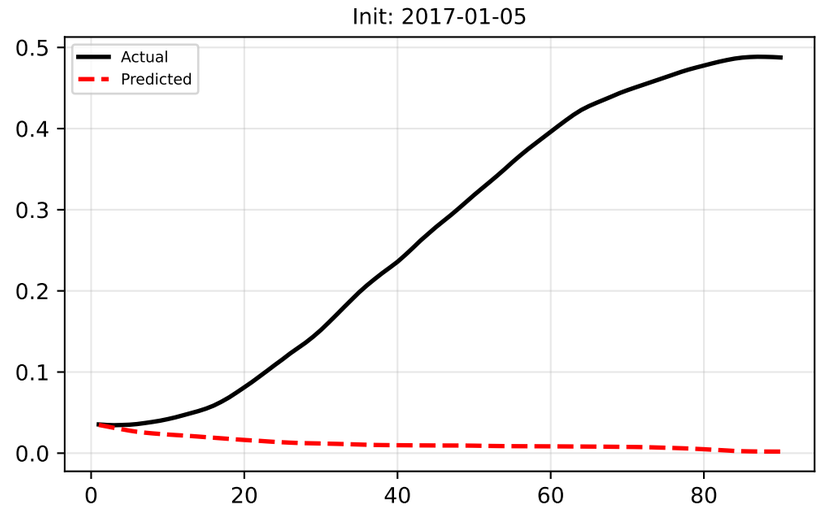


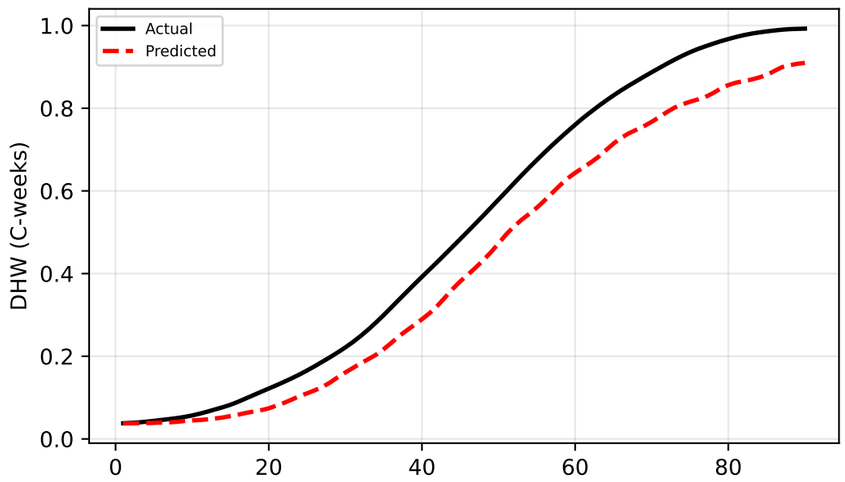


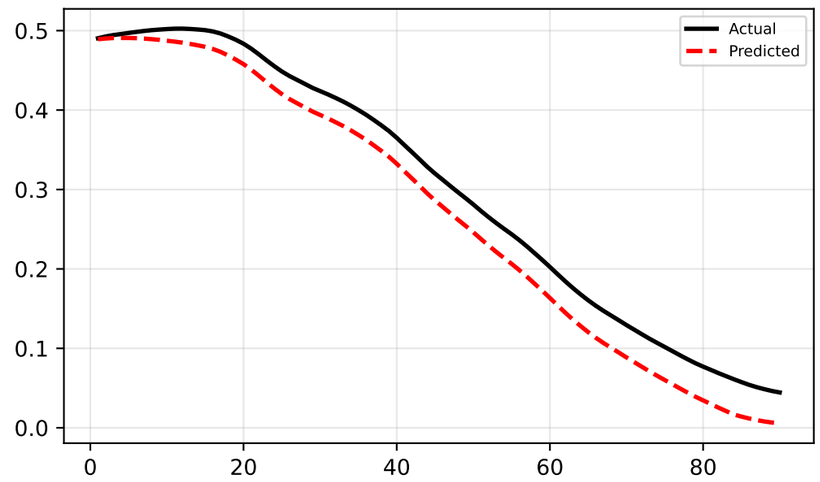
Global-Mean DHW - Val (C-weeks)

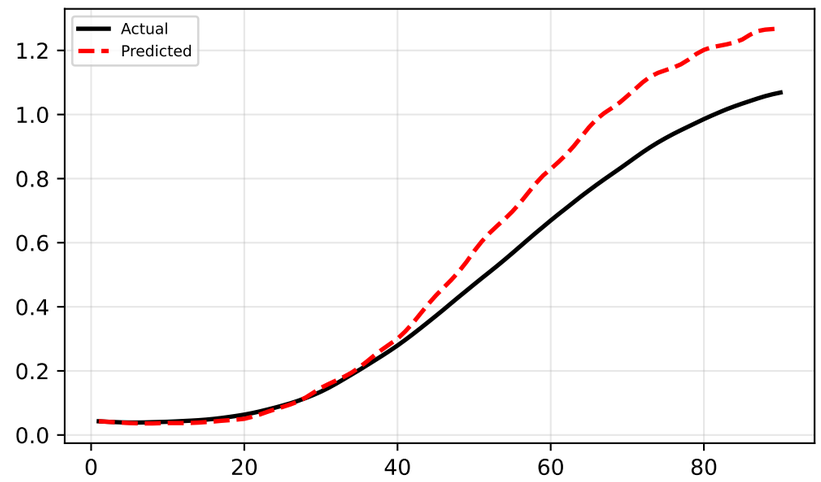


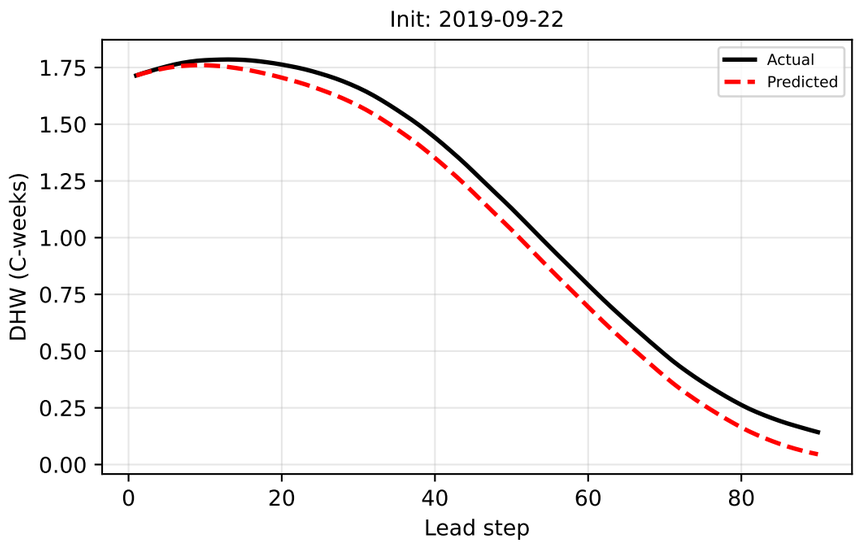




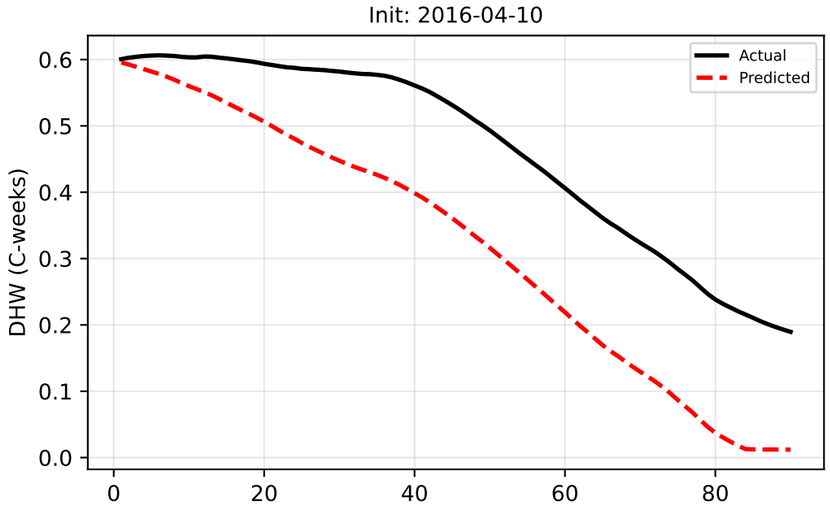


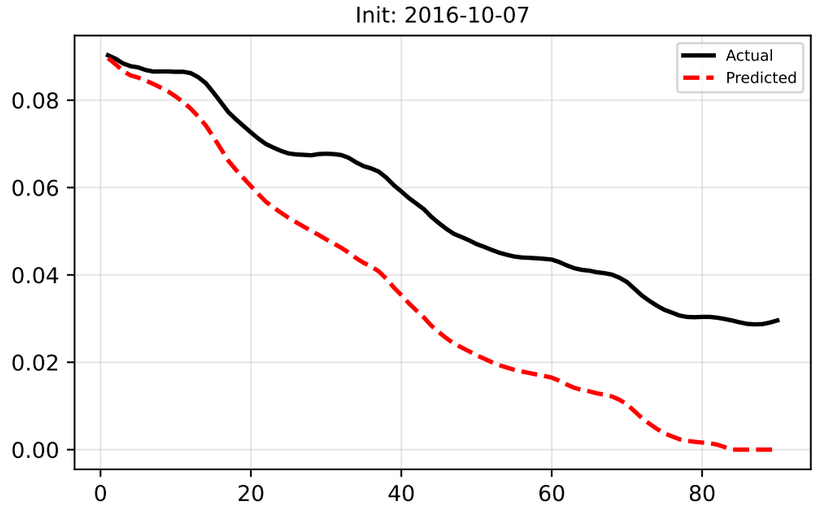


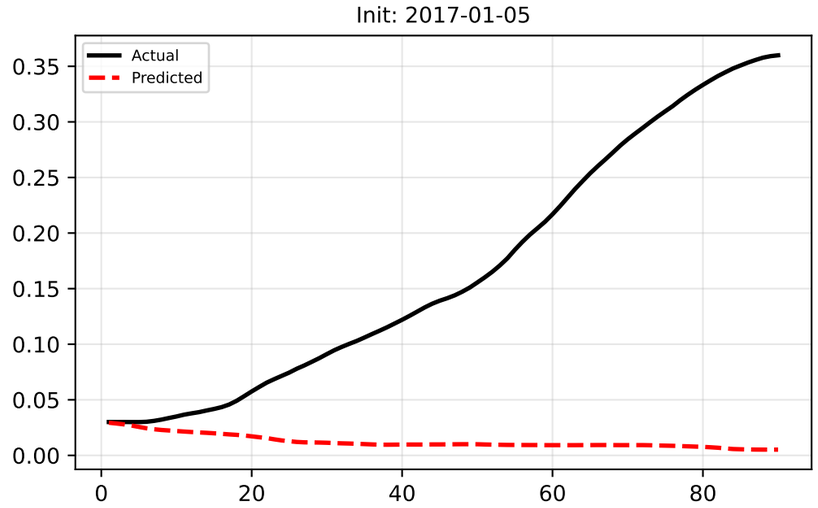


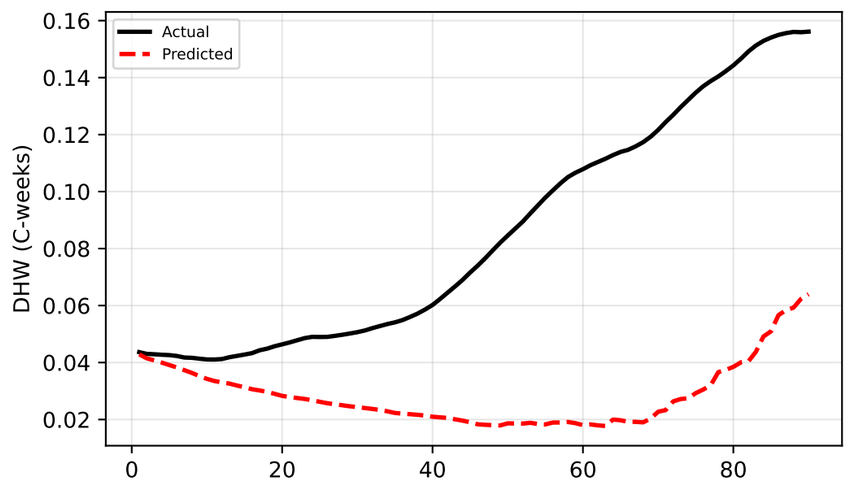


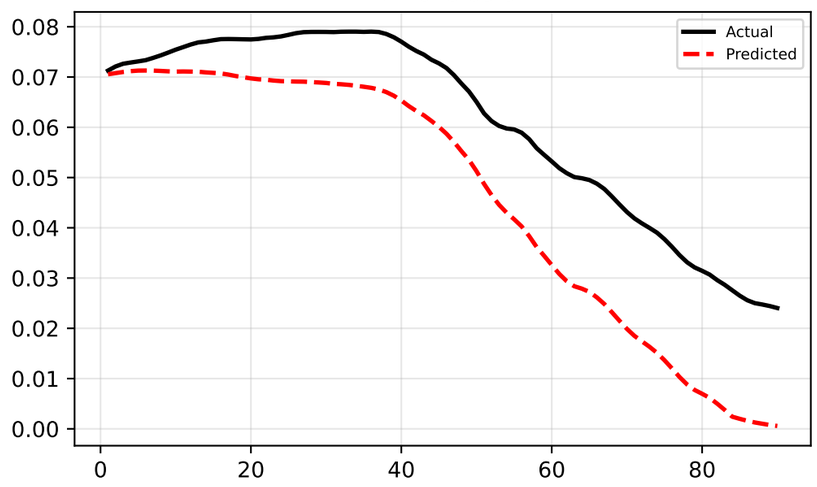
Tropical-Mean DHW (|lat|<=30) - Val (C-weeks)

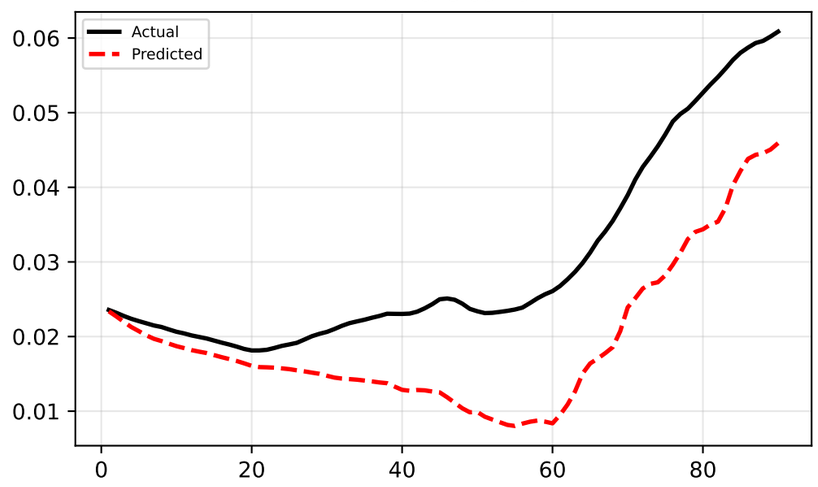


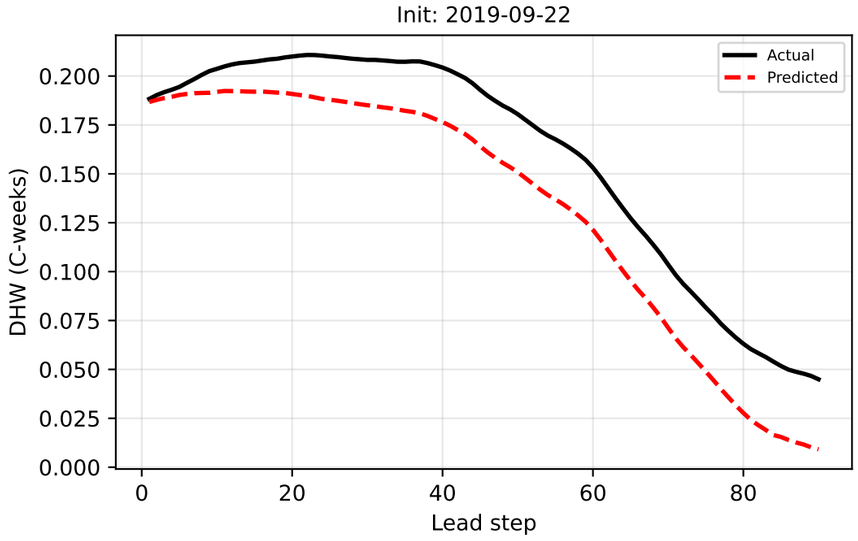


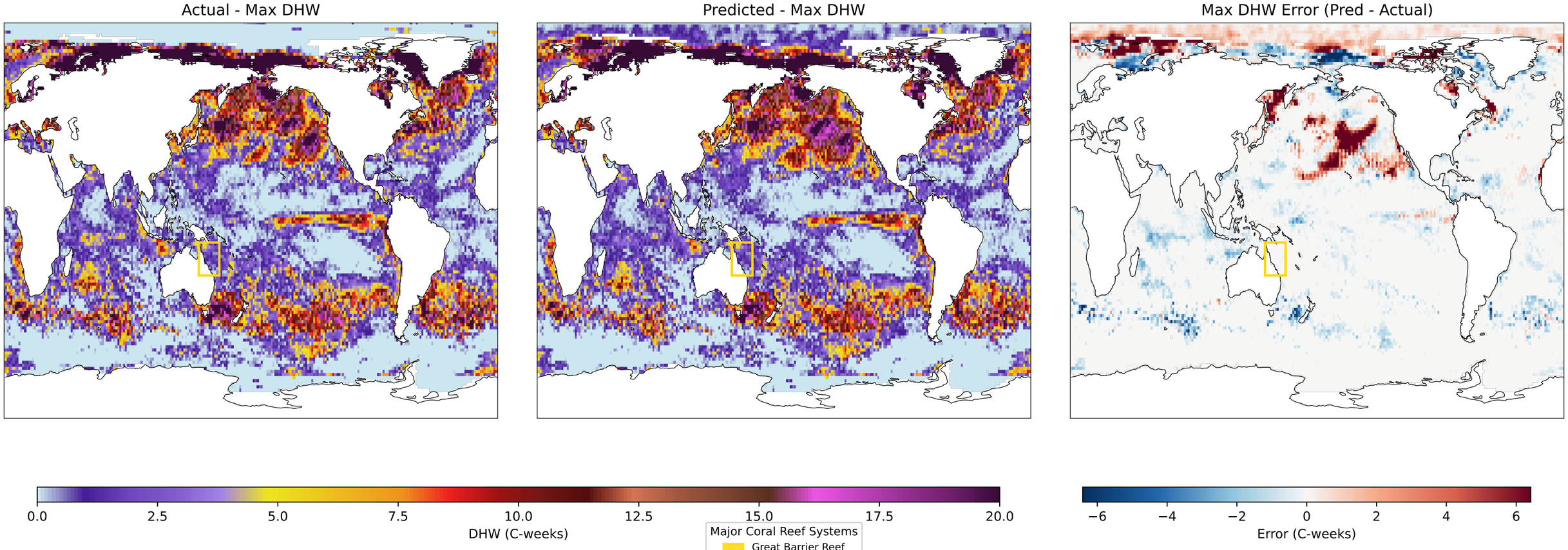


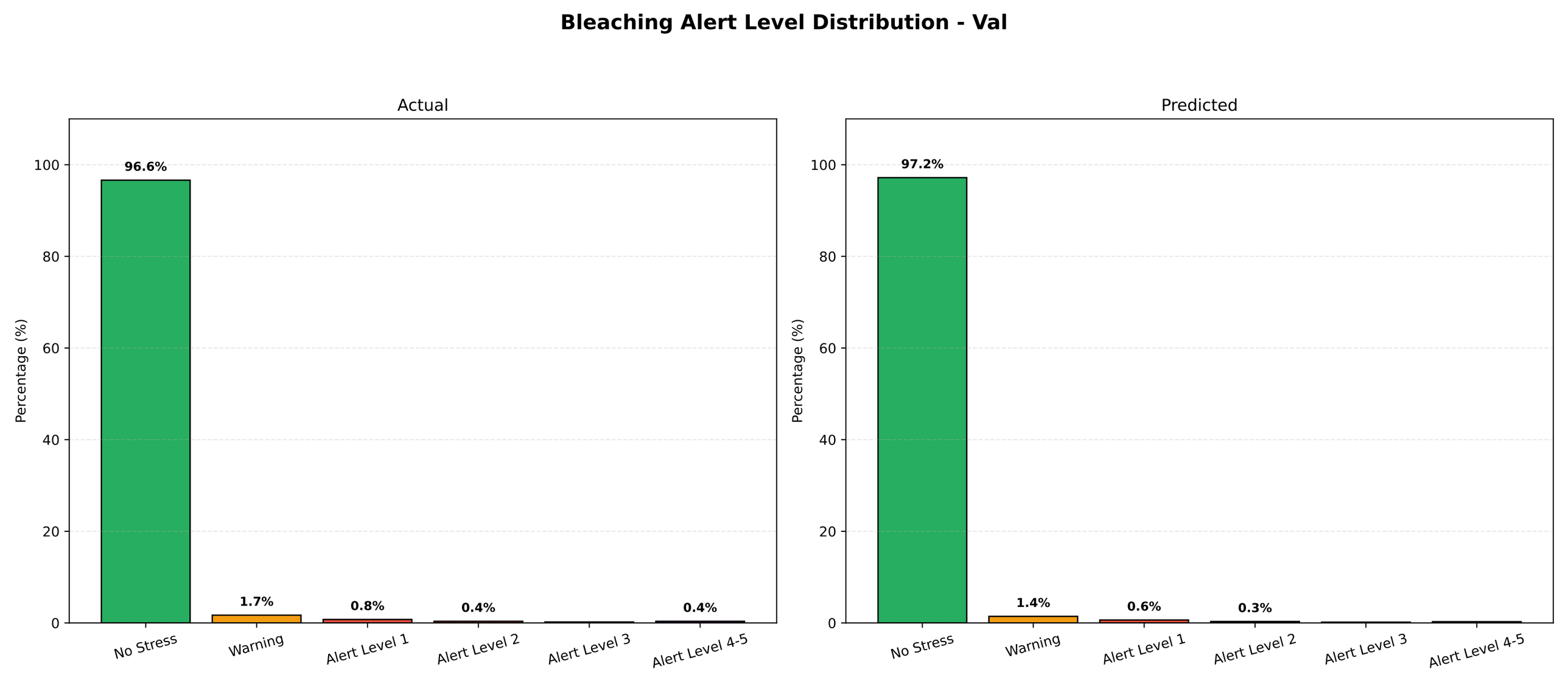






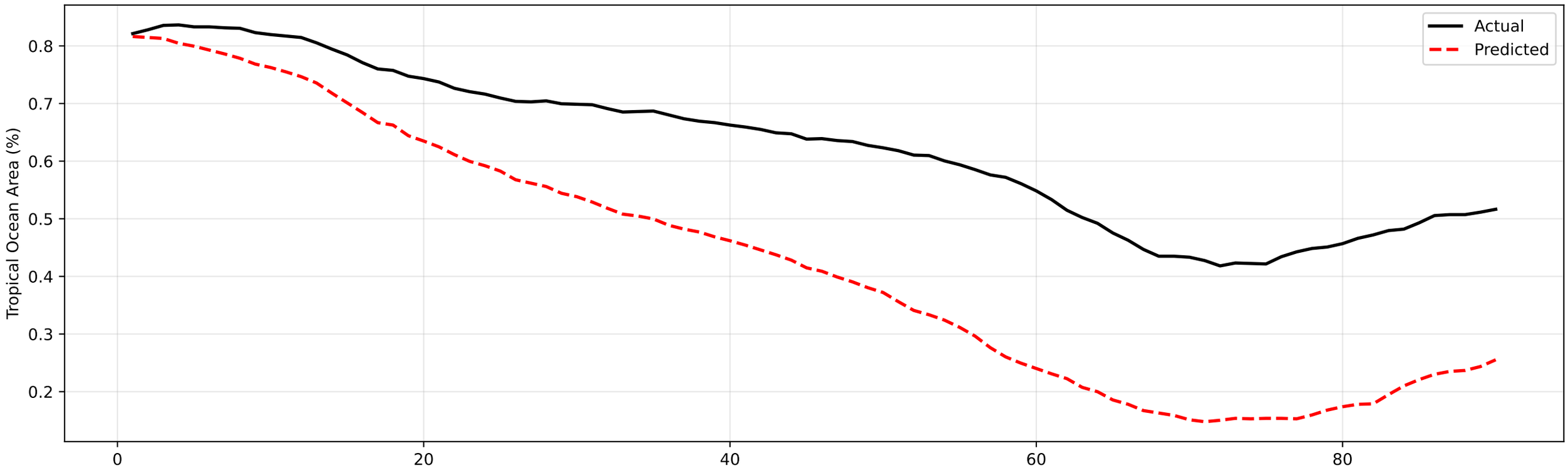



Peak DHW (max over all chains & leads) - Val (C-weeks)
Actual - Max DHW
Predicted - Max DHW
Max DHW Error (Pred - Actual)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
DHW (C-weeks)
Major Coral Reef Systems
Great Barrier Reef
−6
−4
−2
0
2
4
6
Error (C-weeks)

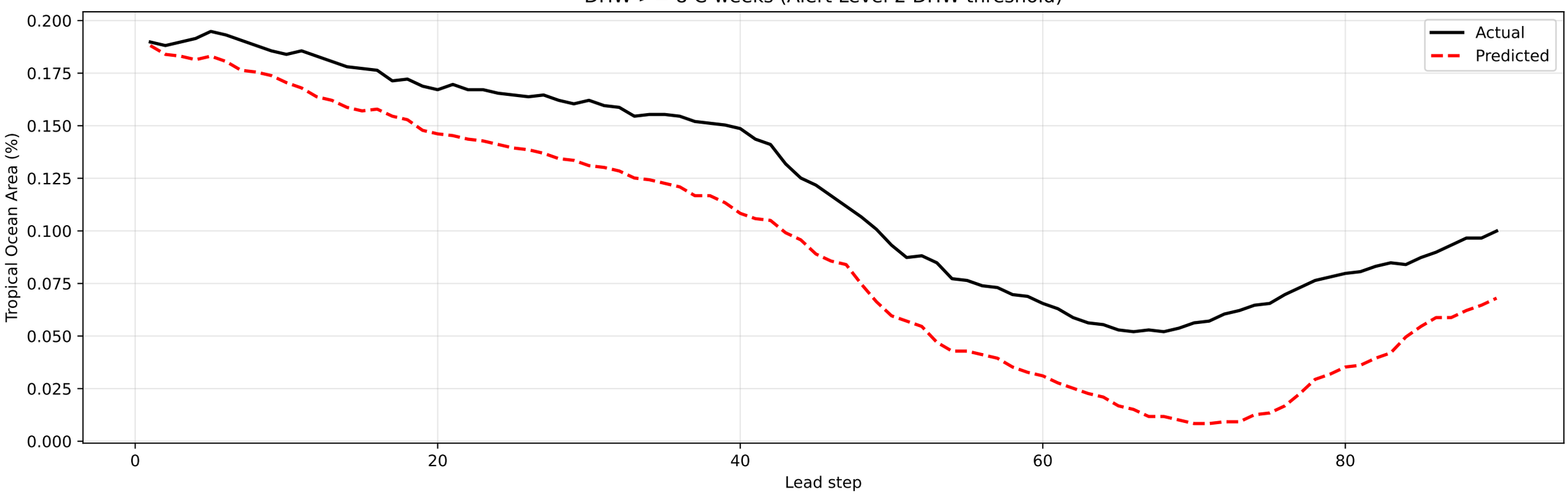

Bleaching Alert Level Distribution - Val
Actual
Predicted
Percentage (%)
0
20
40
60
80
100
96.6%
1.7%
0.8%
0.4%
0.4%
97.2%
1.4%
0.6%
0.3%
No Stress
Warning
Alert Level 1
Alert Level 2
Alert Level 3
Alert Level 4-5

Tropical Ocean Area Under Bleaching Stress - Val

DHW >= 4 C-weeks (Alert Level 1 DHW threshold)



DHW >= 8 C-weeks (Alert Level 2 DHW threshold)



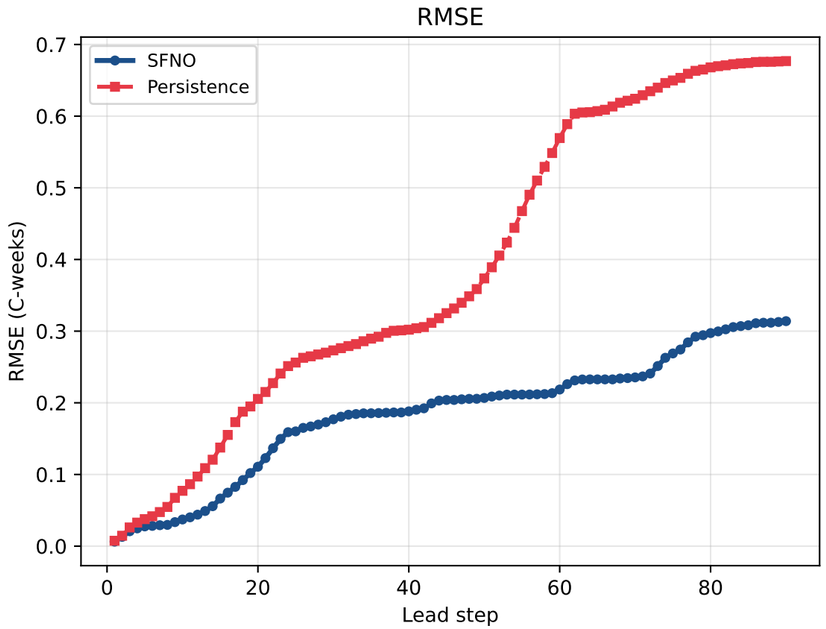
Great Barrier Reef Analysis - Val

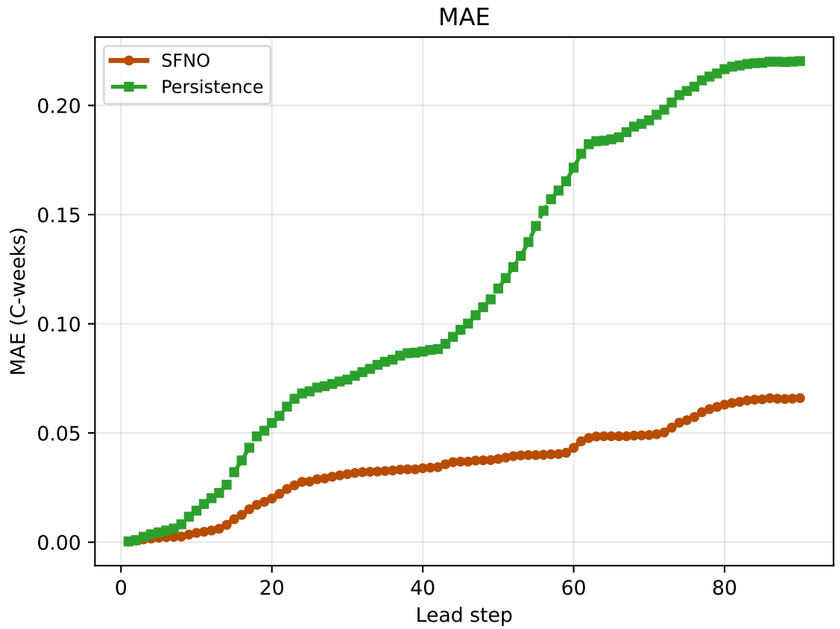
```
                    GREAT BARRIER REEF - VAL
Lat -25.0 deg to -10.0 deg  |  Lon 142.0 deg to 157.0 deg  (0-360)
                    Grid: 11 lat x 11 lon points
```

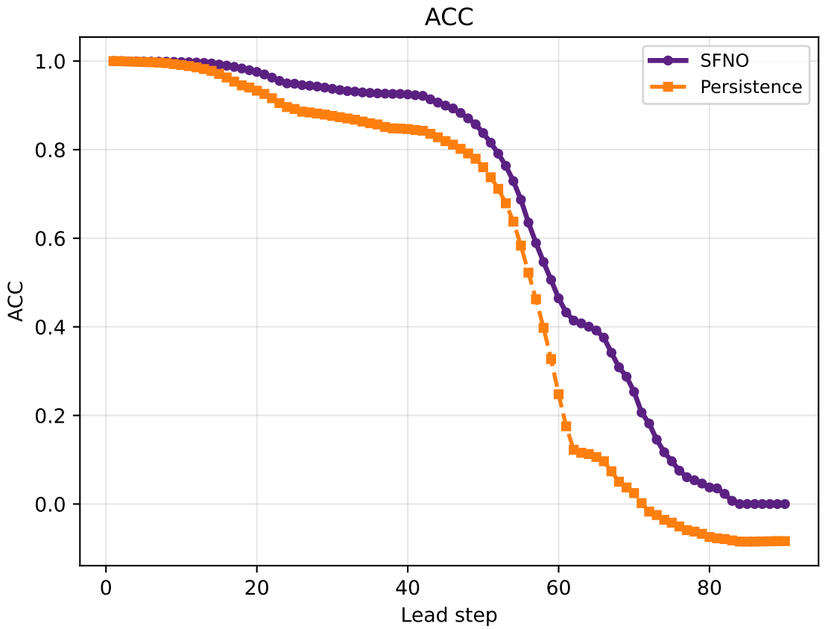
DHW Metrics by Lead Step - Val - GBR (15 chains scored at every lead)

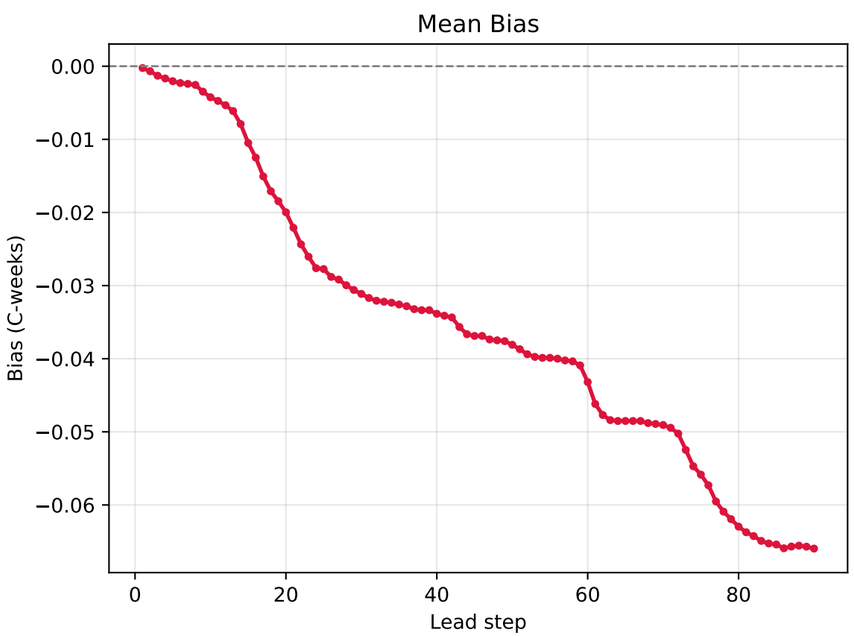
Lead Step	SFNO RMSE	Persist RMSE	Skill %	SFNO MAE	Persist MAE	SFNO Bias	SFNO ACC	Persist ACC	SFNO R²
1	0.0063	0.0075	16.0%	0.0002	0.0004	-0.0002	0.9999	0.9999	0.9999
2	0.0130	0.0145	10.7%	0.0007	0.0009	-0.0007	0.9997	0.9997	0.9995
3	0.0208	0.0261	20.3%	0.0013	0.0025	-0.0013	0.9993	0.9989	0.9987
4	0.0248	0.0328	24.6%	0.0017	0.0036	-0.0017	0.9990	0.9983	0.9982
5	0.0275	0.0377	27.0%	0.0020	0.0045	-0.0020	0.9988	0.9978	0.9977
6	0.0283	0.0417	32.2%	0.0023	0.0053	-0.0023	0.9987	0.9973	0.9976
7	0.0292	0.0476	38.6%	0.0024	0.0062	-0.0024	0.9986	0.9965	0.9974
8	0.0297	0.0547	45.8%	0.0026	0.0082	-0.0026	0.9986	0.9954	0.9973
9	0.0335	0.0674	50.3%	0.0035	0.0117	-0.0035	0.9981	0.9931	0.9965
10	0.0372	0.0772	51.8%	0.0042	0.0144	-0.0042	0.9977	0.9910	0.9956
11	0.0402	0.0864	53.5%	0.0047	0.0175	-0.0047	0.9973	0.9887	0.9948
12	0.0439	0.0971	54.8%	0.0053	0.0201	-0.0053	0.9967	0.9857	0.9937
13	0.0490	0.1088	55.0%	0.0061	0.0225	-0.0061	0.9958	0.9820	0.9920
14	0.0557	0.1206	53.8%	0.0079	0.0263	-0.0079	0.9945	0.9778	0.9896
15	0.0664	0.1375	51.7%	0.0105	0.0320	-0.0105	0.9920	0.9710	0.9849
16	0.0746	0.1552	51.9%	0.0125	0.0374	-0.0125	0.9896	0.9629	0.9803
17	0.0827	0.1730	52.2%	0.0151	0.0432	-0.0151	0.9868	0.9536	0.9751
18	0.0922	0.1876	50.9%	0.0171	0.0484	-0.0171	0.9832	0.9451	0.9684
19	0.1020	0.1949	47.7%	0.0185	0.0510	-0.0185	0.9794	0.9403	0.9612
20	0.1108	0.2054	46.0%	0.0200	0.0546	-0.0200	0.9755	0.9330	0.9539
21	0.1229	0.2151	42.9%	0.0221	0.0578	-0.0221	0.9698	0.9258	0.9435
22	0.1366	0.2275	39.9%	0.0244	0.0621	-0.0244	0.9627	0.9161	0.9304
23	0.1496	0.2409	37.9%	0.0261	0.0657	-0.0261	0.9552	0.9051	0.9169
24	0.1590	0.2512	36.7%	0.0276	0.0681	-0.0276	0.9495	0.8962	0.9067
25	0.1601	0.2562	37.5%	0.0277	0.0691	-0.0277	0.9488	0.8919	0.9053
26	0.1650	0.2628	37.2%	0.0288	0.0708	-0.0288	0.9457	0.8859	0.8998
27	0.1670	0.2649	36.9%	0.0292	0.0714	-0.0292	0.9443	0.8840	0.8974
28	0.1696	0.2675	36.6%	0.0300	0.0725	-0.0300	0.9426	0.8816	0.8943
29	0.1729	0.2700	36.0%	0.0306	0.0736	-0.0306	0.9401	0.8792	0.8900
30	0.1770	0.2732	35.2%	0.0311	0.0745	-0.0311	0.9374	0.8762	0.8852
31	0.1806	0.2762	34.6%	0.0317	0.0762	-0.0317	0.9345	0.8732	0.8802
32	0.1834	0.2792	34.3%	0.0321	0.0779	-0.0321	0.9322	0.8703	0.8763
33	0.1843	0.2819	34.6%	0.0322	0.0794	-0.0322	0.9311	0.8676	0.8743
34	0.1853	0.2859	35.2%	0.0323	0.0812	-0.0323	0.9292	0.8635	0.8711
35	0.1854	0.2895	36.0%	0.0326	0.0826	-0.0326	0.9281	0.8597	0.8690
36	0.1857	0.2922	36.4%	0.0328	0.0836	-0.0328	0.9271	0.8569	0.8672
37	0.1861	0.2976	37.5%	0.0332	0.0854	-0.0332	0.9264	0.8512	0.8662
38	0.1865	0.3005	37.9%	0.0334	0.0866	-0.0334	0.9259	0.8481	0.8652
39	0.1865	0.3011	38.0%	0.0334	0.0868	-0.0334	0.9258	0.8475	0.8651
40	0.1881	0.3020	37.7%	0.0339	0.0873	-0.0339	0.9247	0.8465	0.8632
41	0.1903	0.3041	37.4%	0.0341	0.0880	-0.0341	0.9229	0.8443	0.8601
42	0.1923	0.3057	37.1%	0.0344	0.0884	-0.0344	0.9214	0.8425	0.8575
43	0.1992	0.3116	36.1%	0.0357	0.0909	-0.0357	0.9136	0.8357	0.8438
44	0.2031	0.3180	36.1%	0.0367	0.0940	-0.0367	0.9063	0.8279	0.8312
45	0.2040	0.3251	37.3%	0.0369	0.0972	-0.0369	0.8995	0.8193	0.8196
46	0.2040	0.3317	38.5%	0.0369	0.1001	-0.0369	0.8930	0.8114	0.8085
47	0.2049	0.3397	39.7%	0.0374	0.1039	-0.0374	0.8828	0.8019	0.7914
48	0.2055	0.3486	41.0%	0.0375	0.1076	-0.0375	0.8706	0.7915	0.7710
49	0.2056	0.3587	42.7%	0.0376	0.1112	-0.0376	0.8572	0.7796	0.7489
50	0.2068	0.3735	44.7%	0.0381	0.1161	-0.0381	0.8375	0.7603	0.7172
51	0.2087	0.3891	46.4%	0.0387	0.1209	-0.0387	0.8157	0.7376	0.6828
52	0.2102	0.4055	48.2%	0.0394	0.1260	-0.0394	0.7910	0.7115	0.6451
53	0.2114	0.4235	50.1%	0.0397	0.1311	-0.0397	0.7634	0.6790	0.6043
54	0.2115	0.4441	52.4%	0.0399	0.1374	-0.0399	0.7290	0.6376	0.5564
55	0.2115	0.4675	54.8%	0.0399	0.1448	-0.0399	0.6870	0.5834	0.5008
56	0.2116	0.4904	56.8%	0.0400	0.1517	-0.0400	0.6353	0.5223	0.4359
57	0.2119	0.5101	58.5%	0.0402	0.1570	-0.0402	0.5892	0.4623	0.3801
58	0.2122	0.5292	59.9%	0.0404	0.1610	-0.0404	0.5464	0.3978	0.3309
59	0.2135	0.5485	61.1%	0.0409	0.1653	-0.0409	0.5061	0.3272	0.2869
60	0.2185	0.5694	61.6%	0.0432	0.1715	-0.0432	0.4643	0.2479	0.2427
61	0.2260	0.5888	61.6%	0.0462	0.1779	-0.0462	0.4325	0.1755	0.2101
62	0.2313	0.6033	61.7%	0.0477	0.1822	-0.0477	0.4141	0.1224	0.1930
63	0.2326	0.6051	61.6%	0.0484	0.1836	-0.0484	0.4078	0.1157	0.1877
64	0.2327	0.6055	61.6%	0.0485	0.1838	-0.0485	0.4004	0.1129	0.1811
65	0.2327	0.6070	61.7%	0.0485	0.1845	-0.0485	0.3918	0.1062	0.1731
66	0.2327	0.6089	61.8%	0.0485	0.1854	-0.0485	0.3756	0.0965	0.1592
67	0.2327	0.6135	62.1%	0.0485	0.1878	-0.0485	0.3416	0.0739	0.1321
68	0.2339	0.6187	62.2%	0.0488	0.1903	-0.0488	0.3088	0.0504	0.1073
69	0.2345	0.6215	62.3%	0.0489	0.1916	-0.0489	0.2875	0.0374	0.0921
70	0.2355	0.6242	62.3%	0.0491	0.1932	-0.0491	0.2533	0.0247	0.0697
71	0.2368	0.6292	62.4%	0.0494	0.1958	-0.0494	0.2065	0.0020	0.0418
72	0.2408	0.6347	62.1%	0.0502	0.1981	-0.0502	0.1816	-0.0171	0.0290
73	0.2513	0.6397	60.7%	0.0525	0.2013	-0.0525	0.1454	-0.0248	0.0124
74	0.2628	0.6462	59.3%	0.0547	0.2047	-0.0547	0.1169	-0.0357	0.0008
75	0.2689	0.6498	58.6%	0.0559	0.2066	-0.0559	0.0967	-0.0421	-0.0072
76	0.2743	0.6534	58.0%	0.0573	0.2086	-0.0573	0.0753	-0.0506	-0.0161
77	0.2843	0.6591	56.9%	0.0595	0.2115	-0.0595	0.0606	-0.0589	-0.0216
78	0.2923	0.6632	55.9%	0.0609	0.2132	-0.0609	0.0537	-0.0623	-0.0237
79	0.2943	0.6651	55.7%	0.0619	0.2146	-0.0619	0.0463	-0.0673	-0.0271
80	0.2973	0.6680	55.5%	0.0630	0.2166	-0.0630	0.0374	-0.0744	-0.0304
81	0.2996	0.6697	55.3%	0.0637	0.2178	-0.0637	0.0353	-0.0774	-0.0312
82	0.3025	0.6709	54.9%	0.0643	0.2183	-0.0643	0.0230	-0.0790	-0.0366
83	0.3057	0.6725	54.6%	0.0649	0.2190	-0.0649	0.0071	-0.0822	-0.0439
84	0.3071	0.6736	54.4%	0.0653	0.2194	-0.0653	0.0000	-0.0846	-0.0473
85	0.3082	0.6741	54.3%	0.0654	0.2195	-0.0654	-0.0000	-0.0849	-0.0471
86	0.3111	0.6755	53.9%	0.0659	0.2200	-0.0659	0.0000	-0.0848	-0.0470
87	0.3117	0.6758	53.9%	0.0657	0.2200	-0.0657	0.0000	-0.0843	-0.0465
88	0.3118	0.6758	53.9%	0.0656	0.2199	-0.0656	0.0000	-0.0841	-0.0463
89	0.3128	0.6763	53.8%	0.0657	0.2200	-0.0657	-0.0000	-0.0838	-0.0462
90	0.3139	0.6768	53.6%	0.0660	0.2203	-0.0660	0.0000	-0.0838	-0.0462

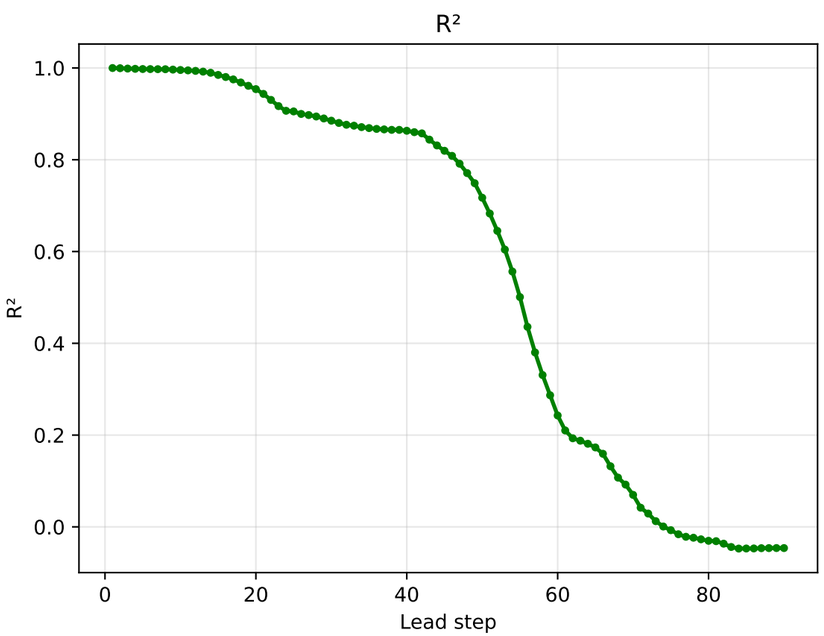
DHW Forecast Skill vs Lead Step - Val - GBR

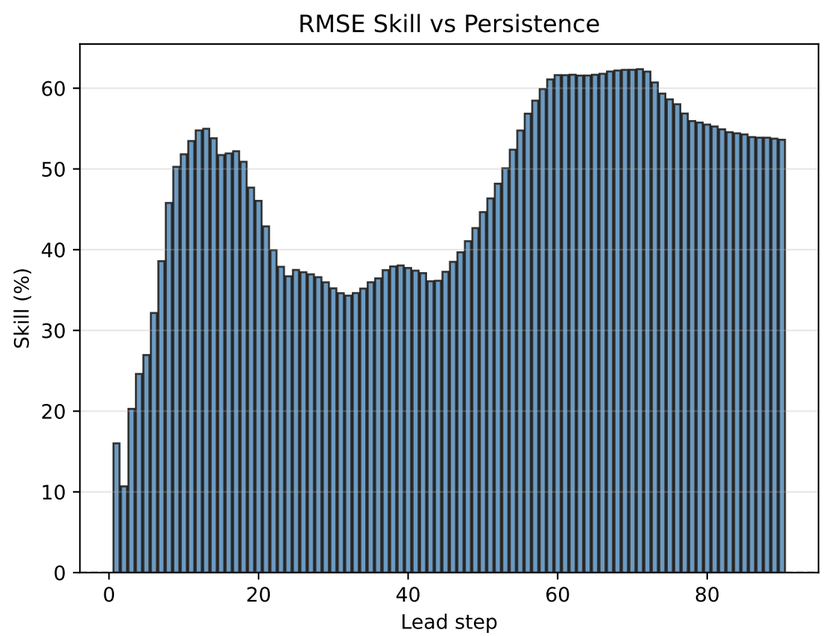


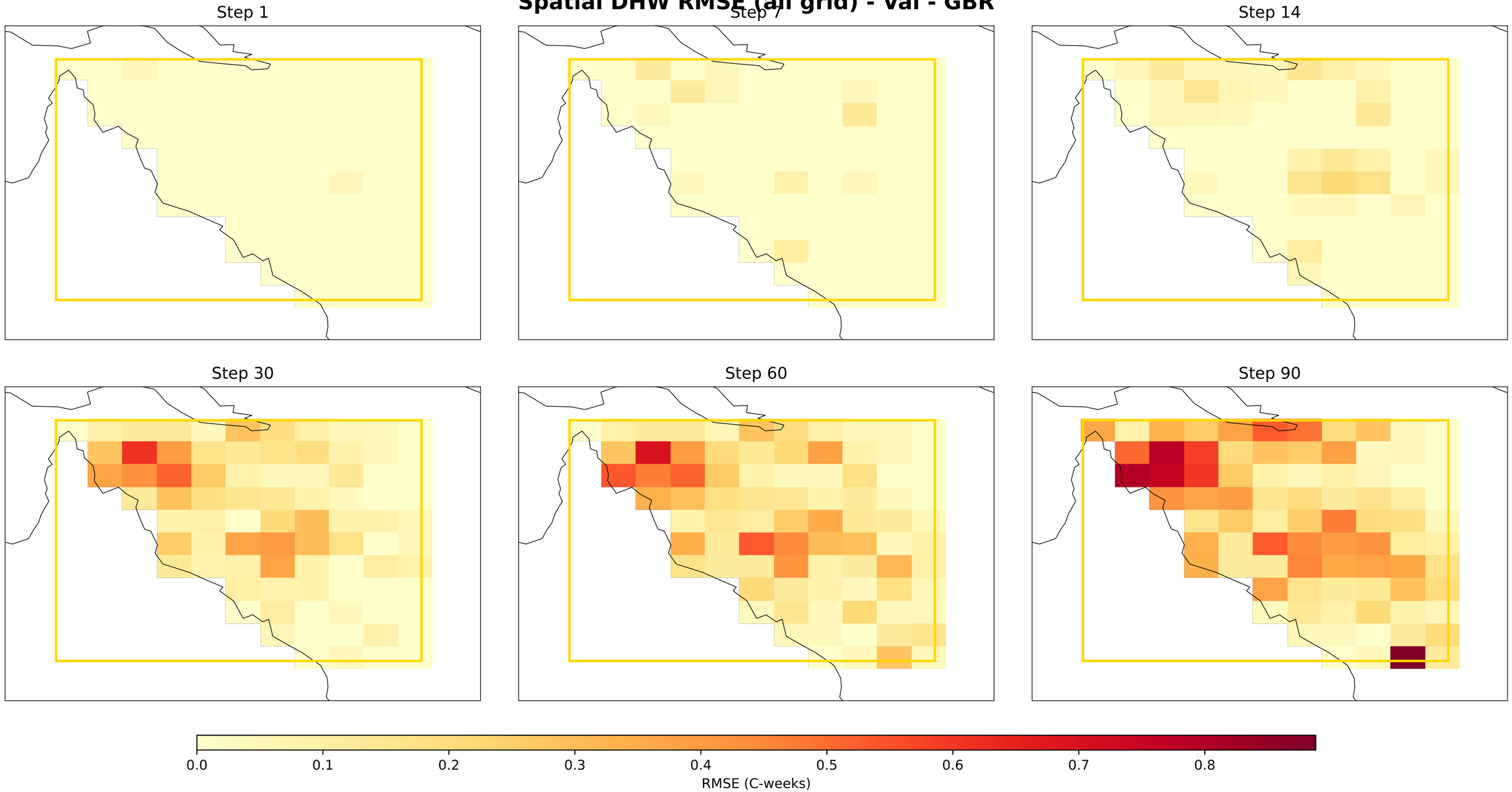


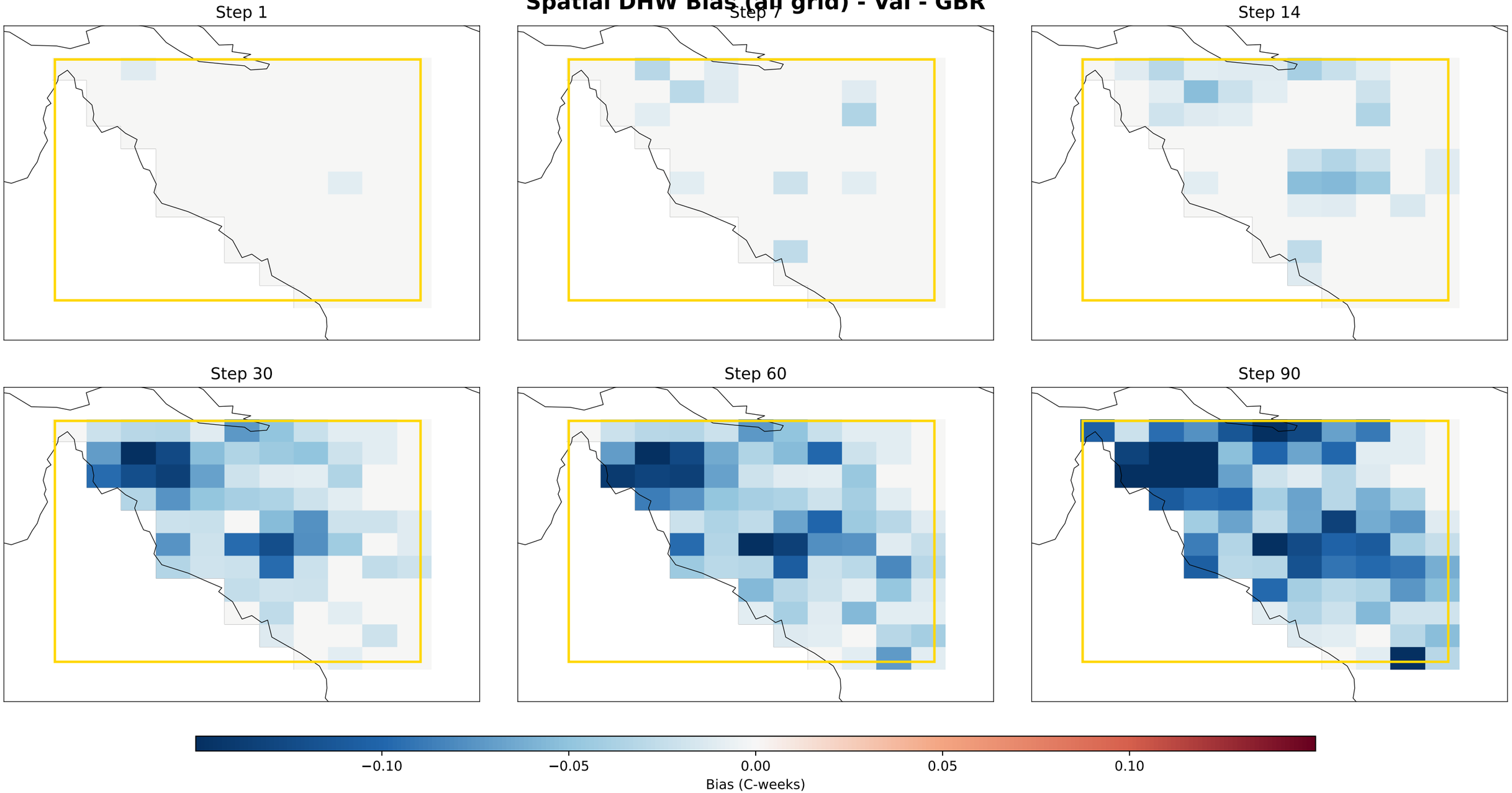


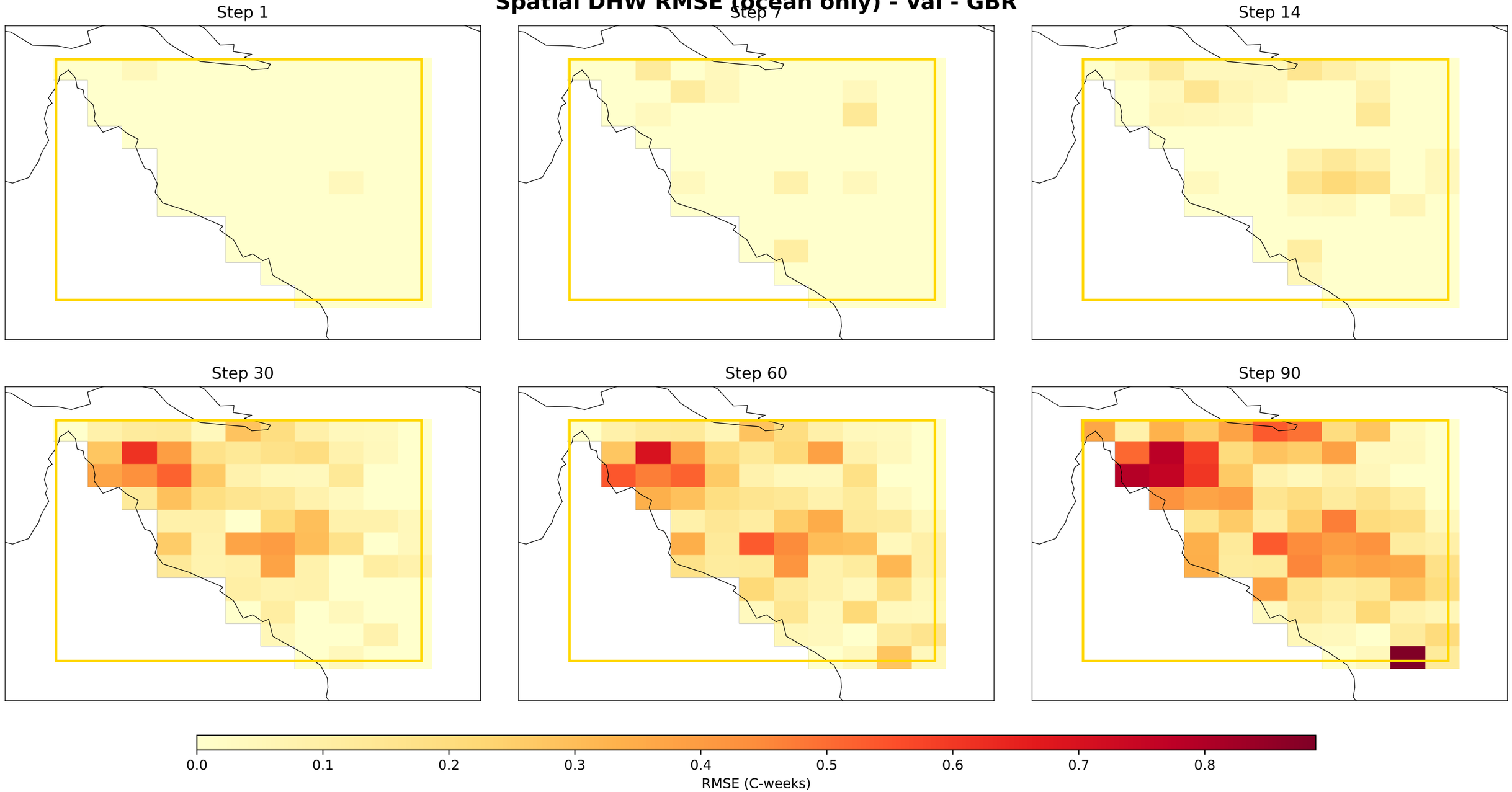


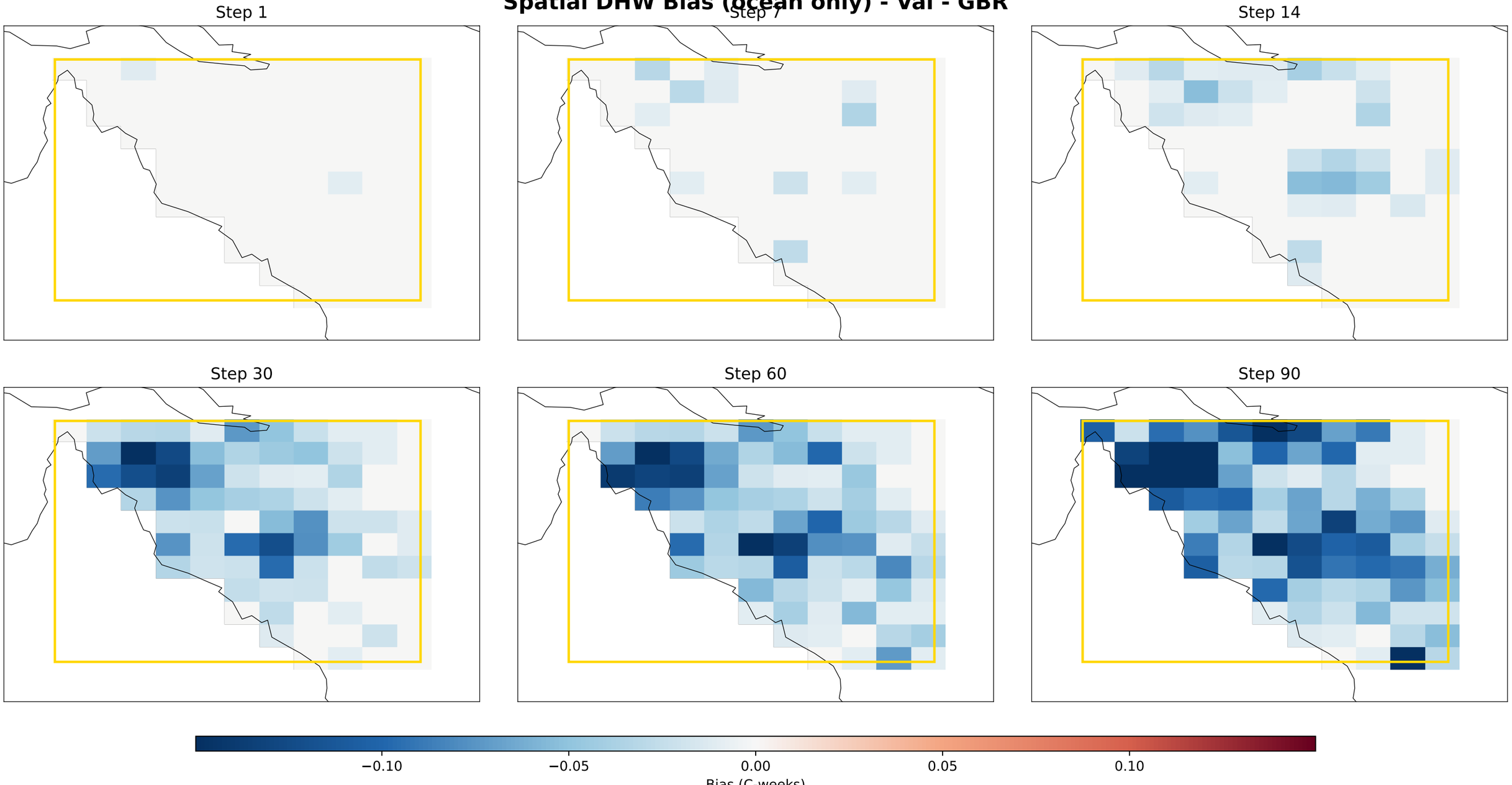





Spatial DHW RMSE (all grid) - Val - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
RMSE (C-weeks)

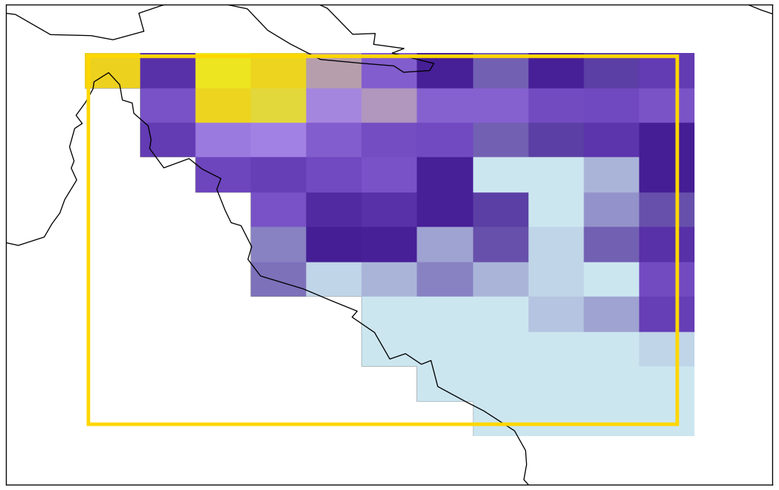

Spatial DHW Bias (all grid) - Val - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−0.10
−0.05
0.00
0.05
0.10
Bias (C-weeks)


Spatial DHW RMSE (ocean only) - Val - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
RMSE (C-weeks)

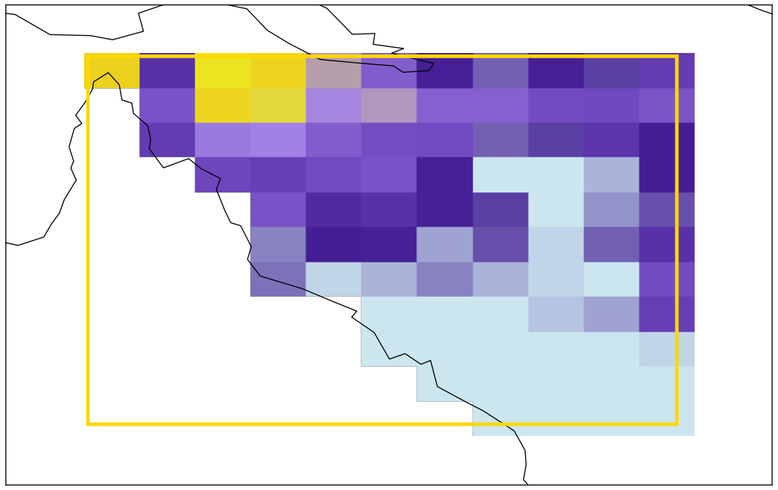

Spatial DHW Bias (ocean only) - Val - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−0.10
−0.05
0.00
0.05
0.10
Bias (C-weeks)

DHW Chain init: 2016-04-10 - Val - GBR (C-weeks)

Actual DHW - Step 1



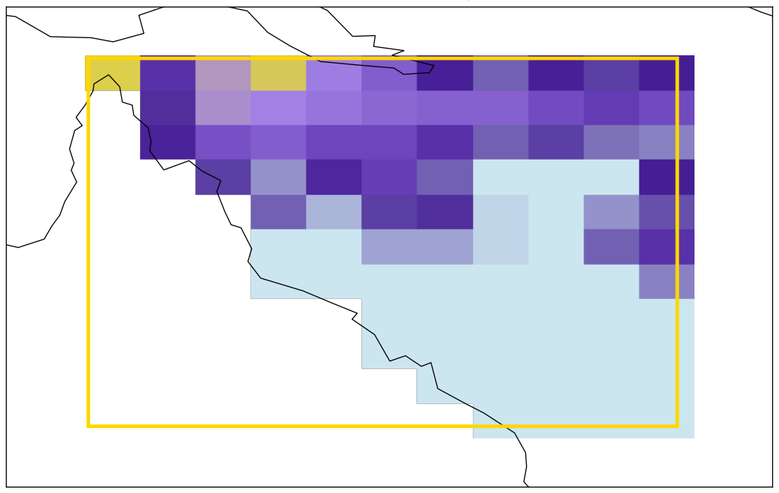
Predicted DHW - Step 1



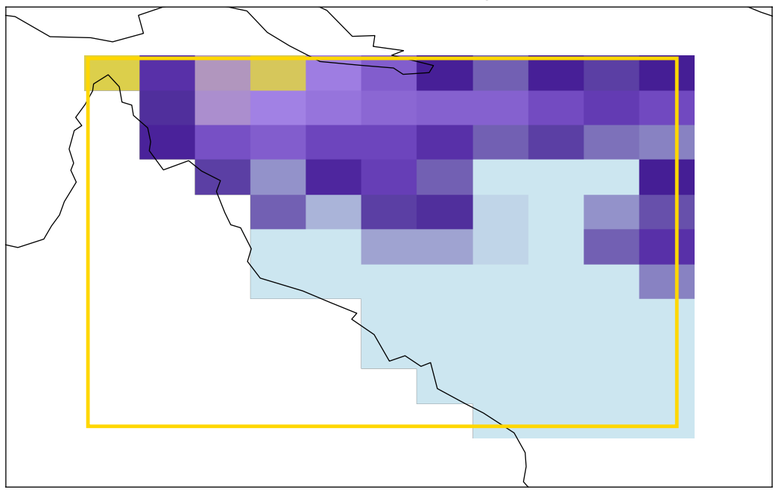
Error (Pred - Actual) - Step 1 [scale ±8]



Actual DHW - Step 46



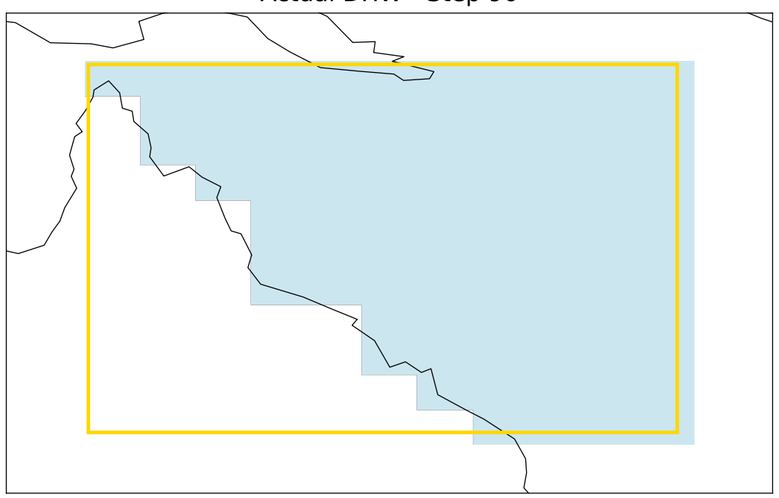
Predicted DHW - Step 46



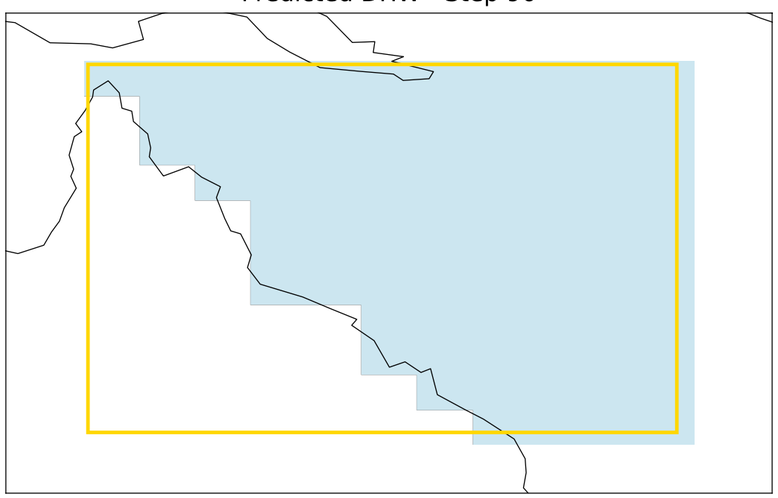
Error (Pred - Actual) - Step 46 [scale ±8]



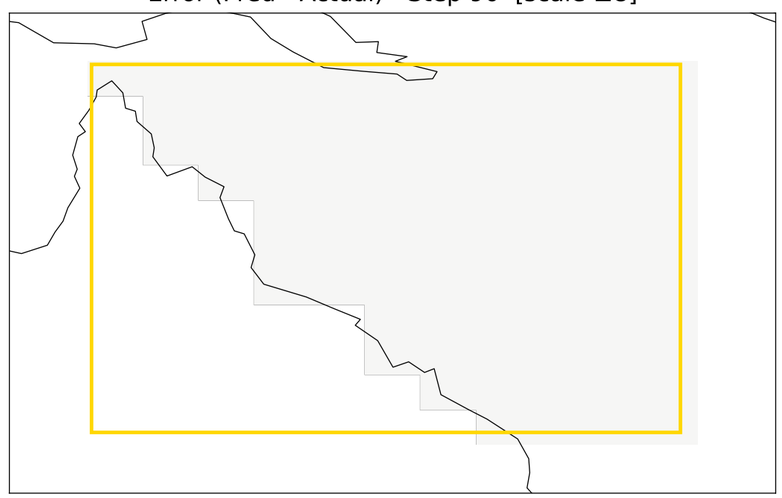
Actual DHW - Step 90

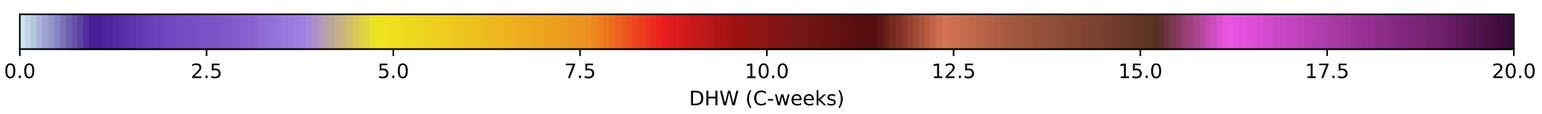


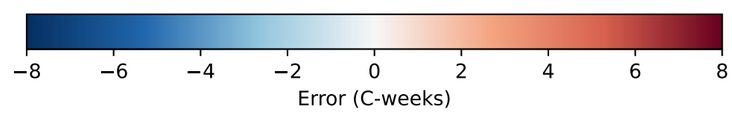
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

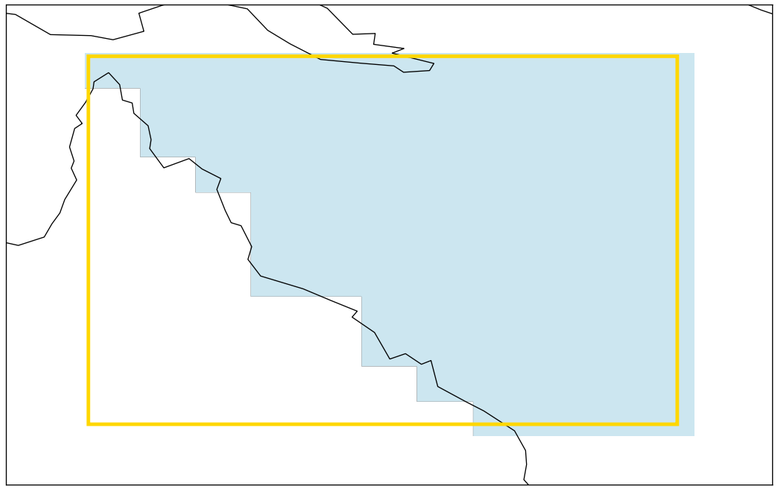




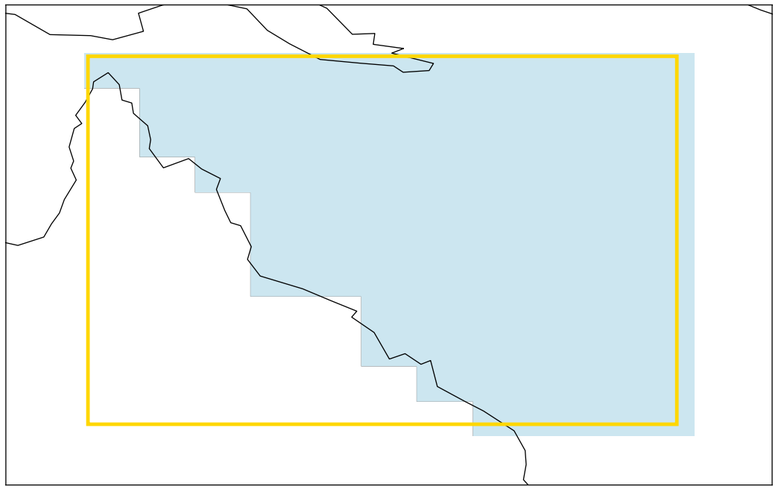


DHW Chain init: 2016-10-07 - Val - GBR (C-weeks)

Actual DHW - Step 1



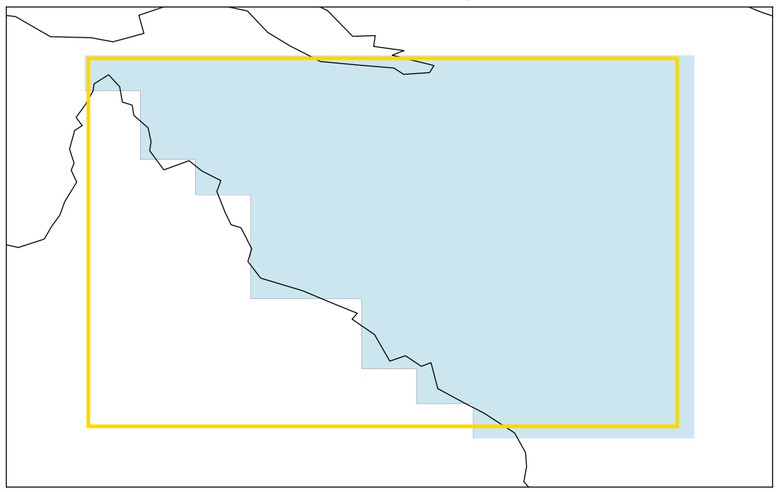
Predicted DHW - Step 1



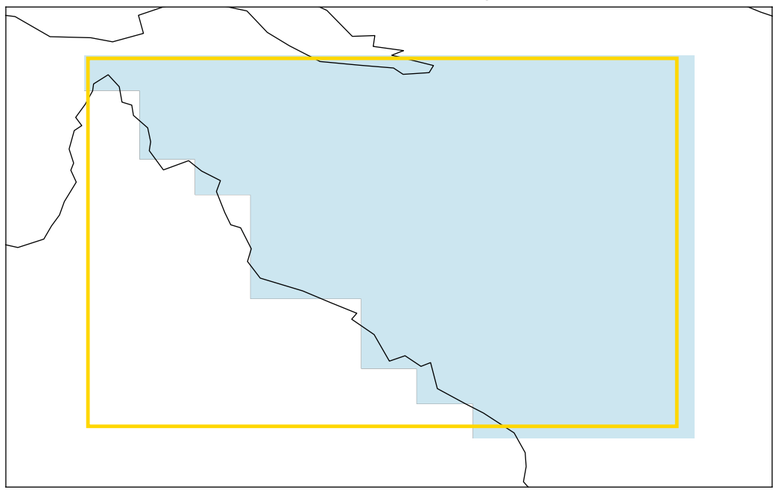
Error (Pred - Actual) - Step 1 [scale ±8]



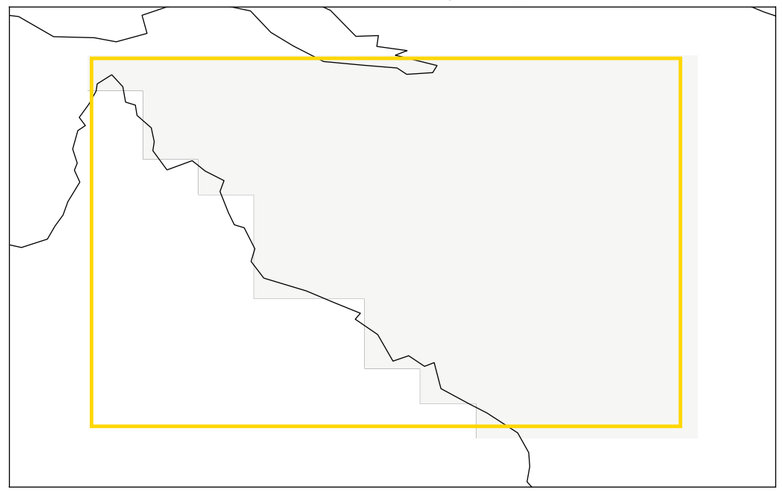
Actual DHW - Step 46



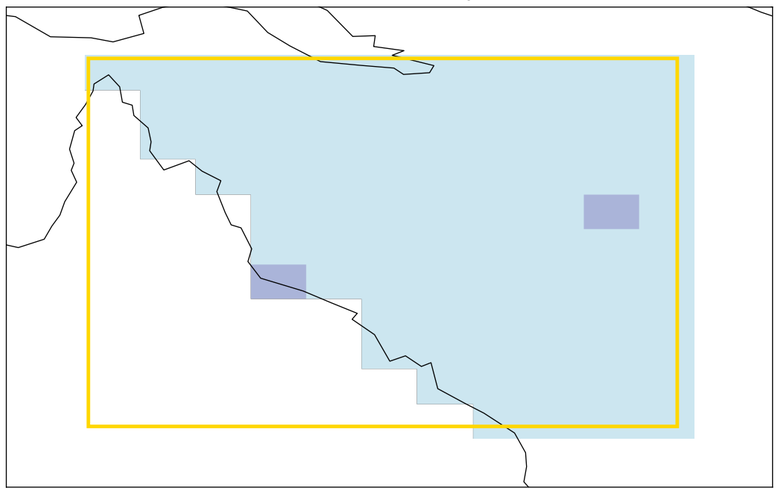
Predicted DHW - Step 46



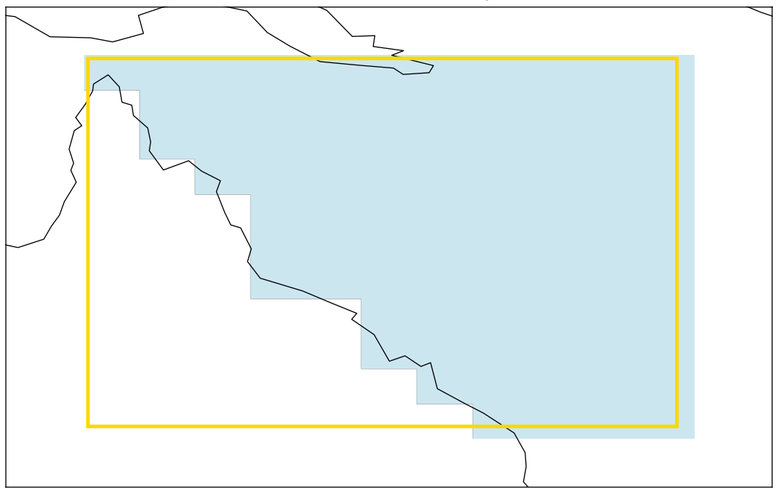
Error (Pred - Actual) - Step 46 [scale ±8]



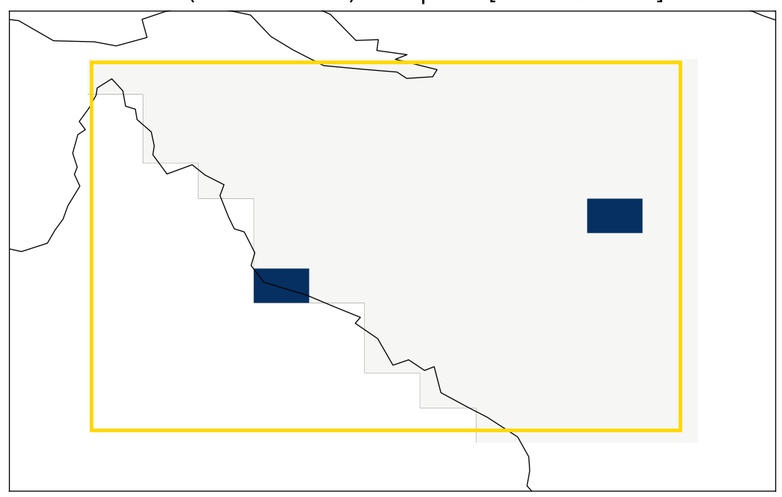
Actual DHW - Step 90

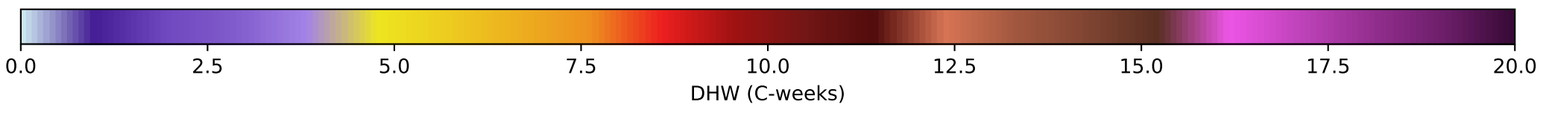


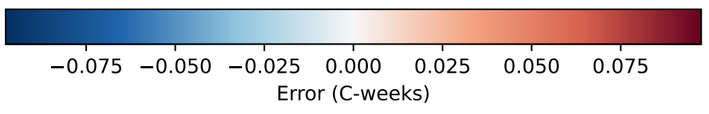
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±0.098]

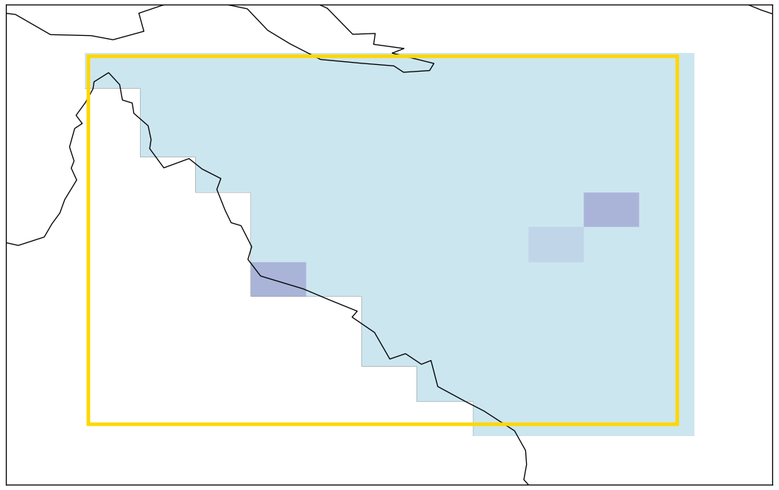




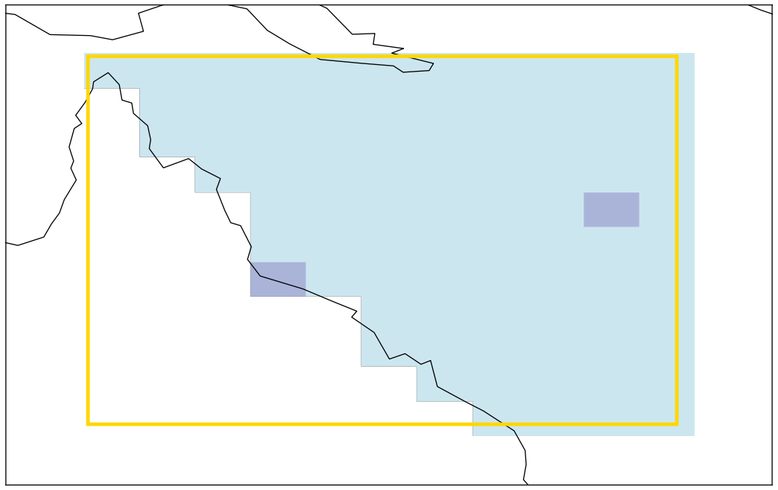


DHW Chain init: 2017-01-05 - Val - GBR (C-weeks)

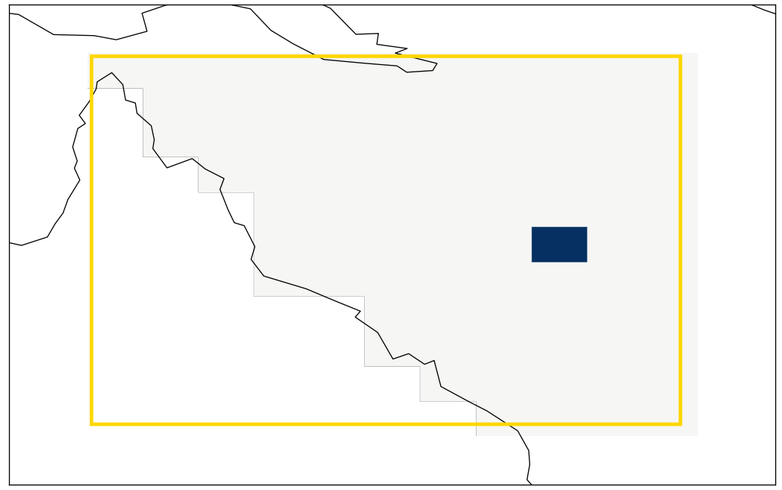
Actual DHW - Step 1



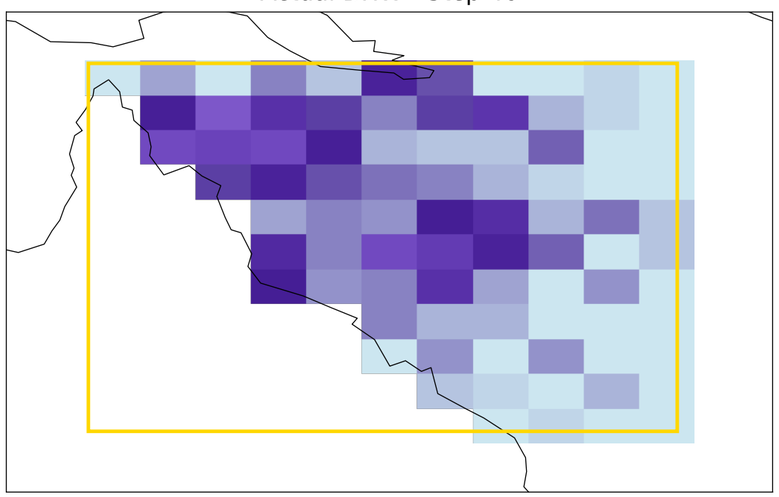
Predicted DHW - Step 1



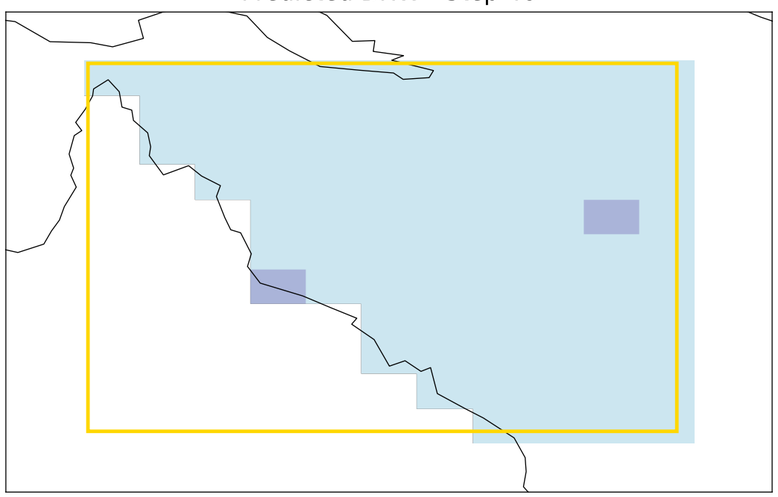
Error (Pred - Actual) - Step 1 [scale ±0.15]



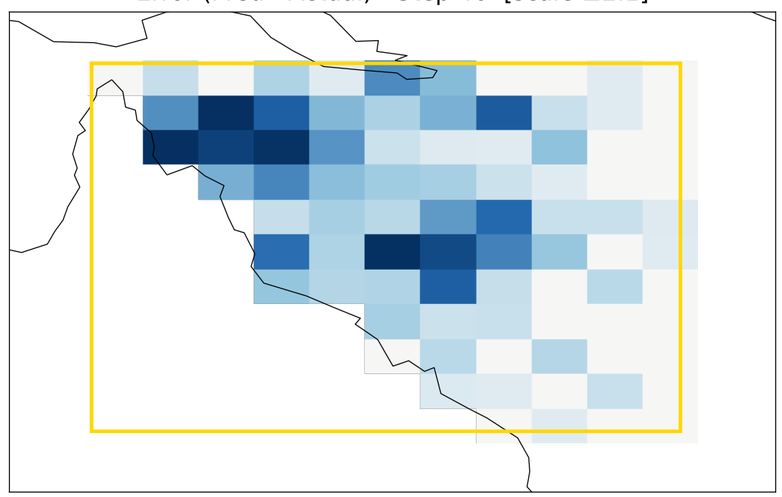
Actual DHW - Step 46



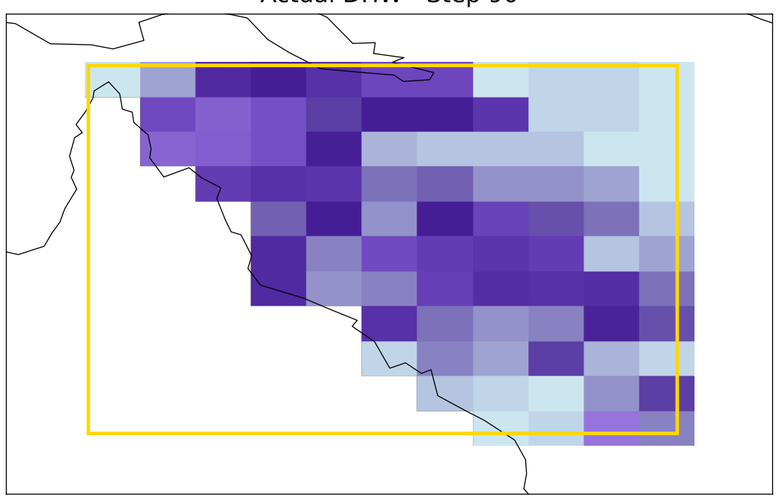
Predicted DHW - Step 46



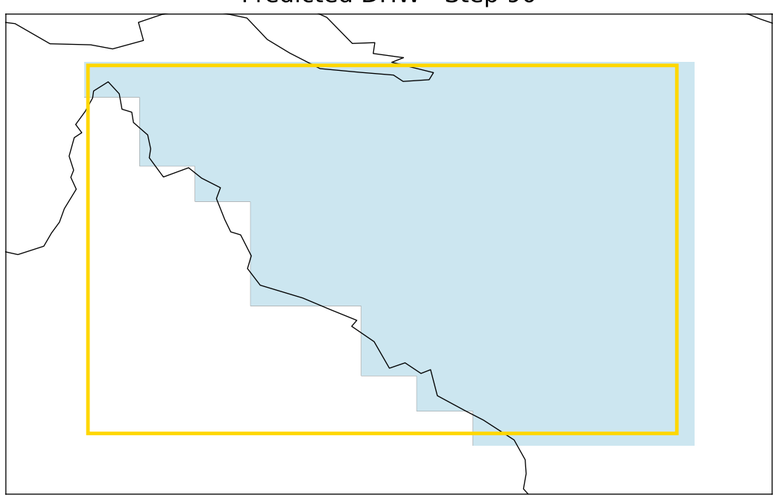
Error (Pred - Actual) - Step 46 [scale ±2.1]



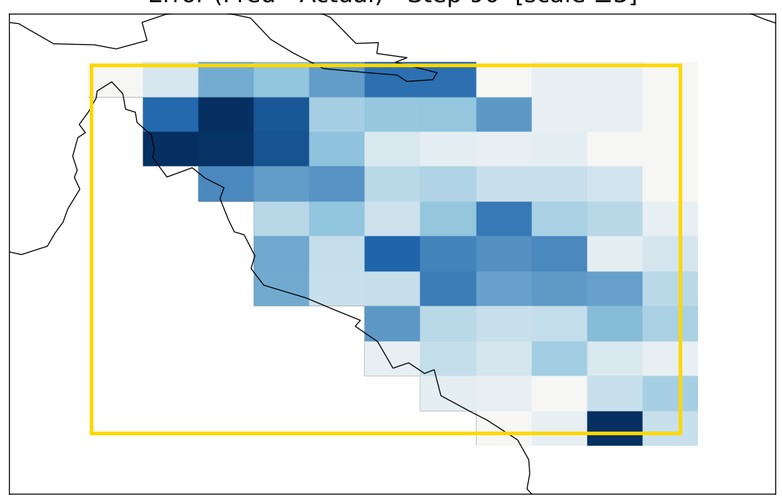
Actual DHW - Step 90

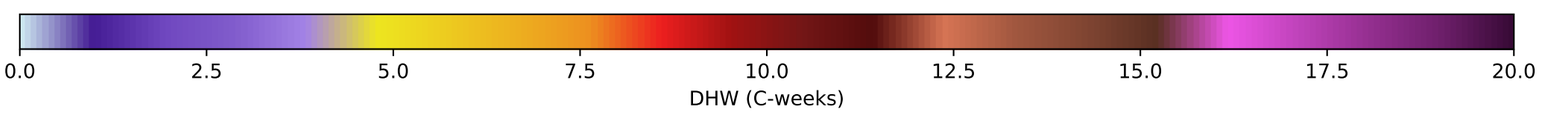


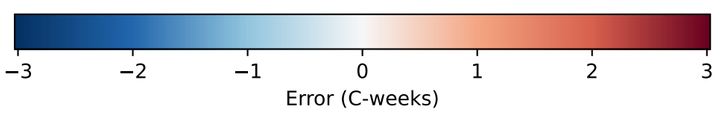
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±3]

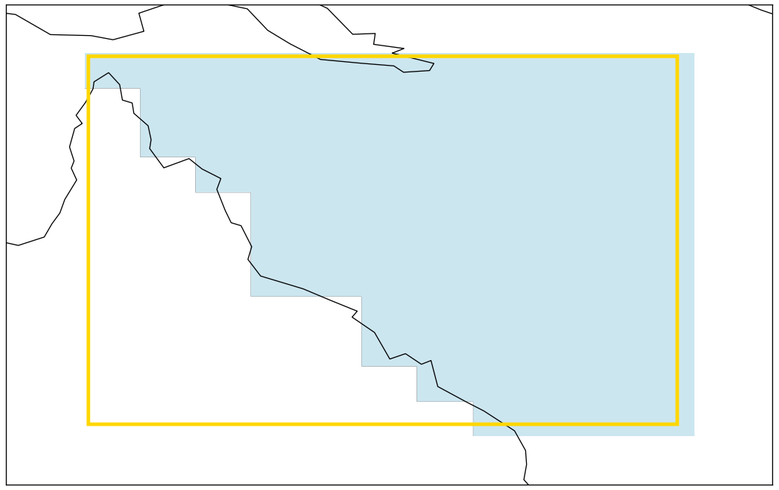




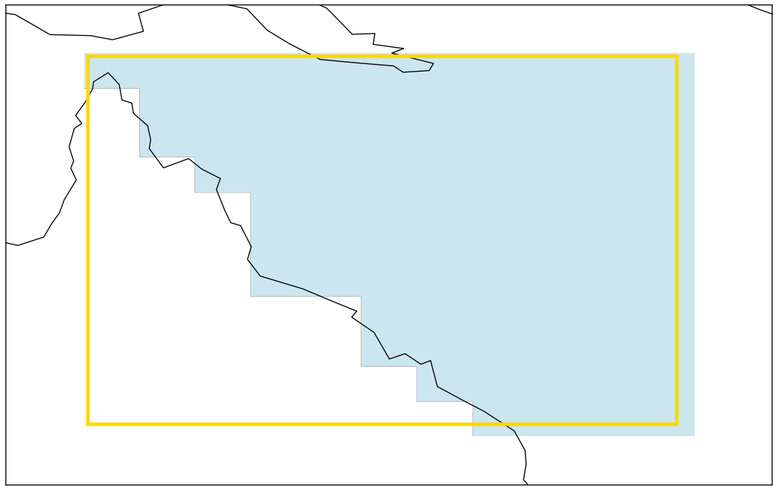


DHW Chain init: 2017-07-04 - Val - GBR (C-weeks)

Actual DHW - Step 1



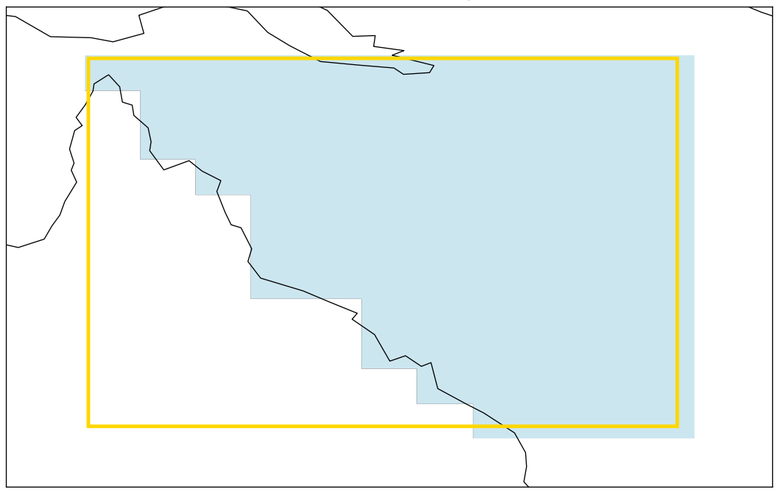
Predicted DHW - Step 1



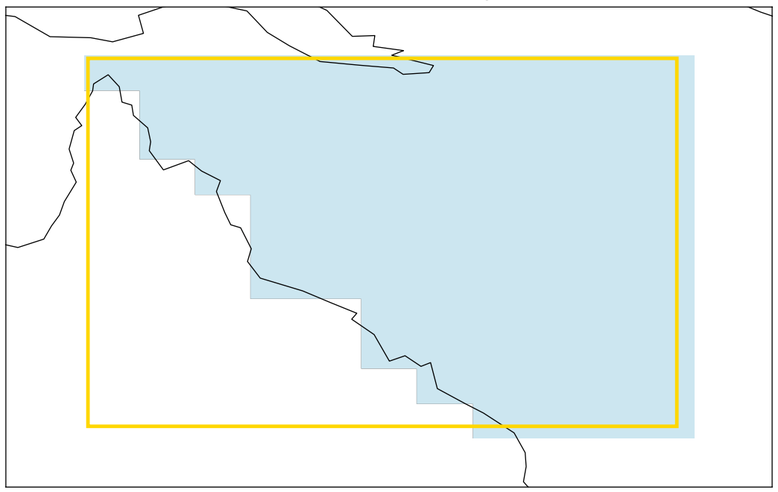
Error (Pred - Actual) - Step 1 [scale ±8]



Actual DHW - Step 46



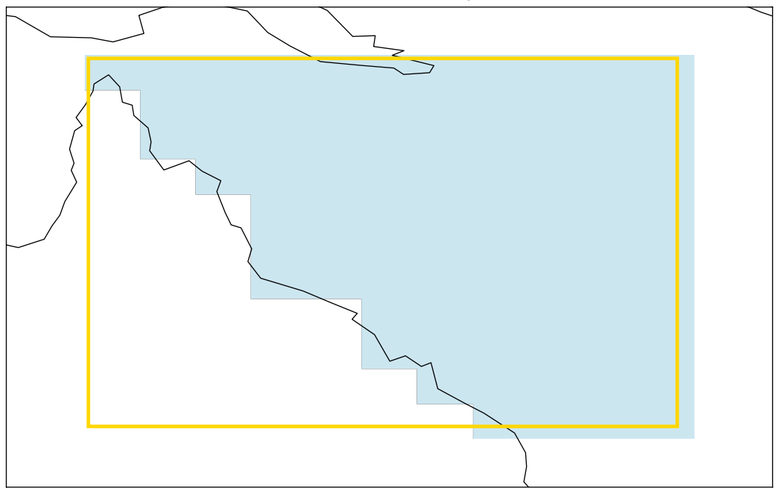
Predicted DHW - Step 46



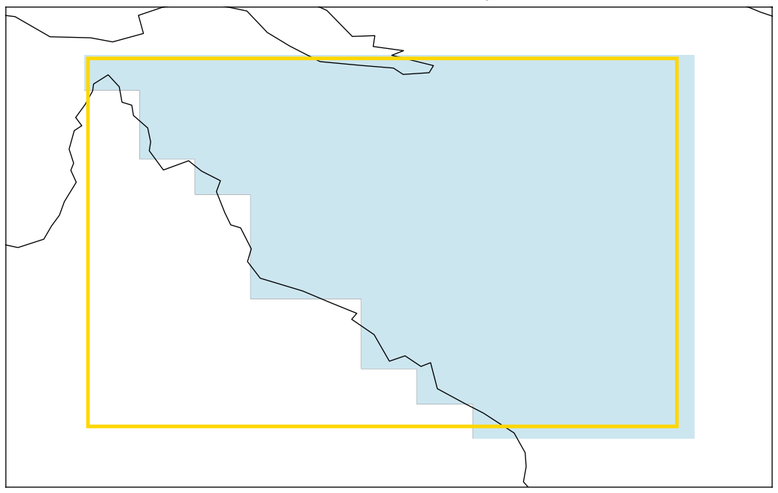
Error (Pred - Actual) - Step 46 [scale ±8]



Actual DHW - Step 90

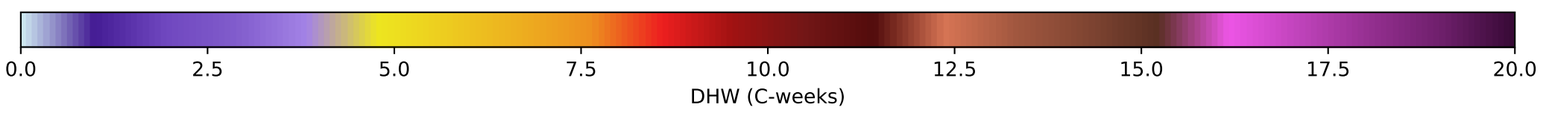


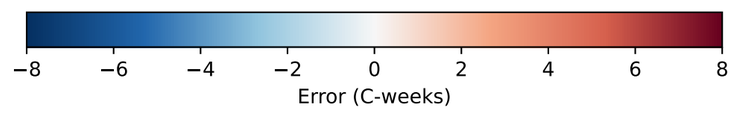
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

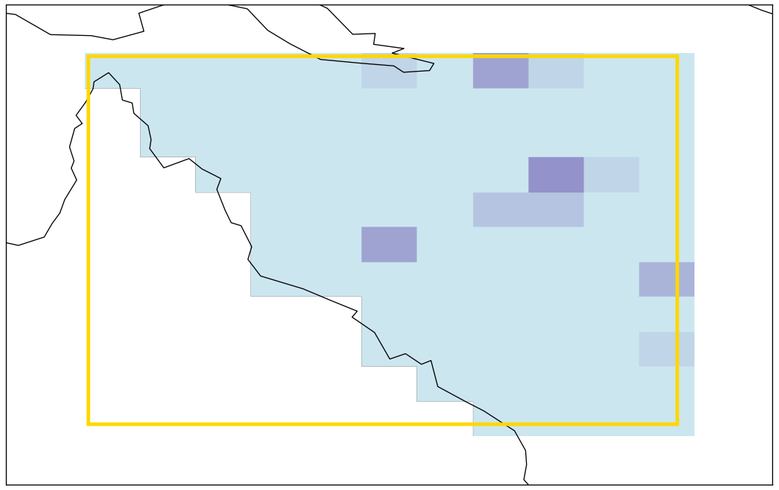




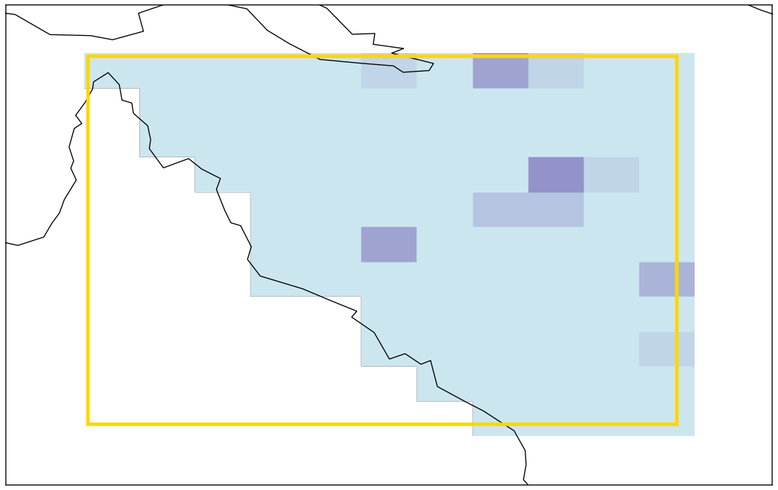


DHW Chain init: 2018-03-31 - Val - GBR (C-weeks)

Actual DHW - Step 1



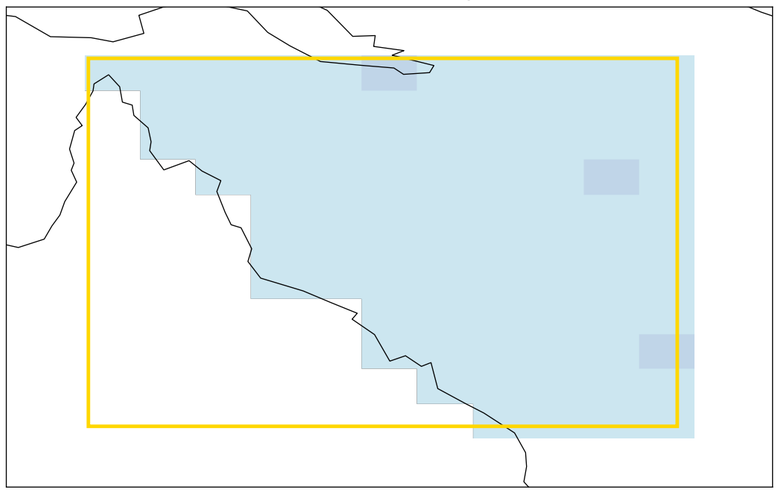
Predicted DHW - Step 1



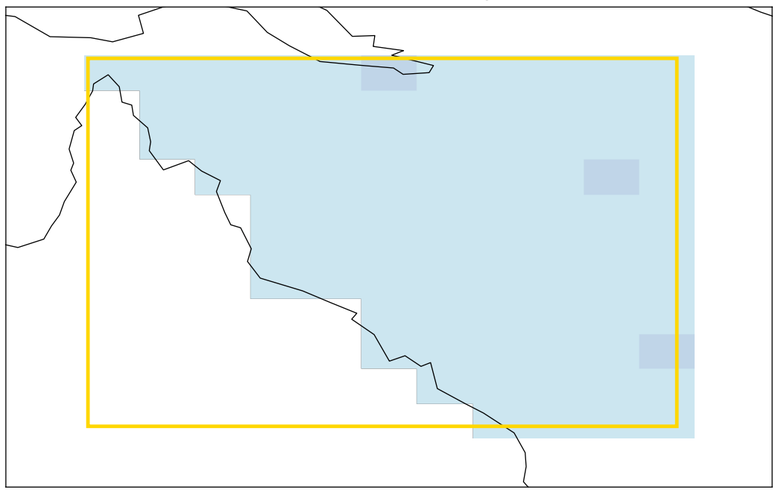
Error (Pred - Actual) - Step 1 [scale ±8]



Actual DHW - Step 46



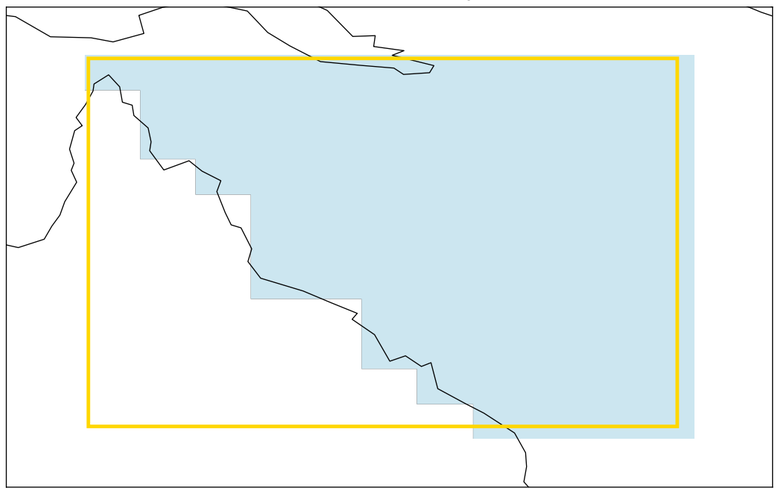
Predicted DHW - Step 46



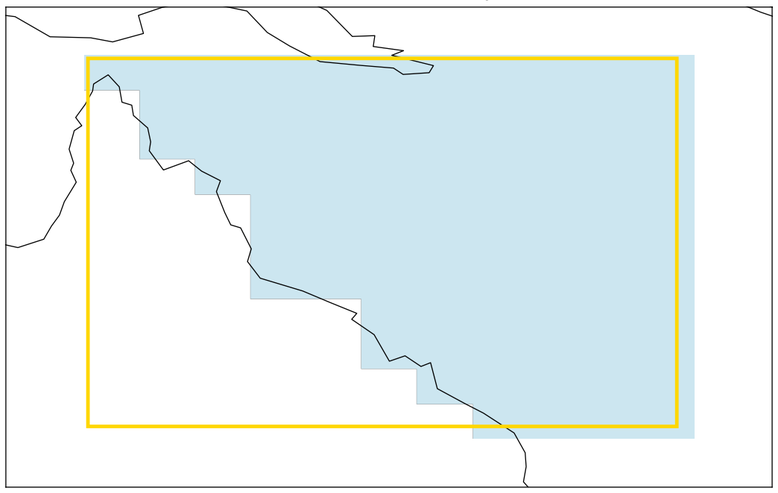
Error (Pred - Actual) - Step 46 [scale ±8]



Actual DHW - Step 90

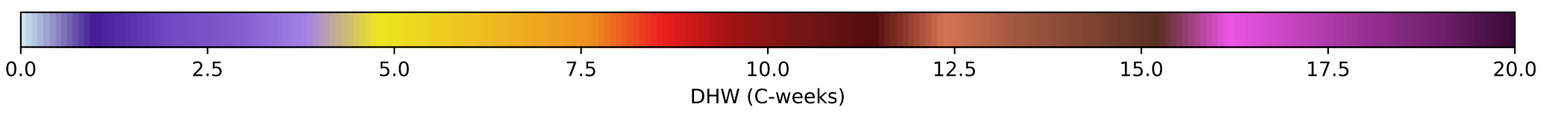


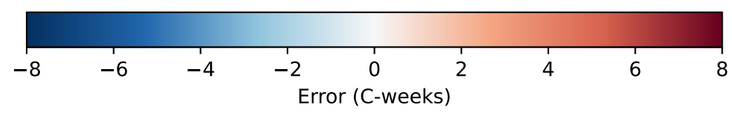
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

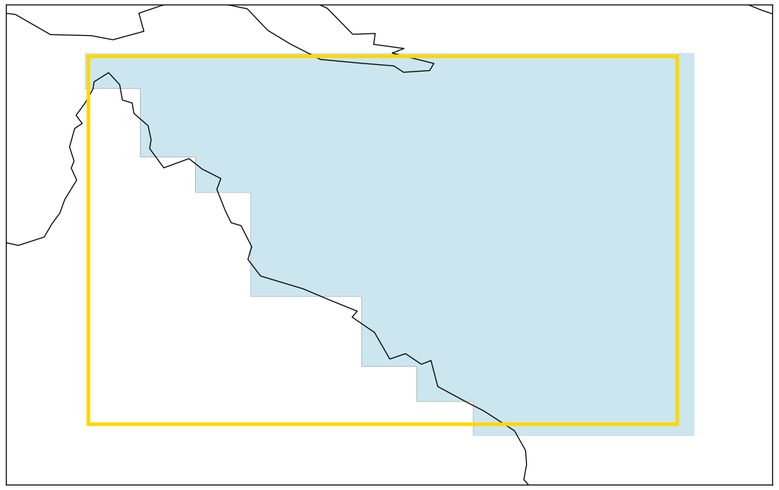




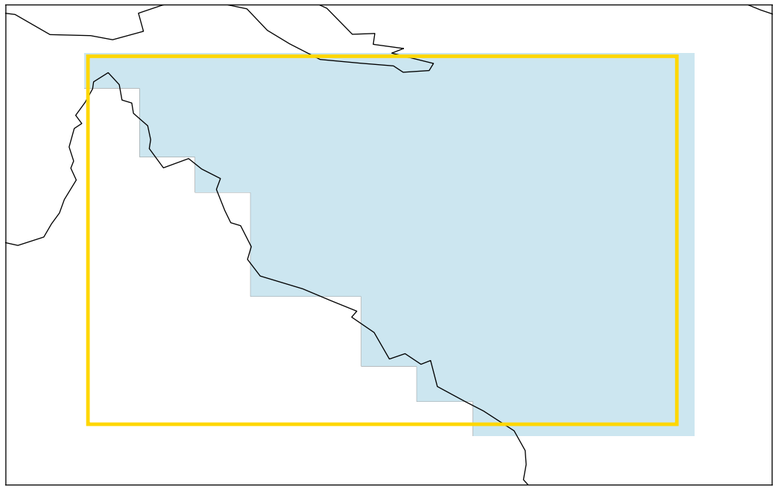


DHW Chain init: 2018-06-29 - Val - GBR (C-weeks)

Actual DHW - Step 1



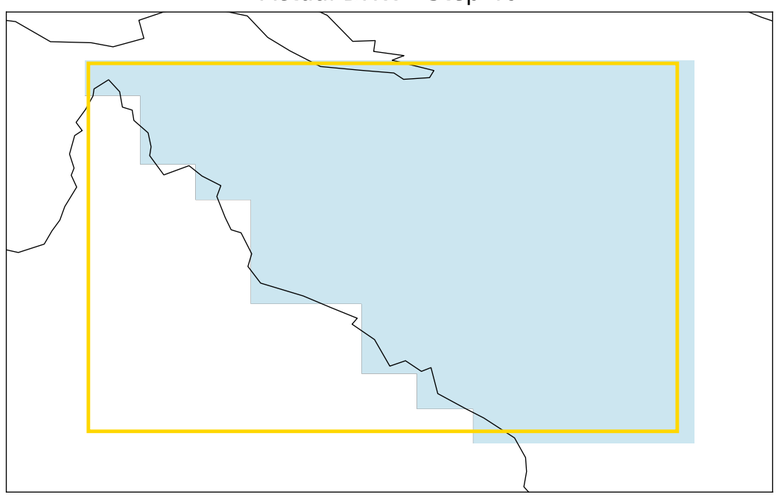
Predicted DHW - Step 1



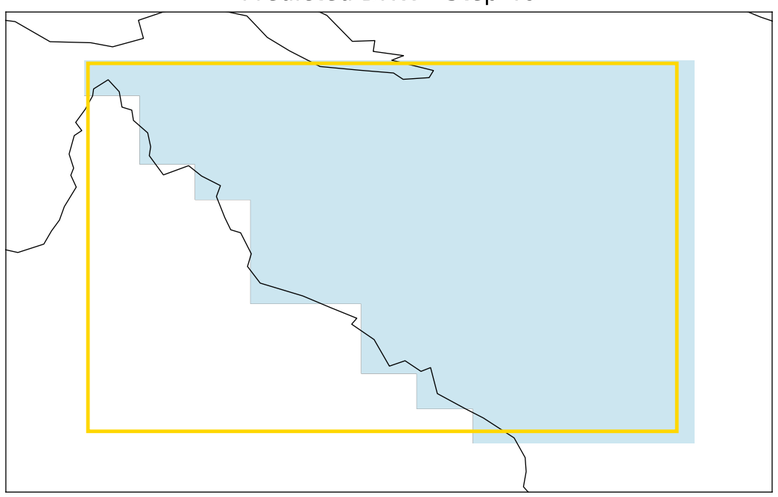
Error (Pred - Actual) - Step 1 [scale ±8]



Actual DHW - Step 46



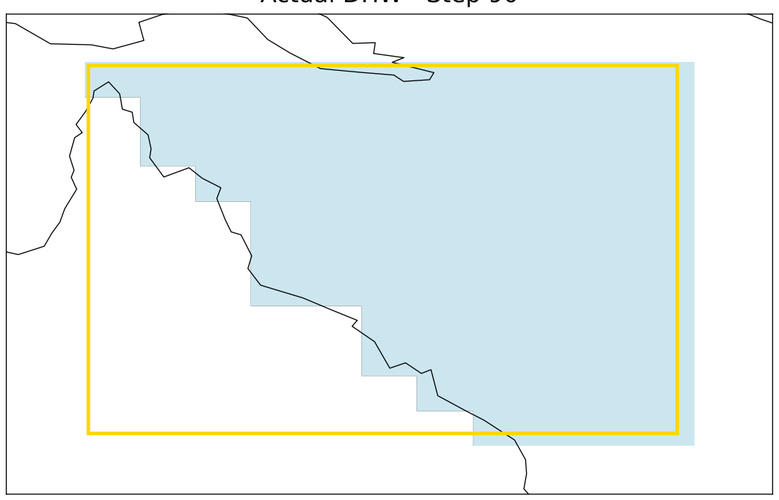
Predicted DHW - Step 46



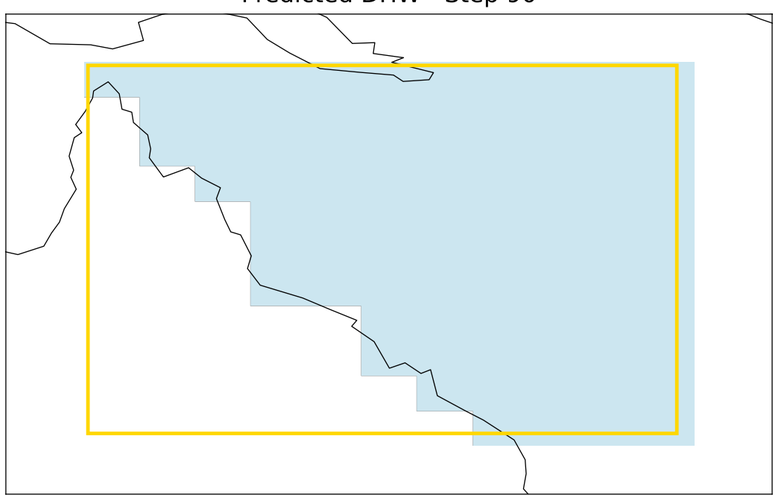
Error (Pred - Actual) - Step 46 [scale ±8]



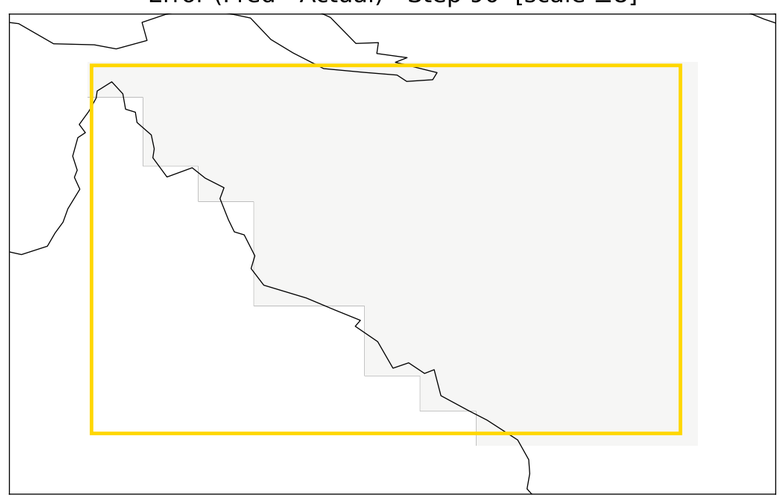
Actual DHW - Step 90

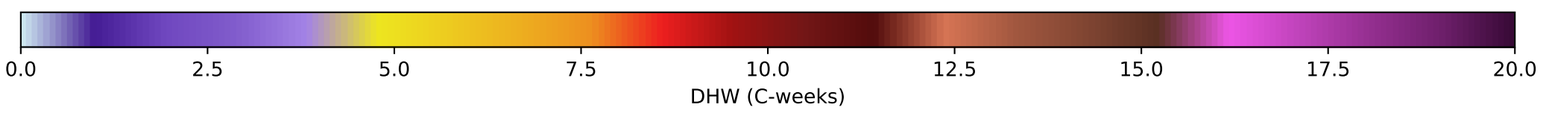


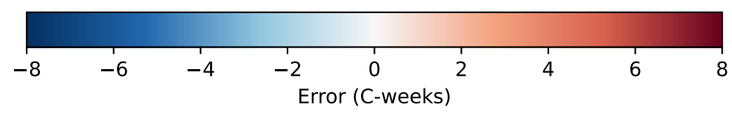
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

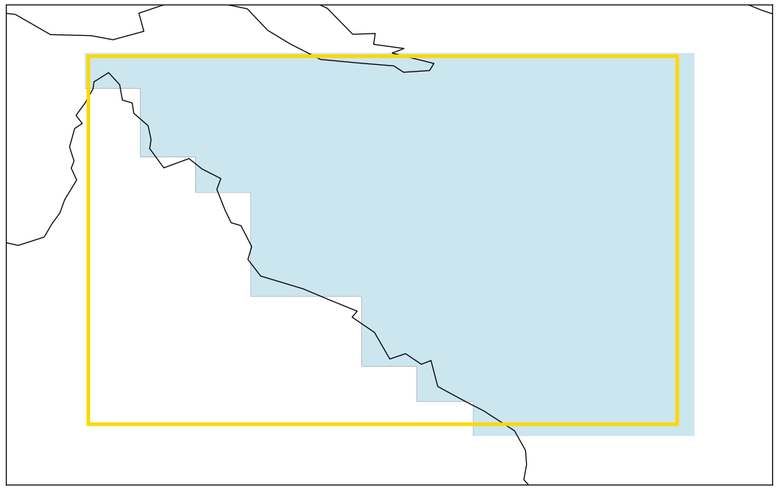




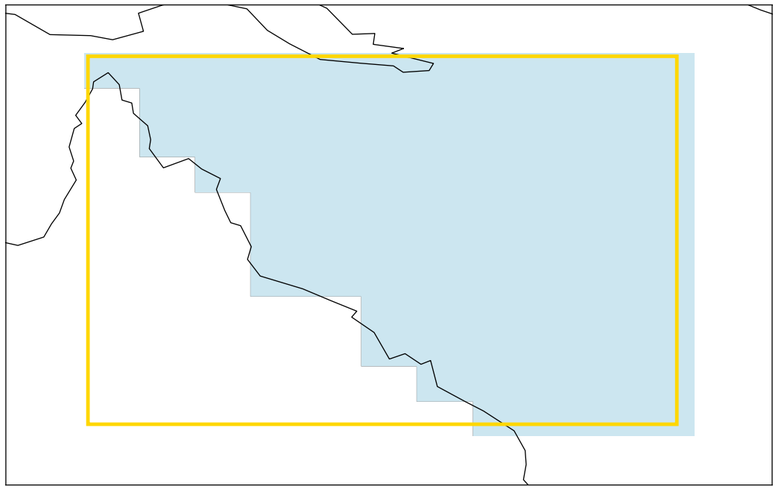


DHW Chain init: 2019-09-22 - Val - GBR (C-weeks)

Actual DHW - Step 1



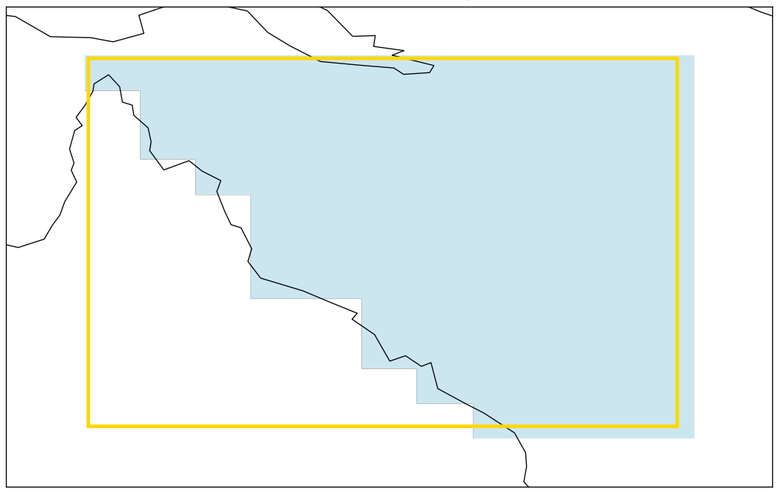
Predicted DHW - Step 1



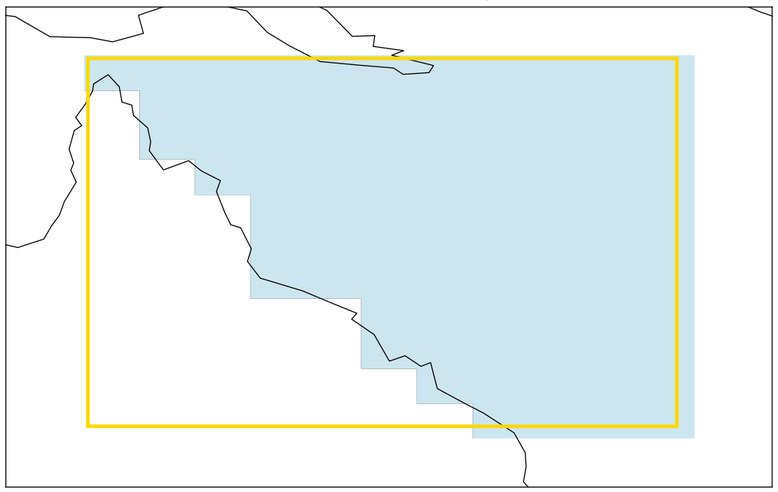
Error (Pred - Actual) - Step 1 [scale ±8]



Actual DHW - Step 46



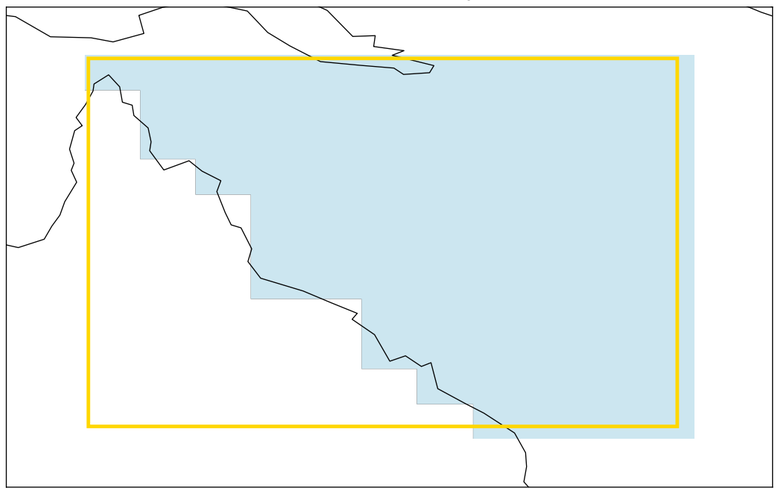
Predicted DHW - Step 46



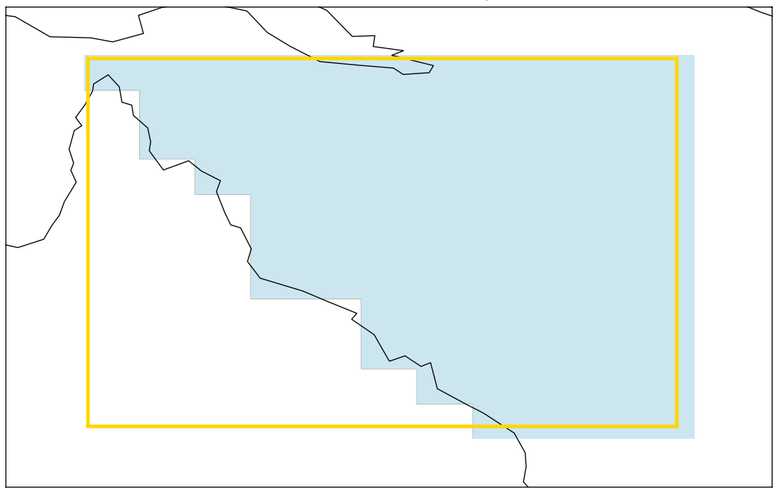
Error (Pred - Actual) - Step 46 [scale ±8]



Actual DHW - Step 90

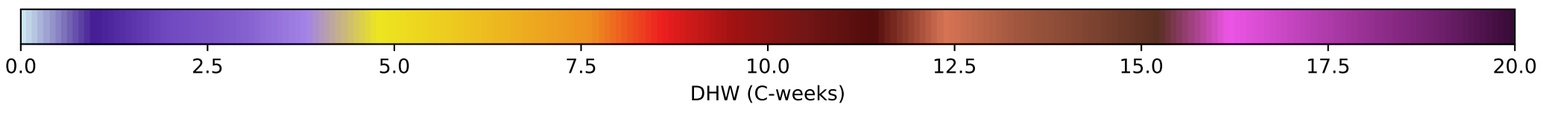


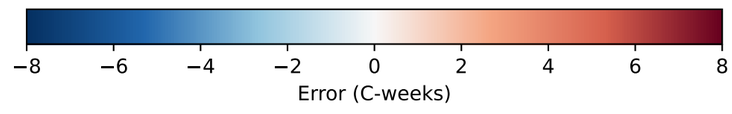
Predicted DHW - Step 90



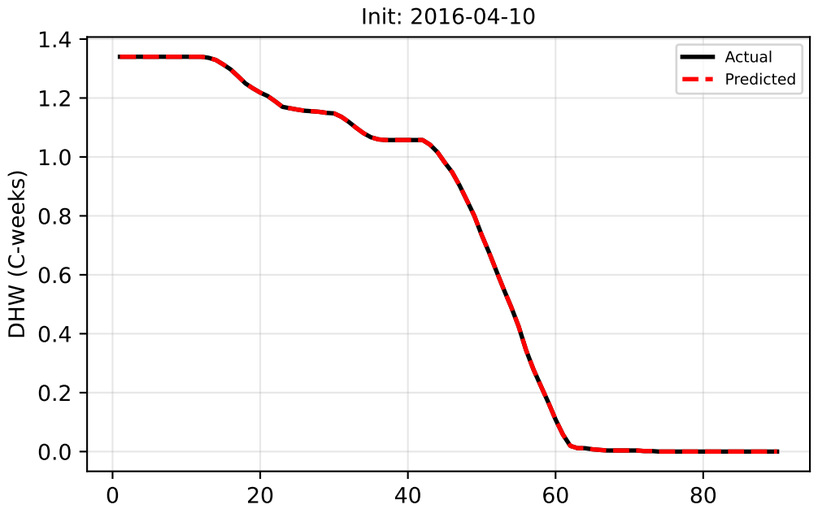
Error (Pred - Actual) - Step 90 [scale ±8]

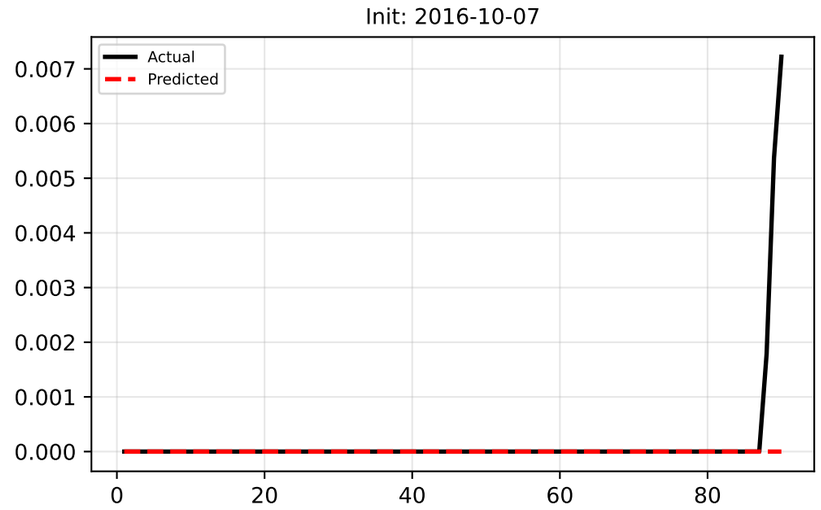


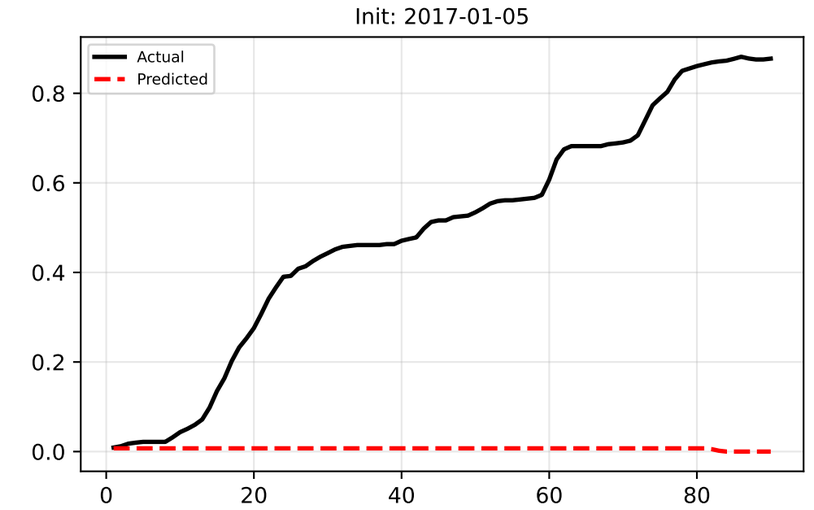


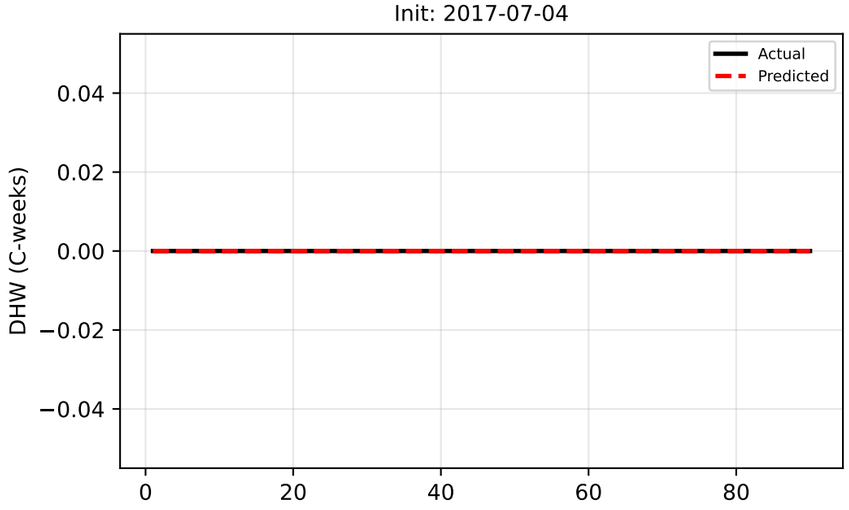


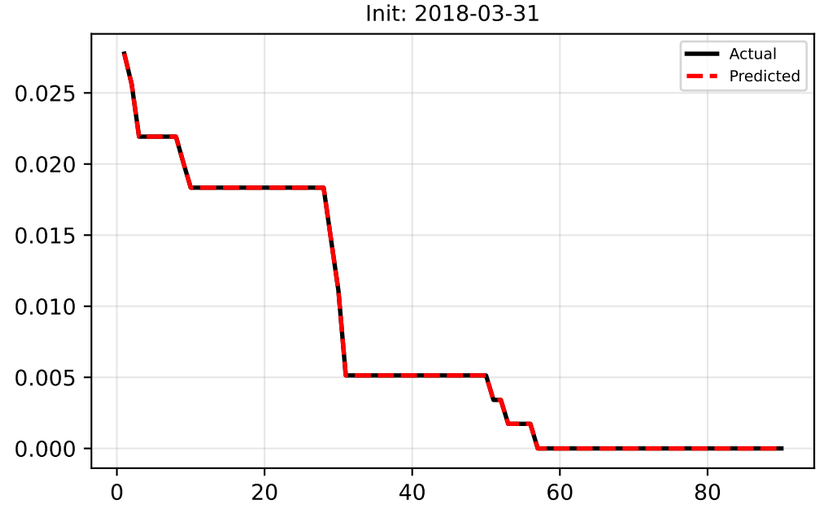
Global-Mean DHW - Val - GBR (C-weeks)

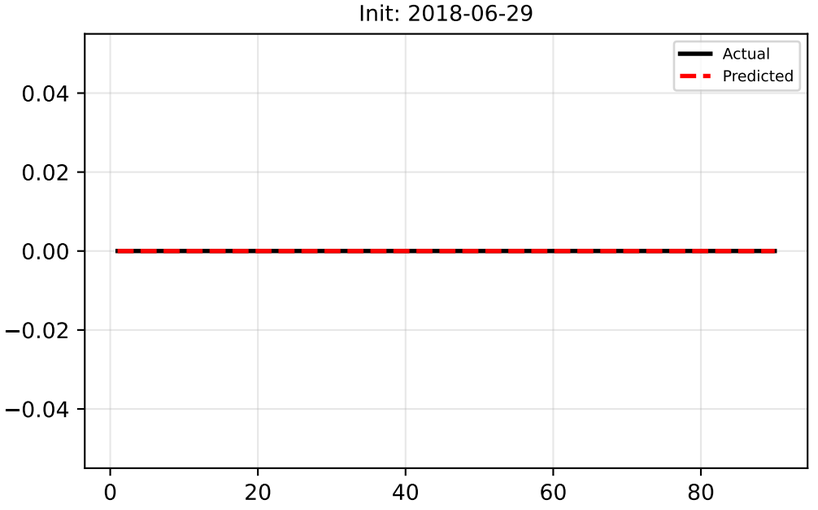


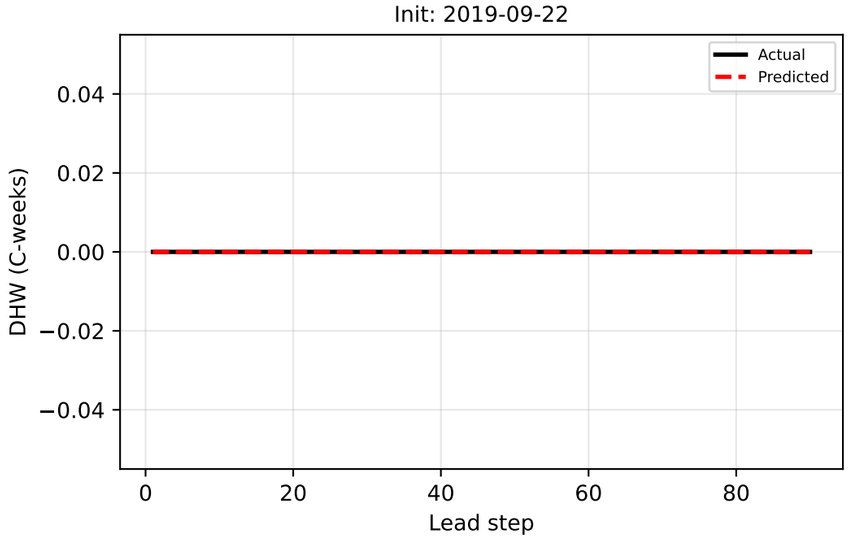


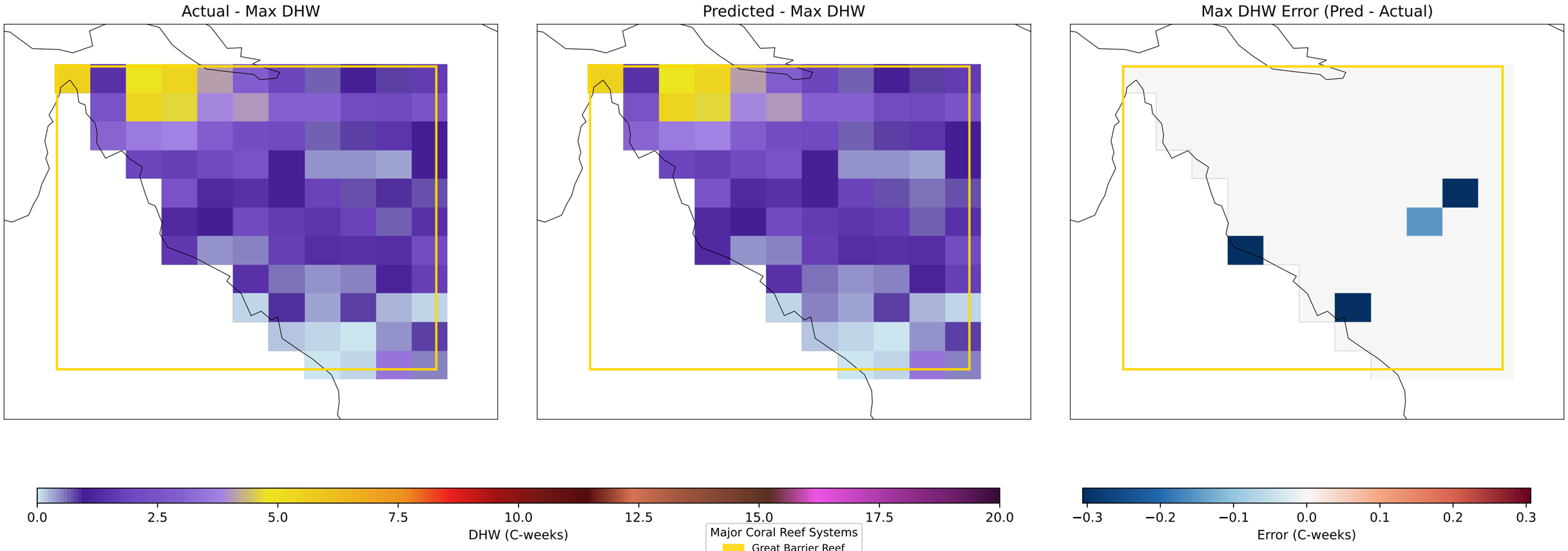


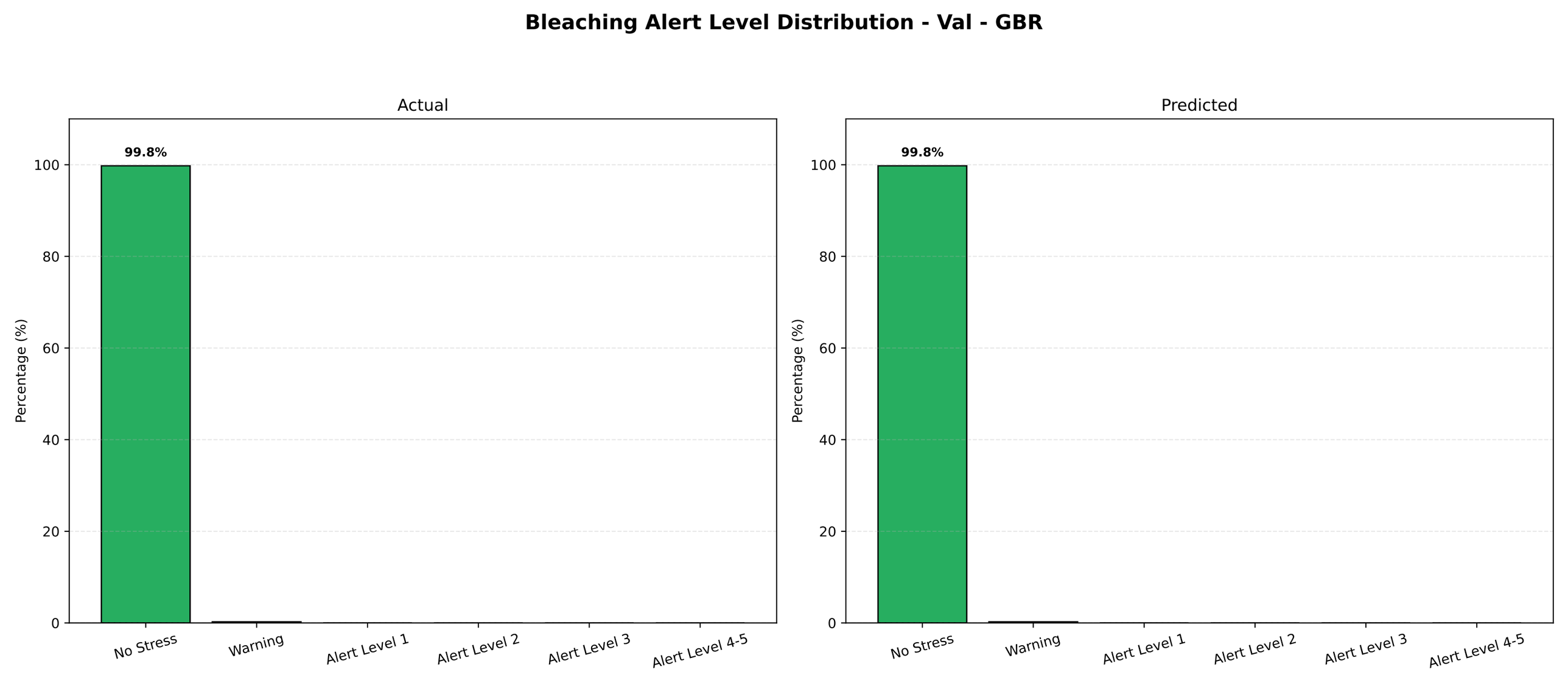






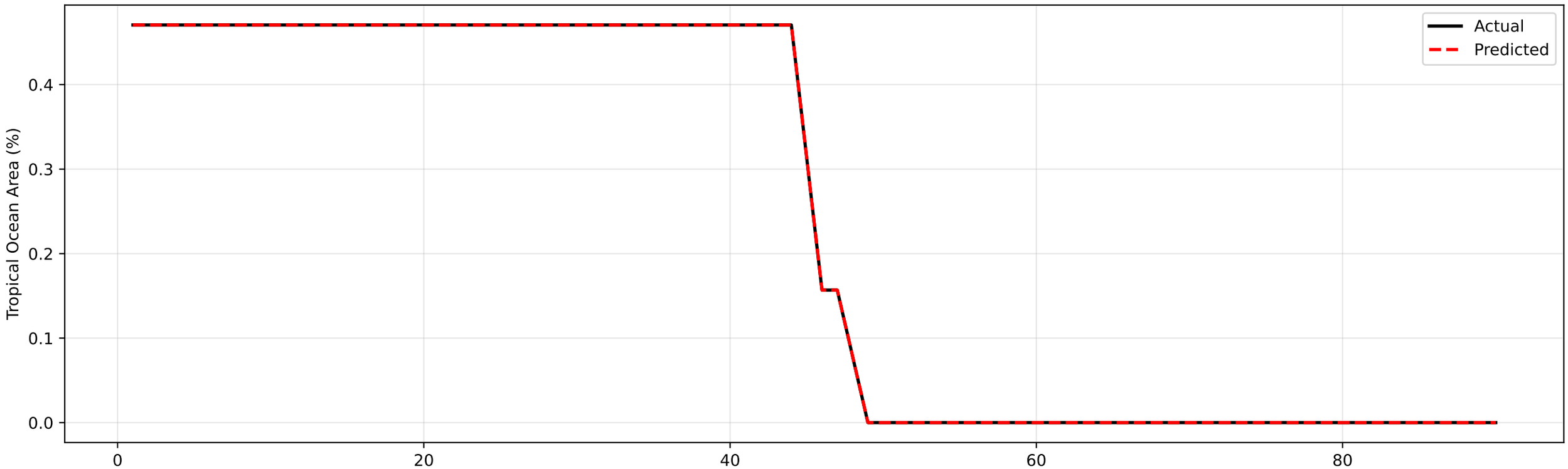



Peak DHW (max over all chains & leads) - Val - GBR (C-weeks)
Actual - Max DHW
Predicted - Max DHW
Max DHW Error (Pred - Actual)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
DHW (C-weeks)
Major Coral Reef Systems
Great Barrier Reef
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3
Error (C-weeks)


Bleaching Alert Level Distribution - Val - GBR
Actual
Predicted
Percentage (%)
0
20
40
60
80
100
99.8%
No Stress
Warning
Alert Level 1
Alert Level 2
Alert Level 3
Alert Level 4-5

Tropical Ocean Area Under Bleaching Stress - Val - GBR

DHW >= 4 C-weeks (Alert Level 1 DHW threshold)



DHW >= 8 C-weeks (Alert Level 2 DHW threshold)

